

“MediGen-AI”

A

Dissertation Work

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CERTIFICATE

This is to certify that the work embodies in this dissertation entitled, “**MediGen-AI**” being submitted by **Ravi Ranjan Kumar** in partial fulfillment of the requirement for the award of “ ” to **CIMAGE Center of Digital Technology & Entrepreneurship** during the academic year is a record of bonafide piece of work, carried out by him under our/my supervision and guidance in the “**Department of Computer Science**”, **CIMAGE Center of Digital Technology & Entrepreneurship**.

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I **Ravi Ranjan Kumar** hereby declare that the work, which is being presented in the dissertation entitled **“MediGen-AI”** partial fulfillment of the requirements for the award of degree of “ ” submitted in the Department of **Computer Science** is an authentic record of my own work carried under the guidance of. I have not submitted the matter embodied in this report for the award of any other degree.

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“A journey is easier when you travel together. Interdependence is certainly more valuable than independence.”

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Project Title:- MediGen-AI

ABSTRACT

MediGen-AI is a secure hybrid healthcare portal designed to modernize hospital operations by integrating Rule-Based Artificial Intelligence with smart prescription management and E-Pharmacy services. The system aims to address major healthcare challenges such as overcrowded OPDs, long patient waiting times, handwritten prescription errors, and difficulty in obtaining authentic medicines.

The project provides a comprehensive digital healthcare ecosystem consisting of three major modules: the OPD Management Module, Doctor Module, and Patient Dashboard. The OPD module enables intelligent queue management and appointment handling, while the Doctor Module assists physicians with rule-based digital prescription generation. The Patient Dashboard allows users to securely access medical records, download prescriptions in PDF format, and order medicines through integrated licensed E-Pharmacy services.

Unlike traditional black-box AI systems, MediGen-AI uses transparent and auditable rule-based decision support, ensuring that doctors remain in complete control of medical decisions. The Human-in-the-Loop AI approach enhances clinical accuracy while maintaining patient safety and professional accountability.

The system is developed using modern web technologies including Python Flask, SQLite, Bootstrap 5, SQLAlchemy, and secure authentication mechanisms such as Bcrypt encryption. The platform follows a three-tier client-server architecture to ensure scalability, security, and efficient data management.

MediGen-AI significantly improves healthcare efficiency by reducing prescription errors, minimizing OPD waiting times, enabling secure digital record management, and ensuring access to verified medicines through E-Pharmacy integration. The project demonstrates how intelligent healthcare systems can transform patient care delivery while maintaining transparency, reliability, and medical safety standards.

Keywords: Rule-Based AI, Healthcare Portal, Smart Prescription System, E-Pharmacy Integration, Clinical Decision Support System, Digital Healthcare, Human-in-the-Loop AI.

The rapid growth of digital healthcare technologies has created new opportunities to improve the quality, accessibility, and efficiency of medical services. However, many hospitals and healthcare centers still rely on traditional manual processes for patient registration, OPD management, prescription writing, and medicine procurement. These outdated systems often lead to overcrowded outpatient departments, long waiting times, prescription errors caused by illegible handwriting, poor record management, and difficulties in obtaining authentic medicines. To address these critical healthcare challenges, the project “MediGen-AI: A Secure Hybrid Healthcare Portal with Rule-Based Smart Prescription & E-Pharmacy Integration” has been developed.

MediGen-AI is an advanced web-based healthcare management platform that combines Rule-Based Artificial Intelligence, secure digital healthcare services, and E-Pharmacy integration into a unified ecosystem. The project is designed to streamline hospital workflows, improve doctor–patient interaction, and provide a secure and efficient healthcare experience for patients, doctors, and hospital administrators.

TOPIC OF PROJECT



**A Secure Hybrid Healthcare Portal with Rule-Based Smart Prescription
& E-Pharmacy Integration**

SUBMITTED FOR THE PARTIAL FULFILLMENT OF THE
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CHAPTER 1: INTRODUCTION

1.1 Introduction

The healthcare industry is one of the most important sectors of modern society because it directly affects human life, public health, and medical services. In today's world, hospitals and healthcare organizations are facing many challenges such as overcrowded OPDs, long patient waiting times, shortage of medical staff, handwritten prescription errors, inefficient hospital workflows, and poor healthcare record management systems. Many traditional healthcare systems still depend heavily on manual operations including paper-based records, handwritten prescriptions, offline registration systems, and manual queue management. These outdated approaches often create delays in treatment, increase administrative workload, reduce healthcare efficiency, and negatively affect patient experience.

One of the major issues in conventional healthcare systems is inefficient OPD management. Patients are often required to wait for long periods before consultation due to manual appointment handling and poor queue management systems. In many hospitals, staff members manually maintain patient information and medical records, making it difficult to retrieve previous healthcare history quickly. This increases the possibility of data duplication, record loss, and delayed medical treatment.

Handwritten prescriptions are another serious problem in traditional healthcare systems. In many cases, unclear handwriting creates confusion in medicine names and dosage instructions, which may lead to medication errors and incorrect treatment. Patients also face inconvenience while searching for prescribed medicines after consultation, especially during emergencies or in overcrowded urban healthcare environments.

With the rapid advancement of Information Technology, Artificial Intelligence, and digital healthcare infrastructure, modern hospitals are now adopting intelligent healthcare systems to automate hospital operations and improve healthcare accessibility. Artificial Intelligence has become an important technology in healthcare because it can assist in symptom analysis, decision support, patient management, prescription assistance, and workflow optimization. Intelligent healthcare systems

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can significantly reduce operational delays, improve accuracy, and support healthcare professionals during diagnosis and treatment processes.

MediGen-AI is a Secure Hybrid Healthcare Portal designed to modernize healthcare operations using Rule-Based Artificial Intelligence, digital prescription systems, smart queue management, and E-Pharmacy integration. The system combines online and offline healthcare workflows into a single integrated platform capable of handling patient registration, OPD management, doctor consultation, prescription generation, digital healthcare records, and medicine ordering services efficiently.

The term “Hybrid Healthcare Portal” refers to the system’s capability to support both online and offline patient workflows. Online users can register themselves through the website, create accounts, log into the system, and generate OPD slips digitally. Offline patients visiting hospitals physically can be registered directly by medical staff through the hospital dashboard. This hybrid workflow increases flexibility and allows hospitals to manage both digital and walk-in patients efficiently. A major feature of MediGen-AI is the integration of a Rule-Based AI Prescription Assistance System. Unlike black-box AI systems that generate unpredictable outputs, the proposed system follows a transparent and explainable Rule-Based AI architecture. The AI engine uses predefined healthcare rules, symptom matching techniques, and medicine recommendation logic to assist doctors during patient consultation. The system provides intelligent prescription suggestions based on symptoms entered by the doctor.

The project follows the Human-in-the-Loop AI approach where Artificial Intelligence assists doctors rather than replacing them. Doctors remain fully responsible for final medical decisions and can review, modify, approve, or reject AI-generated suggestions before generating prescriptions. This approach ensures transparency, medical safety, accountability, and ethical AI implementation in healthcare environment. The system also includes Digital Prescription Management where prescriptions are generated in PDF format and stored securely within the patient dashboard. Patients can access and download prescription history whenever required. This reduces paperwork and improves long-term healthcare record management.

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Another important feature of the project is E-Pharmacy Integration. After prescription generation, patients can directly access E-Pharmacy services from their dashboard to order medicines online. This reduces delays between diagnosis and medicine procurement while improving access to authentic medicines and healthcare. MediGen-AI is implemented using modern web technologies including Python Flask Framework, HTML5, CSS3, Bootstrap 5, JavaScript, SQLite Database, SQLAlchemy ORM, and secure authentication mechanisms. Flask provides a lightweight and scalable backend architecture while Bootstrap enables responsive user interfaces compatible with mobile devices, tablets, and desktop systems.

The project contains multiple integrated modules such as Patient Registration Module, Doctor Dashboard Module, Medical Staff Dashboard, Rule-Based AI Engine, OPD Queue Management System, Digital Prescription Generator, Patient Dashboard, E-Pharmacy Integration Module, and Admin Management System. These modules work together to create a secure, scalable, and efficient healthcare .

The importance of digital healthcare systems increased significantly after the COVID-19 pandemic, where hospitals experienced excessive patient loads and healthcare workflow challenges. Digital healthcare platforms became essential for online consultation, digital prescriptions, patient record management, and contactless healthcare services. MediGen-AI addresses these modern healthcare challenges by providing a smart healthcare automation platform capable of improving treatment efficiency and healthcare accessibility.

The project also supports the broader vision of healthcare digitization and smart healthcare infrastructure development. Government healthcare initiatives promoting digital health systems and electronic healthcare records highlight the growing importance of intelligent healthcare technologies. MediGen-AI contributes toward this vision by creating a secure and scalable healthcare platform capable of future expansion into telemedicine, wearable healthcare integration, multilingual support systems, advanced healthcare analytics, and cloud-based healthcare infrastructure. Therefore, MediGen-AI is not only a hospital management application but also a research-oriented intelligent healthcare platform that demonstrates the integration

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of Rule-Based Artificial Intelligence, healthcare workflow automation, secure digital healthcare infrastructure, and E-Pharmacy services into a unified modern healthcare. The healthcare sector is one of the most important parts of society because it directly affects human life and medical treatment. In many hospitals and clinics, healthcare management is still performed manually using paper-based records, handwritten prescriptions, and offline registration systems. These traditional methods create several problems such as long patient waiting times, overcrowded OPDs, prescription errors, poor record management, and inefficient hospital workflows. Patients often face difficulties during registration, consultation, and medicine purchasing processes. MediGen-AI is a Secure Hybrid Healthcare Portal developed to modernize hospital operations through Rule-Based Artificial Intelligence and E-Pharmacy Integration. The system provides both online and offline healthcare services under a single platform. Online patients can register themselves, log into the portal, and generate OPD slips digitally, while offline patients visiting hospitals can be registered by medical staff. This hybrid workflow helps hospitals manage both digital and walk-in patients efficiently.

The project uses a Rule-Based AI system that assists doctors during prescription generation. The AI engine analyzes patient symptoms and provides medicine.

The project is developed using Python Flask Framework, HTML, CSS, Bootstrap, JavaScript, SQLite Database, and secure authentication mechanisms. MediGen-AI aims to reduce patient waiting time, eliminate handwritten prescription errors, improve healthcare accessibility, and create a secure digital healthcare ecosystem for modern hospitals and clinics.

To overcome these challenges, the MediGen-AI Hospital Management System was developed as an intelligent and automated healthcare management solution. The system integrates modern technologies such as Artificial Intelligence (AI), database management systems, web technologies, and secure authentication mechanisms to simplify hospital operations and improve healthcare services. Although certain challenges such as AI dependency, scalability limitations, and infrastructure requirements still exist, the project establishes a strong foundation for future smart healthcare systems and AI-powered hospital management solutions.

1.2 Need for the Project

In the present healthcare environment, hospitals and clinics are facing several operational and management-related challenges due to the increasing number of patients and the growing demand for efficient medical services. Most traditional healthcare systems still rely on manual processes such as paper-based patient records, handwritten prescriptions, offline appointment handling, and manual queue management. These conventional methods consume more time, increase administrative workload, and reduce the efficiency of healthcare services. As a result, patients often experience long waiting times, delayed treatment, prescription errors, and difficulty in accessing healthcare records.

One of the major problems in traditional healthcare systems is overcrowding in Out Patient Departments (OPDs). Patients are required to stand in long queues for registration and consultation because hospitals often use manual appointment and queue management systems. In many cases, medical staff maintain patient details manually in registers or physical files, making healthcare management slow and inefficient. During emergencies, retrieving patient records from paper documents becomes difficult and time-consuming, which may negatively affect treatment quality and patient safety.

Another significant issue in conventional healthcare systems is handwritten prescriptions. Doctors usually write prescriptions manually, and sometimes the handwriting may not be clear or easily understandable. This creates confusion regarding medicine names, dosage instructions, and treatment procedures. Incorrect interpretation of prescriptions can lead to medication errors and improper treatment. Maintaining physical prescription records is also difficult because documents may be lost, damaged, or duplicated over time.

In addition to these challenges, many healthcare systems do not provide centralized digital healthcare record management. Patient history is often scattered across multiple files and departments, making it difficult for doctors to access previous medical records quickly during follow-up consultations. This lack of centralized healthcare management reduces coordination between patients, doctors, medical staff, and pharmacies.

The rapid growth of Information Technology and Artificial Intelligence has created new opportunities for modernizing healthcare systems. Intelligent healthcare platforms can improve patient management, reduce manual workload, automate repetitive tasks, and provide better coordination between different healthcare departments. Digital healthcare systems can also improve transparency, accuracy, and accessibility in medical services.

MediGen-AI is developed to address these healthcare challenges through a Secure Hybrid Healthcare Portal integrated with Rule-Based Artificial Intelligence and E-Pharmacy services. The project provides both online and offline healthcare facilities under a single platform. Online patients can register themselves digitally through the website, generate OPD slips, and access healthcare services remotely. Offline patients visiting hospitals physically can be registered by medical staff through the hospital dashboard. This hybrid healthcare workflow improves operational flexibility and allows hospitals to manage both digital and walk-in patients efficiently.

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The project also introduces Smart Queue Management to reduce overcrowding and improve OPD workflow efficiency. Through automated patient token generation and queue handling, the system minimizes waiting time and improves hospital management. This helps healthcare staff manage patient flow more effectively.

A major need for the project arises from the requirement for transparent and secure Artificial Intelligence in healthcare systems. Many modern AI systems operate as black-box models where decision-making processes are not easily understandable. In healthcare environments, transparency and doctor supervision are extremely important because incorrect medical decisions can directly affect patient safety. Therefore, MediGen-AI uses a Rule-Based Artificial Intelligence system instead of fully automated AI models.

The Rule-Based AI engine analyzes patient symptoms and provides medicine suggestions based on predefined healthcare rules and symptom-matching logic. However, the system follows the Human-in-the-Loop AI approach, where doctors remain fully responsible for final medical decisions. Doctors can review, modify, approve, or reject AI-generated recommendations before generating final prescriptions. This approach improves transparency, accountability, medical safety, and ethical healthcare implementation.

The project also provides Digital Prescription Management where prescriptions are generated in PDF format and stored securely in the patient dashboard. Patients can access prescription history whenever required and download prescriptions digitally. This reduces paperwork and improves long-term healthcare record maintenance.

Another important need for the project is medicine accessibility. After consultation, many patients face difficulties while searching for prescribed medicines from different pharmacies. This process increases treatment delays and creates inconvenience for patients. MediGen-AI solves this problem through integrated E-Pharmacy services where patients can directly order medicines online through the patient dashboard. This improves convenience and ensures faster medicine availability.

The importance of digital healthcare systems increased significantly after the COVID-19 pandemic. During the pandemic, hospitals experienced excessive patient loads and increased demand for contactless healthcare services. Digital healthcare platforms became essential for online registration, appointment scheduling, digital prescriptions, and smart healthcare management. The pandemic highlighted the importance of intelligent healthcare automation and centralized digital healthcare infrastructure.

The proposed project also supports the vision of Digital India and modern healthcare digitization initiatives. Government healthcare programs promoting electronic healthcare records and digital healthcare services have increased the demand for secure and scalable healthcare technologies. MediGen-AI contributes toward this vision by providing a modern healthcare platform capable of supporting future healthcare innovations.

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Major Reasons for the Need of the Project

1. To Reduce Patient Waiting Time

- **Manual queue systems increase delays in hospitals.**
- **Patients spend long hours waiting for consultation.**
- **Smart queue management improves patient flow efficiency**

2. To Eliminate Handwritten Prescription Errors

- **Handwritten prescriptions may be unclear.**
- **Incorrect interpretation creates medication risks.**
- **Digital prescriptions improve readability and accuracy.**

3. To Improve Healthcare Record Management

- **Paper records are difficult to maintain.**
- **There is risk of record loss and duplication.**
- **Digital healthcare records improve accessibility and storage**

4. To Automate Hospital Workflow

- **Manual systems increase staff workload.**
- **Automation improves operational efficiency.**
- **Faster patient processing becomes possible.**

5. To Assist Doctors Through Rule-Based AI

- **AI helps in symptom analysis and prescription assistance.**
- **Rule-Based AI provides explainable suggestions.**
- **Doctors remain responsible for final decisions.**

6. To Improve Medicine Accessibility

- **Patients face delays while searching for medicines.**
- **E-Pharmacy integration improves convenience.**
- **Faster medicine procurement becomes possible.**

7. To Improve Healthcare Transparency

- **Rule-Based AI ensures explainable healthcare decisions.**
-

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- **Human-in-the-Loop AI maintains accountability.**
- **Transparent systems improve patient trust.**

8. To Support Digital Healthcare Transformation

- **Smart hospitals require digital infrastructure.**
- **Healthcare digitization improves efficiency.**
- **Modern healthcare depends on intelligent technologies.**

9. To Improve Administrative Management

- **Centralized systems improve coordination.**
- **Administrative workload is reduced.**
- **Better hospital management becomes possible.**

10. To Create a Secure Healthcare Ecosystem

- **Secure authentication protects patient data.**
- **Digital systems improve healthcare security.**
- **Centralized healthcare platforms improve scalability**

11. To Improve Accessibility of Healthcare Services

- **Many patients cannot access healthcare services quickly.**
- **Rural and remote areas face healthcare management problems.**
- **Digital healthcare systems improve accessibility for patients.**
- **Online registration and digital consultation support faster treatment.**
- **Integrated healthcare platforms provide better patient convenience.**

Therefore, the MediGen-AI project is needed to modernize healthcare operations, improve hospital workflow efficiency, reduce patient waiting time, eliminate handwritten prescription errors, provide secure digital healthcare infrastructure, and support intelligent healthcare automation using Rule-Based Artificial Intelligence and E-Pharmacy integration. The project demonstrates how healthcare technology can improve patient care, operational management, and healthcare accessibility in modern hospitals and clinics.

The MediGen-AI Hospital Management System was developed to address the growing need for intelligent, automated, secure, and efficient healthcare management solutions. The project solves many problems associated with traditional hospital systems such as manual paperwork, inefficient workflow, prescription errors, poor data management, and delayed healthcare services.

1.3 Aim of the Project

The main aim of this project is to develop a Secure Hybrid Healthcare Portal integrated with Rule-Based Artificial Intelligence and E-Pharmacy services for improving healthcare management and hospital workflow efficiency. The system is designed to modernize traditional healthcare processes by providing intelligent digital healthcare solutions capable of managing patient registration, OPD workflow, doctor consultation, digital prescriptions, and medicine ordering through a single integrated platform. The project aims to reduce patient waiting time, eliminate handwritten prescription errors, improve healthcare record management, and provide better coordination between patients, doctors, medical staff, and hospital administrators.

Another important aim of the project is to implement Rule-Based Artificial Intelligence for assisting doctors during diagnosis and prescription generation. The AI engine analyzes patient symptoms and provides medicine suggestions using predefined healthcare rules and symptom-matching techniques. However, the system follows the Human-in-the-Loop AI approach where doctors remain fully responsible for final medical decisions. This ensures transparency, accountability, safety, and ethical implementation of Artificial Intelligence in healthcare environments. The system also provides Digital Prescription Management where prescriptions are generated in PDF format and stored securely in the patient dashboard for future access and healthcare record management.

The system also aims to integrate E-Pharmacy services into the healthcare workflow so that patients can directly order prescribed medicines online after consultation. This feature reduces delays in medicine procurement and improves healthcare accessibility. The system is developed to support healthcare digitization and smart healthcare infrastructure development by combining Rule-Based AI, secure authentication systems, digital healthcare management, and intelligent hospital workflow automation into a secure and scalable healthcare ecosystem suitable for modern hospitals and clinics.

Main Aims

1. To Digitize Healthcare Management

- Replace paper-based healthcare systems.
- Improve hospital workflow efficiency.
- Provide centralized healthcare management.

2. To Reduce Patient Waiting Time

- Implement smart queue management.
 - Improve OPD workflow handling.
 - Reduce overcrowding in hospitals
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3. To Implement Rule-Based AI Assistance

Assist doctors during diagnosis.

Provide symptom-based medicine suggestions.

Improve healthcare decision support.

4. To Maintain Human Supervision in AI

- **Follow Human-in-the-Loop AI approach.**
- **Keep doctors responsible for final decisions.**
- **Ensure ethical healthcare AI implementation.**

5. To Eliminate Handwritten Prescription Errors

- **Generate digital prescriptions in PDF format.**
- **Improve readability and prescription accuracy.**
- **Reduce medication-related risks.**

6. To Improve Healthcare Record Management

- **Maintain secure digital patient records.**
- **Provide easy access to healthcare history.**
- **Improve long-term medical record management.**

7. To Support Online and Offline Patients

- **Allow online patient registration.**
- **Allow offline registration by medical staff.**
- **Provide hybrid healthcare workflow management.**

8. To Integrate E-Pharmacy Services

- **Provide online medicine ordering facilities.**
- **Improve medicine accessibility for patients.**
- **Reduce delays in medicine procurement.**

Therefore, the aim is to create a secure, scalable, transparent, and intelligent healthcare platform capable of improving patient care, hospital workflow efficiency, digital healthcare management, and healthcare accessibility through Rule-Based Artificial Intelligence and E-Pharmacy integration.

CHAPTER 2: PROBLEM STATEMENT

The healthcare sector is one of the most essential parts of society because it directly affects human health, medical treatment, and public well-being. Despite the rapid growth of technology in many industries, a large number of hospitals and clinics still depend on traditional healthcare management systems that use paper-based records, handwritten prescriptions, manual patient registration, and offline appointment handling. These traditional methods create several operational challenges and reduce the overall efficiency of healthcare services.

One of the major problems in conventional healthcare systems is overcrowding in Out Patient Departments (OPDs). Patients often spend a significant amount of time waiting for registration, token generation, consultation, and medicine purchasing because hospital workflows are managed manually. In many hospitals, patient records and appointment details are maintained in physical files or registers, which increases administrative workload and slows down healthcare operations. During emergency situations, delays in retrieving patient records may negatively affect treatment quality and patient safety.

Another important issue in traditional healthcare systems is handwritten prescriptions. Doctors usually write prescriptions manually, and in many cases the handwriting may be difficult to understand. Unclear prescriptions create confusion regarding medicine names, dosage instructions, and treatment procedures. This may lead to medication errors, incorrect medicine dispensing, delayed treatment, and increased healthcare risks. Handwritten prescriptions also make long-term healthcare record management difficult because paper documents may be lost, damaged, or duplicated over time.

Most hospitals also lack centralized digital healthcare record management systems. Patient information is often stored separately in different departments, making it difficult for doctors and medical staff to access complete medical history quickly during follow-up consultations. This lack of coordination reduces operational efficiency and creates problems in maintaining accurate healthcare records. Traditional systems are therefore unable to provide efficient communication and data sharing between patients, doctors, medical staff, pharmacies, and hospital administrators.

Another major problem in healthcare systems is inefficient queue management. In conventional hospital environments, patients are generally managed manually without any intelligent workflow automation. As the number of patients increases, managing OPD queues becomes more difficult, leading to overcrowding, treatment delays, and poor patient experience. Hospitals require smart healthcare systems capable of automating patient flow and improving hospital workflow management.

The healthcare sector also faces challenges in medicine accessibility. After consultation, many patients must visit multiple pharmacies to search for prescribed medicines. This process wastes time and delays treatment, especially during emergencies or in areas where medicine availability is limited. Lack of integration

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between healthcare systems and pharmacies creates inconvenience for patients and reduces the overall efficiency of healthcare services.

Modern healthcare systems require intelligent digital solutions capable of improving workflow efficiency, reducing manual workload, maintaining secure healthcare records, and supporting healthcare professionals during diagnosis and treatment processes. Although Artificial Intelligence has become an important technology in healthcare, many AI systems depend heavily on black-box models or external APIs that do not provide transparent decision-making processes. In healthcare environments, transparency, explainability, and doctor supervision are extremely important because incorrect medical decisions can directly affect patient safety.

Many existing healthcare systems do not provide proper Human-in-the-Loop AI implementation where doctors remain fully responsible for final medical decisions. Fully automated AI systems may generate suggestions without proper medical accountability, creating ethical and safety concerns. Therefore, healthcare systems require AI-assisted solutions that support doctors while maintaining human control over medical decisions.

Another significant problem is the lack of secure and centralized digital prescription management. Traditional paper prescriptions are difficult to maintain for future reference, and patients often struggle to preserve medical documents safely. Without digital prescription systems, healthcare history management becomes inefficient and time-consuming.

The importance of healthcare digitization increased significantly after the COVID-19 pandemic. During the pandemic, hospitals experienced excessive patient loads, overcrowded healthcare facilities, and increased demand for contactless healthcare services. Traditional healthcare management systems were unable to handle these challenges efficiently. The pandemic highlighted the urgent need for digital healthcare infrastructure capable of supporting online registration, digital prescriptions, healthcare automation, and smart hospital management.

The absence of integrated healthcare platforms also creates coordination problems between patients, doctors, medical staff, pharmacies, and hospital administrators. Most existing systems are fragmented and do not provide a unified healthcare environment where all healthcare services can operate together efficiently. Hospitals require centralized healthcare ecosystems capable of integrating patient management, doctor consultation, prescription handling, healthcare records, and medicine ordering services into a single platform.

To address these problems, there is a strong need for a secure and intelligent healthcare system capable of automating healthcare workflows while maintaining transparency, medical safety, and operational efficiency. The system should support both online and offline patient management, intelligent queue handling, digital prescription generation, centralized healthcare records, and medicine accessibility through integrated E-Pharmacy services.

2.1 Accessibility Issues

Accessibility is one of the major challenges in traditional healthcare systems, especially in developing regions where healthcare resources and digital infrastructure are limited. Many patients face difficulties in accessing proper healthcare services due to long waiting times, overcrowded hospitals, inefficient management systems, lack of digital healthcare platforms, and poor coordination between healthcare departments. These accessibility issues reduce the quality of healthcare services and negatively affect patient experience.

One of the most common accessibility issues is the difficulty patients face during hospital registration and appointment processes. In traditional healthcare systems, patients are often required to visit hospitals physically for registration and consultation. Long queues and manual workflow management create inconvenience, especially for elderly patients, disabled individuals, and people living in rural areas. Delays in patient handling reduce treatment efficiency and increase pressure on hospital staff.

Another major accessibility issue is the lack of centralized healthcare information systems. In many hospitals, patient records are stored manually in paper files, making it difficult to retrieve medical history quickly during emergencies or follow-up consultations. Patients may need to carry physical prescriptions and medical documents repeatedly, which creates inconvenience and increases the risk of document loss or damage.

Healthcare accessibility is also affected by poor communication and coordination between patients, doctors, pharmacies, and hospital administration. Traditional healthcare systems often operate through disconnected workflows where patients must move from one department to another for registration, consultation, prescription collection, and medicine purchasing. This fragmented process consumes time and reduces overall healthcare efficiency.

Medicine accessibility is another important issue in healthcare systems. After consultation, many patients face difficulties while searching for prescribed medicines at nearby pharmacies. In emergency situations or remote areas, lack of medicine availability may delay treatment and negatively affect patient health. Patients often spend additional time and resources visiting multiple pharmacies to find required medicines.

Digital accessibility is also a challenge in many healthcare environments. Several hospitals lack modern healthcare management systems that support online registration, digital prescriptions, appointment scheduling, and patient dashboards. As a result, patients cannot access healthcare services remotely and must depend entirely on offline processes. This became a major problem during the COVID-19 pandemic when contactless healthcare services became essential.

The MediGen-AI Hospital Management System was developed to address the growing need for intelligent, automated, secure, and efficient healthcare management solutions.

2.2 Manual Prescription Errors

Manual prescription errors are one of the most serious problems in traditional healthcare systems. In many hospitals and clinics, doctors still write prescriptions manually on paper, which increases the possibility of mistakes and misunderstandings during treatment. Handwritten prescriptions often become difficult to read because of unclear handwriting, incomplete instructions, or incorrect medicine details. These problems may lead to medication errors, delayed treatment, and healthcare risks for patients.

One of the major causes of manual prescription errors is poor handwriting clarity. In many cases, pharmacists and patients are unable to understand medicine names or dosage instructions correctly. Similar-looking medicine names and unclear abbreviations may create confusion during medicine dispensing. As a result, patients may receive incorrect medicines or wrong dosage quantities, which can negatively affect treatment outcomes and patient safety.

Another important issue is the absence of standardized prescription formats in traditional healthcare systems. Different doctors may follow different writing styles and prescription structures, making it difficult for pharmacists and medical staff to interpret instructions consistently. Missing information such as dosage timing, treatment duration, medicine strength, or patient details may create additional complications during treatment.

Manual prescriptions also increase the administrative burden on hospitals and healthcare providers. Physical prescriptions are difficult to maintain, organize, and retrieve for future reference. Paper documents may be lost, damaged, or duplicated over time, creating difficulties in healthcare record management. During follow-up consultations, doctors may not have quick access to previous prescriptions, reducing the efficiency of medical decision-making.

Medication errors caused by manual prescriptions may result in serious healthcare consequences. Incorrect medicine dispensing, overdose, underdose, or wrong treatment instructions can affect patient recovery and increase healthcare risks. In severe cases, prescription errors may create life-threatening situations, especially for elderly patients, children, or patients with chronic diseases.

Another major limitation of manual prescriptions is the lack of integration with digital healthcare systems and pharmacies. Patients often need to carry paper prescriptions physically while purchasing medicines from pharmacies. If prescriptions are lost or damaged, patients may face difficulties obtaining medicines again. This process also reduces healthcare accessibility and convenience.

Manual prescription errors are a major challenge in traditional healthcare systems and can negatively affect patient safety and treatment quality. Problems such as illegible handwriting, incorrect dosage, delayed processing, and poor record management increase healthcare risks and operational inefficiency.

The MediGen-AI Hospital Management System addresses these issues through AI-assisted digital prescription generation, standardized healthcare workflows, and secure digital record management. Overall, the system improves prescription accuracy, enhances healthcare efficiency, and supports safer and smarter healthcare management practices.

2.3 Delayed Medicine Procurement

Delayed medicine procurement is one of the major problems in traditional healthcare systems. After consultation and prescription generation, patients often face difficulties while searching for prescribed medicines at nearby pharmacies. In many cases, medicines may not be available at a single pharmacy, forcing patients to visit multiple medical stores to complete their treatment requirements. This process consumes time, increases inconvenience, and delays the beginning of proper medical treatment.

In conventional healthcare environments, there is usually no direct coordination between hospitals and pharmacies. After receiving prescriptions, patients must manually search for medicines, compare availability, and purchase required drugs from different locations. This fragmented healthcare process reduces operational efficiency and creates additional stress for patients and their family members. During emergencies, delays in medicine procurement can negatively affect patient health and treatment outcomes.

Another important issue is the lack of digital integration between healthcare systems and medicine providers. Traditional paper-based prescriptions do not provide direct connectivity with pharmacy services, making the medicine purchasing process slow and inefficient. Patients living in rural or remote areas often face even greater difficulties because nearby pharmacies may not maintain proper medicine stock or healthcare supply chains. As a result, patients may experience delays in receiving important medicines and treatment support.

Delayed medicine procurement also creates problems for elderly patients, physically disabled individuals, and patients suffering from chronic diseases who require immediate access to prescribed medicines. Repeated travel to different pharmacies increases physical and financial burden on patients. In many situations, patients may discontinue treatment temporarily because medicines are not easily available, which may worsen medical conditions and reduce treatment effectiveness.

Another major challenge is the absence of centralized digital healthcare platforms capable of connecting doctors, patients, and pharmacies within a single healthcare ecosystem. Traditional healthcare systems focus mainly on consultation and prescription generation but fail to provide post-consultation medicine accessibility solutions. This gap between diagnosis and medicine procurement reduces.

The problem became more serious during the COVID-19 pandemic when movement restrictions, overcrowded pharmacies, and medicine shortages created difficulties in obtaining medicines quickly. Many patients faced delays in treatment because they could not access medicines on time. This situation highlighted the importance of digital healthcare systems integrated with online pharmacy services and medicine delivery.

Delayed medicine procurement is a major problem in traditional healthcare systems and can negatively affect hospital efficiency, patient treatment, and healthcare workflow management. Manual inventory tracking, lack of real-time monitoring, inefficient communication, and paper-based procurement processes are some of the major causes of procurement delays.

2.4 Lack of Digital Records

Lack of digital healthcare records is one of the major challenges faced by traditional healthcare systems. In many hospitals and clinics, patient information, prescriptions, medical reports, appointment details, and treatment history are still maintained manually using paper files and physical documents. This traditional method of record management creates several operational problems and reduces the overall efficiency of healthcare services.

One of the primary problems associated with paper-based healthcare records is difficulty in maintaining and retrieving patient information. Physical files require proper storage space and manual organization, which becomes increasingly difficult as the number of patients grows. In large hospitals, managing thousands of paper records consumes significant time and administrative effort. During emergencies or follow-up consultations, retrieving patient history from physical files may become slow and inefficient, delaying treatment and affecting patient care.

Another important issue is the risk of record loss and damage. Paper documents can easily be misplaced, destroyed, or damaged due to improper handling, environmental conditions, or human error. Lost medical records create serious difficulties for both patients and healthcare providers because previous treatment history, prescriptions, and diagnostic information may no longer be available. This lack of accurate historical data can negatively affect medical decision-making and treatment quality. Traditional healthcare systems also face problems related to duplication and inconsistency of patient records. Since records are maintained manually, the same patient information may be entered multiple times in different departments, creating duplicate files and inaccurate healthcare data. Lack of centralized healthcare record management reduces coordination between doctors, medical staff, laboratories, and pharmacies. As a result, hospitals struggle to maintain efficient communication and integrated healthcare workflows.

Another major limitation of non-digital healthcare systems is poor accessibility of medical records. Patients often need to carry physical prescriptions, reports, and treatment documents during every hospital visit. If documents are forgotten or lost, doctors may not have access to complete medical history during consultation. This creates inconvenience for patients and reduces the effectiveness of diagnosis and treatment procedures.

Healthcare data security is also a serious concern in traditional record management systems. Paper documents can be accessed by unauthorized individuals if proper storage and monitoring are not maintained. Manual record systems provide limited security mechanisms, increasing the risk of data leakage and confidentiality issues. Protecting sensitive patient information becomes difficult in such environments.

Lack of digital records is a major challenge in traditional healthcare systems and negatively affects healthcare efficiency, patient care, workflow management, and data security. Manual paper-based record handling creates problems such as data loss, slow information retrieval, increased workload, and poor coordination between hospital departments.

The MediGen-AI Hospital Management System addresses these issues by implementing centralized digital healthcare record management, secure database.

2.5 Workflow Inefficiency

Workflow inefficiency is one of the most common problems in traditional healthcare systems. In many hospitals and clinics, healthcare operations are still managed manually using paper-based records, handwritten prescriptions, offline registration systems, and non-automated patient handling methods. These traditional workflows increase operational complexity, reduce hospital efficiency, and create delays in healthcare services. As the number of patients continues to increase, healthcare institutions face greater difficulty in managing daily operations effectively.

One of the major causes of workflow inefficiency is manual patient registration and appointment handling. In conventional healthcare systems, patients are required to stand in long queues for registration, token generation, and consultation. Medical staff manually record patient details in registers or physical files, which consumes time and increases administrative workload. During peak hours, overcrowded OPDs create pressure on hospital staff and reduce the speed of healthcare service delivery.

Another important issue is the lack of coordination between different healthcare departments. In traditional systems, patient information is often maintained separately by doctors, medical staff, laboratories, pharmacies, and administration departments. Since these systems are not connected digitally, communication between departments becomes slow and inefficient. This fragmented workflow creates delays in patient processing and reduces operational productivity.

Manual handling of healthcare records also contributes significantly to workflow inefficiency. Maintaining paper-based patient records requires continuous filing, searching, updating, and storage management. Retrieving previous healthcare history during follow-up consultations becomes time-consuming and difficult. In emergency situations, delayed access to patient records may negatively affect treatment quality and patient safety.

Another major problem is the absence of intelligent queue management systems. Traditional healthcare environments usually do not provide automated patient flow monitoring or smart appointment scheduling. As a result, patients often wait for long periods before consultation, while hospital staff struggle to manage overcrowded waiting areas. Inefficient queue handling reduces patient satisfaction and increases stress for both healthcare providers and patients.

Workflow inefficiency is also visible during prescription management and medicine procurement processes. Handwritten prescriptions require manual interpretation by pharmacists, increasing the possibility of delays and medication errors. Patients may need to visit multiple pharmacies to search for prescribed medicines because healthcare systems and pharmacies are not digitally connected. This fragmented Lack of digital records is a major challenge in traditional healthcare systems and negatively affects healthcare efficiency, patient care, workflow management, and data security. Manual paper-based record handling creates problems such as data loss, slow information retrieval, increased workload, and poor coordination between hospital departments.

CHAPTER 3: OBJECTIVES OF THE PROJECT

The main objective of this system is to develop a Secure Hybrid Healthcare Portal integrated with Rule-Based Artificial Intelligence and E-Pharmacy services for improving healthcare management, hospital workflow efficiency, and patient care services. The system is designed to modernize traditional healthcare operations by transforming manual hospital processes into a centralized digital healthcare environment capable of managing patient registration, OPD workflow, doctor consultation, prescription generation, healthcare records, and medicine ordering services efficiently.

One of the primary objectives of the system is to reduce patient waiting time and improve OPD management. In traditional healthcare systems, patients often face delays during registration, consultation, and queue handling because hospital workflows are managed manually. The proposed system introduces smart patient management and digital OPD handling to improve operational efficiency and reduce overcrowding in hospitals and clinics.

Another important objective of the system is to eliminate handwritten prescription errors through digital prescription generation. Traditional paper-based prescriptions are often difficult to understand due to unclear handwriting, which may create confusion regarding medicine names and dosage instructions. The system generates digital prescriptions in a standardized PDF format, improving readability, accuracy, and healthcare transparency while reducing medication-related risks.

The system also aims to implement Rule-Based Artificial Intelligence for assisting doctors during diagnosis and prescription generation. The AI engine analyzes patient symptoms using predefined healthcare rules and symptom-matching techniques to provide medicine suggestions and clinical assistance. However, the system follows the Human-in-the-Loop AI approach where final medical decisions always remain under doctor supervision. This objective ensures transparency, accountability, safety, and ethical implementation of Artificial Intelligence in healthcare systems.

Another major objective of the system is to maintain secure digital healthcare records for improving healthcare accessibility and treatment continuity. Traditional paper-based records are difficult to maintain and may be lost or damaged over time.

Another objective of the system is to create a future-ready healthcare ecosystem capable of supporting advanced healthcare technologies such as telemedicine, cloud healthcare systems, multilingual healthcare support, wearable device integration, and advanced healthcare analytics. The scalable architecture of the system allows future expansion and integration of additional healthcare services according to evolving medical and technological requirements.

Overall, the objective of the system is to create a secure, scalable, transparent, and intelligent healthcare platform capable of improving patient care, reducing operational delays, supporting doctors through Rule-Based Artificial Intelligence, improving medicine accessibility, maintaining secure digital healthcare records, and

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modernizing hospital workflow management through intelligent healthcare automation and E-Pharmacy integration.

3.1 PRIMARY OBJECTIVE OF THE PROJECT

The primary objective of this system is to develop a Secure Hybrid Healthcare Portal integrated with Rule-Based Artificial Intelligence and E-Pharmacy services for improving healthcare management, patient care, and hospital workflow efficiency. The system aims to modernize traditional healthcare operations by replacing manual healthcare processes with an intelligent digital healthcare environment capable of managing patient registration, OPD workflow, doctor consultation, digital prescriptions, healthcare records, and medicine ordering services through a centralized platform.

Another major objective of the system is to reduce patient waiting time and improve hospital workflow management through smart queue handling and automated patient processing. The system is designed to support both online and offline healthcare workflows where online patients can register digitally while offline patients can be managed through hospital staff dashboards. This hybrid healthcare approach improves accessibility, flexibility, and operational efficiency in hospitals and clinics.

The system also aims to eliminate handwritten prescription errors by generating digital prescriptions in a standardized PDF format. This improves prescription readability, treatment accuracy, and healthcare transparency while reducing medication-related risks. In addition, the project focuses on implementing Rule-Based Artificial Intelligence for assisting doctors during diagnosis and prescription generation. The AI engine analyzes patient symptoms using predefined healthcare rules and provides intelligent medicine suggestions while maintaining doctor supervision through the Human-in-the-Loop AI approach.

Another important objective of the system is to maintain secure digital healthcare records and improve coordination between patients, doctors, medical staff, pharmacies, and hospital administration. The project also integrates E-Pharmacy services to improve medicine accessibility by allowing patients to order prescribed medicines online after consultation.

The project also aims to provide a secure healthcare environment through role-based authentication and protected healthcare data management. Since patient information is highly sensitive, the system focuses on maintaining confidentiality, data security, and controlled system access. Secure authentication mechanisms help prevent unauthorized access and improve the reliability of the healthcare ecosystem.

Another major objective of the system is to improve patient convenience and healthcare accessibility. Patients can access healthcare services digitally through online registration, OPD slip generation, prescription downloads, and E-Pharmacy

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integration. This reduces unnecessary hospital visits, improves treatment accessibility, and creates a more patient-friendly healthcare experience.

Overall, the primary objective of the system is to create a secure, scalable, transparent, and intelligent healthcare ecosystem capable of improving healthcare accessibility, reducing operational delays, supporting healthcare professionals through Rule-Based Artificial Intelligence, and modernizing healthcare management through digital healthcare automation and E-Pharmacy integration

3.4 Secondary Objective of Project

The secondary objective of this system is to improve the overall efficiency, transparency, and accessibility of healthcare services through intelligent healthcare automation and centralized digital management. Along with improving primary healthcare operations, the system focuses on supporting doctors, patients, medical staff, and hospital administration through secure and integrated healthcare workflows.

One of the secondary objectives of the system is to improve healthcare record management by maintaining centralized digital patient records. Traditional paper-based records are difficult to organize, retrieve, and maintain for long-term healthcare monitoring. The proposed system stores patient information, consultation history, prescriptions, and medical details securely within a digital database, improving healthcare accessibility and treatment continuity.

Another important secondary objective is to improve coordination between different healthcare departments. In conventional healthcare systems, patient management, prescription handling, pharmacy services, and administration often operate separately, reducing workflow efficiency. The proposed system integrates these services into a unified healthcare platform, improving communication and operational management.

The system also aims to improve medicine accessibility through integrated E-Pharmacy services. Patients can directly order prescribed medicines online after consultation, reducing delays in medicine procurement and improving treatment efficiency. This feature also improves convenience for elderly patients, physically challenged individuals, and patients living in remote areas.

Another secondary objective of the system is to reduce administrative workload within hospitals and clinics. Manual healthcare management requires continuous paperwork, record maintenance, and physical documentation, which consumes time and increases operational complexity. The proposed digital platform automates several healthcare processes, improving productivity and reducing manual effort for hospital staff.

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The project further aims to support transparency and accountability in healthcare systems through Rule-Based Artificial Intelligence and Human-in-the-Loop AI implementation. The AI engine provides explainable medicine suggestions based on predefined healthcare rules while keeping doctors responsible for final medical decisions. This ensures ethical AI implementation and improves trust in intelligent healthcare systems.

The system also focuses on improving patient convenience and healthcare accessibility through online registration, digital OPD slip generation, downloadable prescriptions, and centralized patient dashboards. These features reduce unnecessary hospital visits and create a more patient-friendly healthcare environment.

Another secondary objective is to provide secure healthcare data management using authentication and role-based access control mechanisms. Since healthcare information is highly sensitive, the system focuses on protecting patient confidentiality and preventing unauthorized access to medical records.

CHAPTER 4: SCOPE OF THE PROJECT

The scope of this system is focused on developing a Secure Hybrid Healthcare Portal integrated with Rule-Based Artificial Intelligence and E-Pharmacy services for improving healthcare management, patient care services, and hospital workflow efficiency. The system is designed to provide a centralized digital healthcare environment capable of managing patient registration, OPD workflow, doctor consultation, prescription generation, healthcare record management, and medicine ordering services through a secure and scalable platform.

The project covers both online and offline healthcare workflows. Online patients can register through the healthcare portal, generate OPD slips digitally, and access healthcare services remotely. Offline patients visiting hospitals physically can be managed through the medical staff dashboard. This hybrid healthcare workflow improves accessibility and allows hospitals and clinics to manage both digital and walk-in patients efficiently.

The scope of the system also includes smart patient management and queue handling. Traditional healthcare systems often face problems related to overcrowded OPDs and inefficient patient flow management. The proposed system introduces intelligent patient registration and automated workflow management to reduce waiting time and improve hospital operational efficiency.

Another important scope of the system is digital prescription management. Doctors can generate prescriptions digitally in PDF format instead of handwritten prescriptions. This improves readability, prescription accuracy, and healthcare transparency while reducing medication-related errors. Patients can access and download prescriptions securely from their dashboard whenever required.

The system also includes Rule-Based Artificial Intelligence for assisting doctors during diagnosis and prescription generation. The AI engine analyzes patient

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symptoms using predefined healthcare rules and symptom-matching techniques to provide medicine suggestions and clinical assistance. However, the final medical decision always remains under doctor supervision through the Human-in-the-Loop AI approach. This ensures transparency, accountability, and ethical implementation of Artificial Intelligence in healthcare environments.

The project further covers centralized digital healthcare record management. Patient details, consultation history, prescriptions, and healthcare information are stored securely within a centralized database. This improves healthcare accessibility, treatment continuity, and coordination between doctors, patients, and medical staff.

Another important scope of the system is E-Pharmacy integration. After consultation and prescription generation, patients can directly access online medicine ordering services through the healthcare platform. This improves medicine accessibility, reduces delays in medicine procurement, and creates a more convenient healthcare experience for patients.

The system also supports role-based healthcare management where different users such as patients, doctors, medical staff, and administrators can access separate dashboards according to their responsibilities. This improves workflow management, operational control, and healthcare coordination within hospitals and clinics.

The project includes secure authentication and healthcare data protection mechanisms to maintain confidentiality and prevent unauthorized access to patient information. Since healthcare data is highly sensitive, the system focuses on providing secure digital healthcare management through protected login systems and role-based access control.

The scope of the project is not limited only to current healthcare requirements. The system is designed with scalable architecture so that future technologies and advanced healthcare services can be integrated easily. Future expansion may include telemedicine services, cloud healthcare systems, multilingual support, wearable healthcare device integration, online video consultation, healthcare analytics, and mobile healthcare applications.

The project is mainly intended for hospitals, clinics, healthcare centers, and medical institutions that require intelligent healthcare management systems capable of improving operational efficiency and patient care services. The system can also be adapted for small healthcare organizations and future smart healthcare infrastructures.

The proposed healthcare platform contributes toward healthcare digitization and modernization by combining healthcare automation, centralized digital healthcare records, Rule-Based Artificial Intelligence, smart workflow management, and E-Pharmacy integration within a single platform. The system demonstrates how modern healthcare technologies can improve healthcare accessibility, reduce operational complexity, support healthcare professionals, and create a secure and intelligent healthcare ecosystem suitable for modern hospitals and clinics.

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Additional Scope of the Project

- **The system supports centralized healthcare management for hospitals and clinics.**
 - **The platform improves coordination between patients, doctors, medical staff, and administrators.**
 - **The system reduces paperwork and manual healthcare operations through digital automation.**
 - **The project improves OPD workflow efficiency using smart patient management.**
 - **The system supports both online and offline patient registration processes.**
 - **The platform provides secure digital prescription generation in PDF format.**
 - **The project improves prescription readability and reduces medication-related errors.**
 - **The system maintains centralized digital healthcare records for future reference.**
 - **The platform improves healthcare accessibility through online healthcare services.**
 - **The project supports intelligent queue management for reducing patient waiting time.**
 - **The system provides secure authentication and role-based access control.**
 - **The platform protects sensitive healthcare information and patient confidentiality.**
 - **The project integrates E-Pharmacy services for online medicine ordering.**
 - **The system improves medicine accessibility and reduces procurement delays.**
 - **The platform supports Human-in-the-Loop AI implementation for ethical healthcare management.**
 - **The project provides Rule-Based Artificial Intelligence assistance during diagnosis.**
 - **The system supports transparent and explainable AI-based healthcare recommendations.**
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- **The platform improves healthcare workflow automation and operational efficiency.**
 - **The project reduces administrative workload for healthcare staff and hospitals.**
 - **The system supports long-term healthcare record maintenance and treatment continuity.**
 - **The platform improves communication between different healthcare departments.**
 - **The project contributes toward smart healthcare infrastructure development.**
 - **The system supports future integration of telemedicine and cloud healthcare services.**
 - **The platform can be expanded with multilingual healthcare support systems.**
 - **The project supports future mobile healthcare application integration.**
 - **The system can be integrated with wearable healthcare devices in the future.**
 - **The platform supports scalable healthcare architecture for future expansion.**
 - **The project contributes toward digital healthcare transformation and modernization.**
 - **The system improves patient convenience through centralized healthcare services.**
 - **The platform creates a secure, scalable, and intelligent healthcare ecosystem.**

The proposed healthcare platform contributes toward healthcare digitization and modernization by combining healthcare automation, centralized digital healthcare records, Rule-Based Artificial Intelligence, smart workflow management, and E-Pharmacy integration within a single platform. The system demonstrates how modern healthcare technologies can improve healthcare accessibility, reduce operational complexity, support healthcare professionals, and create a secure and intelligent healthcare ecosystem suitable for modern hospitals and clinics.

The Patient Module is one of the most important components of the MediGen-AI Hospital Management System. This module is designed to manage all patient-related activities efficiently through digital healthcare management and automated workflows.

CHAPTER 5: LITERATURE REVIEW

The healthcare industry is rapidly adopting digital technologies to improve patient care, hospital management, and medical data handling. Traditional healthcare systems often suffer from issues such as manual prescription errors, delayed medicine procurement, lack of digital records, and inefficient workflows. To overcome these challenges, intelligent healthcare management systems integrated with Artificial Intelligence (AI) are becoming increasingly popular.

MediGen-AI is designed as an AI-powered healthcare management system that automates patient management, prescription generation, medicine inventory handling, and medical record storage. This literature review discusses previous research, existing healthcare systems, and technologies related to digital healthcare management and AI-based medical systems.

Review of Existing Systems

1. Hospital Management Systems (HMS)

Hospital Management Systems are software platforms developed to manage hospital operations such as patient registration, doctor appointments, billing, prescriptions, and staff management. Researchers have highlighted that HMS improves efficiency, reduces paperwork, and enhances data accessibility.

However, many traditional HMS platforms lack advanced AI capabilities and intelligent automation features. They mainly focus on record storage rather than predictive healthcare support and smart decision-making.

Key Findings

- Reduces manual paperwork.
- Improves patient data management.
- Enhances hospital workflow.
- Limited AI integration.
- Weak automation in medicine procurement and prescriptions.
- Manual working and slow work

2. Electronic Health Record (EHR) Systems

Electronic Health Record systems digitally store patient medical histories, diagnoses, prescriptions, and treatment details. Studies show that EHR systems help doctors access patient records quickly and improve healthcare accuracy. The MediGen-AI Hospital Management System was developed to address the growing need for intelligent, automated, secure, and efficient healthcare management solutions. The project solves many problems associated with traditional hospital systems.

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Key Findings

- Enables centralized medical data storage.
- Improves patient history tracking.
- Supports faster medical decision-making.
- Security and privacy concerns remain significant.

3. AI in Healthcare

Artificial Intelligence is increasingly used in healthcare for disease prediction, diagnosis support, medical image analysis, and prescription assistance. AI-based healthcare systems can analyze large amounts of medical data and assist doctors in making accurate decisions.

Research indicates that AI can significantly reduce human errors and improve healthcare efficiency. Machine Learning and Natural Language Processing technologies are widely used in modern healthcare applications.

Key Findings

- Reduces manual errors in prescriptions.
- Assists in disease prediction and diagnosis.
- Improves healthcare automation.
- Requires high-quality datasets for better accuracy.

4. Pharmacy and Medicine Inventory Systems

Medicine inventory management systems help hospitals and pharmacies track medicine availability, expiry dates, procurement, and stock levels. Existing research shows that digital inventory systems reduce medicine shortages and improve procurement efficient

The Agile model is an iterative, flexible approach to software development that focuses on continuous feedback, collaboration, and delivering small, working increments of software rather than building the entire system at once.

However, many systems are not integrated with patient treatment workflows, leading to delays in medicine availability.

Key Findings

- Improves medicine stock tracking.
 - Reduces procurement delays.
 - Helps monitor medicine expiry.
 - Helps monitor healthcare.
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Comparative Analysis of Existing Systems

<u>Existing System</u>	<u>Advantages</u>	<u>Limitations</u>
Hospital Management System	Efficient record management	Limited AI features
Electronic Health Record System	Digital patient history storage	Security concerns
AI-Based Healthcare Systems	Smart decision support	High complexity
Pharmacy Inventory Systems	Better stock management	Weak workflow integration

Research Gap

Although many healthcare systems are available, several important issues still exist:

1. Lack of intelligent prescription generation.
2. Poor integration between patient management and medicine inventory.
3. Delays in medicine procurement.
4. Manual errors in patient records and prescriptions.
5. Limited AI support in traditional healthcare software.
6. Workflow inefficiencies due to disconnected systems.

These gaps create the need for an integrated AI-powered healthcare management platform like MediGen-AI.

Proposed Solution: MediGen-AI

MediGen-AI aims to combine patient management, doctor consultation, prescription handling, medicine inventory management, and AI-powered healthcare assistance into a single platform.

- The proposed system provides:
- Digital patient registration.
- AI-assisted prescription support.
- Secure medical record storage.
- Doctor and staff management.

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Several technologies are commonly used in healthcare management systems:

<u>Technology</u>	<u>Purpose</u>
Python	Backend develop AI integration
Flask/Django	Web application framework
MySQL	Database management
Machine Learning	Prediction and automation
HTML/CSS/JavaScript	Frontend user interface
Cloud Storage	Secure medical data storage

Advantages of the Proposed System

The proposed MediGen-AI system offers several advantages over traditional healthcare management systems:

- 1. Reduces manual paperwork and documentation**
 - 2. Improves prescription accuracy using AI assistance.**
 - 3. Enables centralized digital medical records**
 - 4. Enhances communication between doctors, staff, and patients.**
 - 5. Provides faster medicine inventory tracking.**
 - 6. Reduces delays in medicine procurement.**
 - 7. Improves overall hospital workflow efficiency**
 - 8. Supports secure data management and accessibility.**
 - 9. Saves time for healthcare professionals.**
 - 10. Improves patient care and treatment quality.**
 - 11. Improves hospital management.**
 - 12. Reduces delays in hospital management.**
 - 13. Improve doctor prescription decision.**
 - 14. improve doctor decision.**
-

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Limitations of Existing Research

Although previous healthcare management systems provide useful functionalities, several limitations still remain:

- Most systems focus only on data storage.
- Limited integration of Artificial Intelligence.
- Lack of smart prescription recommendation systems.
- Weak interoperability between departments.
- Security and privacy issues in cloud-based systems.
- High maintenance and implementation costs.
- Limited support for real-time analytics.

These limitations motivated the development of MediGen-AI as an integrated healthcare platform.

Future Scope

The future scope of MediGen-AI can be expanded by integrating advanced healthcare technologies and intelligent automation features.

Possible Future Enhancements

1. AI-Based Disease Prediction

Machine Learning algorithms can be used to predict diseases based on patient symptoms, medical history, and diagnostic reports.

2. Telemedicine Integration

The system can support online doctor consultations and virtual healthcare services.

3. Mobile Application Support

A mobile application can be developed for patients and doctors to access healthcare services remotely.

4. Cloud-Based Healthcare System

Cloud integration can improve data accessibility, scalability, and backup management.

5. Real-Time Analytics Dashboard.

Advanced dashboards can provide hospital administrators with real-time healthcare analytics and performance monitoring.

6. Integration with ABHA and Digital Health IDs

The system can be integrated with Indian digital healthcare initiatives such as ABHA for secure patient identification

7. Voice-Based Prescription Generation

Voice recognition technology can help doctors generate prescriptions quickly using speech commands.

8. AI Chatbot for Patient Assistance

An AI chatbot can assist patients with appointment booking, medicine reminders, and healthcare queries.

The MediGen-AI Hospital Management System was developed to address the growing need for intelligent, automated, secure, and efficient healthcare management

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Summary of Literature Review

The literature review highlights the importance of digital transformation in the healthcare sector. Existing systems such as Hospital Management Systems, Electronic Health Records, AI-based healthcare applications, and pharmacy inventory systems have improved healthcare management to some extent.

However, issues such as workflow inefficiency, lack of intelligent automation, delayed procurement, and manual prescription errors still exist. MediGen-AI is proposed as an integrated healthcare management solution that combines AI assistance, digital records, inventory management, and workflow automation.

The study of previous research and technologies provides a strong foundation for the development of the proposed system

The literature review demonstrates that digital healthcare systems have significantly improved hospital operations and patient management. However, existing systems still face challenges related to workflow inefficiency, manual prescription errors, delayed procurement, and lack of intelligent automation.

MediGen-AI addresses these limitations by integrating AI-powered healthcare assistance, digital medical records, inventory management, and workflow automation into a unified system. The proposed platform has the potential to enhance healthcare efficiency, reduce human errors, and provide better patient care services.

The Literature Review provided valuable insights into traditional hospital management systems, digital healthcare technologies, Artificial Intelligence applications, and healthcare security mechanisms. The study revealed the limitations of manual healthcare processes and emphasized the need for intelligent, automated, secure, and centralized healthcare management systems.

Based on the findings of the literature review, the MediGen-AI Hospital Management System was designed to improve healthcare workflow efficiency, reduce manual errors, enhance patient care, support AI-assisted prescription generation, and promote digital healthcare transformation.

Overall, the literature review established a strong theoretical foundation for the development of the project and helped in creating a modern, scalable, and intelligent healthcare management solution.

The MediGen-AI Hospital Management System was successfully designed and developed as a modern healthcare management solution that combines Artificial Intelligence, automation, secure database systems, and digital healthcare technologies into a unified platform. The project aimed to solve major challenges faced by traditional hospital systems such as manual paperwork, delayed patient management, prescription errors, poor record maintenance, and inefficient workflow coordination. The implementation of the system demonstrated how modern software technologies can improve healthcare administration and patient care significantly. By digitizing hospital operations, the project reduced manual effort, improved data accuracy, enhanced healthcare accessibility, and increased overall hospital efficiency.

The integration of AI-based prescription support and symptom analysis proved to be one of the most innovative aspects of the project. The AI module assists healthcare professionals in analyzing symptoms and generating intelligent prescription suggestions, thereby reducing the chances of manual prescription errors and improving treatment efficiency. This integration highlights the growing importance

CHAPTER 6: EXISTING SYSTEM

The existing healthcare management system used in hospitals and clinics is mainly based on manual operations and partially digital platforms. These systems are designed to manage patient registration, doctor consultations, prescriptions, billing, and medicine inventory. Although these systems improve hospital operations to some extent, they still face several limitations related to automation, data management, workflow efficiency, and Artificial Intelligence integration.

Most healthcare organizations still rely on paperwork and disconnected software systems, which increases administrative workload and reduces operational efficiency.

Working of the Existing System

1. Patient Registration

In the existing system, hospital staff manually register patients by entering personal details, contact information, and medical history into registers or basic software systems.

Problems

- Time-consuming process
- Duplicate record generation
- Human data entry errors
- Difficulty in searching old records

2. Doctor Consultation and Prescription

Doctors examine patients and write prescriptions manually on paper.

Problems

- Handwriting may be unclear
- Risk of incorrect medicine interpretation
- Difficult to maintain prescription history
- No intelligent prescription assistance

3. Medical Record Management

- Patient medical records are stored physically or in separate digital systems.
- Problems
- Records can be lost or damaged
- Slow retrieval of patient history
- Weak data security
- Lack of centralized database

The integration of AI-based prescription support and symptom analysis proved to be one of the most innovative aspects of the project. The AI module assists healthcare professionals in analyzing symptoms and generating intelligent prescription suggestions, thereby reducing the chances of manual prescription errors and improving treatment efficiency. This integration highlights the growing importance of Artificial Intelligence in healthcare systems and its potential to support smart medical decision-making.

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4. Medicine Inventory Management

Medicine stock management is usually handled separately by pharmacies or inventory departments

Problems

- Delayed medicine procurement
- Difficulty in tracking stock availability
- Manual stock updates
- No real-time inventory monitoring

5. Appointment and Workflow Management

Appointments and workflow coordination are mostly managed manually.

Problems

- Appointment scheduling conflicts
- Long waiting times
- Poor communication between departments
- Increased administrative workload
- Limitations of Existing System
- Manual Prescription Errors

Handwritten prescriptions may lead to incorrect medicine interpretation and treatment errors.

Delayed Medicine Procurement

Lack of integration between inventory and patient treatment systems causes delays in medicine availability.

Lack of Digital Records

Many healthcare organizations still depend on paper-based records.

Workflow Inefficiency

Departments work independently, causing communication gaps and slower operations.

Limited AI Integration

Traditional systems do not provide intelligent healthcare assistance or automation.

Data Security Risks

Manual and outdated systems are more vulnerable to data loss and unauthorized access.

The Existing System mainly relied on manual paperwork and traditional healthcare management methods, which created several operational challenges such as prescription errors, delayed services, inefficient workflow management, poor data security, and difficulty in maintaining healthcare records.

The analysis of the existing system highlighted the need for a modern, intelligent, secure, and automated healthcare management solution. These limitations motivated the development of the MediGen-AI Hospital Management System, which introduces Artificial Intelligence, centralized database management, digital healthcare workflows, and secure automation to improve hospital efficiency and patient care significantly.

6.1 Limitations of Existing System

- **Manual queue management**
- **Paper-based prescriptions**
- **No AI supports**
- **Lack of integration with medicine delivery**
- **Time-consuming workflow**
- **Increased human error**
- **Poor record management**

Need for Proposed System

The limitations of the existing system create the need for a modern healthcare management platform like MediGen-AI.

The proposed system helps to:

- **Reduce manual errors**
- **Improve healthcare efficiency**
- **Automate workflow management**
- **Maintain secure digital records**
- **Improve medicine inventory tracking**
- **Provide AI-assisted prescription support**

Enhance communication between departments

Conclusion:

The existing healthcare management system suffers from several operational and technical challenges such as manual paperwork, workflow inefficiency, prescription errors, and poor inventory integration. These limitations reduce healthcare service quality and increase administrative complexity.

The proposed MediGen-AI system is designed to overcome these issues through Artificial Intelligence, digital healthcare management, centralized records, and workflow automation.

The Existing System mainly relied on manual paperwork and traditional healthcare management methods, which created several operational challenges such as prescription errors, delayed services, inefficient workflow management, poor data security, and difficulty in maintaining healthcare records.

The analysis of the existing system highlighted the need for a modern, intelligent, secure, and automated healthcare management solution. These limitations motivated the development of the MediGen-AI Hospital Management System, which introduces Artificial Intelligence, centralized database management, digital healthcare.

CHAPTER 7: PROPOSED SYSTEM

The proposed system, MediGen-AI, is an AI-powered healthcare management platform designed to improve hospital and clinic operations through automation, digital record management, and intelligent healthcare assistance.

The system integrates patient registration, doctor consultation, prescription generation, medicine inventory management, appointment scheduling, and medical record storage into a single centralized platform. The main objective of the proposed system is to reduce manual errors, improve workflow efficiency, and provide secure and fast healthcare services.

MediGen-AI uses modern technologies such as Artificial Intelligence, database management systems, and web-based applications to simplify healthcare operations for doctors, medical staff, administrators, and patients.

The proposed system helps healthcare organizations by:

- Digitizing hospital operations
- Reducing paperwork
- Automating healthcare workflows
- Improving patient data management
- Providing AI-assisted prescription support
- Enhancing medicine inventory tracking

The system is designed to provide a faster, smarter, and more efficient healthcare.

7.2 Features of Proposed System

1. Digital Patient Registration

The system allows hospital staff to register patients digitally and maintain secure patient records.

Benefits

- Faster registration process
- Reduced paperwork
- Easy patient data retrieval
- Centralized patient database

2. AI-Assisted Prescription System

Doctors can generate digital prescriptions with AI assistance for better accuracy and efficiency.

Benefits

- Reduces manual prescription errors
- Improves prescription readability
- Maintains prescription history
- Supports intelligent healthcare assistance
- Support healthcare for Diagnostics.

3. Electronic Medical Record Management

The system stores patient medical history, diagnosis, reports, and treatment details digitally.

Benefits

- Secure record storage
-

Project Title:- MediGen-AI

- **Fast access to medical history**
- **Reduced risk of data loss**
- **Better patient monitoring**

4. Doctor and Staff Management

Administrators can manage doctor profiles, medical staff details, and department information.

Benefits

- **Improved hospital administration**
- **Better workflow coordination**
- **Easy staff monitoring**

5. Medicine Inventory Management

The system tracks medicine stock, availability, expiry dates, and procurement processes.

Benefits

- **Prevents medicine shortages**
- **Reduces procurement delays**
- **Improves stock monitoring**
- **Supports real-time inventory updates**

6. Appointment Management System

Patients and staff can manage appointments digitally.

Benefits

- **Reduces appointment conflicts**
- **Minimizes patient waiting time**
- **Improves scheduling efficiency**

7. Secure Login and Authentication

The system provides role-based login access for administrators, doctors, and staff.

Benefits

- **Protects sensitive medical data**
- **Improves system security**
- **Prevents unauthorized access**
- **Improves healthcare security**

8. Report Generation

The system can generate healthcare reports, patient summaries, and inventory reports automatically.

Reports generation is the process of using a tool for creating reports primarily for business users. To generate the report, you need to create a report definition, including what data to retrieve, where to get it, and how to display it.

You may also find that people more often refer to automatic report generation or sometimes refer to semi-report generation. What do they mean? I will talk about the difference later.

7.3 Advantages of Proposed System

The proposed MediGen-AI system provides several advantages over traditional healthcare management systems.

1. Reduces Manual Errors

AI-assisted digital processes minimize prescription and data entry mistakes.

2. Improves Workflow Efficiency

The integration of different healthcare modules improves coordination between departments.

3. Faster Access to Records

Doctors and staff can quickly access patient history and medical information.

4. Centralized Healthcare Management

All healthcare operations are managed from a single platform.

5. Enhanced Data Security

Digital storage and authentication mechanisms improve medical data security

6. Better Inventory Control

Medicine stock and procurement processes are monitored efficiently.

7. Saves Time and Cost

Automation reduces administrative workload and operational costs.

8. Improves Patient Care

Faster services and accurate records help improve treatment quality.

9. Supports Digital Healthcare Transformation

The system promotes modern healthcare technology adoption.

10. Scalable and Future-Ready

The system can be expanded with future technologies such as telemedicine, AI analytics, and cloud integration.

Conclusion

The system overcomes the limitations of traditional healthcare management systems and provides an efficient, secure, and intelligent environment for hospitals, clinics, doctors, and patients.

The Proposed System, MediGen-AI Hospital Management System, was designed to overcome the limitations of traditional healthcare management systems by introducing Artificial Intelligence, automation, secure database management, and digital healthcare workflows.

The system successfully improves hospital efficiency, reduces manual workload, enhances patient care, increases healthcare accessibility, and provides secure.

The proposed system offers a robust, automated solution that eliminates the errors and inefficiencies of traditional manual workflows. By streamlining operations, the system guarantees higher productivity, superior data accuracy, and secure, centralized information management that scales to meet future demands.

A proposed system is a documented framework for a new project or software solution. It outlines a project's purpose, scope, and expected outcomes, typically serving as a pitch to secure stakeholder approval or funding by highlighting exactly how the new solution will fix problems present in an older process

CHAPTER 8: FEASIBILITY STUDY

8.1 Technical Feasibility

Technical feasibility refers to the evaluation of the technical resources and technologies required for the successful development and implementation of the proposed system. The proposed MediGen-AI system is technically feasible because it uses modern and widely available technologies such as Python, Flask, MySQL, HTML, CSS, and JavaScript.

The system can be developed using standard hardware and software resources available in educational institutions and healthcare organizations. The required development tools, frameworks, and database systems are reliable, scalable, and easy to maintain.

Technical Requirements

- **Computer system with internet connectivity**
- **Web browser support**
- **Database management system**
- **Backend and frontend development tools**
- **Secure authentication mechanisms**
- **Advantages of Technical Feasibility**
- **Easy system development and deployment**
- **Availability of modern technologies**
- **Scalable and flexible architecture**
- **Secure and reliable system performance**

The technical analysis shows that the proposed system can be successfully implemented using existing technologies and infrastructure.

8.2 Economic Feasibility

Economic feasibility determines whether the proposed system is financially beneficial and cost-effective. The development cost of the proposed system is relatively low because open-source technologies and software tools are used.

The implementation of the system can reduce paperwork, administrative workload, and operational inefficiencies, which helps save time and resources for healthcare organizations.

Project Title:- MediGen-AI

Economic Benefits

- **Reduced paperwork costs**
- **Lower administrative expenses**
- **Reduced manual errors**
- **Faster healthcare operations**
- **Improved resource management**
- **Estimated Costs**
- **Software development cost**
- **System maintenance cost**
- **Hardware and hosting cost**
- **Training and implementation cost**

Since the long-term benefits are greater than the implementation cost, the proposed system is economically feasible.

8.3 Operational Feasibility

Operational feasibility evaluates whether the proposed system can operate efficiently within the healthcare environment and whether users can easily adopt the system.

The proposed MediGen-AI system is designed with a simple and user-friendly interface for doctors, administrators, medical staff, and patients. The system automates several healthcare operations, making it easier for users to manage daily tasks.

- **Operational Advantages**
- **Easy-to-use interface**
- **Faster patient management**
- **Improved workflow coordination**
- **Better communication between departments**
- **Reduced manual workload**

The system can be integrated into hospital operations without major difficulties, making it operationally feasible.

8.4 Legal Feasibility

Legal feasibility ensures that the proposed system follows legal requirements, healthcare regulations, and data protection policies.

Legal feasibility is an evaluation to determine if a proposed project, policy, or business model aligns with existing laws and regulations. It assesses required licenses, intellectual property, contracts, and compliance constraints to ensure sustainable operations and prevent future litigation or regulatory shutdowns.

Ignoring legal feasibility can lead to millions in unplanned costs, project delays, or business shutdowns. Conducting this analysis early guarantees that your initiative operates securely and avoids legal conflicts that could destroy its timeline.

The proposed system is designed to maintain patient privacy and secure medical data through authentication and controlled access mechanisms. Healthcare information is sensitive and must be protected from unauthorized access.

- Legal Considerations
- Protection of patient medical records
- Secure user authentication
- Data privacy and confidentiality
- Controlled system access
- Compliance with healthcare regulations

Legal feasibility in software determines if a software product can be legally developed, distributed, and operated without violating laws, regulations, or third-party rights. Failure to assess this early can result in forced shutdowns, expensive lawsuits, or the inability to launch your product.

Legal feasibility in software engineering is the process of evaluating a proposed project for potential legal and regulatory risks before development begins. It ensures the software complies with all applicable laws—such as data protection and intellectual property regulations—to prevent costly litigation, fines, or operational

The system aims to follow ethical and legal standards related to digital healthcare management and medical information security.

The Feasibility Study of the MediGen-AI Hospital Management System confirmed that the project is technically, economically, operationally, legally, organizationally, and schedule-wise feasible. The analysis demonstrated that the proposed system can successfully improve hospital operations, healthcare workflow efficiency, patient management, prescription handling, and healthcare data security.

Conclusion

The feasibility study shows that the proposed MediGen-AI system is technically, economically, operationally, and legally feasible. The system can be successfully implemented using available technologies and resources while providing secure, efficient, and cost-effective healthcare management solutions.

CHAPTER 9: REQUIREMENT ANALYSIS

9.1 Hardware Requirements

- **Processor: Intel i3 or higher**
- **RAM: Minimum 4GB**
- **Storage: 20GB free space**
- **Internet Connection**

9.2 Software Requirements

- **Operating System: Windows/Linux**
- **Python 3.x**
- **Flask Framework**
- **SQLite Database**
- **VS Code**
- **Bootstrap 5**
- **Browser: Chrome/Firefox**

9.3 Additional Software Tools

- **XAMPP or WAMP Server**
- **Git and GitHub**
- **Database Management Tools**

9.4 Functional Requirements

Functional requirements define the main features and functionalities that the proposed system must perform.

Main Functional Requirements

1. User Authentication

- **Secure login system for administrators, doctors, and staff.**
 - **Password protection and access control.**
-

Project Title:- MediGen-AI

2. Patient Registration

- **Add new patient details.**
- **Update patient information.**
- **Search patient records.**

3. Doctor Management

- **Add and manage doctor profiles**
- **Assign doctors to patients**
- **Manage specialization details**

4. Prescription Management

- **Generate digital prescriptions**
- **Store prescription history**
- **AI-assisted prescription support**

5. Medical Record Management

- **Store patient medical history.**
- **Upload reports and treatment details.**
- **Access previous records.**

6. Medicine Inventory Management

- **Track medicine stock**
 - **Monitor expiry dates**
 - **Manage procurement records**
-

Project Title:- MediGen-AI

7. Appointment Management

- **Schedule appointments**
- **Update appointment status**
- **Manage patient queues**

8. Report Generation

- **Generate patient reports**
- **Generate inventory reports**
- **Generate healthcare analytics**

These functionalities help automate healthcare operations and improve hospital workflow efficiency.

9.4 Non-Functional Requirements

Non-functional requirements describe the quality, performance, security, and usability aspects of the system.

1. Performance Requirements

- **Fast system response time**
- **Efficient database operations**
- **Smooth user interaction**

2. Security Requirements

- **Secure user authentication**
- **Data encryption and protection**

3. Reliability Requirements

- **Continuous system availability**
 - **Accurate data processing**
-

Project Title:- MediGen-AI

- **Reliable database management**

4. Usability Requirements

- **User-friendly interface**
- **Easy navigation**
- **Simple healthcare workflow management**

5. Scalability Requirements

- **Ability to add new features**
- **Support for increasing users and data**

6. Maintainability Requirements

- **Easy software maintenance**
- **Simple debugging and updates**
- **Modular system architecture**

7. Compatibility Requirements

- **Support for multiple operating systems**
- **Browser compatibility**
- **Internet-based accessibility**

Conclusion

Requirement analysis is the foundational blueprint of the Software Development Life Cycle (SDLC) that directly dictates a project's ultimate success or failure. It acts as a critical communication bridge, translating ambiguous stakeholder expectations into structured, actionable, and testable engineering specifications. Skipping or rushing this phase invariably results in severe budget overruns, missed deadlines, and a final product that fails to satisfy market needs.

Defines Project Scope: It establishes a clear baseline that effectively prevents scope creep. **Mitigates Development Risks:** Identifying inconsistencies early dramatically slashes the financial cost of rewriting code. **Aligns Misaligned Stakeholders:** Resolves conflicting demands through rigorous negotiation and deliberate feature prioritization. **Ensures Verifiable Quality:** Outputs clear metrics that enable QA teams to build precise test scenarios.

Requirement analysis is an important step in the development of the proposed MediGen-AI system. The analysis identifies the hardware, software, functional, and non-functional requirements necessary for successful implementation.

CHAPTER 10: SYSTEM ARCHITECTURE

10.1 Presentation Layer

The Presentation Layer is the front-end layer of the proposed MediGen-AI system. It is responsible for providing interaction between users and the system through graphical user interfaces.

This layer allows administrators, doctors, medical staff, and patients to access the system functionalities using web browsers or digital devices.

Main Functions of Presentation Layer

User login and authentication

- Patient registration interface
- Doctor dashboard
- Appointment management interface
- Prescription generation interface
- Inventory management screens
- Report viewing and printing

Technologies Used

- HTML
- CSS
- JavaScript
- Bootstrap

Advantages

- User-friendly interface
- Easy navigation
- Fast interaction with the system
- Improved user experience

The Presentation Layer acts as the communication bridge between users and the application system.

The System Architecture of the MediGen-AI Hospital Management System was designed to provide a secure, scalable, intelligent, and efficient healthcare management platform. The layered architecture approach helped integrate frontend interfaces, backend processing, AI workflows, database management, and security mechanisms into a unified system.

The architecture successfully supports digital healthcare transformation by improving workflow efficiency, healthcare automation, AI-assisted prescription generation, and secure healthcare data management. Overall, the architecture provides a strong foundation for future smart healthcare system expansion and modernization.

System Architecture represents the overall structural design of the MediGen-AI Hospital Management System. It defines how different components, modules, technologies, databases, and users interact with each other to perform healthcare management operations efficiently.

10.2 Application Layer

The Application Layer contains the main business logic and processing functionalities of the system. It processes user requests, performs healthcare operations, and manages communication between the Presentation Layer and the Data Layer.

This layer controls the major functionalities of the healthcare management system.

Main Functions of Application Layer

- User authentication and authorization
- Patient data processing
- Prescription management
- Appointment scheduling
- Doctor and staff management
- Inventory processing
- Report generation

Technologies Used

- Python
- Flask/Django Framework

Advantage

- Efficient healthcare workflow processing
- Centralized system control
- Improved data handling
- Faster request processing

The Application Layer ensures smooth execution of healthcare operations and business logic.

The Application Layer is the core functional layer of the MediGen-AI Hospital Management System. It acts as an intermediate layer between the Presentation Layer (User Interface) and the Database Layer. This layer contains the business logic, processing mechanisms, workflow management, and communication handling required for system operations.

The Application Layer processes user requests, manages healthcare workflows, handles authentication, controls data processing, and coordinates interactions between different modules of the system. It is responsible for implementing all major functionalities of the hospital management system.

The MediGen-AI system uses the Python Django Framework for implementing the application layer because Django provides secure, scalable, and efficient backend development support.

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Project Title:- MediGen-AI

10.4 Data Layer

The Data Layer is responsible for storing, managing, and retrieving healthcare information from the database system.

All patient records, prescriptions, medicine details, appointment data, and user information are securely stored in this layer.

Main Functions of Data Layer

- Store patient records
- Manage doctor and staff information
- Store prescriptions and treatment history
- Manage medicine inventory data
- Store appointment details
- Generate reports from stored data

Technologies Used

- MySQL Database

Advantages

- Centralized data storage
- Secure medical record management
- Fast data retrieval
- Reduced data redundancy

The Data Layer ensures data consistency, integrity, and secure healthcare information management.

The Data Layer is one of the most important components of the MediGen-AI Hospital Management System. It is responsible for storing, managing, retrieving, and securing all healthcare-related data used by the system. The data layer acts as the foundation of the application because all hospital operations depend on accurate and organized data management.

The MediGen-AI system uses a centralized database structure to maintain patient records, doctor information, appointments, prescriptions, OPD details, reports, and administrative data efficiently. The data layer ensures fast data access, data consistency, secure storage, and reliable healthcare information management.

The Data Layer plays a critical role in the MediGen-AI Hospital Management System by managing healthcare information securely, efficiently, and systematically. It provides centralized database management, fast data retrieval, secure healthcare storage, and reliable support for AI-assisted healthcare workflows.

The use of relational database systems, normalization techniques, and secure database management practices helped create a scalable, maintainable, and efficient healthcare data management system. Overall, the data layer serves as the backbone for reliable healthcare information processing and digital hospital management.

The Data Layer is the backbone of the MediGen-AI Hospital Management System because it manages all healthcare information in a secure, organized, and centralized manner. Through relational database management, normalization techniques, secure authentication, and efficient database communication, the system successfully supports digital healthcare transformation and intelligent hospital workflow management.

10.4 AI Layer

The AI Layer is one of the most important components of the proposed MediGen-AI system. This layer provides intelligent healthcare assistance and automation using Artificial Intelligence technologies.

The AI Layer helps improve healthcare decision-making and reduces manual errors.

Main Functions of AI Layer

- AI-assisted prescription support
- Healthcare recommendation system
- Intelligent data analysis
- Medical pattern recognition
- Automated healthcare assistance

Technologies Used

- Machine Learning
- Artificial Intelligence Algorithms
- Python AI Libraries

Advantages

- Reduces prescription errors
- Improves healthcare efficiency
- Supports intelligent decision-making
- Enhances automation

The AI Layer makes the system smarter and more efficient compared to traditional healthcare systems.

10.5 E-Pharmacy integration

The E-Pharmacy Integration module connects the healthcare management system with digital pharmacy services and medicine inventory management.

This integration allows doctors, medical staff, and pharmacies to manage medicine availability, procurement, and prescription processing efficiently.

Main Functions of E-Pharmacy Integration

- Medicine stock monitoring
- Prescription-based medicine management
- Real-time medicine availability tracking
- Medicine procurement support
- Expiry date monitoring

Advantages

- Faster medicine procurement
- Reduced medicine shortages
- Improved inventory management
- Better coordination between hospitals and pharmacies

Benefits for Patients

- Faster medicine access
 - Improved prescription accuracy
 - Reduced waiting time
-

Project Title:- MediGen-AI

The E-Pharmacy integration improves healthcare workflow and ensures efficient medicine management within the system.

Conclusion

The system architecture of the proposed MediGen-AI system is designed using multiple integrated layers that work together to provide secure, efficient, and intelligent healthcare management services.

The Presentation Layer, Application Layer, Data Layer, AI Layer, and E-Pharmacy Integration collectively ensure smooth healthcare operations, better patient management, improved workflow automation, and secure medical data handling.

System architecture in software engineering is the high-level conceptual blueprint that defines a system's structure, components, and behavior. It acts as the foundational framework, detailing how hardware, software, data, and user interfaces interact to ensure the application is scalable, secure, and maintainable.

System architecture in software engineering serves as the foundational blueprint for a software application. It determines a system's long-term scalability, maintainability, and adaptability.

Ultimately, system architecture bridges the gap between abstract business requirements and actionable technical implementation. It acts as a single source of truth that aligns developers, stakeholders, and operations teams, ensuring the final product is reliable, efficient, and resilient.

System architecture in software engineering serves as the foundational blueprint for a software project, establishing how hardware, software, data, and user interfaces.

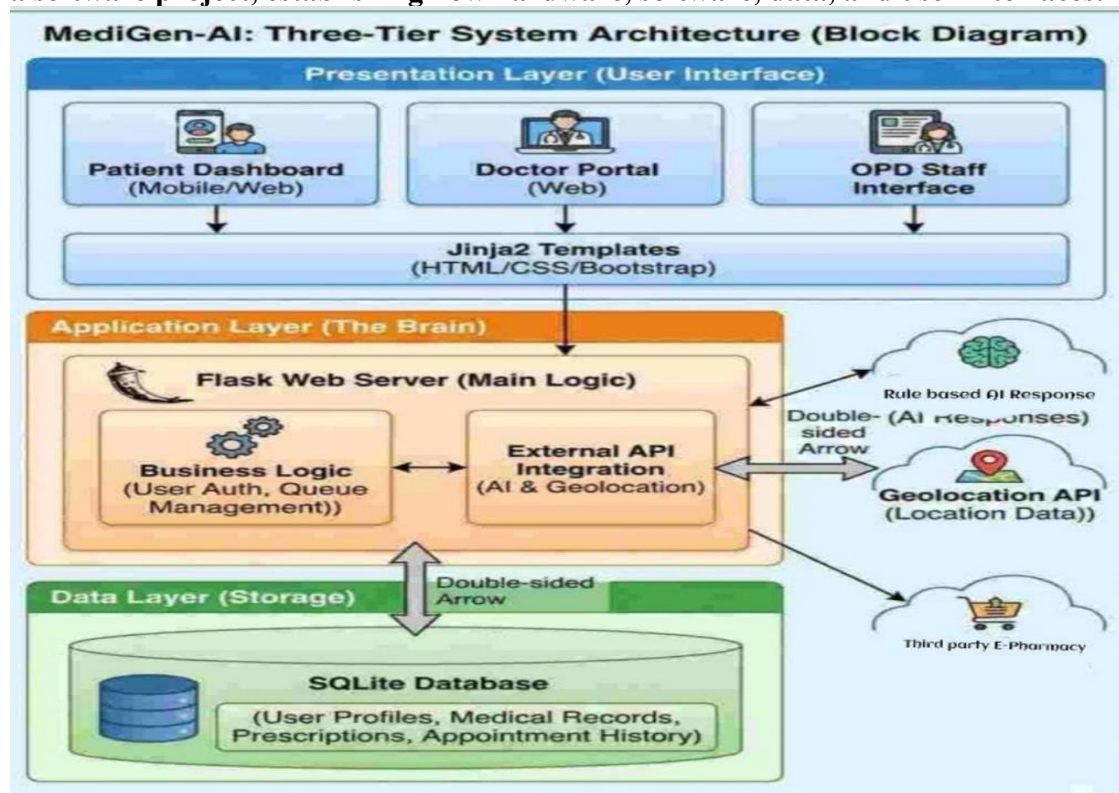


Figure 1: System Architecture of MediGen-AI

CHAPTER 11: MODULES DESCRIPTION

11.1 OPD Module

The OPD (Out Patient Department) Module is one of the main modules of the proposed MediGen-AI system. This module manages outpatient services such as patient registration, appointment scheduling, doctor consultation, and prescription handling.

The OPD module helps hospitals and clinics manage patient flow efficiently and reduces manual paperwork.

Main Functions of OPD Module

- **Patient registration**
- **Appointment scheduling**
- **OPD token management**
- **Doctor assignment**
- **Prescription generation**
- **OPD report generation**
- **Symptom entry**
- **Queue management**
- **Appointment handling**

Advantages

- **Faster outpatient management**
- **Reduced patient waiting time**
- **Improved workflow coordination**
- **Better patient record maintenance**
- **Better patient management**

The OPD module improves the efficiency of outpatient healthcare services.

- **Patient registration**
 - **Symptom entry**
 - **Queue management**
-

11.2 Doctor Module

The Doctor Module is designed to help doctors manage patient consultations, prescriptions, treatment records, and medical history digitally.

Doctors can securely access patient information and provide accurate treatment using AI-assisted healthcare support.

Objectives of Doctor Module

- **To simplify doctor-patient interaction**
- **To generate digital prescriptions**
- **To maintain patient treatment history**
- **To improve healthcare efficiency**
- **To reduce manual paperwork**
- **To support intelligent healthcare assistance**

Main Functions of Doctor Module

- **Doctor login and authentication**
- **View patient records**
- **Generate digital prescriptions**
- **Update treatment details**
- **Access medical history**
- **Appointment management**
- **AI-generated prescription review**
- **Prescription editing**
- **Digital signature**

Advantages

- **Improved prescription accuracy**
 - **Faster access to patient history**
 - **Reduced paperwork**
 - **Better treatment management**
-

11.3 Patient Module

The Patient Module allows patients to access healthcare services digitally through the system.

Patients can view appointment details, prescriptions, treatment history, and medical reports using their patient accounts.

The Patient Module plays a vital role in the MediGen-AI Hospital Management System by managing patient information digitally and efficiently. The module improves healthcare workflow, reduces paperwork, enhances patient data accessibility, and supports secure healthcare record management.

The implementation of centralized patient management and automated workflows significantly improves hospital efficiency and patient care quality. Overall, the Patient Module provides a secure, scalable, and intelligent solution for modern digital healthcare management.

The Patient Module is one of the most important components of the MediGen-AI Hospital Management System. This module is designed to manage all patient-related activities efficiently through digital healthcare management and automated workflows.

The Patient Module helps hospitals maintain organized patient records, manage appointments, store medical histories, and improve communication between patients, doctors, and hospital staff. It reduces manual paperwork, improves data accessibility, and supports faster healthcare services.

Objectives of Patient Module

- **To simplify patient registration and management**
- **To provide digital access to healthcare services**
- **To maintain secure medical records**
- **To improve patient-doctor communication**
- **To reduce paperwork and waiting time**
- **To enhance patient experience**
- **Prescription viewing**
- **PDF download**
- **Medicine ordering**
- **Medical history tracking**

Main Functions of Patient Module

- **Patient registration and login**
 - **View appointment status**
 - **Access digital prescriptions**
 - **View medical history**
 - **Download healthcare reports**
 - **Receive healthcare notifications advantages**
 - **Easy access to healthcare services**
 - **Improved communication with doctors**
-

CHAPTER 12: TECHNOLOGY STACK

The proposed MediGen-AI system uses modern web technologies, database systems, and Artificial Intelligence tools to provide efficient and secure healthcare management.

1. Frontend Technologies

Frontend technologies are used to design the user interface and improve user interaction with the system.

Technologies Used

- HTML, CSS, JavaScript, Bootstrap

Purpose

- Designing web pages
- Creating responsive user interfaces
- Improving user experience
- Supporting interactive healthcare dashboard

2. Backend Technologies

Backend technologies handle business logic, server-side processing, and communication between the frontend and database.

Technologies Used

- Python
- Flask/Django Framework

Purpose

- Processing healthcare operations
- Managing system functionalities
- Handling user authentication

3. Database Technologies

Database technologies are used to store and manage healthcare information securely.

Technologies Used

- MySQL

Purpose

- Storing patient records
- Managing doctor and staff information
- Maintaining prescriptions and inventory data
- Supporting secure data retrieval

4. Artificial Intelligence Technologies

AI technologies are used to provide intelligent healthcare assistance and automation.

Technologies Used

- Rule based AI Python AI Libraries

Purpose

- AI-assisted prescription support
 - Intelligent healthcare recommendations
 - Healthcare workflow automation
-

Project Title:- MediGen-AI

5. Development Tools

Development tools are used for coding, testing, debugging, and project management.

Tools Used

- Visual Studio Code
- PyCharm
- Git and GitHub
- XAMPP/WAMP Server

Purpose

- Software development
- Version control
- Database management
- Local server hosting

6. Operating System Support

The proposed system supports multiple operating systems.

Supported Platforms

- Window, Linux, MacOS

7. Browser Support

The web application can run on modern web browsers.

Supported Browsers

- Google Chrome Mozilla Firefox Microsoft Edge

Advantages of the Selected Technology Stack

- Easy system development and maintenance
- Secure and scalable architecture
- Fast processing and data management
- Cross-platform compatibility

Conclusion

The technology stack selected for the proposed MediGen-AI system is modern, reliable, scalable, and suitable for healthcare management applications. The combination of frontend technologies, backend frameworks, database systems, and Artificial Intelligence tools helps provide an efficient, secure, and intelligent healthcare management solution.

The ideal technology stack balances innovation with stability, developer experience with performance, and current needs with long-term scalability. The right combination directly influences your application's speed, maintenance costs, and ability to handle user growth.

Ultimately, technology is just a means to an end. Choose tools that empower your team to build efficiently and maintainable while remaining adaptable to future industry shifts.

A technology stack is the foundational combination of tools, programming languages, and frameworks used to build a web or mobile application. It is divided into three main layers: the Front-End (user interface), Back-End (server logic), and Data Layer (databases).

Project Title:- MediGen-AI

Category	Technology	Purpose
Frontend	HTML5	Structure
Frontend	CSS3	Styling
Frontend	JavaScript ES6	Interactivity
Frontend	Bootstrap 5	Responsive UI
Backend	Python	Core programming
Backend	Flask	Web framework
Backend	Jinja2	Templating
Database	SQLite	Data storage
Database	SQLAlchemy	ORM
AI Suggested	Rule Based AI	AI prescription generation
Security	Bcrypt	Password hashing
Security	Flask-Login	Authentication
Integration	Geolocation API	Nearby pharmacy detection

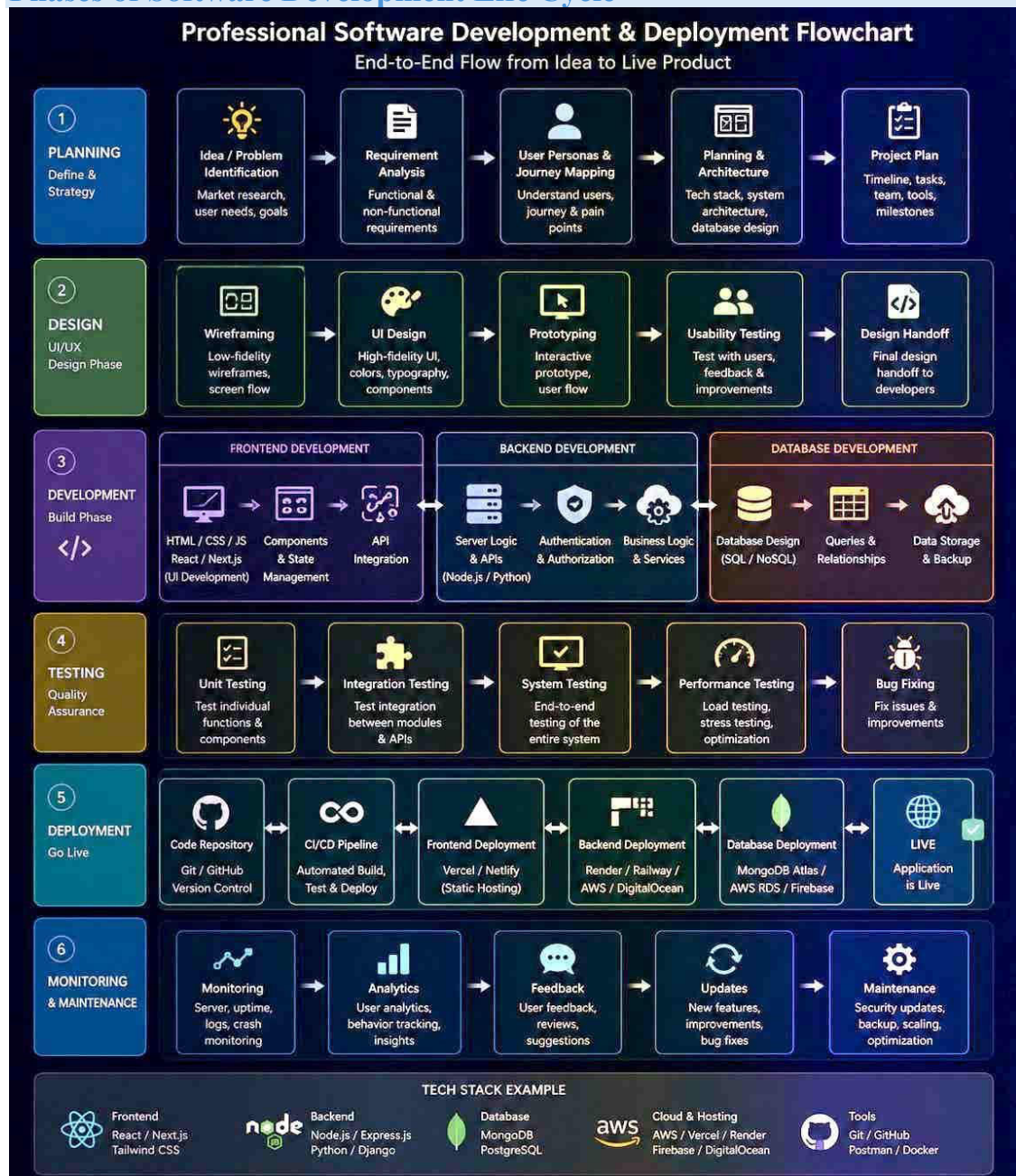
Project Title:- MediGen-AI

CHAPTER 13: SOFTWARE DEVELOPMENT LIFE CYCLE

The Software Development Life Cycle (SDLC) is a systematic process used for planning, designing, developing, testing, deploying, and maintaining software systems. The SDLC helps ensure that the proposed MediGen-AI system is developed in a structured, efficient, and reliable manner.

The SDLC model provides proper guidelines for software development and helps improve software quality, security, and maintainability.

Phases of Software Development Life Cycle



Project Title:- MediGen-AI

1. Requirement Analysis

Requirement Analysis is the first phase of the SDLC process. In this phase, the requirements of the healthcare management system are identified and analyzed.

Activities Performed

- **Understanding healthcare workflow**
- **Identifying system requirements**
- **Gathering user requirements**
- **Defining project objectives**

Output

Requirement Specification Document

The Requirement Specification Document (RSD) is a formal document that describes the functional and non-functional requirements of the proposed MediGen-AI system. This document defines the system objectives, features, hardware requirements, software requirements, and operational functionalities required for successful system development and implementation.

The Requirement Specification Document acts as a communication bridge between developers, users, administrators, and project stakeholders.

Objectives of Requirement Specification Document

- **To define system requirements clearly**
- **To provide guidance for system development**
- **To identify functional and non-functional requirements**
- **To improve project planning and implementation**
- **To ensure system quality and reliability**
- **To ensure system scalability.**
- **To ensure system stable.**

Functional Requirements

Functional requirements describe the major operations and functionalities performed by the system.

Main Functional Requirements

1. User Authentication

- **Secure login system**
- **Role-based access control**
- **Password management**

2. Patient Registration

Project Title:- MediGen-AI

- **Add patient details**
- **Update patient records**
- **Search patient information**

3. Doctor Management

- **Manage doctor profiles**
- **Assign doctors to patients**
- **Maintain specialization records**

4. Prescription Management

- **Generate digital prescriptions**
- **Store prescription history**
- **AI-assisted prescription support**

5. Appointment Management

- **Schedule appointments**
- **Manage appointment status**
- **View appointment history**

7. Report Generation

- **Generate healthcare reports**
- **Generate inventory reports**
- **Generate patient summaries**

Non-Functional Requirements

Non-functional requirements describe the quality and performance standards of the system.

1. Performance Requirements

- **Fast system response**
- **Efficient database management**
- **Smooth healthcare workflow processing**

2. Security Requirements

- **Secure user authentication**
-

Poject Title:- MediGen-AI

- **Data protection and privacy**
- **Role-based access control**

3. Reliability Requirements

- **Continuous system availability**
- **Accurate data processing**
- **Secure data storage**

4. Usability Requirements

- **User-friendly interface**
- **Easy navigation**
- **Simple healthcare management**

5. Scalability Requirements

- **Ability to add future features**
- **Support for increasing users and data**
- **Hardware Requirements**

Minimum Hardware Requirements

- **Intel Core i3 Processor or higher**
- **4 GB RAM**
- **500 GB Hard Disk**

Recommended Hardware Requirements

- **Intel Core i5 Processor or higher**
- **8 GB RAM**
- **SSD Storage**

Software Requirements development Software

- **Python, Flask/Django, MySQL .HTML, CSS, JavaScript**

Operating Systems

- **Windows, Linux, macOS**

Development Tools

- **Visual Studio Code, PyCharm, GitHub, XAMPP/WAMP Server**
-

Project Title:- MediGen-AI

System Users

The proposed MediGen-AI system supports multiple users.

Types of Users

- **Admin, Doctors, Medical Staff, Patients**

Advantages of Requirement Specification Document

- **Improves project planning**
- **Reduces development confusion**
- **Helps in proper system implementation**
- **Ensures software quality**
- **Improves communication among stakeholders**
- **Improves communication among patient and doctor**

Conclusion

The Requirement Specification Document (RSD) provides detailed information about the proposed MediGen-AI system requirements, functionalities, and technical specifications.

The document helps ensure that the system is developed according to user needs and healthcare management objectives while maintaining software quality, security, and efficiency.

A Software Requirements Specification (SRS) is a foundational document that outlines the purpose, behavior, features, and constraints of a software application. It serves as a clear roadmap and contract between stakeholders, developers, and QA teams, detailing exactly what the software needs to do and how it will perform by establishing exactly what needs to be built before coding begins. Aligns Teams: Keeps business sponsors, developers, and testers on the same page regarding project scope and expectations. Provides a Benchmark: Acts as the reference point for Quality Assurance (QA) teams to build test plans and verify success.

The System Requirements Document (SRD) provides a detailed overview of the hardware, software, security, database, and operational requirements of the MediGen-AI Hospital Management System. It acts as a foundational document for successful system development, implementation, testing, deployment, and future maintenance.

Overall, the SRD ensures that the system remains secure, scalable, reliable, and capable of supporting intelligent healthcare management operations efficiently.

The System Requirements Document (SRD) describes the functional and non-functional requirements of the MediGen-AI Hospital Management System. It defines the hardware, software, security, database, and operational requirements necessary for the successful implementation of the system.

13.2 Planning Phase

The Planning Phase is the second and one of the most important phases in the Software Development Life Cycle (SDLC) of the proposed MediGen-AI system.

In this phase, the project objectives, system requirements, resources, timeline, and development strategies are identified and planned properly. The planning phase helps ensure smooth project execution and reduces development risks.

The main purpose of this phase is to understand the healthcare problems, define project goals, and prepare a roadmap for successful system development.

Objectives of Planning Phase

- To identify healthcare management problems
- To define project goals and objectives
- To analyze project feasibility
- To estimate project cost and resources
- To prepare development strategies
- To create project timelines and schedules
- Activities Performed in Planning Phase

1. Problem Identification

The first step of planning is identifying the problems faced in traditional healthcare systems.

- Identified Problems
- Manual prescription errors
- Delayed medicine procurement
- Lack of digital medical records
- Workflow inefficiency
- Poor communication between departments
- Limited healthcare automation

2. Requirement Gathering

In this step, the system requirements are collected from healthcare processes.

This process is gathering user information and gathering requirement.

Gathering Requirement from software, it is useful for made a software made.

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Requirements Collected From

- **Doctors**
- **Medical Staff**
- **Administrators**
- **Patients**
- **Hospital Workflow**

Purpose

- **Understand user expectations**
- **Define system functionalities**
- **Improve healthcare workflow**

3. Feasibility Study

A feasibility study is performed to determine whether the proposed system can be developed successfully.

Types of Feasibility

- **Technical Feasibility**
- **Economic Feasibility**
- **Operational Feasibility**
- **Legal Feasibility**

Purpose

- **Analyze project viability**
- **Reduce implementation risks**
- **Ensure resource availability**

4. Resource Planning

Resource planning identifies the hardware, software, and human resources required for the project.

- **Resources Required**
- **Computers and internet connectivity**
- **Development software and tools**

Purpose

Project Title:- MediGen-AI

- **Ensure proper resource allocation**
- **Improve project management**

5. Time Planning

Time planning defines the project schedule and development timeline.

Activities Included

Requirement analysis timeline

- **Design phase schedule**
- **Development schedule**
- **Testing and deployment planning**

Purpose

- **Complete the project on time**
- **Improve task management**
- **Monitor project progress**

6. Risk Analysis

Risk analysis identifies possible challenges and development risks.

- **Possible Risks**
- **Data security issues**
- **Development delays**
- **System integration challenges**
- **Technical limitations**

Purpose

- **Prepare risk management strategies**
 - **Reduce project failure chances**
 - **Importance of Planning Phase**
 - **Improves project organization**
 - **Reduces development confusion**
 - **Helps manage project resources**
-

Project Title:- MediGen-AI

- Ensures better decision-making
- Supports successful project completion

Advantages of Planning Phase

- Better project management
- Clear understanding of objectives
- Reduced project risks
- Improved workflow efficiency
- Proper resource utilization
- Faster software development process

Conclusion

The Planning Phase plays an important role in the successful development of the proposed MediGen-AI system.

It helps identify healthcare challenges, define project goals, analyze feasibility, allocate resources, and prepare development strategies. Proper planning ensures efficient software development, better healthcare workflow management, and successful project implementation

The Planning Phase is a fundamental step in the development of the proposed MediGen-AI system. It helps identify healthcare management problems, define project objectives, gather system requirements, and prepare proper development strategies.

Through activities such as feasibility analysis, resource planning, time scheduling, and risk analysis, the planning phase ensures that the project can be developed efficiently and successfully. Proper planning reduces development risks, improves workflow management, and supports better decision-making throughout the software development process.

Overall, the Planning Phase provides a strong foundation for building a secure, efficient, and intelligent healthcare management system.

The Planning Phase is an essential part of the Software Development Life Cycle (SDLC) for the proposed MediGen-AI system. This phase provides a clear direction for the development process by identifying system requirements, healthcare challenges, available resources, and project objectives.

During the planning phase, various important activities such as requirement gathering, feasibility analysis, resource allocation, time scheduling, and risk management are performed. These activities help ensure that the project is technically possible, economically beneficial, operationally efficient, and legally secure.

The Patient Module is one of the most important components of the MediGen-AI Hospital Management System. This module is designed.

13.3 System Design

System Design is an important phase of software development that defines the architecture, components, modules, interfaces, and data flow of the proposed MediGen-AI system.

The purpose of system design is to convert system requirements into a structured and organized design that helps developers build the software efficiently. It provides a complete blueprint of how the healthcare management system will function.

The system design of MediGen-AI focuses on automation, digital healthcare management, secure data handling, and Artificial Intelligence integration.

Objectives of System Design

- **To design a structured healthcare management system**
- **To improve workflow efficiency**
- **To ensure secure data management**
- **To provide modular system architecture**
- **To support future scalability and maintenance**
- **Types of System Design**

1. Logical Design

Logical design describes the flow of information, system functionalities, and processing operations without focusing on physical implementation.

Logical Design Includes

- **Data flow diagrams**
- **System processes**
- **Functional modules**
- **User interactions**

Objectives

- **Define healthcare workflow**
- **Improve process understanding**
- **Support software planning**

2. Physical Design

Physical design describes the actual implementation of hardware, software, database systems, and network components.

Physical Design Includes

- **Hardware configuration**
 - **Database implementation**
 - **Server setup**
-

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System design document

The System Design Document (SDD) is a detailed document that describes the architecture, components, modules, database structure, interfaces, and design specifications of the proposed MediGen-AI system.

The purpose of the System Design Document is to provide a complete blueprint for software development and implementation. It explains how the system will function and how different modules will interact with each other.

Objectives of System Design Document

- To define system architecture and structure
- To describe system modules and functionalities
- To provide database and interface design
- To guide software development and implementation
- To ensure proper workflow management
- Components of System Design Document

1. System Architecture Design

The system architecture defines the overall structure of the healthcare management system.

Main Layers of Architecture

- Presentation Layer
- Application Layer
- Data Layer
- Rule based AI
- E-Pharmacy Integration Layer

Purpose

- Improves system organization
- Ensures secure communication between modules
- Supports efficient healthcare workflow

2. Database Design

Database design defines how healthcare data will be stored and managed within the system

- Main Database Tables
 - Patient Table
 - Doctor Table
 - Staff Table
-

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- **Appointment Table**
- **Prescription Table**
- **Inventory Table**
- **Admin Table**

Purpose

- **Secure medical record storage**
- **Fast data retrieval**
- **Centralized healthcare information management**

3. User Interface Design

The user interface design defines the screens, forms, and dashboards used by system users.

- **Main Interfaces**
- **Login Page**
- **Admin Dashboard**
- **Doctor Dashboard**
- **Patient Dashboard**
- **OPD Management Screen**
- **Inventory Management Screen**
- **Purpose**
- **User-friendly interaction**
- **Easy healthcare management**
- **Better user experience**

4. Module Design

The proposed system is divided into multiple modules for better management and workflow automation.

- **Main Modules**
- **OPD Module**
- **Doctor Module**
- **Patient Module**
- **Admin Module**
- **Inventory Module**
- **E-Pharmacy Module**

Modular design is a methodology that structures systems, products, or software into smaller, independent, and reusable units called "modules". By focusing on

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standardized interfaces, this approach enables easy interchangeability, scalability, and accelerated development across multiple industries

Purpose

- **Modular system development**
- **Easy maintenance and scalability**
- **Better workflow management**

5. Input Design

Input design defines how data is entered into the system.

- **Input Methods**
- **Patient registration forms**
- **Appointment forms**
- **Prescription entry forms**
- **Inventory update forms**

Objectives

- **Reduce data entry errors**
- **Improve data accuracy**
- **Simplify healthcare operations**

6. Output Design

Output design defines the reports and results generated by the system.

Main Outputs

- **Digital prescriptions**
- **Patient reports**
- **Inventory reports**
- **Appointment summaries**
- **Healthcare analytics**

Objectives

- **Fast report generation**
- **Improved healthcare analysis**
- **Better decision-making support**

7. Security Design

Security design ensures protection of healthcare information and user privacy.

Security is important for protect the patient data.

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Security design ensures all domain data.

Security Features

- Secure login authentication
- Password encryption
- Role-based access control
- Secure database management

Objectives

- Prevent unauthorized access
- Protect medical records
- Ensure healthcare data privacy
- Advantages of System Design Document
- Provides clear development guidelines
- Improves software quality
- Supports proper system planning
- Reduces development complexity
- Improves communication among developers

Conclusion

The System Design Document (SDD) provides a complete structural and functional overview of the proposed MediGen-AI system.

The document helps developers understand the architecture, database structure, modules, interfaces, inputs, outputs, and security mechanisms required for successful healthcare management system implementation.

The system design presented above provides a structured and scalable approach to solving the defined problem while balancing performance, reliability, and cost. By using a modular architecture (such as microservices or well-separated components), the system ensures flexibility for future enhancements and easier maintenance.

From a performance perspective, the inclusion of caching, load balancing, and horizontal scaling strategies allows the system to handle increasing user demand without significant degradation in response time. Reliability is strengthened through fault tolerance mechanisms like retries, circuit breakers, and backup strategies, ensuring minimal downtime and data loss.

From a performance perspective, the inclusion of caching, load balancing, and horizontal scaling strategies allows the system to handle increasing user demand without significant degradation in response time. Reliability is strengthened through fault tolerance mechanisms like retries, circuit breakers, and backup strategies, ensuring minimal downtime and data loss.

System design is the foundational blueprint that translates business requirements into scalable, reliable, and maintainable software architecture. It is a continuous process of making architectural trade-offs to ensure systems operate efficiently under growing demands

13.4 Coding and Development

Coding and Development is one of the most important phases in software engineering. It involves transforming system requirements and design specifications into a fully functional software application using programming languages, frameworks, databases, and development tools. In this phase, all modules of the project are implemented according to the system architecture and user requirements.

For the MediGen-AI Hospital Management System, the coding and development phase focuses on building an intelligent, secure, and scalable healthcare management platform capable of handling patient records, doctor management, OPD services, AI-based prescription generation, medicine inventory, and administrative operations.

The development process follows a modular and structured approach to ensure maintainability, reliability, scalability, and high system performance.

2. Objectives of Coding and Development

The major objectives of the coding and development phase are:

- To convert system design into executable software.**
- To develop user-friendly interfaces for different users.**
- To implement secure database connectivity.**
- To automate hospital management activities.**
- To reduce manual paperwork and human errors.**
- To integrate Artificial Intelligence for prescription assistance.**
- To ensure efficient communication between modules.**
- To provide real-time access to hospital records.**
- To maintain data security and confidentiality.**

3. Development Methodology

The project follows the Software Development Life Cycle (SDLC) methodology for systematic software development.

Phases of Development

3.1 Requirement Analysis

In this phase, all project requirements were collected and analyzed. Functional and non-functional requirements were identified based on hospital workflow and user needs.

3.2 System Design

The architecture, database schema, user interfaces, data flow diagrams, and module structures were designed before implementation.

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3.3 Coding Phase

The actual software implementation was performed using Python, Django, HTML, CSS, JavaScript, and SQLite/MySQL.

3.4 Testing Phase

Each module was tested individually and collectively to identify and remove errors.

3.5 Deployment Phase

The project was deployed on a local server for execution and performance testing.

3.6 Maintenance Phase

Future updates, bug fixing, and performance optimization are included in this phase.

4. Technologies Used in Development

4.1 Frontend Technologies

HTML (HyperText Markup Language)

HTML is used for creating the structure of web pages.

CSS (Cascading Style Sheets)

CSS is used for designing and styling the user interface.

JavaScript

JavaScript provides interactivity and dynamic functionality.

4.2 Backend Technologies

Python

Python is the primary programming language used for backend logic and AI functionalities.

Django Framework

Django is used for rapid web application development, authentication, database operations, and routing.

4.3 Database Technology

SQLite / MySQL

The database stores patient records, doctor details, prescriptions, OPD reports, medicines, appointments, and administrative information.

5. Development Environment

The following tools and software were used during development:

Poject Title:- MediGen-AI

Component	Technology Used
Programming Language	Python
Framework	Django
Frontend	HTML, CSS, JavaScript
Database	SQLite/MySQL
IDE	VS Code
Operating System	Windows
Version Control	Git
Browser	Chrome

6. System Architecture During Development

The project follows a layered architecture:

- **Presentation Layer**

Handles user interaction through web pages and dashboards.

- **Application Layer**

Processes business logic, authentication, AI processing, and request handling.

- **Database Layer**

Stores and manages all hospital-related information securely.

7. Module-Wise Development

7.1 Patient Module

The Patient Module allows registration and management of patient information.

Features

- **Patient Registration**
-

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- **ABHA Number Management**
- **Patient History**
- **Appointment Booking**
- **Prescription Viewing**

7.2 Doctor Module

The Doctor Module provides doctors with facilities to manage patients and prescriptions.

Features

- **Doctor Dashboard**
- **Patient Diagnosis**
- **AI Prescription Generation**
- **OPD Management**
- **Report Access**

7.3 OPD Module

This module manages outpatient department activities.

Features

- **OPD Registration**
- **Doctor Assignment**
- **Patient Queue Management**
- **Visit History**

7.4 Admin Module

The Admin Module controls overall system activities.

Features

- **User Management**
 - **Doctor Management**
 - **Staff Monitoring**
 - **System Reports**
 - **Database Management**
-

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8. AI Integration in Development

Artificial Intelligence plays a major role in the MediGen-AI system.

AI Functionalities

- **Smart Prescription Suggestions**
- **Symptom Analysis**
- **Medicine Recommendations**
- **Error Reduction in Prescriptions**

The AI logic is implemented using Python-based intelligent processing techniques

9. Database Development

The database was designed using relational database concepts.

Main Tables

- **Patient Table**
- **Doctor Table**
- **Prescription Table**
- **Appointment Table**
- **OPD Table**
- **Medicine Table**
- **Admin Table**
- **Database Features**
- **Data Integrity**
- **Fast Retrieval**
- **Secure Storage**

10. User Interface Development

The user interface was designed to be:

- **Responsive**
 - **Simple**
 - **User-Friendly**
-

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- **Interactive**
- **Mobile Compatible**

Bootstrap and CSS styling techniques were used to improve appearance and usability.

11. Security Implementation

Security mechanisms implemented in the project include:

- **User Authentication**
- **Password Encryption**
- **Session Management**
- **Role-Based Access Control**
- **Secure Database Connectivity**

These features help protect sensitive medical information.

12. Testing During Development

Different testing techniques were applied:

- **Unit Testing**

Testing individual modules separately.

- **Integration Testing**

Testing communication between modules.

- **System Testing**

Testing complete application functionality.

- **User Acceptance Testing**

Ensuring the system meets user requirements.

Ensuring the system is stable.

Ensuring software and system is stable.

13. Challenges Faced During Development

- **Some challenges encountered include:**
 - **Database synchronization issues**
 - **UI responsiveness problems**
-

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- **AI prescription accuracy**
- **Session management errors**
- **Data validation handling**

These issues were resolved through debugging and optimization techniques.

14. Advantages of the Developed System

- **Reduces manual paperwork**
- **Improves hospital efficiency**
- **Faster patient management**
- **Accurate record maintenance**
- **AI-assisted prescription generation**
- **Improved healthcare workflow**
- **Better administrative control**

15. Future Enhancement

- **Future improvements may include:**
- **Cloud Database Integration**
- **Mobile Application Support**
- **Video Consultation**
- **Online Payment Gateway**
- **Advanced AI Diagnosis**
- **Multi-Hospital Connectivity**

16. Conclusion

The Coding and Development phase successfully transformed the system design into a fully functional hospital management application. The MediGen-AI system provides an intelligent, secure, and efficient healthcare management platform that automates hospital operations and improves overall productivity. Proper coding standards, modular development, database management, and AI integration helped. The proposed system offers a robust, automated solution that eliminates the errors and inefficiencies of traditional manual workflows. By streamlining operations, the system guarantees higher productivity, superior data accuracy, and secure, centralized information management that scales to meet future demands.

13.5 Testing

Testing is one of the most important phases in software development. It is the process of evaluating and verifying whether the developed system meets the specified requirements and functions correctly without errors. The primary objective of testing is to identify bugs, ensure software quality, improve performance, and verify that the application works according to user expectations.

In the MediGen-AI Hospital Management System, testing was performed to ensure that all modules such as patient registration, doctor management, OPD handling, AI prescription generation, and database operations function accurately, securely, and efficiently.

The testing process helps improve system reliability, usability, security, and overall performance before deployment.

Objectives of Testing

The major objectives of software testing are:

- To identify and remove software bugs and errors.
- To verify that the system meets functional requirements.
- To ensure data accuracy and consistency.
- To improve system reliability and stability.
- To validate security mechanisms.
- To ensure proper integration among modules.
- To evaluate system performance under different conditions.
- To increase user satisfaction and software quality.

3. Importance of Testing

- Testing is essential because it:
- Reduces software failures.
- Prevents data loss and security risks.
- Ensures smooth system performance.
- Improves user confidence.
- Enhances software maintainability.
- Detects errors before deployment.
- Ensures system compatibility.

In healthcare systems like MediGen-AI, testing is critical because incorrect data or system failures may affect patient care and hospital operations.

4. Types of Testing Used

Several testing techniques were used during the development of the project.

4.1 Unit Testing

Unit Testing is performed to test individual modules or components independently.

Purpose

To verify that each module performs correctly according to its functionality.

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Modules Tested

- **Patient Registration Module**
- **Doctor Login Module**
- **Prescription Module**
- **OPD Management Module**
- **Admin Dashboard**

Advantages

- **Easy bug identification**
- **Improves code quality**
- **Simplifies debugging**
- **Bug finds easy.**
- **And easy to debug**

4.2 Integration Testing

Integration Testing verifies the communication and interaction between different modules.

Example

- **Patient data transferring to Doctor Dashboard**
- **Prescription data storing in database**
- **OPD records updating automatically**

Objectives

- **Ensure smooth data flow**
- **Detect interface-related errors**
- **Validate module connectivity**

4.3 System Testing

System Testing evaluates the complete application as a whole.

Testing Areas

- **Complete workflow execution**
- **Database operations**
- **Authentication process**
- **AI prescription generation**
- **User dashboard functionality**

Outcome

The entire system worked according to project requirements

4.4 User Acceptance Testing (UAT)

User Acceptance Testing is performed to verify whether the system satisfies user requirements.

Users Involved

- **Admin**
 - **Doctors**
 - **Medical Staff**
 - **Objectives**
 - **Verify usability**
 - **Ensure user satisfaction**
 - **Validate practical functionality**
-

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4.5 Validation Testing

Validation Testing ensures that the software performs according to user expectations.

Examples

Required field validation

- Correct data input checking
- Email and phone validation
- Duplicate record prevention

4.6 Security Testing

Security Testing was performed to protect sensitive hospital data.

Security Features Tested

- Login authentication
- Password protection
- Session handling
- Role-based access control
- Unauthorized access prevention

4.7 Performance Testing

Performance Testing checks system efficiency under different workloads.

Parameters Tested

- Page loading speed
- Database response time
- Multi-user handling
- System stability

5. Testing Environment

The project was tested in the following environment:

Component	Environment
Operating System	Windows
Browser	Google Chrome
Backend Framework	Django
Programming Language	Python
Database	SQLite/MySQL
IDE	VS Code

6. Test Cases

6.1 Patient Registration Test Case

Test Case ID TC001

- Module Patient Registration
 - Objective Verify patient registration
 - Input Valid patient details
 - Expected Result Patient registered successfully
 - Actual Result Successfully registered
 - Status Pass
-

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6.2 Prescription Generation Test Case

- Test Case ID TC003
- Module AI Prescription
- Objective Verify prescription generation
- Input Symptoms and diagnosis
- Expected Result Prescription generated
- Actual Result Generated successfully
- Status Pass

6.3 Doctor Assignment Test Case

- Test Case ID TC004
- Module OPD Management
- Objective Verify doctor assignment
- Input Patient and doctor details
- Expected Result Doctor assigned successfully
- Actual Result Assignment successful
- Status Pass

7. Errors Identified During Testing

Some errors found during testing were:

- Duplicate patient entries
- Session expiration issues
- Incorrect form validations
- Database connection delays
- UI responsiveness problems

These errors were fixed using debugging and optimization techniques.

8. Debugging Process

- Debugging was performed using:
- Django error logs
- Browser developer tools
- Console debugging
- Database query analysis

The debugging process helped improve software performance and reliability.

9. Testing Tools Used

The following tools were used during testing:

Tool	Purpose
Django Testing Tools	Backend testing
Browser Developer Tools	Frontend debugging
SQLite/MySQL Tools	Database testing
Manual Testing	Functional verification

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10. Results of Testing

- After performing all testing activities:
- The system operated correctly.
- Modules communicated successfully.
- Security features functioned properly.
- Database operations were accurate.
- User interfaces worked efficiently.
- AI prescription generation performed successfully.
- Overall system performance was satisfactory.

11. Advantages of Proper Testing

- Improves software quality
- Reduces system failures
- Enhances security
- Increases system reliability
- Ensures accurate functionality
- Improves user experience

12. Limitations of Testing

- Although extensive testing was performed, some limitations remain:
- Testing was conducted in a controlled environment.
- Real-world hospital workload simulation was limited.
- AI prediction accuracy may vary depending on input data.

13. Future Improvements in Testing

Future testing enhancements may include:

- Automated Testing
- Cloud-based Testing
- AI Model Accuracy Testing
- Load Testing for Large Hospitals
- Cross-platform Compatibility Testing

14. Conclusion

Testing played a vital role in ensuring the reliability, efficiency, security, and functionality of the MediGen-AI Hospital Management System. Different testing methods such as Unit Testing, Integration Testing, System Testing, and Security Testing were successfully performed to identify and eliminate errors. The testing process confirmed that the system operates according to user requirements and provides a secure and efficient healthcare management solution.

The testing phase for the MediGen-AI project has been successfully completed, validating both the functional integrity and the core AI capabilities of the healthcare portal. Testing focused on ensuring accurate response generation, seamless user navigation, secure handling of data inputs, and system stability under varying simulated user loads.

The testing phase in the Software Development Life Cycle (SDLC) is the critical stage where software is evaluated to ensure it meets requirements and is free of tool.

13.6 Deployment

Deployment is the final phase of the Software Development Life Cycle (SDLC) in which the developed software application is installed, configured, and made available for end users. It is the process of transferring the tested application from the development environment to the production environment where users can access and utilize the system effectively.

In the MediGen-AI Hospital Management System, deployment ensures that all hospital management functionalities such as patient registration, doctor management, OPD services, AI-based prescription generation, appointment handling, and administrative operations are accessible in a real-time environment.

The deployment phase also includes server setup, database configuration, security implementation, application hosting, and performance monitoring.

2. Objectives of Deployment

The primary objectives of deployment are:

- To make the application available for real-time use.
- To configure the software in a production environment.
- To ensure smooth execution of all modules.
- To establish secure database connectivity.
- To provide accessibility to authorized users.
- To monitor system performance after installation.
- To ensure system stability and reliability.
- To minimize deployment-related risks and failures.

3. Importance of Deployment

Deployment is an essential phase because it:

- Converts the project into a usable application.
- Allows real users to access the system
- Ensures system availability and reliability.
- Helps identify real-world operational issues.
- Improves organizational workflow efficiency.
- Provides centralized healthcare data management.

In healthcare systems, proper deployment is very important because uninterrupted access to patient data and hospital operations is critical.

4. Deployment Architecture

The MediGen-AI system follows a multi-layer deployment architecture.

4.1 Client Layer

This layer includes users who access the application through web browsers or mobile devices.

- Admin
 - Medical Staff
 - Doctors, Patient
-

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4.2 Application Layer

This layer handles:

- Business logic
- AI processing
- Authentication
- Request handling
- Data processing

The application is developed using the Django framework.

4.3 Database Layer

The database layer stores:

- Patient records
- Doctor details
- Prescriptions
- OPD records
- Appointment data
- User credentials

SQLite/MySQL is used for database management.

5. Deployment Environment

The deployment environment used for the project includes:

Component	Technology
Operating System	Windows/Linux
Backend Framework	Django
Programming Language	Python
Database	SQLite/MySQL
Web Browser	Google Chrome
Web Server	Localhost/Django Server
IDE	VS Code

6. Deployment Process

The deployment process was completed in several stages.

6.1 Application Preparation

Before deployment:

- All modules were tested.
- Bugs and errors were removed.
- Database connectivity was verified.

Security configurations were checked.

6.2 Database Configuration

Database tables were created for:

- Patients
- Doctors
- Prescriptions
- OPD Management, Admin

Database migration commands were executed successfully.

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6.3 Server Configuration

The Django development server was configured for deployment.

Server Responsibilities

- Request processing
- Data communication
- User authentication
- Session management

6.4 Application Deployment

The project application was deployed on a local server environment.

Deployment Steps

1. Install required dependencies.
2. Configure database settings.
3. Run migration commands.
4. Create admin credentials.
5. Start Django server
6. Access application through browser.

7. Security During Deployment

Security mechanisms implemented during deployment include:

- Authentication System
- Users must log in with valid credentials.
- Password Protection
- Passwords are securely stored and encrypted.
- Session Management
- Sessions prevent unauthorized access.
- Database Security

Sensitive hospital information is securely maintained.

8. Access Control

- Different users have different permissions.
- Different users have different role.
- Different users have separate access.
- Access control different.

9. Deployment Challenges

Some challenges faced during deployment include:

Database connection configuration errors

Server startup issues

- Static file handling problems
- Authentication configuration issues
- Session timeout handling
- Browser compatibility issues

These problems were resolved through debugging and server optimization.

Project Title:- MediGen-AI

10. Deployment Testing

After deployment, the following testing activities were performed:

Testing Type	Purpose
Functional Testing	Verify module functionality
Security Testing	Verify secure access
Performance Testing	Check response speed
Integration Testing	Verify module communication
User Acceptance Testing	Validate user satisfaction

11. Advantages of Successful Deployment

- Real-time hospital management
- Faster patient handling
- Centralized record management
- Improved administrative control

12. Maintenance After Deployment

Maintenance activities performed after deployment include:

- Bug fixing
- System updates
- Database backup
- Performance optimization
- Security updates
- Feature enhancements

Regular maintenance ensures long-term system stability and reliability.

13. Future Deployment Enhancements

Future deployment improvements may include:

- Cloud Hosting Deployment
- Mobile Application Integration
- Multi-Hospital Connectivity
- Backup and Disaster Recovery System

14. Limitations of Current Deployment

The current deployment has some limitations:

- Local server-based deployment
- Limited scalability
- No cloud infrastructure
- Limited remote accessibility

These limitations can be overcome in future versions.

15. Conclusion

Deployment is a crucial phase that transforms the developed software into a fully operational system for real-world usage. The MediGen-AI Hospital Management System was successfully deployed in a controlled environment with proper database configuration, server setup, authentication mechanisms, and security implementation. The deployment process ensured smooth functionality of all modules and provided an efficient, secure, and reliable healthcare management solution suitable for modern hospital operations.

13.7 Maintenance

Maintenance is the final phase of the Software Development Life Cycle (SDLC) that focuses on modifying, updating, improving, and supporting the software application after deployment. The purpose of maintenance is to ensure that the system continues to function efficiently, securely, and reliably according to user requirements and changing organizational needs.

In the MediGen-AI Hospital Management System, maintenance activities are essential to ensure uninterrupted hospital operations, accurate patient data management, improved system performance, and security enhancement. Since healthcare systems continuously evolve, regular maintenance helps keep the application updated with modern technologies, medical requirements, and user expectations.

Maintenance also helps identify and fix errors that may appear after deployment and improves the overall quality and usability of the system.

2. Objectives of Maintenance

- The major objectives of software maintenance are:**
- To fix software bugs and errors.**
- To improve system performance and efficiency.**

To update system features according to user requirements.

- To ensure system security and data protection.**
- To maintain database consistency and reliability.**
- To improve user experience and interface design.**
- To enhance compatibility with new technologies.**
- To provide uninterrupted healthcare services.**

3. Importance of Maintenance

- Maintenance is important because:**
- Software systems require continuous improvement.**
- User requirements may change over time.**
- Security vulnerabilities must be updated regularly.**
- Performance optimization improves system speed.**
- New features may need to be integrated.**

In healthcare management systems, maintenance is highly critical because system failure or outdated information may affect patient care and hospital administration.

4. Types of Software Maintenance

Software maintenance is generally divided into four categories.

4.1 Corrective Maintenance

Corrective maintenance is performed to identify and fix bugs, errors, and defects found after deployment.

Corrective maintenance in the SDLC is a reactive phase that occurs after software is deployed. Its primary goal is to identify, isolate, and fix bugs, residual errors, or system failures reported by users, ensuring the software returns to its optimal working condition.

Project Title:- MediGen-AI

Examples

- **Fixing login issues**
- **Resolving database connectivity errors**
- **Correcting form validation problems**
- **Fixing AI prescription generation errors**

Benefits

- **Improves software stability**
- **Reduces system crashes**
- **Enhances reliability**

4.2 Adaptive Maintenance

Adaptive maintenance is performed to make the software compatible with new environments, technologies, or requirements.

Examples

- **Updating Django or Python versions**
- **Supporting new browsers**
- **Database migration**
- **Compatibility with new operating systems**
- **Operating systems is useful for adaptive maintained**
- **Upgrade system when need.**

Benefits

- **Improves system adaptability**
- **Ensures long-term usability**
- **Improve system long -term usability**

4.3 Perfective Maintenance

Perfective maintenance focuses on improving system performance and adding new features.

Examples

- **Enhancing dashboard design**
- **Improving response speed**
- **Adding AI functionalities**
- **Optimizing database queries**

Benefits

- **Enhances user experience**
- **Increases software efficiency**

4.4 Preventive Maintenance

Preventive maintenance is performed to prevent future problems before they occur.

Examples

- **Regular database backup**
- **Security patch updates**
- **Performance monitoring**
- **Code optimization**

Benefits

- **Prevents major failures**
-

Project Title:- MediGen-AI

-
- Improves software lifespan
 - Reduces maintenance cost

5. Maintenance Activities in MediGen-AI

The following maintenance activities were performed in the project:

System Monitoring

Continuous monitoring of application performance and server status.

Bug Fixing

Identifying and resolving software errors and exceptions.

Database Maintenance

Managing data backup, recovery, and optimization.

Security Updates

Updating authentication and protection mechanisms.

Feature Enhancement

Adding new functionalities according to hospital requirements.

6. Maintenance Environment

The maintenance activities were performed using the following environment:

Component	Technology Used
Programming Language	Python
Framework	Django
Database	SQLite/MySQL
Operating System	Windows/Linux
IDE	VS Code
Browser	Chrome

7. Database Maintenance

Database maintenance is essential for protecting sensitive healthcare information.

Activities Performed

- Database backup
- Data recovery
- Query optimization
- Removing duplicate records
- Data validation

Benefits

- Prevents data loss
-

Project Title:- MediGen-AI

- Improves retrieval speed
- Ensures data integrity

8. Security Maintenance

Security maintenance ensures protection against unauthorized access and cyber threats.

Security Measures

- Password encryption updates
- Session management improvements
- Access control verification
- Authentication enhancement
- Security patch implementation
- Role-based Authentication
- Security is all the level
- Encryption is every level

9. Performance Optimization

Performance optimization activities include:

- Reducing page loading time
- Optimizing database queries
- Removing unnecessary code
- Improving server response
- Enhancing frontend responsiveness

These activities improve overall application performance and user experience.

10. Challenges Faced During Maintenance

- Some challenges encountered during maintenance include:
- Managing large patient records
- Database synchronization issues
- Compatibility with updated technologies
- AI model improvement
- Security vulnerability handling

These challenges were resolved through continuous monitoring and optimization.

11. Backup and Recovery

Backup and recovery mechanisms are critical in healthcare systems.

- Backup Activities
- Regular database backup
- Secure data storage
- Backup scheduling
- Recovery Activities
- Data restoration
- Error recovery
- System recovery after failures

These mechanisms help prevent permanent data loss.

Project Title:- MediGen-AI

12. Future Maintenance Enhancements

Future improvements planned for maintenance include:

- **Cloud database backup**
- **Automated monitoring system**
- **AI model training updates**
- **Real-time error detection**
- **Automated security scanning**
- **Mobile application support**
- **Real time monitoring your health**
- **IOT based application support**
- **Enhance user experience in medical procurement**
- **Hand watch application support**
- **Wearable in use IOT Support**

13. Advantages of Proper Maintenance

- **Improves software quality**
- **Enhances system reliability**
- **Reduces operational errors**
- **Ensures continuous availability**
- **Improves healthcare workflow**
- **Protects sensitive patient data**
- **Increases user satisfaction**

14. Limitations of Maintenance

- **Despite regular maintenance, some limitations exist:**
- **Requires continuous monitoring**
- **Time-consuming process**
- **Maintenance cost increases over time**
- **Advanced security updates may require additional resources**

15. Conclusion

Maintenance is an essential phase that ensures the long-term functionality, reliability, and efficiency of the MediGen-AI Hospital Management System. Through corrective, adaptive, preventive, and perfective maintenance activities, the system remains secure, updated, and optimized for healthcare operations. Regular maintenance helps improve system performance, enhance security, and ensure uninterrupted hospital management services. Proper maintenance practices contribute significantly to the stability and success of the overall healthcare management system.

The Maintenance phase plays a crucial role in ensuring the continuous performance, reliability, security, and efficiency of the MediGen-AI Hospital Management System after deployment. Regular maintenance activities such as bug fixing, performance optimization, database management, security updates, and feature enhancement help the system operate smoothly in a real-world healthcare environment.

Through corrective, adaptive, perfective, and preventive maintenance, the system remains updated according to changing hospital requirements and technological advancements. Proper maintenance also helps protect sensitive patient information.

CHAPTER 14: AGILE MODEL IN USE MY PROJECT

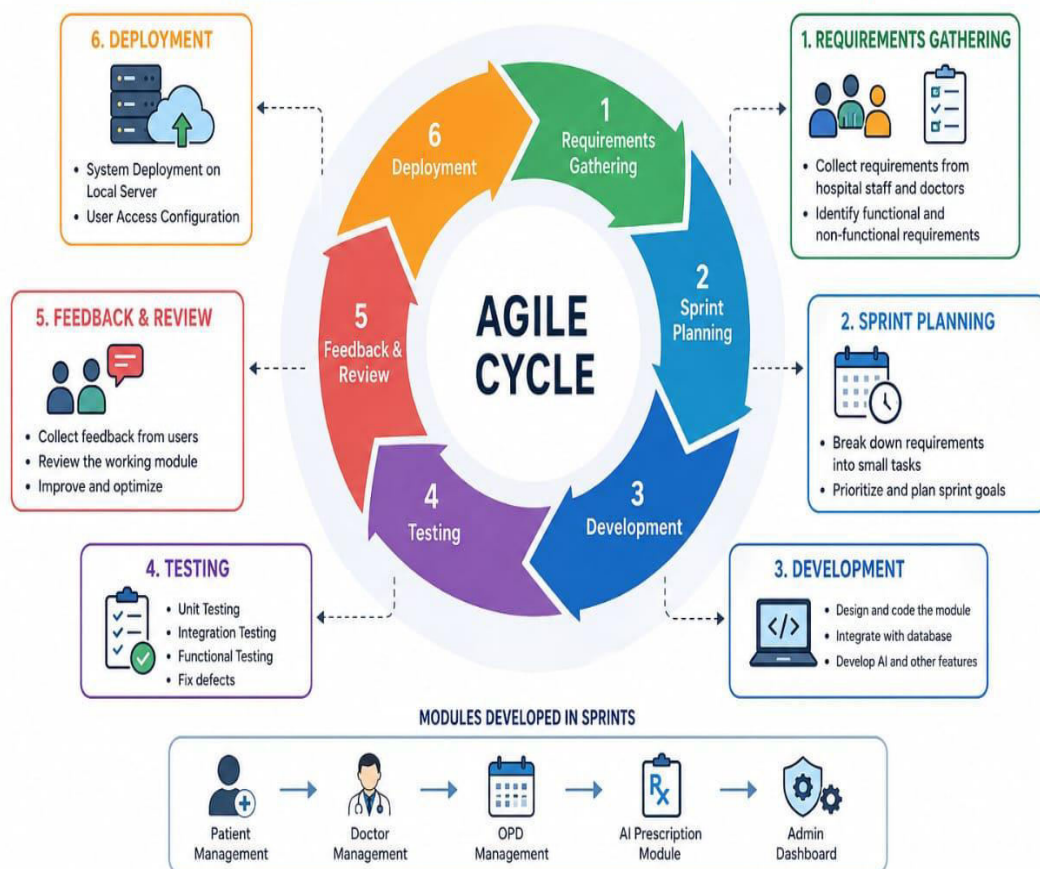
1. Introduction

The Agile Model is a modern Software Development Life Cycle (SDLC) methodology that focuses on iterative development, continuous improvement, customer feedback, and rapid delivery of software components. Unlike traditional models, Agile development divides the project into smaller modules called iterations or sprints, where each module is developed, tested, and improved continuously.

In the MediGen-AI Hospital Management System, the Agile Model was adopted to ensure flexibility, faster development, continuous testing, and regular improvements throughout the project lifecycle. Since healthcare systems require frequent modifications, user feedback, and feature enhancements, Agile methodology provided an efficient and adaptive development approach.

The Agile Model helped the development process become more interactive, collaborative, and user-oriented while ensuring high software quality and faster problem resolution.

AGILE MODEL IN MEDIGEN-AI PROJECT



Project Title:- MediGen-AI

2. Need for Agile Model in the Project

The MediGen-AI project contains multiple interconnected modules such as:

- Patient Management
- Doctor Management
- OPD System
- AI Prescription Generation
- Appointment Handling
- Admin Dashboard
- Medical Record Management

Due to changing requirements, UI improvements, security updates, and AI feature enhancements, a flexible development methodology was required. Therefore, the Agile Model was implemented in the project.

3. Features of Agile Methodology

The major features of Agile methodology include:

- Incremental development
- Continuous feedback
- Faster delivery
- Flexible requirement handling
- Continuous testing
- Team collaboration
- Frequent updates and improvements
- Customer-oriented development
- Modern Software Development Methodology
- Iterative and incremental Methodology
- Customer feedback more required

4. Agile Process Followed in the Project

The development of the MediGen-AI Hospital Management System was divided into small iterations or sprints. Each sprint focused on developing a specific module of the system.

The Agile model is an iterative, flexible approach to software development that focuses on continuous feedback, collaboration, and delivering small, working increments of software rather than building the entire system at once.

Agile Workflow

- 1. Requirement Gathering**
 - 2. Sprint Planning**
 - 3. Module Development**
 - 4. Testing**
 - 5. Feedback Collection**
 - 6. Improvement and Optimization**
 - 7. Final Deployment**
-

Project Title:- MediGen-AI

4.1 Agile Process Followed in the Project

The development of the MediGen-AI Hospital Management System was divided into small iterations or sprints. Each sprint focused on developing a specific module of the system.

Agile Workflow

1. Requirement Gathering
2. Sprint Planning
3. Module Development
4. Testing
5. Feedback Collection
6. Improvement and Optimization
7. Final Deployment

5. Agile Phases in MediGen-AI Project

5.1 Requirement Gathering Phase

In this phase:

Hospital workflow requirements were collected.

- Functional and non-functional requirements were identified.
- User expectations were analyzed.
- Requirements Identified
- Patient Registration
- Doctor Dashboard
- OPD Management
- Prescription Generation
- Admin Controls
- Secure Data Handling

5.2 Sprint Planning Phase

The entire project was divided into smaller development tasks.

- Sprint Planning Activities
- Module prioritization
- Time estimation
- Resource allocation
- Task scheduling
- Example Sprints
- Sprint Module

Sprint 1	Patient Module
Sprint 2	Doctor Module
Sprint 3	OPD Module
Sprint 4	Admin Dashboard
Sprint 5	AI Prescription Module

Project Title:- MediGen-AI

5.3 Development Phase

During each sprint, coding and implementation were performed.

Technologies Used

- Python
- Django
- HTML
- CSS
- JavaScript
- SQLite/MySQL

Development Activities

- Frontend development
- Backend logic implementation
- Database integration
- UI enhancement
- AI feature implementation

5.4 Testing Phase

Testing was performed continuously after every sprint.

Testing Techniques Used

- Unit Testing
- Integration Testing
- Functional Testing
- Security Testing

Benefits

- Early bug detection
- Faster issue resolution
- Improved software quality

5.5 Feedback and Improvement Phase

User feedback and testing observations were used to improve the system continuously.

Improvements Made

- Enhanced dashboard design
- Improved patient registration workflow
- Optimized AI prescription generation
- Improved UI responsiveness
- Fixed authentication issues

5.6 Deployment Phase

After successful completion of all sprints and testing activities, the project was deployed on the local server environment.

The development phase is the core stage where abstract designs and requirements are turned into a tangible product. Depending on the industry, it involves coding software, building prototypes, or engineering physical systems, and concludes with readiness for evaluation.

Project Title:- MediGen-AI

6. Agile Team Collaboration

Agile development encourages teamwork and communication.

- Team Activities
- Requirement discussions
- Sprint reviews
- Error tracking
- Feature enhancement discussions
- Testing collaboration

This collaboration improved development speed and software quality.

7. Advantages of Agile Model in the Project

The Agile Model provided several benefits during project development:

- Faster software development
- Flexible requirement handling
- Continuous improvement
- Better software quality
- Improved user satisfaction
- Efficient bug fixing
- Early testing and validation

8. Challenges Faced While Using Agile

Some challenges encountered during Agile implementation include:

- Frequent requirement changes
- Managing multiple modules simultaneously
- Time management during sprint execution
- AI module optimization
- Continuous testing workload

These challenges were managed through proper sprint planning and regular communication.

9. Comparison Between Traditional and Agile Model

Traditional Model	Agile Model
Sequential process	Iterative process
Less flexibility	Highly flexible
Testing at the end	Continuous testing
Slow feedback	Fast feedback
Difficult modifications	Easy modifications

10. Future Scope Using Agile

Agile methodology supports future system enhancement and scalability.

- Future Improvements
 - Cloud deployment
 - Mobile application integration
 - Online consultation
 - AI-based disease prediction
 - Multi-hospital connectivity
-

Project Title:- MediGen-AI

Real-time analytics

Agile development will help implement these features efficiently in future versions.

11. Importance of Agile in Healthcare Systems

Agile methodology is highly suitable for healthcare projects because:

- **Healthcare requirements change frequently.**
- **Security updates are continuously needed.**
- **User feedback is important.**
- **Patient data management requires continuous improvement.**
- **Real-time problem-solving is necessary.**

The Agile Model helped ensure that the MediGen-AI system remained adaptable, secure, and efficient.

The Agile approach also enhanced team collaboration, improved communication, and allowed faster implementation of new features and user requirements.

12. Conclusion

The Agile Model played an important role in the successful development of the MediGen-AI Hospital Management System. Its iterative and flexible approach allowed continuous development, testing, feedback collection, and improvement throughout the project lifecycle. Agile methodology improved software quality, enhanced collaboration, reduced development risks, and enabled faster issue resolution.

By dividing the project into manageable sprints, the system was developed efficiently with continuous enhancements and user-focused improvements. Overall, Agile methodology helped create a reliable, scalable, secure, and user-friendly healthcare management system suitable for modern digital hospital operations.

The Agile Model proved to be highly effective in the development of the MediGen-AI Hospital Management System because it provided a flexible, iterative, and user-centered approach to software development. By dividing the project into smaller sprints and continuously testing and improving each module, the development process became more efficient, manageable, and adaptable to changing requirements. Through Agile methodology, important modules such as Patient Management, Doctor Management, OPD Handling, AI Prescription Generation, and Admin Dashboard were developed systematically with continuous feedback and enhancement. Regular testing and review after every sprint helped in early bug detection, improved software quality, and reduced development risks.

The Agile approach also enhanced team collaboration, improved communication, and allowed faster implementation of new features and user requirements. Continuous integration and incremental development ensured that the system remained stable, secure, and scalable throughout the project lifecycle.

Furthermore, Agile methodology supported continuous improvement in user interface design, database management, system performance, and AI functionalities. The iterative nature of Agile made it easier to optimize healthcare workflows and provide a better user experience for doctors, administrators, medical staff, and patients.

CHAPTER 14: DATABASE DESIGN

1. Introduction

Database Design is one of the most important phases in software development. It is the process of organizing, structuring, and managing data efficiently to ensure accurate storage, retrieval, and processing of information. A well-designed database improves system performance, data consistency, security, scalability, and maintainability.

In the MediGen-AI Hospital Management System, the database plays a crucial role in managing hospital-related information such as patient records, doctor details, OPD reports, prescriptions, appointments, medicines, and administrative data. The database was designed using relational database concepts to ensure efficient communication between different modules of the system.

The database structure was carefully planned to reduce data redundancy, maintain integrity, and support secure healthcare data management.

2. Objectives of Database Design

The primary objectives of database design are:

- To store hospital data efficiently.
- To reduce data redundancy.
- To maintain data integrity and consistency.
- To provide fast data retrieval.
- To ensure secure data storage.
- To establish relationships between modules.
- To support multi-user access.
- To improve system performance and reliability.

3. Need for Database in the Project

The MediGen-AI system handles large amounts of healthcare data daily. Therefore, a structured database system is required for:

- Managing patient record
- Maintaining doctor information
- Storing prescriptions
- Tracking appointments
- Handling OPD reports
- Managing medicine inventory
- Generating administrative reports

Without a proper database, managing hospital operations manually would be difficult, time-consuming, and error-prone.

A database is essential for storing, organizing, and retrieving your project's information safely and efficiently. It eliminates scattered files, prevents duplicate data, and allows multiple users to interact with the system simultaneously without conflicts.

Project Title:- MediGen-AI

4. Database Model Used

The project uses the Relational Database Model.

In the relational model:

- **Data is stored in tables.**
- **Tables are connected using relationships.**
- **Each table contains rows and columns.**
- **Primary keys and foreign keys maintain relationships.**
- **The relational model was selected because it:**
- **Provides structured data management**
- **Supports data consistency**
- **Allows easy querying**
- **Improves data security**
- **Supports scalability**

5. Database Management System (DBMS)

The MediGen-AI project uses:

- **SQLite**

SQLite was used during development because:

- **It is lightweight**
- **Easy to configure**
- **Suitable for academic projects**
- **Requires minimal setup**
- **MySQL (Optional Future Enhancement)**

MySQL can be used for large-scale deployment because:

- **It supports multi-user access**
- **Provides better scalability**
- **Handles large databases efficiently**

6. Database Architecture

The database architecture consists of:

- **Data Storage Layer**
- **Stores all hospital-related records.**
- **Data Access Layer**
- **Handles database queries and transactions.**
- **Application Layer**

Communicates with the database using Django ORM

An Object-Relational Mapper (ORM) is a software layer that lets you interact with a database using the programming language you already know (like Python or JavaScript) instead of writing raw SQL. It maps database tables to "objects" in your code, making data manipulation feel like working with standard variables and classes

Project Title:- MediGen-AI

7. Main Database Tables

The following major tables were created in the system:

Table Name	Purpose
Patient Table	Stores patient details
Doctor Table	Stores doctor information
Appointment Table	Stores appointment records
Prescription Table	Stores prescription data
OPD Table	Stores OPD visit records
Admin Table	Stores admin credentials
Medicine Table	Stores medicine details

8. Patient Table Design

The Patient Table stores patient-related information.

Field Name	Data Type	Description
Patient	Integer	Unique patient ID
Patient Name	Varchar	Patient full name
Age	Integer	Patient age
Gender	Varchar	Gender
Phone	Varchar	Contact number
Address	Text	Patient address
ABHA_Number	Varchar	ABHA identification

Features

- Unique patient identification
- Easy retrieval of patient records
- Secure storage of healthcare data

9. Doctor Table Design

The Doctor Table stores doctor information.

Field Name	Data Type	Description
Doctor_ID	Integer	Unique doctor ID
Doctor Name	Varchar	Doctor name
Specialization	Varchar	Medical specialization
Contact	Varchar	Contact details
Email	Varchar	Email address

10. Appointment Table Design

This table stores appointment details between patients and doctors.

Project Title:- MediGen-AI

Field Name	Data Type	Description
Appointment_ID	Integer	Unique appointment ID
Patient_ID	Integer	Reference to patient
Doctor_ID	Integer	Reference to doctor
Appointment_Date	Date	Appointment date
Status	Varchar	Appointment status

11. Prescription Table Design

This table stores prescription information generated by doctors or AI.

Field Name	Data Type	Description
Prescription_ID	Integer	Unique prescription ID
Patient_ID	Integer	Patient reference
Doctor_ID	Integer	Doctor reference
Symptoms	Text	Patient symptoms
Diagnosis	Text	Diagnosis details
Medicines	Text	Prescribed medicines

12. OPD Table Design

The OPD table manages outpatient records.

Field Name	Data Type	Description
OPD_ID	Integer	Unique OPD ID
Patient_ID	Integer	Patient reference
Doctor_ID	Integer	Doctor reference
Visit_Date	Date	Date of visit
OPD_Status	Varchar	Visit status

13. Relationships Between Tables

Relationships help connect different tables.

- One-to-Many Relationships
- One doctor can handle many patients.
- One patient can have multiple appointments.
- One patient can have multiple prescriptions.

Foreign Keys Used

- Patient_ID
- Doctor_ID
- Appointment_ID

These relationships ensure data consistency and integrity.

Table relationships are associations between different tables in a database. They are established using Primary Keys (unique identifiers for a row in one table) and Foreign Keys (references to that primary key in another table). These connections allow databases to combine data without duplication

In a relational database, table relationships are the connections established between tables that allow data to be organized efficiently and retrieved using SQL JOIN.

14. Entity Relationship (ER) Design

The ER Design represents relationships between entities such as:

Project Title:- MediGen-AI

- Patient
- Doctor
- Prescription
- Appointment
- OPD
- Admin

The ER diagram helps visualize:

Table relationships

- Data flow
- Primary keys
- Foreign keys

15. Normalization

Normalization was applied to improve database efficiency.

Types of Normalization Used

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)

Benefits

- Reduces data redundancy
- Improves consistency
- Enhances database performance

16. Database Security

Database security mechanisms implemented include:

- User authentication
- Password encryption
- Access control
- Secure database connectivity
- Session management

These features help protect sensitive healthcare information.

17. Database Connectivity

The Django framework uses ORM (Object Relational Mapping) to connect the application with the database.

Advantages of ORM

- Simplifies database operations
- Reduces SQL complexity

18. Database Backup and Recovery

Backup mechanisms are important in healthcare systems.

Backup Features

- Regular database backup
- Data restoration
- Recovery after system failure

These mechanisms prevent permanent loss of patient data.

19. Advantages of Database Design

- Efficient data management
-

Project Title:- MediGen-AI

- Faster record retrieval
- Reduced manual work
- Improved data security
- Better hospital workflow
- Centralized information management
- Improved reporting system

20. Challenges Faced During Database Design

Some challenges encountered include:

- Managing relationships between tables
- Avoiding duplicate records
- Ensuring secure data storage
- Query optimization
- Handling large datasets

These issues were resolved through normalization and optimization techniques.

21. Future Enhancements

- Future database improvements may include:
- Cloud database integration
- Real-time synchronization
- Big data analytics
- Multi-hospital database management
- Advanced backup systems

22. Conclusion

The Database Design of the MediGen-AI Hospital Management System was developed using relational database principles to ensure efficient, secure, and reliable healthcare data management. Proper table structures, relationships, normalization, and security mechanisms helped improve system performance and maintain data integrity. The database successfully supports all major modules of the system, including patient management, doctor management, OPD handling, prescriptions, and administrative operations. Overall, the database design provides a strong foundation for an intelligent and scalable hospital management system.

Design is a critical process of creating a structured framework to store and manage data efficiently. The phases of database design cover everything from conceptual modeling to physical implementation. Understanding schema refinement in database design, logical database design, and how to avoid pitfalls in relational database design is crucial to building a robust, scalable database system.

By following the various steps of database design and leveraging techniques like ER diagrams, normalization, and proper indexing, you can create a database that meets business requirements and is optimized for performance, scalability, and security.

Database design is the foundational blueprint of your application. A successful design balances data integrity, minimal redundancy, and scalability. It is not a one-time setup, but an evolving architecture that dictates long-term performance, maintenance ease, and application responsiveness

CHAPTER 14: UML DIAGRAMS

1. Introduction

Unified Modeling Language (UML) is a standardized modeling language used in software engineering to visualize, design, and document the structure and behavior of a software system. UML diagrams help developers, designers, and stakeholders understand system functionality, workflow, module interaction, and data communication in a clear and organized manner.

In the MediGen-AI Hospital Management System, UML diagrams were used to represent different components of the system such as users, modules, database interactions, workflows, and system architecture. These diagrams helped simplify system understanding and improved software planning and development.

UML modeling also supports better communication between developers and users by providing graphical representations of system operations.

2. Objectives of UML Diagrams

The major objectives of UML diagrams are:

- To visualize system architecture and workflow.
- To represent interactions between modules.
- To simplify software design understanding.
- To improve communication among developers and users.
- To document system functionality properly.
- To analyze system behavior before implementation.
- To reduce development complexity.

3. Importance of UML Diagrams

UML diagrams are important because they:

- Provide a visual representation of the system.
- Improve software planning and documentation.
- Help detect design issues early.
- Simplify complex system structures.
- Improve communication between stakeholders.

In healthcare projects like MediGen-AI, UML diagrams help in understanding patient flow, doctor interaction, prescription management, and administrative processes effectively.

4. Types of UML Diagrams Used in the Project

The following UML diagrams were used in the MediGen-AI Hospital Management System:

- 1. Use Case Diagram**
 - 2. Class Diagram**
 - 3. Sequence Diagram**
 - 4. Activity Diagram**
 - 5. State Diagram**
 - 6. Component Diagram**
 - 7. Deployment Diagram**
-

5. USE CASE DIAGRAM

1. Introduction

A Use Case Diagram represents the interaction between users (actors) and the system functionalities.

Actors in the System

- Admin
- Doctor
- Patient
- Medical Staff
- Main Use Cases
- Patient Registration
- Doctor Login
- Appointment Management
- Prescription Generation
- OPD Handling
- Report Generation
- Admin Management

Advantages

- Easy understanding of system functionality
- Clear representation of user interactions
- Improves requirement analysis

6. Class Diagram

1. Introduction

A Class Diagram represents the static structure of the system including classes, attributes, methods, and relationships.

Main Classes

- Patient
- Doctor
- Prescription
- Appointment
- OPD
- Admin
- Relationships
- One doctor handles multiple patients.
- One patient can have multiple prescriptions.
- Appointments connect patients and doctors.

Benefits

- Improves object-oriented design
 - Represents database structure clearly
 - Simplifies coding and development
-

7. Sequence Diagram

1. Introduction

A Sequence Diagram shows the sequence of interactions between objects over time.

Example Workflow

- 1. Patient registers into the system.**
- 2. Staff assigns doctor.**
- 3. Doctor checks patient details.**
- 4. AI generates prescription.**
- 5. Prescription stored in database.**

Benefits

- **Shows communication flow**
- **Helps understand process execution**
- **Improves system interaction design**

8. Activity Diagram

Introduction

An Activity Diagram represents the workflow of activities in the system.

- **Activities Represented**
- **Patient registration process**
- **Appointment booking process**
- **OPD workflow**
- **Prescription generation process**

Benefits

- **Simplifies workflow understanding**
- **Represents decision-making processes**
- **Helps identify process improvements**
- **Helps identify process workflow**
- **It helps data flow.**
- **Easy to understand workflow**

9. State Diagram

Introduction

A State Diagram represents different states of an object during its lifecycle.

Example States

- **Appointment States**
- **Pending**
- **Approved**
- **Completed**
- **Cancelled**
- **Patient States**
- **Registered**
- **Under Treatment**
- **Discharged**

Benefits

Project Title:- MediGen-AI

-
- Shows state transitions clearly
 - Helps manage object behavior

10. Component Diagram

Introduction

A Component Diagram represents the physical components of the software system and their relationships.

Components Included

- Frontend Interface
- Backend Server
- Database
- AI Module
- Authentication Module

Advantages

- Shows system modularity
- Improves system organization
- Simplifies deployment planning

11. Deployment Diagram

Introduction

A Deployment Diagram represents the physical deployment architecture of the system.

- Deployment Components
- Client Device
- Web Server
- Application Server
- Database Server
- Technologies Used
- Django Server
- SQLite/MySQL
- Web Browser

Benefits

- Represents hardware and software deployment
- Helps understand server architecture
- Helps identify process workflow
- It helps data flow.
- Easy to understand workflow

12. UML Diagram Tools Used

The following tools were used to create UML diagrams:

Tool Name	Purpose
Draw.io	UML diagram creation
Lucid chart	System modeling
Canva	Diagram design
StarUML	UML modeling

13. Advantages of UML Diagrams in the Project

Project Title:- MediGen-AI

- Better system visualization
- Improved project documentation
- Easier software development
- Better communication among team members
- Simplified debugging and maintenance
- Improved understanding of workflows

14. Challenges Faced While Designing UML Diagram

- Some challenges encountered include:
- Managing complex relationship
- Designing accurate workflow
- Maintaining diagram clarity
- Synchronizing diagrams with development changes

These challenges were resolved through proper planning and continuous diagram updates.

15. Future Enhancements in UML Modeling

- Future UML improvements may include:
- Cloud architecture diagrams
- AI workflow diagram
- Microservices-based architecture diagrams
- Advanced security modeling

16. Conclusion

UML diagrams played a significant role in the design and development of the MediGen-AI Hospital Management System. They provided a clear visual representation of system architecture, workflows, user interactions, database relationships, and module communication. Different UML diagrams such as Use Case, Class, Sequence, Activity, Component, and Deployment diagrams helped simplify software planning, improve development efficiency, and enhance system understanding.

Overall, UML modeling contributed significantly to building a structured, scalable, secure, and efficient healthcare management system suitable for modern digital hospital operations.

Structural Diagrams (e.g., Class, Component Diagrams): Focus: System architecture, object relationships, and software data models. Conclusion snippet: "In summary, the structural UML diagrams successfully mapped the static architecture of the application. The Class Diagram established explicit object-oriented hierarchies and multiplicities, effectively mapping code dependencies before implementation to prevent structural bottlenecks."

Behavioral Diagrams (e.g., Sequence, Use Case, Activity Diagrams) Focus: User interactions, processes, and dynamic state changes over time. Conclusion snippet: "Conclusively, the behavioral UML models captured the dynamic, runtime processes of the system. The Sequence and Use Case diagrams accurately validated user-to-system interactions. This ensured that all functional workflow variations and logic flows were accounted for prior to any physical coding"

CHAPTER 15: DATA FLOW DIAGRAM

1. Introduction

A Data Flow Diagram (DFD) is a graphical representation used to illustrate the flow of data within a software system. It shows how data enters the system, how it is processed, stored, and transferred between different modules or entities. DFD helps in understanding system functionality, workflow, and communication between users and processes.

In the MediGen-AI Hospital Management System, the Data Flow Diagram was used to represent the movement of data between different modules such as Patient Management, Doctor Management, OPD Handling, Prescription Generation, Appointment System, and Admin Control.

The DFD provides a clear understanding of how information flows throughout the hospital management system and helps in analyzing system processes efficiently.

2. Objectives of Data Flow Diagram

The major objectives of using DFD are:

- To represent system workflow visually.
- To show movement of data between processes.
- To simplify system understanding.
- To identify input and output processes.
- To improve communication between developers and users.
- To analyze system functionality before implementation.
- To reduce system complexity

3. Importance of DFD in the Project

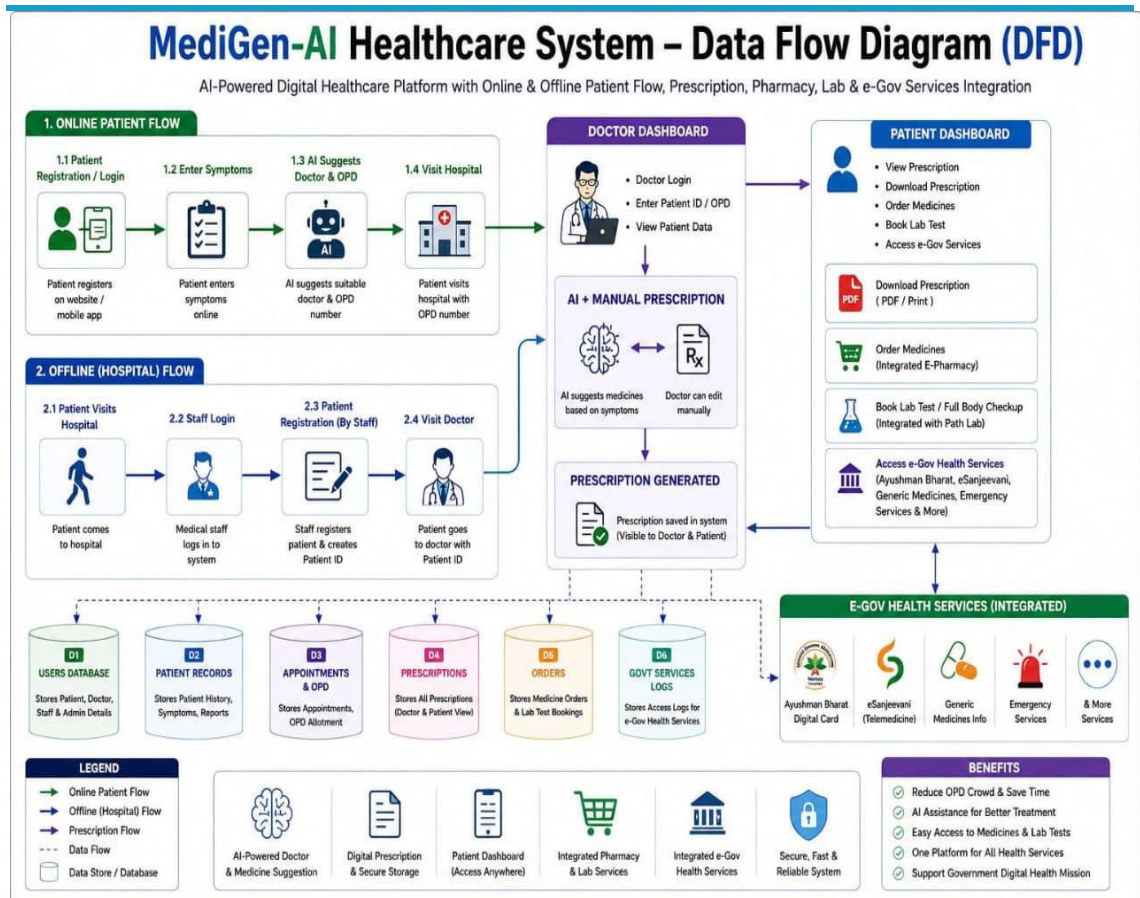
Data Flow Diagrams are important because they:

- Provide a clear visualization of the system.
- Simplify complex healthcare workflows.
- Help identify process dependencies.
- Improve system documentation.
- Assist in database and module design.
- Support efficient software development.

In healthcare management systems like MediGen-AI, DFD helps in understanding patient registration flow, doctor interaction, prescription management, and administrative operations.

A data flow diagram uses graphical symbols to illustrate the paths, processes and storage repositories for data from the point it enters a system until it exits. This visual model helps professionals identify ways to improve the efficiency and effectiveness of existing systems and processes and create new ones.

Project Title:- MediGen-AI



4. Components of Data Flow Diagram

A DFD mainly consists of four components

4.1 External Entity

External entities are users or systems that interact with the application.

Entities in MediGen-AI

- Patient
- Doctor
- Admin
- Medical Staff

4.2 Process

Processes represent operations performed on data.

Processes in the System

- Patient Registration
- Doctor Authentication
- Appointment Management
- Prescription Generation
- OPD Management
- Report Generation

4.3 Data Store

Project Title:- MediGen-AI

Data stores are locations where information is stored.

- **Data Stores Used**
- **Patient Database**
- **Doctor Database**
- **Prescription Database**
- **Appointment Records**
- **OPD Records**

4.4 Data Flow

Data flow represents movement of information between entities, processes, and databases.

Examples

- **Patient details sent to registration module**
- **Prescription stored in database**
- **Appointment information sent to doctor dashboard**

5. Levels of Data Flow Diagram

The MediGen-AI project uses multiple levels of DFD for better understanding.

5.1 Level 0 DFD (Context Diagram)

Introduction

Level 0 DFD provides a high-level overview of the entire system. It represents the complete system as a single process and shows interactions between external entities and the system.

External Entities

- **Patient**
- **Doctor**
- **Admin**
- **Medical Staff**
- **Main System Process**

MediGen-AI Hospital Management System

Data Flow

- **Patient sends registration details**
- **Doctor receives patient records**
- **Admin manages hospital operations**
- **Staff updates appointment information**

Advantages

- **Simple system overview**
- **Easy understanding of system boundaries**

5.2 Level 1 DFD

Introduction

Level 1 DFD breaks the main system into multiple sub-processes.

Project Title:- MediGen-AI

Major Processes

1. Patient Registration Process

Handles patient information and record creation.

2. Doctor Management Process

Manages doctor details and authentication.

3. Appointment Process

Handles appointment scheduling and management.

4. Prescription Process

Generates and stores prescriptions.

5. Admin Management Process

Controls overall system activities.

- **Data Stores Connected**
- **Patient Database**
- **Doctor Database**
- **Prescription Database**
- **Appointment Database**

5.3 Level 2 DFD

Introduction

Level 2 DFD provides detailed internal workflow of individual modules.

Example: Patient Registration Workflow

Steps

- 1. User enters patient details.**
- 2. System validates information.**
- 3. Data stored in patient database.**
- 4. Registration confirmation generated.**

Example: AI Prescription Workflow

Steps

- 1. Doctor enters symptoms.**
- 2. AI analyzes patient data.**
- 3. Prescription generated.**
- 4. Prescription stored in database.**

6. DFD of Major Modules

6.1 Patient Registration DFD

Input

- **Patient Name**
 - **Age**
 - **Gender**
 - **Contact Details**
 - **ABHA Number**
 - **Process**
 - **Validation**
 - **Record Creation**
-

Project Title:- MediGen-AI

- Database Storage
- Output
- Registered Patient Record

6.2 Doctor Module DFD

Input

- Doctor Login Credentials
- Patient Details
- Process
- Authentication
- Diagnosis
- Prescription Creation

Output

- Updated Patient Prescription

6.3 Appointment Management DFD

Input

- Appointment Request
- Doctor Availability
- Process
- Appointment Scheduling
- Doctor Assignment

Output

- Appointment Confirmation
- Patient listing

6.4 Admin Module DFD

Input

- Administrative Commands
- User Information

Process

- User Management
- Report Generation
- Database Monitoring

Output

- System Reports
- User Updates

7. Symbols Used in DFD

Symbol	Meaning
Circle	Process
Rectangle	External Entity
Arrow	Data Flow
Open Rectangle	Data Store

8. Advantages of DFD

Project Title:- MediGen-AI

-
- Simplifies complex workflows
 - Improves system understanding
 - Enhances communication
 - Supports efficient development
 - Helps identify errors early
 - Improves documentation

9. Limitations of DFD

- Does not represent system timing
- Cannot show detailed logic
- Difficult for very large systems
- Requires regular updates during development

10. Tools Used for Creating DFD

The following tools were used for designing DFD diagrams:

Tool Name	Purpose
Draw.io	Diagram creation
Canva	Visual design
Lucidchart	System flow design
StarUML	UML and DFD modeling

11. Challenges Faced During DFD Design

- Managing complex data flow
- Maintaining diagram clarity
- Representing multiple modules together

These issues were resolved through modular design and simplified workflow representation.

12. Future Improvements in DFD

- Future improvements may include:
- Cloud-based data flow representation
- AI workflow diagrams
- Real-time data synchronization models
- Multi-hospital workflow diagrams

13. Conclusion

The Data Flow Diagram (DFD) played an important role in analyzing and designing the MediGen-AI Hospital Management System. It provided a clear graphical representation of how data moves between users, processes, and databases within the system. Different DFD levels helped simplify complex healthcare workflows and improved understanding of patient management, doctor interaction, appointment handling, and prescription generation processes.

Overall, the DFD contributed significantly to proper system planning, efficient development, improved communication, and successful implementation of the healthcare management system.

CHAPTER 16: ENTITY RELATIONSHIP DIAGRAM

1. Introduction

An Entity Relationship Diagram (ER Diagram) is a graphical representation used to design and visualize the structure of a database system. It illustrates the entities, attributes, and relationships between different components of a system. ER diagrams help developers understand how data is stored, connected, and managed within a database.

In the MediGen-AI Hospital Management System, the ER Diagram was used to represent the relationship between major entities such as Patients, Doctors, Prescriptions, Appointments, OPD Records, Medicines, and Admin. The ER model helped in designing a structured, efficient, and reliable database system for managing healthcare operations.

The ER Diagram provides a clear overview of database architecture and supports efficient database implementation and maintenance.

An Entity-Relationship Diagram (ERD) is a visual blueprint that illustrates how different data "entities" (such as people, objects, or concepts) relate to one another within a database or information system. Most frequently used by software engineers and business analysts, ERDs bridge the gap between abstract business requirements and technical database deployment.

Entities: The "nouns" or core objects of the system, represented visually as rectangles (e.g., Customer, Order, Product).

Attributes: The "adjectives" or specific characteristics describing an entity, represented as ovals (e.g., Email _Address, Order _ID, Price).

Relationships: The "verbs" indicating how entities interact, represented as diamonds or connecting lines (e.g., Customer places an Order)

Key Attribute: Uniquely identifies a record in an entity set (Primary Key). It is drawn as an oval with underlined text.

Composite Attribute: Made up of multiple sub-attributes (e.g., an Address containing Street, City, and Zip Code).

Multivalued Attribute: Can hold more than one value for a single instance (e.g., multiple Phone Numbers), depicted as a double oval.

Derived Attribute: Calculated dynamically from other values (e.g., Age calculated from a Date of Birth), drawn with a dashed oval.

2. Objectives of ER Diagram

The primary objectives of the ER Diagram are:

- To represent database structure visually.
 - To identify entities and their attributes.
 - To define relationships between tables.
 - To reduce data redundancy.
 - To improve database efficiency.
 - To simplify database development.
 - To maintain data consistency and integrity.
 - To maintain data integrity
-

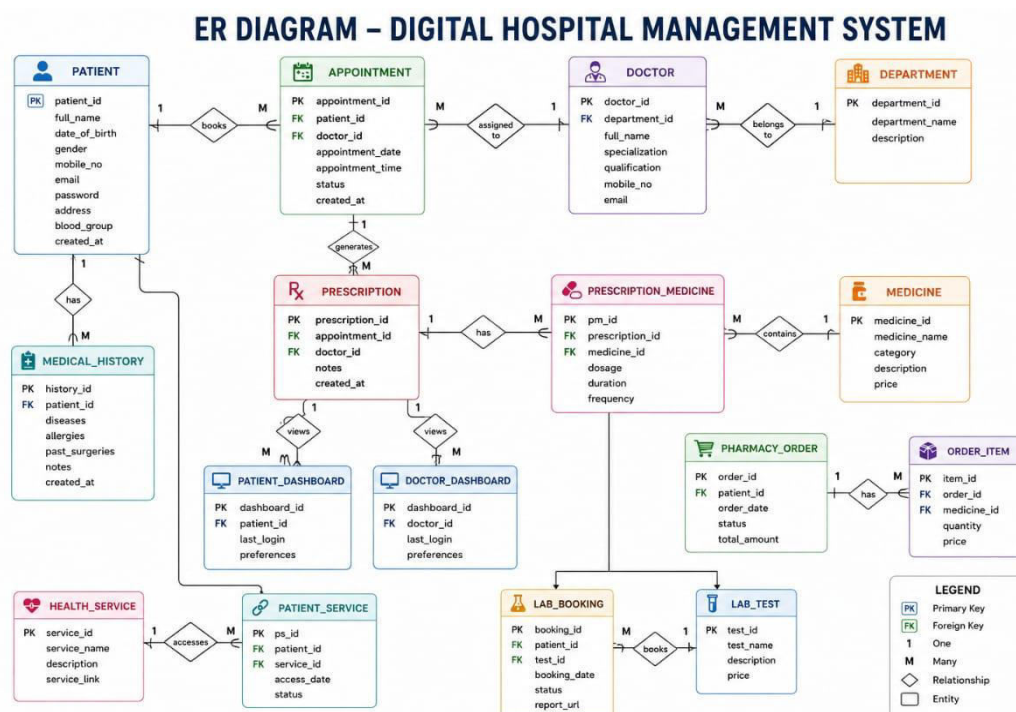
Project Title:- MediGen-AI

3. Importance of ER Diagram

ER diagrams are important because they:

- Provide a clear understanding of database design.
- Simplify complex relationships.
- Help developers during implementation.
- Improve communication among developers and users.
- Support efficient database normalization.
- Reduce database errors.

In healthcare systems like MediGen-AI, ER diagrams help manage patient records, doctor information, prescriptions, appointments, and administrative operations efficiently.



4. Components of ER Diagram

The ER Diagram mainly consists of the following components:

4.1 Entity

An entity represents a real-world object or component.

Main Entities in MediGen-AI

- Patient
- Doctor
- Appointment
- Prescription
- OPD

Project Title:- MediGen-AI

- Admin
- Medicine

4.2 Attributes

Attributes define the properties of an entity.

Example

- Patient Attributes
- Patient_ID
- Patient_Name
- Age
- Gender
- Contact Number
- Address
- ABHA_Number

Doctor Attributes

- Doctor_ID
- Doctor_Name
- Specialization
- Email
- Contact

4.3 Relationship

Relationships define how entities are connected.

Examples

- One Doctor treats many Patients.
- One Patient can have multiple Prescriptions.
- One Appointment connects one Patient and one Doctor.

4.4 Primary Key

A Primary Key uniquely identifies each record in a table.

Examples

- Patient_ID
- Doctor_ID
- Prescription_ID

4.5 Foreign Key

A Foreign Key establishes relationships between tables.

Examples

- Patient_ID in Prescription Table
- Doctor_ID in Appointment Table

5. Main Entities and Relationships

5.1 Patient Entity

The Patient entity stores patient-related information.

Attributes

- Patient_ID
 - Name
 - Age
-

Project Title:- MediGen-AI

- **Gender**
- **Phone**
- **Email**
- **Address**
- **ABHA_Number**

Relationships

- **One patient can book multiple appointments.**
- **One patient can receive multiple prescriptions.**

5.2 Doctor Entity

The Doctor entity stores doctor information.

Attributes

- **Doctor_ID**
- **Doctor_Name**
- **Specialization**
- **Contact**
- **Email**
- **Relationships**

One doctor handles many patients.

One doctor generates multiple prescriptions.

5.3 Appointment Entity

This entity manages appointment scheduling.

Attributes

- **Appointment_ID**
- **Patient_ID**
- **Doctor_ID**
- **Appointment_Date**

Status

Relationships

Connects Patient and Doctor entities.

5.4 Prescription Entity

Stores prescription details generated by doctors or AI.

Attributes

- **Prescription_ID**
- **Patient_ID**
- **Doctor_ID**
- **Symptoms**
- **Diagnosis**
- **Medicines**

5.5 OPD Entity

Stores outpatient department records.

Attributes

- **OPD_ID**
 - **Patient_ID**
-

Project Title:- MediGen-AI

- Doctor_ID
- Visit_Date , OPD_Status
-

5.6 Admin Entity

Stores administrator information.

Attributes

- Admin_ID
- Username
- Password, Role

6. Relationships in the ER Diagram

Relationship Type

Doctor → Patient	One-to-Many
Patient → Appointment	One-to-Many
Doctor → Prescription	One-to-Many
Patient → Prescription	One-to-Many
Appointment → Doctor	Many-to-One

7. ER Diagram Representation

The ER Diagram of the MediGen-AI system includes:

- Rectangles representing entities
- Ellipses representing attributes
- Diamonds representing relationships
- Lines representing entity connections

The diagram visually represents the complete database structure of the hospital management system.

8. ER Diagram Workflow

Step 1

Patient registers into the system.

Step 2

Appointment is created with a doctor.

Step 3

Doctor diagnoses patient.

Step 4

Prescription is generated and stored.

Step 5

Admin monitors and manages system activities.

9. Normalization in ER Design

Normalization techniques were applied to improve database efficiency.

Forms Used

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)

Benefits

- Removes duplicate data
 - Improves consistency
-

Project Title:- MediGen-AI

- Enhances performance

10. Advantages of ER Diagram

- Simplifies database design
- Improves data organization
- Reduces redundancy
- Supports efficient implementation
- Improves data consistency
- Improve data integrity
- Enhances system understanding

11. Limitations of ER Diagram

- Difficult for very large systems
- Requires regular updates
- Complex relationships may reduce readability

12. Tools Used for ER Diagram

The following tools were used:

Tool Name	Purpose
Draw.io	ER Diagram creation
Lucidchart	Database modeling
Canva	Diagram designing
StarUML	System modeling

13. Challenges Faced During ER Design

Some challenges encountered include:

- Managing multiple entity relationships
- Avoiding data redundancy
- Maintaining diagram simplicity

Synchronizing ER design with database implementation

These issues were resolved through normalization and structured database planning.

14. Future Enhancements

Future improvements in ER design may include:

- Cloud database integration
- Multi-hospital database architecture
- AI-based data analytics
- Distributed database systems

15. Conclusion

The Entity Relationship Diagram (ER Diagram) played an important role in designing the database structure of the MediGen-AI Hospital Management System. It helped visualize entities, attributes, and relationships clearly and systematically. The ER model improved database organization, reduced redundancy, maintained data integrity, and supported efficient healthcare data management.

Overall, the ER Diagram provided a strong foundation for developing a secure, scalable, and efficient hospital management database system capable of handling modern healthcare operations effectively.

CHAPTER 17: GANTT CHART DESCRIPTION

1. Introduction

A Gantt Chart is a graphical project management tool used to represent the schedule of tasks and activities involved in software development. It displays project phases, task duration, dependencies, starting dates, and completion timelines in a visual format. Gantt charts help in planning, monitoring, and managing project progress efficiently.

In the MediGen-AI Hospital Management System, the Gantt Chart was used to organize and track all development activities such as requirement analysis, planning, system design, coding, testing, deployment, and maintenance. It helped ensure proper time management and timely completion of the project.

The Gantt Chart provided a clear overview of project workflow and helped in monitoring the progress of each development phase systematically.

2. Objectives of Gantt Chart

The major objectives of using a Gantt Chart are:

- To plan project activities effectively.
- To manage project timelines.
- To track task completion status.
- To allocate resources efficiently.

3. Importance of Gantt Chart in the Project

The Gantt Chart played an important role in the MediGen-AI project because it:

- Helped organize project tasks sequentially.
- Improved time management.
- Simplified project scheduling.
- Assisted in monitoring project progress.
- Reduced development delays.

Helped complete the project within the planned duration.

In healthcare management system development, proper planning and scheduling are important because multiple modules must be developed and integrated.

4. Phases Represented in the Gantt Chart

The Gantt Chart of the MediGen-AI Hospital Management System includes the following phases:

1. Requirement Analysis
 2. Planning
 3. System Design
 4. Database Design
 5. Coding and Development
 6. Testing
 7. Deployment
 8. Maintenance
 9. Documentation
-

Project Title:- MediGen-AI

5. Requirement Analysis Phase

Description

In this phase, all project requirements were collected and analyzed.

Activities

- **Identifying hospital management problems**
- **Collecting user requirements**
- **Defining system objectives**
- **Preparing Software Requirement Specification (SRS)**
- **Duration**
- **Approximately 1–2 weeks.**

6. Planning Phase

Description

This phase focused on project planning and scheduling.

Activities

- **Project scheduling**
- **Resource allocation**
- **Technology selection**
- **Risk analysis**
- **Duration**
- **Approximately 1 week.**

7. System Design Phase

Description

The architecture and design of the system were prepared in this phase.

Activities

- **UML Diagram Design**
- **Data Flow Diagram (DFD)**
- **ER Diagram Design**
- **User Interface Design**
- **System Architecture Design**
- **Duration**
- **Approximately 2 weeks.**

8. Database Design Phase

Description

The database structure and relationships were designed.

Activities

- **Table creation**
 - **Relationship design**
 - **Normalization**
 - **Database connectivity planning**
 - **Duration**
 - **Approximately 1 week.**
-

Project Title:- MediGen-AI

9. Coding and Development Phase

Description

Actual implementation of the system was performed.

Activities

- **Frontend development**
- **Backend development**
- **AI integration**
- **Module implementation**
- **Database integration**
- **Modules Developed**
- **Patient Module**
- **Doctor Module**
- **OPD Module**
- **Admin Module**
- **AI Prescription Module**
- **Duration**
- **Approximately 4–6 weeks.**

10. Testing Phase

Description

Testing was performed to identify and remove software errors.

Activities

- **Unit Testing**
- **Integration Testing**
- **System Testing**
- **Security Testing**
- **User Acceptance Testing**
- **Duration**
- **Approximately 2 weeks.**

11. Deployment Phase

Description

The developed system was deployed into the working environment.

Activities

- **Server configuration**
- **Database setup**
- **Application deployment**
- **User access configuration**
- **Duration**
- **Approximately 1 week.**

12. Maintenance Phase

Description

Maintenance activities were performed after deployment.

Activities

- **Bug fixing**
 - **Performance optimization**
-

Project Title:- MediGen-AI

- Security updates
- Database maintenance
- Duration
- Continuous process after deployment.

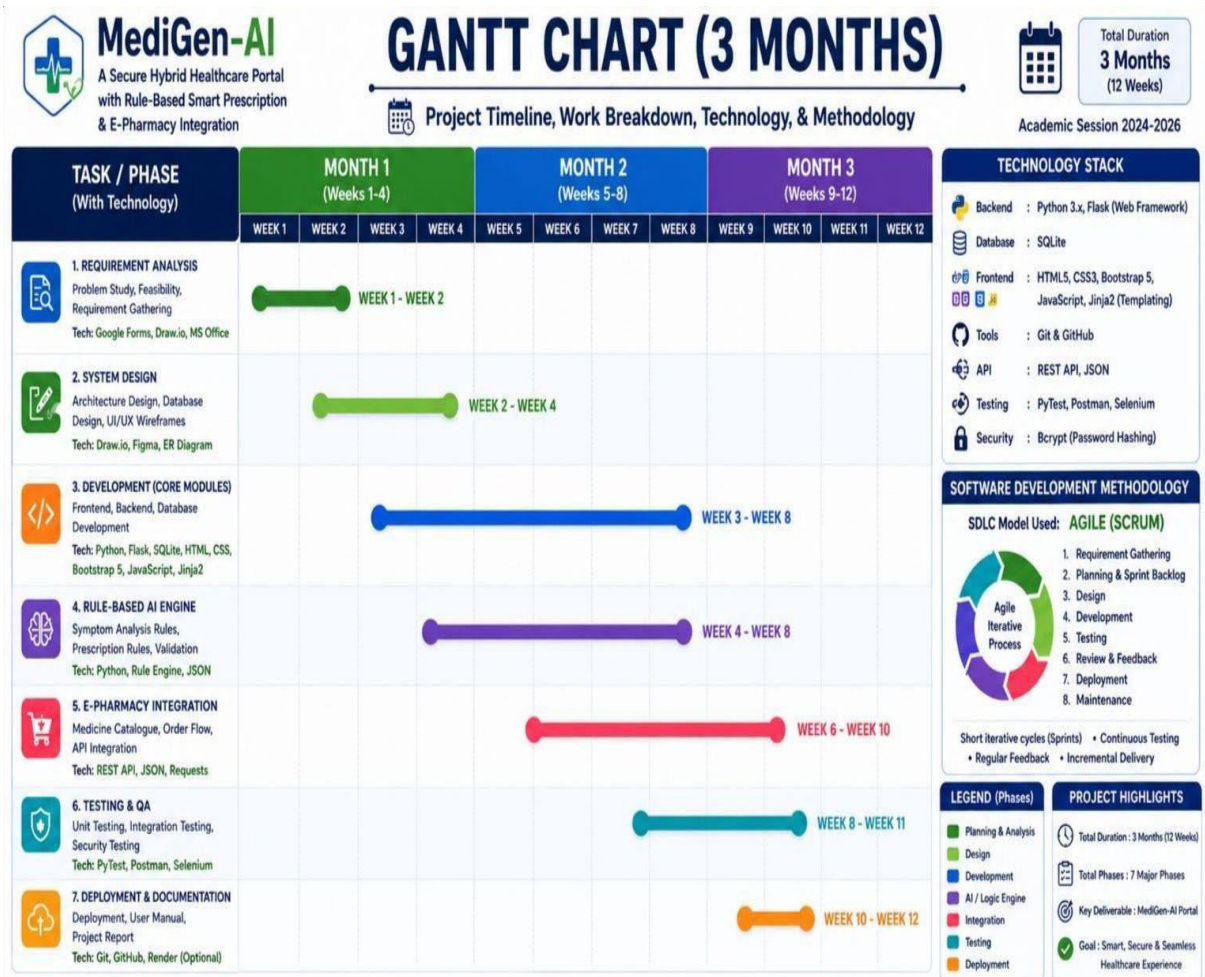
13. Documentation Phase

Description

Project documentation was prepared throughout the development lifecycle.

- Documents Prepared
- SRS Document
- System Design Document
- Testing Report
- Project Report
- User Manual
- Duration
- Continuous process during the project.

14. Sample Gantt Chart Workflow



Project Title:- MediGen-AI

15. Advantages of Gantt Chart

- Easy project visualization
- Better time management
- Improved task tracking
- Efficient resource utilization
- Simplified project scheduling
- Better coordination among team members

16. Limitations of Gantt Chart

- Difficult to manage for very large projects
- Requires regular updates
- Complex dependencies may reduce readability

17. Tools Used for Gantt Chart Design

The following tools were used:

Tool Name	Purpose
Microsoft Excel	Gantt chart creation
Canva	Project timeline design
Draw.io	Workflow planning
Microsoft Project	Project management

18. Challenges Faced During Project Scheduling

- Some challenges encountered include:
- Managing multiple modules simultaneously
- Balancing development and testing time
- Handling requirement changes
- Synchronizing AI module development

These issues were resolved through proper planning and continuous monitoring.

19. Future Improvements

Future project scheduling improvements may include:

- Automated project tracking tools
- Cloud-based project management
- Real-time task monitoring
- Agile sprint management integration
- Iterative software methodology

20. Conclusion

The Gantt Chart played a significant role in the successful planning and execution of the MediGen-AI Hospital Management System. It helped organize project activities, monitor task progress, manage timelines, and ensure systematic development of all modules. The Gantt Chart improved project coordination, reduced delays, and supported timely completion of the healthcare management system.

Overall, the Gantt Chart served as an effective project management tool that contributed significantly to the successful development, testing, deployment, and maintenance of the MediGen-AI Hospital Management System.

CHAPTER 18: AI INTEGRATION WORKFLOW

1. Introduction

Artificial Intelligence (AI) Integration Workflow refers to the systematic process of incorporating AI functionalities into a software application to automate tasks, improve decision-making, enhance efficiency, and reduce human effort. AI workflows define how data is collected, processed, analyzed, and transformed into intelligent outputs within the system.

In the MediGen-AI Hospital Management System, AI integration plays a crucial role in improving healthcare services by assisting doctors in prescription generation, symptom analysis, patient record processing, and decision support. The AI workflow was designed to automate medical processes while ensuring accuracy, speed, and reliability.

The integration of AI into the hospital management system helps improve healthcare workflow efficiency, reduce manual prescription errors, and support intelligent healthcare management.

2. Objectives of AI Integration

The primary objectives of AI integration in the project are:

- To automate prescription generation.
- To analyze patient symptoms intelligently.
- To reduce manual errors in diagnosis and prescriptions.
- To improve healthcare workflow efficiency.
- To provide faster medical assistance.
- To support doctors in decision-making.
- To enhance patient management and monitoring.
- To improve system intelligence and automation.

3. Need for AI Integration in the Project

Traditional hospital management systems mainly focus on record management and administration. However, healthcare systems require intelligent processing for:

- Faster diagnosis support
- Prescription recommendation
- Symptom analysis
- Patient monitoring
- Medical decision support

Therefore, AI integration was implemented in the MediGen-AI system to make the healthcare management process more efficient, intelligent, and automated.

AI-integrated projects combine machine learning, data processing, and large language models (LLMs) to solve practical problems. Popular areas include building RAG (Retrieval-Augmented Generation) systems for document querying, resume optimizers, and multimodal search tools using frameworks like LangChain and Hugging Face

Project Title:- MediGen-AI

4. AI Workflow Architecture

The AI workflow in the MediGen-AI system follows a step-by-step intelligent processing structure.

AI Workflow Stages

1. Data Collection

2. Data Preprocessing

3. Symptom Analysis

4. AI Processing

5. Prescription Recommendation

6. Result Generation

7. Database Storage

5. Data Collection Workflow

Description

The workflow starts with collecting patient-related healthcare data.

Input Data

- Patient symptoms
- Medical history
- Age and gender
- Disease information
- Previous prescriptions
- Data Sources
- Patient registration forms
- Doctor diagnosis forms
- Medical records database

6. Data Preprocessing Workflow

Description

Collected data is cleaned and prepared before AI processing.

Activities Performed

- Removing invalid data
- Formatting medical records
- Data validation
- Text normalization
- Missing value handling

Benefits

- Improves AI accuracy
- Reduces processing errors
- Enhances prediction quality

7. Symptom Analysis Workflow

Description

The AI system analyzes patient symptoms entered by doctors or medical staff.

Project Title:- MediGen-AI

Process

Symptoms are extracted from input.

AI compares symptoms with medical datasets.

Related disease patterns are identified.

Example

- Input Symptoms
- Fever
- Cough
- Headache
- AI Output
- Possible viral infection
- Suggested medicines

8. AI Processing Workflow

Description

The AI engine processes patient information using intelligent algorithms.

AI Processing Activities

- Pattern recognition
- Medical data analysis
- Recommendation generation
- Decision support

Technologies Used

- Python

Machine Learning Logic

- Rule-Based AI Processing

9. Prescription Recommendation Workflow

Description

Based on symptom analysis, the AI system generates prescription suggestions.

Process

1. AI analyzes symptoms.
2. Disease patterns identified.
3. Medicine recommendations generated.
4. Doctor reviews AI suggestions.
5. Final prescription approved.

Benefits

- Faster prescription generation
- Reduced human error
- Improved treatment efficiency

10. Doctor Verification Workflow

Description

Doctors verify and approve AI-generated recommendations.

Activities

- Reviewing diagnosis
 - Modifying prescriptions
 - Approving final treatment plan
-

Project Title:- MediGen-AI

Importance

AI supports doctors but does not replace medical professionals. Final decisions remain under doctor supervision.

It is a rule-based AI Support system

11. Database Storage Workflow

Description

After verification, the generated prescription and medical records are stored securely.

- **Data Stored**
- **Symptoms**
- **Diagnosis**
- **Prescriptions**
- **Patient reports**
- **AI recommendations**

Benefits

- **Future record access**
- **Historical analysis**
- **Improved patient monitoring**

12. AI Integration Workflow Diagram Description

The AI workflow diagram of the MediGen-AI system contains the following flow:

Step 1

Patient enters symptoms and healthcare details.

Step 2

Data is validated and preprocessed.

Step 3

AI engine analyzes symptoms and medical history.

Step 4

Prescription recommendations are generated.

Step 5

Doctor reviews and approves prescription.

Step 6

Final records are stored in the database.

Step 7

Reports and prescriptions are generated for the patient.

13. AI Modules Integrated in the System

The Patient Module is one of the most important components of the MediGen-AI Hospital Management System. This module is designed to manage all patient-related activities efficiently through digital healthcare management and automated workflows.

The Patient Module helps hospitals maintain organized patient records, manage appointments, store medical histories, and improve communication between patients, doctors, and hospital staff. It reduces manual paperwork, improves data accessibility, and supports faster healthcare services.

Project Title:- MediGen-AI

The following AI-related modules were integrated:

AI Module	Purpose
Symptom Analyzer	Analyze patient symptoms
Prescription Generator	Suggest medicines
Decision Support System	Assist doctors
Medical Recommendation Engine	Generate healthcare suggestions

14. Technologies Used for AI Integration

The following technologies were used:

Technology	Purpose
Python	AI processing
Django	Backend integration
SQLite/MySQL	Data storage
Machine Learning Logic	Intelligent prediction
APIs	Data communication

15. Advantages of AI Integration

- Faster diagnosis assistance
- Improved prescription accuracy
- Reduced manual workload
- Enhanced healthcare efficiency
- Better patient monitoring
- Intelligent healthcare management
- Reduced prescription errors

16. Challenges Faced During AI Integration

Some challenges encountered include:

- Managing healthcare datasets
- AI prediction accuracy
- Database synchronization
- Medical recommendation validation
- Performance optimization

These challenges were resolved through continuous testing and optimization.

17. Security in AI Workflow

Security mechanisms implemented include:

- Secure patient data handling
 - Role-based access control
-

Project Title:- MediGen-AI

-
- Encrypted database storage
 - Authentication and authorization
 - Secure API communication

These measures help protect sensitive healthcare information

18. Future Enhancements in AI Workflow

Future AI improvements may include:

- Advanced disease prediction
- Deep learning integration
- Voice-based medical assistance
- Real-time patient monitoring
- AI chatbot support

19. Importance of AI in Healthcare Systems

AI plays a significant role in modern healthcare because it:

- Supports medical professionals
- Improves healthcare efficiency
- Reduces human errors
- Speeds up diagnosis processes
- Improves patient care quality
- Supports intelligent decision-making

The MediGen-AI system demonstrates how AI can enhance digital healthcare management effectively.

20. Conclusion

The AI Integration Workflow significantly enhanced the functionality and intelligence of the MediGen-AI Hospital Management System. The integration of AI technologies helped automate symptom analysis, prescription recommendation, healthcare decision support, and patient data processing. The workflow ensured systematic processing of healthcare data while maintaining accuracy, efficiency, and security.

Overall, AI integration improved hospital workflow automation, reduced manual effort, enhanced prescription accuracy, and provided intelligent healthcare support, making the MediGen-AI system a modern, scalable, and efficient digital healthcare management solution.

AI integration shifts systems from reactive to proactive, data-driven frameworks. Ultimately, AI acts as a collaborative partner rather than a complete replacement. Its successful implementation requires harmonizing human expertise with machine capabilities, prioritizing ethical governance, and focusing on strategic, long-term value creation.

AI integration embeds artificial intelligence directly into existing business workflows, automating routine tasks and enabling data-driven decision making. While it requires significant technological and organizational adaptation, it provides a crucial competitive advantage in fast-moving markets by driving scalable innovation and lowering operational costs.

Ultimately, AI integration is no longer a peripheral experiment but a central strategic mandate for businesses. As emerging technologies streamline how Large Language Models (LLMs) connect to organizational data, the focus is shifting toward.

CHAPTER 19: SECURITY MECHANISMS

1. Introduction

Security Mechanisms are the techniques, protocols, and protective measures implemented in a software system to safeguard data, prevent unauthorized access, and ensure system reliability. In healthcare applications, security is extremely important because sensitive patient information, medical records, prescriptions, and administrative data must be protected from cyber threats and misuse.

In the MediGen-AI Hospital Management System, various security mechanisms were implemented to ensure confidentiality, integrity, and availability of healthcare data. The system was designed with secure authentication, access control, encrypted communication, and database protection techniques to maintain secure hospital operations.

The implementation of security mechanisms helps build a reliable and trustworthy digital healthcare management environment.

2. Objectives of Security Mechanisms

The major objectives of implementing security mechanisms are:

- To protect sensitive patient information.
- To prevent unauthorized system access.
- To secure medical records and prescriptions.
- To ensure safe communication between modules.
- To maintain data confidentiality and integrity.
- To prevent cyber-attacks and data breaches.
- To improve system reliability and trustworthiness.

3. Need for Security in Healthcare Systems

Healthcare systems manage highly sensitive information such as:

- Patient medical records
- Doctor information
- Prescriptions
- Appointment detail
- Personal identification data
- Hospital administrative records

Without proper security mechanisms, the system may face:

- Data theft
- Unauthorized access
- Database attacks
- Identity misuse
- Data corruption
- Data Protection issue
- Data loss by the attack

Encryption: Converts readable data into unreadable code (e.g., using BitLocker or VeraCrypt) that requires a decryption key.

Therefore, strong security measures are essential in the MediGen-AI Hospital Management System.

Project Title:- MediGen-AI

4. Types of Security Mechanisms Implemented

The following security mechanisms were implemented in the project:

1. User Authentication
2. Authorization and Access Control
3. Password Encryption
4. Session Management
5. Database Security
6. Input Validation
7. Secure API Communication
8. Backup and Recovery
9. Error Handling and Logging

5. User Authentication Mechanism

Introduction

Authentication verifies the identity of users before granting system access.

- Authentication Features
- Username and password login
- Secure admin authentication
- Doctor login verification
- Medical staff login system

Workflow

1. User enters credentials.
2. System validates information.
3. Access granted after verification.

Benefits

- Prevents unauthorized access
- Improves system security
- Protects patient data

6. Authorization and Access Control

Introduction

Authorization determines what actions users are allowed to perform.

Role-Based Access Control (RBAC)

Different user roles were implemented:

User Role	Permissions
Admin	Full system control
Doctor	Prescription and patient access
Medical Staff	Patient registration
Patient	Limited personal access

Benefits

- Restricts unauthorized activities
 - Protects confidential information
 - Improves system management
 - Ensure CIA properties
-

Project Title:- MediGen-AI

7. Password Encryption

Introduction

Passwords were encrypted before storage to enhance security.

- **Encryption Techniques**
- **Django password hashing**
- **Secure password storage**
- **Encoded authentication data**

Benefits

- **Prevents password theft**
- **Protects user credentials**
- **Enhances authentication security**

8. Session Management

Introduction

Session management helps maintain secure user activity during login.

- **Security Features**
- **Automatic session timeout**
- **Secure session IDs**
- **Logout functionality**
- **Prevention of session hijacking**

Benefits

- **Improves account security**
- **Prevents unauthorized session access**

9. Database Security

Introduction

Database security protects stored healthcare information.

- **Security Measures Implemented**
- **Restricted database access**
- **Secure database connectivity**
- **Backup and recovery system**
- **Query validation**
- **Data integrity management**
- **Protected Data**
- **Patient records**
- **Doctor information**
- **Prescriptions**
- **Appointment details**

10. Input Validation and Data Sanitization

Introduction

Input validation ensures only valid and safe data enters the system.

Validation Activities

- **Form validation**
 - **Data type checking**
 - **Required field verification**
 - **Prevention of invalid input**
-

Project Title:- MediGen-AI

- **Security Against Attacks**
- **SQL Injection prevention**
- **Cross-Site Scripting (XSS) protection**
- **Malicious input filtering**

11. Secure API Communication

Introduction

Secure APIs ensure safe communication between frontend and backend systems.

- **Security Features**
- **Authentication tokens**
- **Secure request validation**
- **Protected data transmission**

Benefits

- **Safe data exchange**
- **Improved communication security**

12. Backup and Recovery Mechanism

Introduction

Backup systems protect against accidental data loss and system failure.

Backup Activities

- **Regular database backup**
- **Secure backup storage**
- **Recovery management**
- **Benefits**
- **Prevents permanent data loss**
- **Ensures system continuity**

13. Error Handling and Logging

Introduction

Error handling and logging help monitor system activity and detect security issues.

Activities

- **Error monitoring**
- **Activity logging**
- **Exception handling**
- **Security event tracking**

Benefits

- **Faster issue detection**
- **Improved system monitoring**
- **Better debugging support**

14. Security Workflow in MediGen-AI

The security workflow follows these steps:

Step 1

User enters login credentials.

Step 2

Authentication system verifies identity.

Step 3

Access permissions are checked.

Project Title:- MediGen-AI

Step 4

Authorized user accesses modules.

Step 5

All activities are monitored and logged.

Step 6

Data is securely stored in the database.

15. Security Technologies Used

Technology	Purpose
Django Authentication	User login security
Password Hashing	Credential protection
SQLite/MySQL Security	Database protection
Session Management	User session security
Form Validation	Input security

16. Advantages of Security Mechanisms

- Protects sensitive healthcare data
- Prevents unauthorized access
- Reduces cyber threats
- Improves system reliability
- Enhances user trust
- Ensures secure healthcare operations

17. Challenges Faced During Security Implementation

Some challenges encountered include:

- Managing user permissions
- Securing patient records
- Preventing unauthorized database access
- Handling session management securely
- Implementing secure authentication

These challenges were resolved using secure coding practices and continuous testing.

18. Future Security Enhancements

Future improvements may include:

- Two-factor authentication (2FA)
- Biometric authentication
- Cloud security integration
- AI-based threat detection
- Advanced encryption mechanisms
- Blockchain-based medical record security

19. Importance of Security in Healthcare Systems

Security is highly important in healthcare systems because:

- Patient data is confidential.
- Medical records must remain accurate.
- Unauthorized access can affect patient care.

Hospitals require secure digital operations.

CHAPTER 20: ADVANTAGES OF THE SYSTEM

1. Introduction

The MediGen-AI Hospital Management System is an intelligent digital healthcare solution designed to automate and simplify hospital operations. The system integrates modern technologies such as Artificial Intelligence (AI), database management, secure authentication, and digital record handling to improve healthcare services and hospital administration.

The implementation of this system provides several advantages for hospitals, doctors, medical staff, administrators, and patients. It improves operational efficiency, reduces manual workload, enhances data security, and supports faster healthcare management processes.

2. Major Advantages of the System

The major advantages of the MediGen-AI Hospital Management System are:

- 1. Automation of Hospital Operations**
- 2. Faster Patient Management**
- 3. AI-Based Prescription Support**
- 4. Secure Data Management**
- 5. Reduced Manual Errors**
- 6. Improved Healthcare Workflow**
- 7. Better Appointment Management**
- 8. Centralized Database System**
- 9. Time and Cost Efficiency**
- 10. Improved Reporting System**

3. Automation of Hospital Operations

The system automates multiple hospital activities such as:

- **Patient registration**
- **Appointment scheduling**
- **Prescription generation**
- **Doctor management**
- **OPD handling**
- **Report generation**

Benefits

- **Reduces paperwork**
- **Minimizes manual effort**
- **Improves operational efficiency**

4. Faster Patient Management

The system enables quick patient registration and record management.

Features

- **Digital patient records**
 - **Fast patient search**
 - **Instant record retrieval**
 - **ABHA number integration**
-

Project Title:- MediGen-AI

Benefits

- Saves time
- Improves patient handling
- Reduces waiting time

5. AI-Based Prescription Support

One of the major advantages of the system is AI integration for healthcare support.

AI Features

- Symptom analysis
- Prescription recommendation
- Decision support for doctors

Benefits

- Faster diagnosis support
- Reduced prescription errors
- Improved healthcare efficiency

6. Secure Data Management

The system implements strong security mechanisms to protect healthcare data.

Security Features

- User authentication
- Password encryption
- Role-based access control
- Secure database management

Benefits

- Protects patient privacy
- Prevents unauthorized access
- Ensures data confidentiality

7. Reduced Manual Errors

- Manual hospital management often leads to:
- Prescription mistakes
- Record duplication
- Data loss
- Calculation errors

The MediGen-AI system minimizes these issues through automation and validation mechanisms

Benefits

- Improved accuracy
- Reliable healthcare records
- Better patient safety

8. Improved Healthcare Workflow

The system improves coordination between:

- Doctors
 - Medical staff
-

Project Title:- MediGen-AI

- **Administrators**
- **Patients**

Benefits

- **Faster communication**
- **Efficient patient handling**
- **Smooth hospital workflow**

9. Better Appointment Management

- **The appointment management module helps:**
- **Schedule appointments**
- **Track doctor availability**
- **Manage patient queues**

Benefits

- **Organized appointment handling**
- **Reduced scheduling conflicts**
- **Better patient experience**

10. Centralized Database System

All healthcare information is stored in a centralized database.

- **Data Stored**
- **Patient records**
- **Doctor information**
- **Prescriptions**
- **Appointment details**
- **OPD reports**

Benefits

- **Easy data retrieval**
- **Improved data consistency**
- **Better record management**

11. Time Efficiency

- **Automation significantly reduces processing time.**
- **Time-Saving Features**
- **Instant record access**
- **Fast report generation**
- **Quick prescription handling**

Benefits

- **Faster hospital operations**
- **Reduced administrative delays**

12. Cost Efficiency

- **Digital management reduces operational costs.**
 - **Cost Reduction Areas**
 - **Reduced paperwork**
 - **Lower storage requirements**
 - **Reduced manual workforce dependency**
-

Project Title:- MediGen-AI

Benefits

- Improved resource utilization
- Cost-effective hospital management

13. Improved Reporting System

The system generates digital reports efficiently.

- Reports Generated
- Patient reports
- Prescription reports
- Appointment reports
- OPD reports

Benefits

- Better hospital monitoring
- Improved decision-making
- Faster report generation

14. User-Friendly Interface

The system provides an easy-to-use interface for all users.

Features

- Simple navigation
- Dashboard management
- Responsive design

Benefits

- Easy system operation
- Reduced training requirements

15. Scalability and Flexibility

The system can be expanded according to future requirements.

- Future Expansion Possibilities
- Cloud integration
- Mobile application support
- Multi-hospital connectivity
- Online consultation

Benefits

- Long-term usability
- Adaptability to new technologies
- Improved decision-making
- Faster report generation

16. Real-Time Data Access

The system provides real-time access to healthcare information.

Benefits

- Instant patient record access
 - Faster treatment decisions
 - Improved monitoring
-

Project Title:- MediGen-AI

17. Improved Decision-Making

AI support and digital reports help doctors and administrators make better decisions.

Benefits

- Accurate medical analysis
- Better hospital management
- Improved patient care

18. Environment-Friendly System

Digital hospital management reduces paper usage.

Benefits

- Eco-friendly operations
- Reduced physical storage requirements

19. Reliability and Availability

The system is designed for continuous and reliable operation.

Features

- Backup mechanisms
- Error handling
- Secure database management
- Benefits
- Reduced system downtime
- Improved reliability

20. Advantages for Different Users

User	Advantages
Patient	Faster service and better record management
Doctor	AI support and easy prescription handling
Medical Staff	Simplified workflow

Admin Better monitoring and control

21. Future Advantages

Future enhancements may provide:

- AI disease prediction
- Voice-based healthcare assistance
- Cloud-based hospital management
- Smart healthcare analytics

22. Conclusion

The MediGen-AI Hospital Management System provides numerous advantages by integrating automation, AI technology, secure database management, and digital healthcare workflows into a single platform. The system improves healthcare efficiency, reduces manual workload, enhances data security, and supports better.

The Patient Module is one of the most important components of the MediGen-AI Hospital Management System. This module is designed to manage all patient-

Overall, the system offers a reliable, scalable, intelligent, and user-friendly solution for modern hospital management and digital healthcare operations.

CHAPTER 21: LIMITATIONS

1. Introduction

Although the MediGen-AI Hospital Management System provides an efficient, intelligent, and automated healthcare management solution, certain limitations still exist in the current version of the system. These limitations are mainly related to infrastructure, scalability, AI accuracy, security dependency, and technological constraints.

Identifying system limitations is important because it helps developers understand areas that require improvement and supports future enhancement planning. Despite these limitations, the system successfully performs major healthcare management operations effectively.

2. Major Limitations of the System

The major limitations of the MediGen-AI Hospital Management System are:

1. Dependence on Internet and System Availability
2. Limited AI Accuracy
3. Local Server Deployment
4. Limited Scalability
5. Data Security Risks
6. Dependency on Technical Knowledge
7. Limited Real-Time Monitoring
8. Hardware Dependency
9. Maintenance Requirements
10. Limited Multi-Hospital Support

3. Dependence on Internet and System Availability

The system depends on digital infrastructure and server availability.

Limitations

- System access may be interrupted during server failure.
- Poor internet connectivity can affect performance.
- Real-time operations may slow down in low network conditions.
- Impact
- Delays in accessing patient records
- Interrupted healthcare workflow

4. Limited AI Accuracy

The AI module provides intelligent recommendations but may not always guarantee complete medical accuracy.

Limitations

- AI suggestions depend on input quality.
- Incorrect symptom input may produce inaccurate recommendations.
- AI cannot fully replace professional medical judgment.

Impact

Doctor verification is always required.

Complex medical cases may need manual analysis.

Project Title:- MediGen-AI

5. Local Server Deployment

Currently, the system is deployed mainly on a local environment.

Limitations

- Limited remote accessibility
- Restricted scalability
- No cloud-based synchronization

Impact

- Difficult for large-scale hospital networks
- Limited multi-location access

6. Limited Scalability

The current system is designed mainly for small and medium-scale hospital operations.

Limitations

- Large-scale healthcare data may affect performance.
- Increased users may slow database operations.

Impact

- Performance optimization required for larger hospitals

7. Data Security Risks

Although security mechanisms are implemented, healthcare systems are always vulnerable to cyber threats.

- Possible Risks
- Unauthorized access attempts
- Data breaches
- Malware attacks
- Server vulnerabilities
- Impact
- Sensitive patient data may be at risk if security is not updated regularly.

8. Dependency on Technical Knowledge

Users require basic technical understanding to operate the system efficiently.

Limitations

- Staff training may be required.
- Non-technical users may face difficulties initially.

Impact

- Learning curve for hospital staff

9. Limited Real-Time Monitoring

The current version provides basic monitoring features.

Limitations

- No advanced real-time patient tracking
- Limited AI-based monitoring

Impact

- Future enhancements required for advanced healthcare analytics

10. Hardware Dependency

The system depends on proper hardware infrastructure.

Requirements

Project Title:- MediGen-AI

- **Computer systems**
- **Database storage**
- **Server support**
- **Network connectivity**

Limitations

- **Hardware failure may affect operations.**
- **Low-performance systems may reduce speed.**

11. Maintenance Requirements

The system requires regular maintenance and updates.

Maintenance Activities

- **Bug fixing**
- **Database backup**
- **Security updates**
- **Performance optimization**

Impact

- **Continuous monitoring is necessary**

12. Limited Multi-Hospital Support

The current system mainly supports a single hospital environment.

Limitations

- **No centralized cloud integration**
- **Limited cross-hospital communication**

Impact

- **Difficult for large healthcare networks**

13. Database Limitations

The current database configuration has some restrictions.

Limitations

- **SQLite is suitable mainly for smaller systems.**
- **Large datasets may require advanced database solutions.**

Future Requirement

Migration to MySQL/PostgreSQL or cloud database systems

14. Limited Mobile Support

The current project mainly focuses on web-based implementation.

Limitations

- **No dedicated mobile application**
- **Limited mobile optimization**

Impact

- **Reduced accessibility on smartphones**

15. AI Dataset Dependency

AI performance depends heavily on healthcare datasets.

Limitations

- **Limited training data may reduce prediction quality.**
- **Medical datasets require continuous updates.**

Impact

- **AI recommendations may vary in complex scenarios**

16. Integration Limitations

Project Title:- MediGen-AI

The current version has limited third-party integration capabilities.

Limitations

- **No payment gateway integration**
- **Limited telemedicine support**
- **No wearable device integration**

17. User Dependency on System Accuracy

Healthcare staff depend on the accuracy of system-generated data.

Limitation

Incorrect data entry by users may affect:

- **Prescriptions**
- **Patient records**
- **Reports**

Impact

- **Manual verification is still necessary**

18. Challenges Due to Rapid Technological Changes

Healthcare technology changes rapidly.

Limitations

- **Frequent software updates required**
- **AI models require continuous improvement**
- **Security standards evolve regularly**

19. Future Improvements to Overcome Limitations

The following future enhancements can overcome current limitations:

- **Cloud-based deployment**
- **Advanced AI diagnosis system**
- **Mobile application integration**
- **Multi-hospital connectivity**
- **Real-time patient monitoring**
- **Blockchain-based healthcare security**
- **AI chatbot support**
- **Automated backup systems**

20. Conclusion

Although the MediGen-AI Hospital Management System has some limitations related to scalability, AI accuracy, deployment infrastructure, and security dependency, it still provides an efficient and intelligent healthcare management solution. Most limitations can be addressed through future technological improvements, cloud integration, advanced AI implementation, and enhanced security mechanisms.

Overall, the system successfully automates major hospital operations, improves healthcare workflow, and provides a strong foundation for future expansion and modernization in digital healthcare. Medical Generative AI is a powerful clinical assistant but a dangerous independent decision-maker. While it significantly reduces administrative burdens and accelerates data synthesis, its current technical and ethical limitations prevent it from replacing human clinical judgment. For the foreseeable future, Medical GenAI must operate strictly under a "Human-in-the-Loop" framework to ensure patient safety and regulatory compliance.

CHAPTER 22: SOCIAL IMPACT

1. Introduction

The MediGen-AI Hospital Management System is designed not only to improve hospital operations but also to create a positive impact on society and healthcare services. The integration of Artificial Intelligence (AI), digital record management, automation, and secure healthcare processing helps improve the quality, accessibility, and efficiency of medical services for patients, doctors, hospitals, and society.

The system contributes to digital healthcare transformation by reducing manual processes, improving patient care, increasing healthcare accessibility, and supporting modern smart healthcare management practices

2. Objectives of Social Impact

The major objectives of the social impact of the system are:

- To improve healthcare accessibility.
- To provide faster and better medical services.
- To reduce manual healthcare management problems.
- To support digital transformation in healthcare.
- To improve patient safety and healthcare quality.
- To enhance hospital efficiency.
- To reduce healthcare operational burden.
- To promote smart healthcare systems.

3. Improvement in Healthcare Services

The system improves overall healthcare services through automation and digital management.

Improvements

- Faster patient registration
- Quick access to medical records
- Improved prescription handling
- Better appointment management
- Faster healthcare decision-making

Social Benefits

- Reduced patient waiting time
- Better treatment efficiency
- Improved healthcare experience

4. Reduction in Manual Work

Traditional hospital systems involve heavy paperwork and manual operations.

- Problems in Manual Systems
 - Record loss
 - Prescription errors
 - Slow data retrieval
 - Increased workload
-

Project Title:- MediGen-AI

The MediGen-AI system automates these operations digitally.

Social Impact

- **Reduced administrative burden**
- **Improved work efficiency**
- **Better resource utilization**

5. Improved Patient Care

The system helps doctors and medical staff provide better healthcare services.

Features Supporting Patient Care

- **AI-based prescription assistance**
- **Digital patient history**
- **Faster diagnosis support**
- **Organized healthcare records**
- **Benefits to Society**
- **Improved treatment quality**
- **Better patient monitoring**
- **Increased patient satisfaction**

6. Promotion of Digital Healthcare

The project supports digital transformation in hospitals and healthcare institutions.

- **Digital Features**
- **Electronic medical records**
- **Online data management**
- **Automated workflows**
- **AI integration**

Social Contribution

- **Encourages smart healthcare adoption**
- **Supports Digital India initiatives**
- **Modernizes hospital operations**

7. Reduction in Prescription Errors

Manual prescription writing may cause:

- **Incorrect medicine dosage**
- **Illegible handwriting**
- **Wrong medicine selection**

AI-assisted prescription generation reduces these risks.

Social Benefits

- **Improved patient safety**
- **Reduced medical mistakes**
- **Better treatment reliability**

8. Time-Saving Benefits

Automation improves healthcare efficiency.

- **Time-Saving Areas**
 - **Appointment scheduling**
 - **Patient registration**
 - **Report generation**
-

Project Title:- MediGen-AI

- **Prescription handling**
- **Impact on Society**
- **Faster healthcare services**
- **Reduced waiting time**
- **Improved hospital workflow**

9. Better Healthcare Data Management

The system maintains centralized digital healthcare records.

Benefits

- **Easy patient history access**
- **Faster data retrieval**
- **Secure record maintenance**

Social Impact

- **Improved healthcare continuity**
- **Better medical decision-making**

10. Improved Accessibility to Healthcare Information

Digital systems make healthcare data easily accessible to authorized users.

Advantages

- **Instant access to reports**
- **Better communication among staff**
- **Improved coordination between departments**

11. Support for Rural and Small Hospitals

The system can help smaller healthcare centers improve their management process.

Benefits

- **Affordable digital management**
- **Simplified healthcare operations**
- **Reduced paperwork dependency**

Social Contribution

Supports healthcare modernization in rural areas

12. Employment and Skill Development

The adoption of digital healthcare systems increases the demand for technical and healthcare management skills.

Impact

- **Encourages technical learning**
- **Promotes digital literacy**
- **Supports IT and healthcare collaboration**

13. Environmental Impact

The system reduces paper usage through digital management.

Benefits

- **Eco-friendly healthcare operations**
 - **Reduced physical file storage**
 - **Lower paper waste**
 - **Social Contribution**
 - **Supports environmental sustainability**
-

Project Title:- MediGen-AI

14. Improved Hospital Administration

The system helps administrators manage hospital activities efficiently.

Administrative Features

- Staff management
- Report generation
- Database monitoring
- Appointment tracking

Social Benefits

- Better healthcare organization
- Improved hospital productivity

15. Enhanced Security and Privacy

The system protects patient information using secure authentication and database security mechanisms.

- Security Features
- Password encryption
- Access control
- Secure data storage
- Social Impact

Improved trust in digital healthcare systems

Better protection of patient privacy

16. AI Contribution to Society

Artificial Intelligence integration supports intelligent healthcare management.

AI Benefits

- Faster analysis
- Better prescription recommendations
- Healthcare decision support

17. Future Social Impact

Future enhancements may provide:

- Telemedicine support
- Mobile healthcare applications
- AI disease prediction
- Cloud healthcare systems
- Multi-hospital connectivity

These improvements will further strengthen digital healthcare services.

18. Conclusion

The MediGen-AI Hospital Management System creates a significant positive social impact by improving healthcare efficiency, reducing manual workload, enhancing patient care, and supporting digital transformation in hospitals. The integration of AI and automation technologies helps provide faster, safer, and more reliable healthcare services for society.

Overall, the system contributes to modern healthcare development, promotes smart healthcare practices, improves hospital management, and supports better healthcare accessibility and quality for patients and medical professionals.

CHAPTER 23: FUTURE SCOPE

1. Introduction

The MediGen-AI Hospital Management System is a modern digital healthcare solution designed to automate hospital operations and improve healthcare management using Artificial Intelligence (AI), database systems, and secure digital technologies. Although the current system successfully manages major hospital functionalities, there are several opportunities for future enhancement and expansion.

The future scope of the project focuses on improving system intelligence, scalability, accessibility, security, automation, and healthcare analytics. Future technologies can further transform the system into a fully smart, cloud-based, and AI-driven healthcare platform.

2. Objectives of Future Enhancements

The major objectives of future improvements are:

- To improve healthcare automation.
- To enhance AI capabilities.
- To support cloud-based healthcare systems.
- To increase system scalability.
- To improve security and accessibility.
- To provide real-time healthcare monitoring.
- To support multi-hospital connectivity.
- To improve patient care and healthcare analytics.

3. Cloud-Based Deployment

Currently, the system mainly works in a local environment. Future versions can be deployed on cloud platforms.

Future Features

- Cloud database integration
- Online hospital management
- Remote data access
- Multi-location synchronization

Benefits

- Improved scalability
- Better accessibility
- Centralized healthcare management

4. Mobile Application Integration

A dedicated mobile application can be developed for doctors, patients, and administrators.

Features

- Digital prescription access
 - Real-time notifications
 - Patient health monitoring
 - Mobile appointment booking
-

Project Title:- MediGen-AI

Benefits

- **Increased accessibility**
- **Better user experience**
- **Remote healthcare support**

5. Advanced AI and Machine Learning Integration

Future versions can implement advanced AI and Machine Learning algorithms.

- **AI Enhancements**
- **Disease prediction**
- **Smart diagnosis assistance**
- **AI-based healthcare analytics**
- **Predictive healthcare monitoring**

Benefits

- **Improved medical decision-making**
- **Faster diagnosis support**
- **Enhanced healthcare intelligence**

6. Telemedicine Integration

The system can be expanded to support telemedicine services.

Features

- **Online doctor consultation**
- **Video appointment system**
- **Digital healthcare communication**

Benefits

- **Remote patient treatment**
- **Improved rural healthcare access**
- **Reduced hospital crowding**

7. Real-Time Patient Monitoring

Future systems may include IoT-based patient monitoring.

- **Monitoring Features**
- **Heart rate monitoring**
- **Blood pressure tracking**
- **Oxygen level monitoring**
- **Emergency alerts**

Benefits

- **Improved patient safety**
- **Faster emergency response**
- **Better critical care monitoring**

8. Multi-Hospital Connectivity

The current system mainly supports a single hospital environment.

- **Future Improvements**
- **Multi-hospital database integration**
- **Shared patient records**
- **Centralized healthcare network**

Benefits

Project Title:- MediGen-AI

- Better healthcare coordination
- Improved patient record accessibility

9. Blockchain-Based Security

Blockchain technology can improve healthcare data security.

Features

- Tamper-proof medical records
- Secure healthcare transactions
- Decentralized record management

Benefits

- Enhanced data security
- Improved trust and transparency

10. Voice-Based Healthcare Assistance

Voice recognition technology can be integrated into the system.

Features

- Voice-based patient registration
- Voice prescription generation
- Smart healthcare assistants

Benefits

- Faster operations
- Better accessibility for non-technical users

11. AI Chatbot Integration

An AI healthcare chatbot can assist patients and staff.

Features

- Appointment assistance
- Basic symptom guidance
- FAQ support
- Automated patient communication

Benefits

- Reduced workload
- Faster patient support
- Improved user interaction

12. Big Data and Healthcare Analytics

Future systems can analyze large healthcare datasets.

Features

- Disease trend analysis
- Hospital performance analytics
- Patient behavior analysis

Benefits

- Better healthcare planning
- Improved hospital management

13. Online Payment Gateway Integration

The system can include secure online payment features.

Features

Project Title:- MediGen-AI

- Digital billing
- Online payment processing
- Automated invoice generation

Benefits

Cashless healthcare operations

Faster payment management

14. Advanced Security Enhancements

Future security improvements may include:

Features

- Two-factor authentication (2FA)
- Biometric authentication
- AI-based cyber threat detection
- Advanced encryption systems

Benefits

- Stronger healthcare data protection
- Improved system security

15. Integration with Government Healthcare Systems

The system can integrate with national healthcare databases and digital healthcare initiatives.

- Possible Integrations
- ABHA integration
- Government health schemes
- National healthcare databases

Benefits

- Improved healthcare interoperability
- Better patient identification

16. Smart Report Generation

Future systems can generate intelligent reports automatically.

Features

- AI-generated medical summaries
- Automated healthcare statistics
- Smart hospital reports

Benefits

- Faster decision-making
- Improved administrative efficiency

17. Wearable Device Integration

The system can connect with smart healthcare devices.

- Devices
- Smartwatches
- Fitness trackers
- Health monitoring sensors

Benefits

- Continuous patient monitoring
 - Real-time healthcare analysis
-

Project Title:- MediGen-AI

18. Internationalization and Multi-Language Support

Future systems can support multiple languages.

Benefits

- Improved accessibility
- Better communication with diverse users

19. Environmental and Social Benefits

Future digital healthcare systems will:

- Reduce paper usage
- Improve healthcare accessibility
- Support eco-friendly hospital operations

20. Research and Innovation Opportunities

The project can serve as a foundation for future research in:

- AI-based healthcare systems
- Medical data analytics
- Smart hospital technologies
- Healthcare automation

21. Conclusion

The future scope of the MediGen-AI Hospital Management System is extensive and highly promising. By integrating advanced technologies such as cloud computing, artificial intelligence, telemedicine, IoT, blockchain, and healthcare analytics, the system can evolve into a fully intelligent and scalable smart healthcare platform.

Future enhancements will improve healthcare accessibility, patient monitoring, medical decision-making, system security, and hospital efficiency. Overall, the project has strong potential to contribute significantly to the modernization and digital transformation of healthcare management systems in the future.

The MediGen-AI Hospital Management System was developed to provide a smart, secure, and efficient solution for managing modern healthcare operations digitally. The project successfully integrates Artificial Intelligence, database management, automation, and secure healthcare workflows into a unified platform that improves hospital administration and patient care.

The system automates important hospital activities such as patient registration, doctor management, appointment scheduling, OPD handling, prescription generation, and report management. The implementation of AI-based prescription support and intelligent healthcare workflows helped reduce manual workload, minimize prescription errors, and improve operational efficiency.

Throughout the development lifecycle, various software engineering methodologies, database design techniques, UML modeling, DFD analysis, ER diagram planning, testing strategies, deployment mechanisms, and maintenance procedures were implemented successfully. These practices helped ensure the system remained scalable, reliable, secure, and user-friendly.

The MediGen-AI system provides significant advantages for hospitals, doctors, medical staff, administrators, and patients by improving communication, reducing delays, enhancing healthcare accessibility, and supporting better medical decision-

CHAPTER 24: CASE STUDY

1. Introduction

A Case Study is an in-depth analysis of a real-world problem, its causes, and the solution implemented to overcome those challenges. In software engineering, a case study helps explain how a project solves practical issues using modern technologies and systematic methodologies.

The MediGen-AI Hospital Management System case study focuses on improving traditional hospital management processes through digital transformation, Artificial Intelligence (AI), automation, and secure healthcare data management. The system was designed to address common problems faced by hospitals such as manual record management, prescription errors, appointment delays, and inefficient workflow.

This case study demonstrates how intelligent healthcare management systems can improve operational efficiency, patient care, and hospital administration.

2. Background of the Problem

Traditional hospital management systems mainly rely on manual processes such as:

- Paper-based patient records
- Manual prescription writing
- Offline appointment handling
- Physical report storage
- Manual administrative operations

These traditional methods create several challenges:

- Increased paperwork
- Slow data retrieval
- Prescription errors
- Data duplication
- Difficulty in maintaining records
- Delayed patient services

Hospitals require a modern digital solution capable of automating operations and improving healthcare efficiency.

3. Problem Statement

The major problems identified in traditional hospital management systems are:

1. Manual patient record handling
2. Delayed appointment management
3. Prescription writing errors
4. Lack of centralized healthcare data
5. Poor workflow coordination
6. Limited healthcare automation
7. Difficulty in report generation

4. Data security risks

These problems reduce hospital efficiency and affect patient care quality.

4.1 Objective of the Case Study

The objectives of this case study are:

To analyze problems in traditional hospital management.

Project Title:- MediGen-AI

-
- To study the implementation of AI in healthcare systems.
 - To improve hospital workflow automation.
 - To enhance healthcare data management.
 - To reduce manual operational errors.
 - To improve patient care and healthcare efficiency.

5. Proposed Solution

The proposed solution is the development of the MediGen-AI Hospital Management System, which integrates:

- Artificial Intelligence
- Digital healthcare management
- Secure database systems
- Automated workflows
- Role-based authentication

The system provides a centralized digital platform for managing hospital operations efficiently.

6. Technologies Used

The following technologies were used in the project:

Technology	Purpose
Python	Backend programming
Django	Web framework
HTML/CSS/JavaScript	Frontend development
SQLite/MySQL	Database management
AI Logic	Prescription recommendation
Bootstrap	User interface design

7. Modules Included in the System

The MediGen-AI system consists of the following modules:

1. Patient Module
2. Doctor Module
3. OPD Module
4. Admin Module
5. AI Prescription Module
6. Appointment Management Module
8. Patient Module Case Study

Problem

Manual patient registration causes:

- Long queues
- Record duplication
- Slow retrieval

Solution

The digital patient module provides:

- Online registration
 - Unique patient records
 - Instant patient search
-

Project Title:- MediGen-AI

Result

- **Faster patient handling**
- **Reduced paperwork**
- **Improved efficiency**

9. Doctor Module Case Study

Problem

Doctors spend significant time managing prescriptions and patient records manually.

Solution

- **The doctor module provides:**
- **Digital prescription generation**
- **AI-assisted recommendations**
- **Instant patient history access**

Result

- **Faster diagnosis support**
- **Reduced prescription errors**
- **Improved healthcare workflow**

10. Appointment Management Case Study

Problem

Traditional appointment systems create scheduling conflicts and delays.

Solution

- **The appointment module automates:**
- **Appointment scheduling**
- **Doctor assignment**
- **Queue management**

Result

- **Reduced waiting time**
- **Better appointment organization**

11. AI Prescription Module Case Study

Problem

- **Manual prescription writing may lead to:**
- **Medicine errors**
- **Illegible handwriting**
- **Incorrect dosage recommendations**

Solution

The AI module analyzes symptoms and suggests possible prescriptions.

Workflow

- 1. Symptoms entered by doctor.**
- 2. AI analyzes patient information.**
- 3. Suggested medicines generated.**
- 4. Doctor verifies and approves prescription.**

Result

- **Improved prescription efficiency**
 - **Reduced manual workload**
 - **Better treatment support**
-

Project Title:- MediGen-AI

12. Database Management Case Study

Problem

Paper-based healthcare records are difficult to manage securely.

Solution

- **A centralized relational database system was implemented.**
- **Features**
- **Digital data storage**
- **Fast data retrieval**
- **Secure patient information management**

Result

- **Improved data consistency**
- **Better healthcare record management**

13. Security Case Study

Problem

Healthcare data is highly sensitive and vulnerable to unauthorized access.

Solution

- **Security mechanisms implemented:**
- **Authentication system**
- **Password encryption**
- **Role-based access control**
- **Session management**

Result

- **Improved healthcare data security**
- **Better privacy protection**

14. Workflow Analysis

The MediGen-AI workflow operates as follows:

Step 1

Patient registration is completed digitally.

Step 2

Appointment assigned to doctor.

Step 3

Doctor accesses patient information.

Step 4

AI assists in prescription generation.

Step 5

Prescription stored in database.

Step 6

Reports generated automatically.

This workflow improves coordination between all hospital departments.

Hospitals handle highly sensitive patient data (Protected Health Information) and operate critical medical machinery 24/7. Therefore, their IT teams prioritize network uptime and strict data privacy.

Hospital IT networks value stability above everything else. If your college brings in a service-based IT company or a bank, take the job offer to build your confidence and financial security. While working there, you can spend your weekends practicing.

Project Title:- MediGen-AI

15. Benefits Observed

The following improvements were observed after implementing the system:

Area	Improvement
Patient Registration	Faster processing
Prescription Management	Reduced errors
Data Retrieval	Instant access
Appointment Handling	Better scheduling
Hospital Workflow	Improved efficiency
Record Management	Centralized storage

16. Social Impact of the Case Study

The project contributes positively to society by:

- Improving healthcare accessibility
- Reducing patient waiting time

Supporting digital healthcare transformation

Enhancing hospital management efficiency

17. Challenges Faced During Implementation

Some challenges encountered include:

- Managing AI integration
- Handling database relationships
- Securing patient data
- Maintaining workflow synchronization

These issues were resolved through continuous testing and optimization.

18. Limitations Identified

The current system still has some limitations:

- Limited cloud support
- AI accuracy dependency
- Local deployment restrictions
- Limited real-time monitoring

Future improvements can address these limitations.

19. Future Enhancements Suggested

Future improvements may include:

- Cloud deployment
- Telemedicine integration
- Mobile healthcare application
- Advanced AI disease prediction
- IoT-based patient monitoring

20. Conclusion

The case study of the MediGen-AI Hospital Management System demonstrates how Artificial Intelligence and digital healthcare technologies can improve traditional hospital management processes significantly. The system successfully automated hospital operations, improved healthcare workflow, enhanced data security, reduced manual errors, and supported intelligent prescription management.

The implementation of AI-assisted healthcare management provided faster healthcare services, improved patient care, and better hospital administration.

CHAPTER 25: CONCLUSION

1. Introduction

The MediGen-AI Hospital Management System was successfully developed as an intelligent and modern healthcare management solution aimed at improving hospital operations through automation, Artificial Intelligence (AI), and secure digital technologies. The project effectively addressed major challenges faced in traditional hospital systems such as manual record management, prescription errors, delayed appointment handling, and inefficient workflow coordination.

The system integrates multiple modules including:

- Patient Management
- Doctor Management
- OPD Management
- Appointment Scheduling
- AI-Based Prescription Generation
- Admin Management
- Secure Database Handling

Through the implementation of these modules, the project successfully improved healthcare workflow efficiency, reduced paperwork, minimized manual errors, and enhanced communication between doctors, patients, medical staff, and administrators.

The integration of Artificial Intelligence played a significant role in providing intelligent prescription recommendations, symptom analysis, and decision support for doctors. This improved healthcare service quality and demonstrated the importance of AI in modern digital healthcare systems.

During the development process, various software engineering methodologies and technologies were applied, including:

- Agile SDLC Model
- Database Design
- UML Modeling
- Data Flow Analysis
- ER Diagram Design
- Testing and Deployment Techniques
- Security Mechanisms

These practices helped ensure that the system remained scalable, secure, reliable, and user-friendly.

The project also highlighted the importance of:

- Digital healthcare transformation
 - Secure healthcare data management
 - Automation in hospital operations
 - AI-powered medical assistance
 - Secure medical record storage
 - Enhanced healthcare accessibility
-

Project Title:- MediGen-AI

Centralized healthcare information systems

Several advantages were achieved through the system, such as:

- **Faster patient registration**
- **Improved appointment management**
- **Better prescription handling**
- **Secure medical record storage**
- **Enhanced healthcare accessibility**
- **Reduced operational workload**

Although certain limitations still exist, including dependency on infrastructure, AI accuracy limitations, and scalability challenges, the project provides a strong foundation for future healthcare innovations. Future enhancements such as cloud computing, telemedicine, mobile applications, blockchain security, IoT-based monitoring, and advanced AI analytics can further improve the capabilities of the system.

The project also creates a positive social impact by supporting smart healthcare initiatives, reducing paper usage, improving patient care, and promoting digital transformation in healthcare institutions.

Overall, the MediGen-AI Hospital Management System successfully demonstrates how modern technologies can transform traditional hospital management into an intelligent, automated, secure, and efficient healthcare ecosystem. The project represents a valuable contribution toward the future of smart healthcare management and AI-driven medical services.

The development of the MediGen-AI Hospital Management System represents an important step toward the modernization and digital transformation of healthcare management systems. The project was designed with the vision of creating an intelligent, automated, secure, and user-friendly platform capable of improving hospital operations and healthcare services efficiently.

These technologies worked together to create a centralized healthcare management environment capable of handling hospital operations in a faster, safer, and more organized manner.

helped automate routine healthcare processes and significantly reduced manual workload and paperwork. The project improved the efficiency of healthcare operations while maintaining proper data organization and security.

One of the most innovative aspects of the project is the integration of Artificial Intelligence into healthcare workflows. The AI module supports symptom analysis, intelligent prescription suggestions, and healthcare decision support, helping doctors improve diagnosis efficiency and reducing the possibility of manual prescription errors. Although AI does not replace medical professionals, it acts as an intelligent assistant that improves healthcare service quality and operational productivity.

The project also focused strongly on security and privacy protection. Since healthcare systems handle sensitive patient information, several security mechanisms such as: were implemented to ensure secure and reliable healthcare data management.

Throughout the project lifecycle, various software engineering methodologies and development strategies were applied successfully. Proper planning, requirement analysis, system design, database normalization, UML modeling, testing, deployment.

CHAPTER 26: BIBLIOGRAPHY / REFERENCE

1. Introduction

The following books, websites, research materials, and online resources were referred to during the analysis, design, development, and implementation of the MediGen-AI Hospital Management System project.

1. Books References

1. Ian Sommerville,
Software Engineering,
Pearson Education, 10th Edition.
2. Roger S. Pressman
Software Engineering: A Practitioner's Approach,
McGraw Hill Publication.
3. Elmasri & Navathe,
Fundamentals of Database Systems,
Pearson Education.
4. Herbert Schildt,
Python: The Complete Reference,
McGraw Hill Education.
5. Eric Matthes,
Python Crash Course,
No Starch Press.
6. Steven Holzner,
Django Web Development,
Pearson Publication.
7. Thomas H. Cormen,
Introduction to Algorithms,
MIT Press.
8. Nina Godbole & Sunit Belapure,
Cyber Security
Wiley India Publication.
9. Achyut S. Godbole,
Data Communications and Networking,
Tata McGraw Hill
10. Ramez Elmasri,
Database Management Systems,
Pearson Publication.

2. Website References

1. [Python Official Website] (https://www.python.org?utm_source=chatgpt.com)
 2. [Django Documentation] (https://docs.djangoproject.com?utm_source=chatgpt.com)
 3. [SQLite Official Website] (https://www.sqlite.org?utm_source=chatgpt.com)
 4. [MySQL Official Website] (https://www.mysql.com?utm_source=chatgpt.com)
 5. [Bootstrap Official Website] (https://getbootstrap.com?utm_source=chatgpt.com)
 6. [MDN Web Docs] (https://developer.mozilla.org?utm_source=chatgpt.com)
 7. [W3Schools] (https://www.w3schools.com?utm_source=chatgpt.com)
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8. [GeeksforGeeks](https://www.geeksforgeeks.org?utm_source=chatgpt.com)
 9. [Stack Overflow] (https://stackoverflow.com?utm_source=chatgpt.com)
 10. [Canva Official Website] (https://www.canva.com?utm_source=chatgpt.com)
 11. [Draw.io Diagram Tool] (https://app.diagrams.net?utm_source=chatgpt.com)
 12. [Lucid chart Website] (https://www.lucidchart.com?utm_source=chatgpt.com)

3. Research Papers and Articles

1. Research articles related to:

- Artificial Intelligence in Healthcare
- Smart Hospital Management Systems
- AI-Based Prescription Systems
- Digital Healthcare Transformation
- Healthcare Data Security

2. IEEE and Google Scholar publications on:

- Machine Learning in Healthcare
- Hospital Automation Systems
- Healthcare Information Systems

4. Online Learning Resources

1. [Coursera](https://www.coursera.org?utm_source=chatgpt.com)
2. [Udemy](https://www.udemy.com?utm_source=chatgpt.com)
3. [YouTube Tutorials] (https://www.youtube.com?utm_source=chatgpt.com)
4. [freeCodeCamp](https://www.freecodecamp.org?utm_source=chatgpt.com)

5. Tools and Software References

Tool / Software	Purpose
Python	Backend Programming
Django	Web Framework
SQLite/MySQL	Database Management
HTML/CSS/JavaScript	Frontend Development
Bootstrap	UI Design
Canva	Diagram and Presentation Design
Draw.io	UML/DFD/ER Diagram Design
VS Code	Code Editor
Git & GitHub	Version Control

6. Conclusion

The successful development of the MediGen-AI Hospital Management System was made possible through the use of various academic books, online documentation, research papers, development tools, and educational resources. These references provided valuable guidance in understanding software engineering principles, database management, AI integration, web development, security mechanisms, and healthcare system design.

CHAPTER 27: APPENDIX

The Appendix section contains supplementary information, supporting documents, diagrams, screenshots, technical details, and additional materials related to the development of the MediGen-AI Hospital Management System. The appendix provides extra reference material that supports the understanding, implementation, and documentation of the project.

This section includes system screenshots, sample database tables, reports, diagrams, coding references, testing samples, and additional project-related information used during development.

The Appendix section provides supporting technical and functional information related to the MediGen-AI Hospital Management System. It contains additional resources, screenshots, database samples, testing records, diagrams, and implementation details that strengthen the understanding and documentation of the project.

Overall, the appendix serves as an important reference section that complements the complete project report and provides detailed supporting information for future development, maintenance, and research purposes.

Sample Screenshots to Include

1. Login Page
2. Registration Page
3. OPD Dashboard
4. Doctor Dashboard
5. AI Prescription Screen
6. Patient Dashboard
7. PDF Prescription
8. Medicine Order Interface
9. Database Tables
10. Admin Panel

Source Code Snippets

Include:

- Flask routing code
 - Database connection code
 - Authentication logic
 - Prescription generation logic
-

Project Title:- MediGen-AI

🏠 MediGen-AI

A Generative AI-Powered Healthcare Portal

MCA Project | Ravi Ranjan Kumar | Aryabhatta Knowledge University, Patna

⚡ Quick Setup (Local Server)

Step 1 — Install Python

Download Python 3.10+ from: <https://www.python.org/downloads/>

✅ During install, check "Add Python to PATH"

Step 2 — Open Terminal / CMD in this folder

Right-click the `medigen-ai` folder → "Open in Terminal"

Step 3 — Install Dependencies

pip install -r requirements.txt

flask==3.0.0

flask-sqlalchemy==3.1.1

werkzeug==3.0.1

Step 4 — Run the Server

python app.py

Step 5 — Open Browser

Go to: ****http://127.0.0.1:5000****

🔑 Default Login Credentials

| Role | Login ID | Password |

|-----|-----|-----|

| ****Admin**** | ADMIN001 | admin123 |

| Doctor | (given by Admin) | (given by Admin) |

| Staff | (given by Admin) | (given by Admin) |

| Patient | (given by Staff) | (given by Staff) |

📖 How to Use

Admin Workflow:

1. Login with ADMIN001 / admin123

2. Go to "Add Doctor/Staff" → register doctors with name, qualification, mobile, email

3. Login ID & Password are auto-generated — note them for the person

Medical Staff Workflow:

1. Login with Staff credentials

2. Click "Register Patient" → fill all patient details

3. Patient ID & Password are shown after successful registration

4. Give these credentials to the patient

Doctor Workflow:

1. Login with Doctor credentials

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2. See patient list on dashboard
3. Click "Write Prescription" for a patient
4. Enter symptoms → Click "🩺 AI Suggest Medicines" for auto-fill
5. Review & edit medicines if needed
6. Click "Finalize & Sign Prescription"

Patient Workflow:

1. Login with Patient ID & Password given by staff
2. View prescriptions on dashboard
3. Click "View" to see full prescription with doctor's digital signature
4. Click Net Meds/Pharm Easy links to order medicines online
5. Click "Print" to save/print prescription

🧩 Tech Stack

- **Backend**: Python 3 + Flask
- **Database**: SQLite (auto-created on first run)
- **Frontend**: HTML5 + CSS3 + Vanilla JS
- **AI**: Rule-based medicine suggestion engine

📁 Project Structure

...

```
mediGen-AI/
├── __pycache__/
├── app.cpython-312.pyc
├── {templates,static}/
│   ├── {css,js,images},instance}/
├── instance/
│   ├── medigen.db
│   └── online_portal.db
├── static/
│   ├── css/
│   ├── images/
│   └── js/
├── templates/
│   ├── admin/
│   │   ├── dashboard.html
│   │   ├── edit_user.html
│   │   └── register_user.html
│   ├── doctor/
│   │   ├── dashboard.html
│   │   └── write_prescription.html
│   ├── online/
│   │   ├── dashboard.html
│   │   ├── forgot_password.html
│   │   ├── login.html
│   │   ├── opd_slip.html
│   │   └── register.html
│   ├── patient/
│   │   ├── dashboard.html
│   │   └── prescription.html
│   ├── staff/
│   │   ├── dashboard.html
│   │   ├── edit_patient.html
│   │   ├── opd_slip.html
│   │   └── register_patient.html
│   ├── base.html
│   ├── forgot_password.html
│   ├── index.html
│   └── login.html
├── app.py
├── README.md
└── requirements.txt
```

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app.py-----> main file

```
from built-ins import Exception, float, int, Len, min, print, set, str

from flask import Flask, render template, request, redirect, url_for, session, flash, jsonify, make
response
from flask_sqlalchemy import SQLAlchemy
from werkzeug. security import generates _password _hash, check _password _hash
from datetime import datetime
import os
import json
import string
import random
otp_storage = {}

app = Flask(__name__)
app.secret_key = 'medigen-ai-secret-key-2024'
app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///medigen.db'
app.config['SQLALCHEMY_TRACK_MODIFICATIONS'] = False
app.config['SQLALCHEMY_BINDS'] = {'online': 'sqlite:///online_portal.db'}

db = SQLAlchemy(app)

# ===== MODELS =====
import pytz
from datetime import datetime
def get_india_time():
    return datetime.now(pytz.timezone('Asia/Kolkata'))
class User(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    login_id = db.Column(db.String(50), unique=True, nullable=False)
    password_hash = db.Column(db.String(200), nullable=False)
    plain_password = db.Column(db.String(100)) # stored for admin view/reset
    role = db.Column(db.String(20), nullable=False) # admin, doctor, staff, patient
    name = db.Column(db.String(100), nullable=False)
    email = db.Column(db.String(100))
    mobile = db.Column(db.String(15))
    qualification = db.Column(db.String(200))
    created_at = db.Column(db.DateTime, default=get_india_time)
    is_active = db.Column(db.Boolean, default=True)
import pytz
from datetime import datetime
def get_india_time():
    return datetime.now(pytz.timezone('Asia/Kolkata'))
class Patient(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    patient_id = db.Column(db.String(20), unique=True, nullable=False)
    user_id = db.Column(db.Integer, db.ForeignKey('user.id'))
    assigned_doctor_id = db.Column(db.Integer, db.ForeignKey('user.id'))
    name = db.Column(db.String(100), nullable=False)
    gender = db.Column(db.String(20))
    mobile = db.Column(db.String(15))
    email = db.Column(db.String(100))
    abha_address = db.Column(db.String(100))
```

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```
address = db.Column(db.String(300))
age = db.Column(db.Integer)
weight = db.Column(db.Float)
registered_by = db.Column(db.Integer, db.ForeignKey('user.id'))
registered_at = db.Column(db.DateTime, default=get_india_time)

import pytz
from datetime import datetime
def get_india_time():
    return datetime.now(pytz.timezone('Asia/Kolkata'))
class Prescription(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    prescription_id = db.Column(db.String(20), unique=True, nullable=False)
    patient_id = db.Column(db.String(20), nullable=False)
    doctor_id = db.Column(db.Integer, db.ForeignKey('user.id'))
    symptoms = db.Column(db.Text)
    diagnosis = db.Column(db.Text)
    medicines = db.Column(db.Text) # JSON string
    lab_reports = db.Column(db.Text)
    notes = db.Column(db.Text)
    created_at = db.Column(db.DateTime, default=get_india_time)
    is_finalized = db.Column(db.Boolean, default=False)
# ===== ONLINE PRESCRIPTION MODEL =====

class OnlinePrescription(db.Model):

    __bind_key__ = 'online'

    id = db.Column(db.Integer, primary_key=True)

    prescription_id = db.Column(
        db.String(20),
        unique=True,
        nullable=False
    )

    patient_id = db.Column(
        db.String(20),
        nullable=False
    )

    doctor_id = db.Column(
        db.Integer,
        nullable=False
    )

    symptoms = db.Column(db.Text)

    diagnosis = db.Column(db.Text)

    medicines = db.Column(db.Text)

    lab_reports = db.Column(db.Text)

    notes = db.Column(db.Text)
```

Project Title:- MediGen-AI

```
        created_at = db.Column(
            db.DateTime,
            default=get_india_time
        )

        is_finalized = db.Column(
            db.Boolean,
            default=True
        )
# ===== ONLINE PATIENT MODEL =====
class OnlinePatient(db.Model):
    __bind_key__ = 'online'
    id = db.Column(db.Integer, primary_key=True)
    patient_id = db.Column(db.String(20), unique=True)
    name = db.Column(db.String(100))
    gender = db.Column(db.String(20))
    age = db.Column(db.Integer)
    weight = db.Column(db.Float)
    mobile = db.Column(db.String(15))
    email = db.Column(db.String(100))
    abha_address = db.Column(db.String(100))
    address = db.Column(db.String(300))
    symptoms = db.Column(db.Text)
    suggested_department = db.Column(db.String(100))
    suggested_doctor_type = db.Column(db.String(100))
    login_id = db.Column(db.String(50), unique=True)
    password_hash = db.Column(db.String(255))
    created_at = db.Column(db.DateTime, default=get_india_time)
# ===== HELPERS =====

def generate_id(prefix, length=6):
    return prefix + ''.join(random.choices(string.digits, k=length))

def login_required(role=None):
    def decorator(f):
        from functools import wraps
        @wraps(f)
        def decorated(*args, **kwargs):
            if 'user_id' not in session:
                flash('Please login first.', 'error')
                return redirect(url_for('login'))
            if role and session.get('role') != role:
                flash('Access denied.', 'error')
                return redirect(url_for('dashboard'))
            return f(*args, **kwargs)
        return decorated
    return decorator

# ===== ROUTES =====

@app.route('/')
def index():
    return render_template('index.html')
```

Project Title:- MediGen-AI

```
@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        login_id = request.form.get('login_id')
        password = request.form.get('password')
        role = request.form.get('role')

        user = User.query.filter_by(login_id=login_id, role=role, is_active=True).first()

        if user and check_password_hash(user.password_hash, password):
            session['user_id'] = user.id
            session['login_id'] = user.login_id
            session['role'] = user.role
            session['name'] = user.name

            if role == 'admin':
                return redirect(url_for('admin_dashboard'))
            elif role == 'doctor':
                return redirect(url_for('doctor_dashboard'))
            elif role == 'staff':
                return redirect(url_for('staff_dashboard'))
            elif role == 'patient':
                return redirect(url_for('patient_dashboard'))
            else:
                flash('Invalid credentials. Please try again.', 'error')

        return render_template('login.html')

@app.route('/logout')
def logout():
    session.clear()
    return redirect(url_for('index'))

@app.route('/dashboard')
def dashboard():
    role = session.get('role')
    if role == 'admin':
        return redirect(url_for('admin_dashboard'))
    elif role == 'doctor':
        return redirect(url_for('doctor_dashboard'))
    elif role == 'staff':
        return redirect(url_for('staff_dashboard'))
    elif role == 'patient':
        return redirect(url_for('patient_dashboard'))
    return redirect(url_for('login'))

# ===== ADMIN ROUTES =====

@app.route('/admin')
def admin_dashboard():
    if session.get('role') != 'admin':
        return redirect(url_for('login'))

    doctors = User.query.filter_by(role='doctor', is_active=True).all()
    staff = User.query.filter_by(role='staff', is_active=True).all()
```

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```
patients = Patient.query.all()
prescriptions = Prescription.query.all()

return render_template('admin/dashboard.html',
                      doctors=doctors, staff=staff,
                      patients=patients, prescriptions=prescriptions)

@app.route('/admin/register-doctor', methods=['GET', 'POST'])
def admin_register_doctor():
    if session.get('role') != 'admin':
        return redirect(url_for('login'))

    if request.method == 'POST':
        name = request.form.get('name')
        email = request.form.get('email')
        mobile = request.form.get('mobile')
        qualification = request.form.get('qualification')
        role = request.form.get('role', 'doctor')
        login_id = request.form.get('login_id', '').strip()
        password = request.form.get('password', '').strip()

        if not login_id or not password:
            flash('Login ID and Password are required.', 'error')
            return render_template('admin/register_user.html')

        # Check if login_id already exists
        existing = User.query.filter_by(login_id=login_id).first()
        if existing:
            flash(f'Login ID "{login_id}" already exists. Please choose a different one.', 'error')
            return render_template('admin/register_user.html')

        user = User(
            login_id=login_id,
            password_hash=generate_password_hash(password),
            plain_password=password,
            role=role,
            name=name,
            email=email,
            mobile=mobile,
            qualification=qualification
        )
        db.session.add(user)
        db.session.commit()

        flash(f'{"Doctor" if role == "doctor" else "Staff"} registered successfully! Login ID: {login_id}',
              'success')
        return redirect(url_for('admin_dashboard'))

    return render_template('admin/register_user.html')

@app.route('/admin/edit-user/<int:user_id>', methods=['GET', 'POST'])
def admin_edit_user(user_id):
    if session.get('role') != 'admin':
        return redirect(url_for('login'))
```

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```
user = User.query.get_or_404(user_id)

if request.method == 'POST':
    new_login_id = request.form.get('login_id', '').strip()
    user.name = request.form.get('name')
    user.email = request.form.get('email')
    user.mobile = request.form.get('mobile')
    user.qualification = request.form.get('qualification')

    # Check if new login_id is taken by another user
    if new_login_id and new_login_id != user.login_id:
        conflict = User.query.filter_by(login_id=new_login_id).first()
        if conflict:
            flash(f'Login ID "{new_login_id}" is already taken.', 'error')
            return render_template('admin/edit_user.html', user=user)
        user.login_id = new_login_id

    new_password = request.form.get('new_password', '').strip()
    if new_password:
        user.password_hash = generate_password_hash(new_password)
        user.plain_password = new_password

    db.session.commit()
    flash('User updated successfully!', 'success')
    return redirect(url_for('admin_dashboard'))

return render_template('admin/edit_user.html', user=user)
@app.route('/admin/delete-user/<int:user_id>')
def admin_delete_user(user_id):
    if session.get('role') != 'admin':
        return redirect(url_for('login'))
    user = User.query.get_or_404(user_id)
    if user.role == 'admin':
        flash('Admin cannot be deleted!', 'error')
        return redirect(url_for('admin_dashboard'))
    db.session.delete(user)
    db.session.commit()
    flash('User deleted permanently!', 'success')
    return redirect(url_for('admin_dashboard'))

# ===== ONLINE PATIENT ROUTES =====
@app.route('/online-register', methods=['GET', 'POST'])
def online_register():
    if request.method == 'POST':
        name = request.form.get('name')
        gender = request.form.get('gender')
        age = request.form.get('age')
        weight = request.form.get('weight')
        mobile = request.form.get('mobile')
        email = request.form.get('email')
        abha_address = request.form.get('abha_address')
        address = request.form.get('address')
        symptoms = request.form.get('symptoms')
        login_id = request.form.get('login_id').strip()
        password = request.form.get('password').strip()
```

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```
existing = OnlinePatient.query.filter_by(
    login_id=login_id
).first()
if existing:
    flash('Login ID already exists.', 'error')
    return redirect(url_for('online_register'))
# AI Suggestion
symptoms_lower = symptoms.lower()
department = "General Medicine"
doctor_type = "MBBS Doctor"
if 'heart' in symptoms_lower or 'chest' in symptoms_lower:
    department = "Cardiology"
    doctor_type = "Cardiologist"
elif 'skin' in symptoms_lower or 'allergy' in symptoms_lower:
    department = "Dermatology"
    doctor_type = "Skin Specialist"
elif 'child' in symptoms_lower or 'baby' in symptoms_lower:
    department = "Pediatrics"
    doctor_type = "Child Specialist"
elif 'eye' in symptoms_lower:
    department = "Ophthalmology"
    doctor_type = "Eye Specialist"
elif 'bone' in symptoms_lower or 'joint' in symptoms_lower:
    department = "Orthopedic"
    doctor_type = "Orthopedic Doctor"
patient = OnlinePatient(
    patient_id=generate_id('ONL'),
    name=name,
    gender=gender,
    age=int(age) if age else None,
    weight=float(weight) if weight else None,
    mobile=mobile,
    email=email,
    abha_address=abha_address,
    address=address,
    symptoms=symptoms,
    suggested_department=department,
    suggested_doctor_type=doctor_type,
    login_id=login_id,
    password_hash=generate_password_hash(password))
db.session.add(patient)
db.session.commit()
flash('Online Registration Successful!', 'success')
return redirect(url_for('online_register'))
return render_template('online/register.html')
# ===== ONLINE LOGIN =====

@app.route('/online-login', methods=['GET', 'POST'])
def online_login():

    if request.method == 'POST':

        login_id = request.form.get('login_id', '').strip()
        password = request.form.get('password', '').strip()
```

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```
patient = OnlinePatient.query.filter_by(
    login_id=login_id
).first()

if patient and check_password_hash(
    patient.password_hash,
    password
):

    session['online_patient_id'] = patient.id
    session['online_login_id'] = patient.login_id
    session['online_name'] = patient.name

    flash('Login Successful!', 'success')

    return redirect(url_for('online_dashboard'))

else:

    flash('Invalid Login ID or Password', 'error')

return render_template('online/login.html')

# ===== ONLINE DASHBOARD =====

@app.route('/online-dashboard')
def online_dashboard():

    if 'online_patient_id' not in session:
        return redirect(url_for('online_login'))

    patient = OnlinePatient.query.get(
        session['online_patient_id']
    )

    prescriptions = OnlinePrescription.query.filter_by(
        patient_id=patient.patient_id,
        is_finalized=True
    ).order_by(
        OnlinePrescription.created_at.desc()
    ).all()

    for p in prescriptions:

        try:
            p.medicines_list = json.loads(
                p.medicines
            ) if p.medicines else []

        except:
            p.medicines_list = []

    doctor = User.query.get(p.doctor_id)
```

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```
p.doctor_name = doctor.name if doctor else 'Unknown'

p.doctor_qualification = (
    doctor.qualification if doctor else ''
)

return render_template(
    'online/dashboard.html',
    patient=patient,
    prescriptions=prescriptions
)

# ===== STAFF ROUTES =====

@app.route('/staff')
def staff_dashboard():
    if session.get('role') != 'staff':
        return redirect(url_for('login'))

    patients = Patient.query.filter_by(registered_by=session['user_id']).all()
    return render_template('staff/dashboard.html', patients=patients)

@app.route('/staff/register-patient', methods=['GET', 'POST'])
def staff_register_patient():

    if session.get('role') not in ['staff', 'doctor']:
        return redirect(url_for('login'))

    if request.method == 'POST':
        name = request.form.get('name')
        gender = request.form.get('gender')
        mobile = request.form.get('mobile')
        email = request.form.get('email')
        abha_address = request.form.get('abha_address')
        address = request.form.get('address')
        age = request.form.get('age')
        weight = request.form.get('weight')
        assigned_doctor_id = request.form.get('assigned_doctor_id')
        login_id = request.form.get('login_id', '').strip()
        password = request.form.get('password', '').strip()

        if not login_id or not password:
            flash('Login ID and Password are required for patient registration.', 'error')
            return render_template('staff/register_patient.html', doctors=User.query.filter_by(
                role='doctor',
                is_active=True).all())

        # Check uniqueness
        existing = User.query.filter_by(login_id=login_id).first()
        if existing:
            flash(f'Login ID "{login_id}" already exists. Choose a different one.', 'error')
            return render_template('staff/register_patient.html', doctors=User.query.filter_by(
                role='doctor',
                is_active=True).all())
```

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```
# Create user account for patient
user = User(
    login_id=login_id,
    password_hash=generate_password_hash(password),
    plain_password=password,
    role='patient',
    name=name,
    email=email,
    mobile=mobile
)
db.session.add(user)
db.session.flush()
today_count = Patient.query.count() + 1
patient = Patient(
    patient_id=login_id,
    user_id=user.id,
    assigned_doctor_id=int(assigned_doctor_id) if assigned_doctor_id else None,
    name=name,
    gender=gender,
    mobile=mobile,
    email=email,
    abha_address=abha_address,
    address=address,
    age=int(age) if age else None,
    weight=float(weight) if weight else None,
    registered_by=session['user_id']
)
db.session.add(patient)
db.session.commit()

flash(f'Patient registered successfully! Token Number: {today_count}', 'success')
if session.get('role') == 'doctor':
    return redirect(url_for('doctor_dashboard'))
    return redirect(url_for('staff_dashboard'))
return render_template(
    'staff/register_patient.html',
    doctors=User.query.filter_by(
        role='doctor',
        is_active=True
    ).all()
)

@app.route('/edit_patient/<int:id>', methods=['GET', 'POST'])
def edit_patient(id):
    if session.get('role') != 'staff':
        return redirect(url_for('login'))
    patient = Patient.query.get(id)
    if request.method == 'POST':
        patient.name = request.form.get('name')
        patient.gender = request.form.get('gender')
        patient.mobile = request.form.get('mobile')
        patient.email = request.form.get('email')
        patient.abha_address = request.form.get('abha_address')
        patient.address = request.form.get('address')
        patient.age = request.form.get('age')
```

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```
        patient.weight = request.form.get('weight')
        db.session.commit()
        flash('Patient updated successfully!', 'success')
        return redirect(url_for('staff_dashboard'))
    return render_template(
        'staff/edit_patient.html',
        patient=patient)
@app.route('/staff/clear-dashboard')
def clear_staff_dashboard():
    if session.get('role') != 'staff':
        return redirect(url_for('login'))
    patients = Patient.query.filter_by(
        registered_by=session['user_id']
    ).all()
    for patient in patients:
        patient.registered_by = None
    db.session.commit()
    flash('Dashboard cleared successfully!', 'success')
    return redirect(url_for('staff_dashboard'))
# ===== SEARCH ONLINE PATIENT =====
@app.route('/doctor/search-online-patient', methods=['GET', 'POST'])
def search_online_patient():

    if session.get('role') != 'doctor':
        return redirect(url_for('login'))

    patient_id = request.values.get('patient_id')

    online_patient = OnlinePatient.query.filter_by(
        patient_id=patient_id
    ).first()

    patients = Patient.query.filter_by(
        assigned_doctor_id=session['user_id']
    ).filter(
        ~Patient.patient_id.in_(
            db.session.query(Prescription.patient_id)
                .filter_by(doctor_id=session['user_id'])
        )
    ).all()

    prescriptions = Prescription.query.filter_by(
        doctor_id=session['user_id']
    ).all()

    if not online_patient:

        flash('Online patient not found', 'error')

    return render_template(
        'doctor/dashboard.html',
        patients=patients,
        prescriptions=prescriptions
    )
```

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```
return render_template(
    'doctor/dashboard.html',
    patients=patients,
    prescriptions=prescriptions,
    online_patient=online_patient
)

# ===== WRITE ONLINE PRESCRIPTION =====

@app.route('/doctor/write-online-prescription/<patient_id>', methods=['GET', 'POST'])
def write_online_prescription(patient_id):

    if session.get('role') != 'doctor':
        return redirect(url_for('login'))

    patient = OnlinePatient.query.filter_by(
        patient_id=patient_id
    ).first_or_404()

    doctor = User.query.get(session['user_id'])

    if request.method == 'POST':

        symptoms = request.form.get('symptoms')

        diagnosis = request.form.get('diagnosis')

        medicines_json = request.form.get(
            'medicines_json',
            '[]'
        )

        lab_reports = request.form.get('lab_reports')

        notes = request.form.get('notes')

        prescription_id = generate_id('ONRX')

        prescription = OnlinePrescription(
            prescription_id=prescription_id,
            patient_id=patient.patient_id,
            doctor_id=session['user_id'],
            symptoms=symptoms,
            diagnosis=diagnosis,
            medicines=medicines_json,
            lab_reports=lab_reports,
```

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```
        notes=notes,

        is_finalized=True
    )

    db.session.add(prescription)

    db.session.commit()

    flash(
        'Online prescription created successfully!',
        'success'
    )

    return redirect(url_for('doctor_dashboard'))

return render_template(
    'doctor/write_prescription.html',
    patient=patient,
    doctor=doctor
)

# ===== DOCTOR ROUTES =====

@app.route('/doctor')
def doctor_dashboard():
    if session.get('role') != 'doctor':
        return redirect(url_for('login'))
    patients = Patient.query.filter_by(
        assigned_doctor_id=session['user_id']
    ).filter(
        ~Patient.patient_id.in_(
            db.session.query(Prescription.patient_id)
                .filter_by(doctor_id=session['user_id'])
        )
    ).all()

    prescriptions = Prescription.query.filter_by(doctor_id=session['user_id']).all()
    return render_template('doctor/dashboard.html', patients=patients, prescriptions=prescriptions)

@app.route('/doctor/write-prescription/<patient_id>', methods=['GET', 'POST'])
def write_prescription(patient_id):
    if session.get('role') != 'doctor':
        return redirect(url_for('login'))

    patient = Patient.query.filter_by(patient_id=patient_id).first_or_404()
    doctor = User.query.get(session['user_id'])

    if request.method == 'POST':
        symptoms = request.form.get('symptoms')
        diagnosis = request.form.get('diagnosis')
        medicines_json = request.form.get('medicines_json', '[]')
        lab_reports = request.form.get('lab_reports')
        notes = request.form.get('notes')
```

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```
prescription_id = generate_id('RX')

prescription = Prescription(
    prescription_id=prescription_id,
    patient_id=patient_id,
    doctor_id=session['user_id'],
    symptoms=symptoms,
    diagnosis=diagnosis,
    medicines=medicines_json,
    lab_reports=lab_reports,
    notes=notes,
    is_finalized=True
)
db.session.add(prescription)
db.session.commit()

flash(f'Prescription {prescription_id} created successfully!', 'success')
return redirect(url_for('doctor_dashboard'))

return render_template('doctor/write_prescription.html', patient=patient, doctor=doctor)

@app.route('/doctor/ai-suggest', methods=['POST'])
def ai_suggest():
    """AI suggestion endpoint - uses Claude API for medicine suggestions"""
    if session.get('role') != 'doctor':
        return jsonify({'error': 'Unauthorized'}), 401

    symptoms = request.json.get('symptoms', '')

    # Medicine database for suggestions
    medicine_db = {
        'fever': [
            {'name': 'Paracetamol 500mg', 'dosage': '1 tablet', 'frequency': 'TDS', 'duration': '5 days'},
            {'name': 'Dolo 650mg', 'dosage': '1 tablet', 'frequency': 'TDS', 'duration': '5 days'},
            {'name': 'Crocin Advance', 'dosage': '1 tablet', 'frequency': 'TDS', 'duration': '5 days'},
            {'name': 'Calpol 650mg', 'dosage': '1 tablet', 'frequency': 'TDS', 'duration': '5 days'},
            {'name': 'PCM Syrup', 'dosage': '10ml', 'frequency': 'TDS', 'duration': '3 days'},
            {'name': 'Ibuprofen 400mg', 'dosage': '1 tablet', 'frequency': 'BD after food', 'duration': '3 days'}
        ],
        'cold': [
            {'name': 'Cetirizine 10mg', 'dosage': '1 tablet', 'frequency': 'OD (once a day)', 'duration': '5 days'},
            {'name': 'Ambroxol 30mg', 'dosage': '1 tablet', 'frequency': 'TDS', 'duration': '5 days'},
        ],
        'cough': [
            {'name': 'Dextromethorphan 15mg', 'dosage': '10ml syrup', 'frequency': 'TDS', 'duration': '5 days'},
            {'name': 'Guaifenesin 200mg', 'dosage': '1 tablet', 'frequency': 'BD', 'duration': '5 days'},
        ],
        'headache': [
            {'name': 'Aspirin 650mg', 'dosage': '1 tablet', 'frequency': 'BD', 'duration': '3 days'},
            {'name': 'Diclofenac 50mg', 'dosage': '1 tablet', 'frequency': 'BD after food', 'duration': '5 days'}
        ]
    }
```

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```
'stomach': [  
    {'name': 'Pantoprazole 40mg', 'dosage': '1 tablet', 'frequency': 'OD before breakfast',  
'duration': '7 days'},  
    {'name': 'Domperidone 10mg', 'dosage': '1 tablet', 'frequency': 'TDS before meals',  
'duration': '5 days'},  
    {'name': 'ORS Sachets', 'dosage': '1 sachet in 1L water', 'frequency': 'As needed', 'duration':  
'3 days'},  
],  
'infection': [  
    {'name': 'Amoxicillin 500mg', 'dosage': '1 capsule', 'frequency': 'TDS', 'duration': '7 days'},  
    {'name': 'Azithromycin 500mg', 'dosage': '1 tablet', 'frequency': 'OD', 'duration': '5 days'},  
],  
'diabetes': [  
    {'name': 'Metformin 500mg', 'dosage': '1 tablet', 'frequency': 'BD with meals', 'duration': '30  
days'},  
],  
'hypertension': [  
    {'name': 'Amlodipine 5mg', 'dosage': '1 tablet', 'frequency': 'OD morning', 'duration': '30  
days'},  
    {'name': 'Telmisartan 40mg', 'dosage': '1 tablet', 'frequency': 'OD morning', 'duration': '30  
days'},  
],  
'pain': [  
    {'name': 'Tramadol 50mg', 'dosage': '1 capsule', 'frequency': 'BD', 'duration': '5 days'},  
    {'name': 'Aceclofenac 100mg', 'dosage': '1 tablet', 'frequency': 'BD after food', 'duration': '5  
days'},  
],  
'vitamin': [  
    {'name': 'Vitamin D3 60000 IU', 'dosage': '1 sachet', 'frequency': 'Once a week', 'duration': '8  
weeks'},  
    {'name': 'Vitamin B Complex', 'dosage': '1 tablet', 'frequency': 'OD', 'duration': '30 days'},  
    {'name': 'Multivitamin tablet', 'dosage': '1 tablet', 'frequency': 'OD after breakfast',  
'duration': '30 days'},  
],  
'allergy': [  
    {'name': 'Levocetirizine 5mg', 'dosage': '1 tablet', 'frequency': 'OD at night', 'duration': '7  
days'},  
    {'name': 'Montelukast 10mg', 'dosage': '1 tablet', 'frequency': 'OD at night', 'duration': '7  
days'},  
],  
}  
  
symptoms_lower = symptoms.lower()  
suggested_medicines = []  
  
keywords = {  
    'fever': ['fever', 'bukhar', 'temperature', 'temp'],  
    'cold': ['cold', 'sardi', 'runny nose', 'nasal', 'sneezing'],  
    'cough': ['cough', 'khansi', 'throat'],  
    'headache': ['headache', 'sir dard', 'migraine'],  
    'stomach': ['stomach', 'pet dard', 'vomiting', 'diarrhea', 'acidity', 'gastric'],  
    'infection': ['infection', 'bacterial', 'throat infection', 'uti'],  
    'diabetes': ['diabetes', 'sugar', 'blood sugar'],  
    'hypertension': ['bp', 'blood pressure', 'hypertension'],  
    'pain': ['pain', 'dard', 'joint pain', 'body ache'],  
}
```

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```
'vitamin': ['weakness', 'kamzori', 'fatigue', 'vitamin deficiency'],
'allergy': ['allergy', 'rash', 'itching', 'skin allergy'],
}

added = set()
for category, kws in keywords.items():
    for kw in kws:
        if kw in symptoms_lower and category not in added:
            meds = medicine_db.get(category, [])

    if meds:

        random_meds = random.sample(
            meds,
            min(3, len(meds))
        )

        suggested_medicines.extend(random_meds)

    added.add(category)
    break

# Always add some basic supportive care
if not suggested_medicines:
    suggested_medicines = [
        {'name': 'Paracetamol 500mg', 'dosage': '1 tablet', 'frequency': 'TDS if needed', 'duration': '3
days'},
        {'name': 'Vitamin C 500mg', 'dosage': '1 tablet', 'frequency': 'OD', 'duration': '7 days'},
    ]

return jsonify({
    'success': True,
    'medicines': suggested_medicines[:6], # Max 6 suggestions
    'message': f'AI suggested {len(suggested_medicines[:6])} medicines based on symptoms'
})

# ===== PATIENT ROUTES =====

@app.route('/patient')
def patient_dashboard():
    if session.get('role') != 'patient':
        return redirect(url_for('login'))

    patient = Patient.query.filter_by(patient_id=session['login_id']).first()
    prescriptions = Prescription.query.filter_by(
        patient_id=session['login_id'], is_finalized=True
    ).order_by(Prescription.created_at.desc()).all()

    # Parse medicines for each prescription
    for p in prescriptions:
        try:
            p.medicines_list = json.loads(p.medicines) if p.medicines else []
        except:
            p.medicines_list = []
```

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```
        doctor = User.query.get(p.doctor_id)
        p.doctor_name = doctor.name if doctor else 'Unknown'
        p.doctor_qualification = doctor.qualification if doctor else ''

    return render_template('patient/dashboard.html', patient=patient, prescriptions=prescriptions)

# opd_slip rout
@app.route('/patient/opd-slip/<patient_id>')
def patient_opd_slip(patient_id):

    patient = Patient.query.filter_by(patient_id=patient_id).first()

    if not patient:
        flash('Patient not found')
        return redirect('/patient')

    return render_template(
        'patient_opd_slip.html',
        patient=patient
    )
# ===== DOCTOR VIEW PRESCRIPTION =====

@app.route('/doctor/prescription/<prescription_id>')
def doctor_view_prescription(prescription_id):

    if session.get('role') != 'doctor':
        return redirect(url_for('login'))

    prescription = Prescription.query.filter_by(
        prescription_id=prescription_id
    ).first_or_404()

    patient = Patient.query.filter_by(
        patient_id=prescription.patient_id
    ).first()

    doctor = User.query.get(prescription.doctor_id)

    try:
        medicines = json.loads(prescription.medicines) if prescription.medicines else []
    except:
        medicines = []

    return render_template(
        'patient/prescription.html',
        prescription=prescription,
        patient=patient,
        doctor=doctor,
        medicines=medicines
    )
# ===== EDIT PRESCRIPTION =====

@app.route('/doctor/edit-prescription/<prescription_id>', methods=['GET', 'POST'])
def edit_prescription(prescription_id):
```

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```
if session.get('role') != 'doctor':
    return redirect(url_for('login'))

prescription = Prescription.query.filter_by(
    prescription_id=prescription_id,
    doctor_id=session['user_id']
).first_or_404()

patient = Patient.query.filter_by(
    patient_id=prescription.patient_id
).first()

doctor = User.query.get(session['user_id'])

if request.method == 'POST':

    prescription.symptoms = request.form.get('symptoms')
    prescription.diagnosis = request.form.get('diagnosis')
    prescription.medicines = request.form.get('medicines_json', '[]')
    prescription.lab_reports = request.form.get('lab_reports')
    prescription.notes = request.form.get('notes')

    db.session.commit()

    flash('Prescription updated successfully!', 'success')

    return redirect(url_for('doctor_dashboard'))

return render_template(
    'doctor/write_prescription.html',
    patient=patient,
    doctor=doctor,
    prescription=prescription,
    edit_mode=True
)
@app.route('/patient/prescription/<prescription_id>')
def view_prescription(prescription_id):
    if session.get('role') != 'patient':
        return redirect(url_for('login'))

    prescription = Prescription.query.filter_by(
        prescription_id=prescription_id,
        patient_id=session['login_id']
    ).first_or_404()

    patient = Patient.query.filter_by(patient_id=session['login_id']).first()
    doctor = User.query.get(prescription.doctor_id)

    try:
        medicines = json.loads(prescription.medicines) if prescription.medicines else []
    except:
        medicines = []

    return render_template('patient/prescription.html',
        prescription=prescription,
```

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```
        patient=patient,
        doctor=doctor,
        medicines=medicines)
@app.route('/print-opd-slip/<int:id>')
def print_opd_slip(id):

    if session.get('role') != 'staff':
        return redirect(url_for('login'))

    patient = Patient.query.get_or_404(id)

    doctor = None

    if patient.assigned_doctor_id:
        doctor = User.query.get(patient.assigned_doctor_id)

    return render_template(
        'staff/opd_slip.html',
        patient=patient,
        doctor=doctor
    )
# ===== ONLINE VIEW PRESCRIPTION =====

@app.route('/online/prescription/<prescription_id>')
def online_view_prescription(prescription_id):

    if 'online_patient_id' not in session:
        return redirect(url_for('online_login'))

    prescription = OnlinePrescription.query.filter_by(
        prescription_id=prescription_id
    ).first_or_404()

    patient = OnlinePatient.query.filter_by(
        patient_id=prescription.patient_id
    ).first()

    doctor = User.query.get(
        prescription.doctor_id
    )

    try:
        medicines = json.loads(
            prescription.medicines
        ) if prescription.medicines else []

    except:
        medicines = []

    return render_template(
        'patient/prescription.html',
        prescription=prescription,
        patient=patient,
        doctor=doctor,
        medicines=medicines
```

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```
)

# ===== API ROUTES =====

@app.route('/api/patients')
def api_patients():
    if session.get('role') not in ['admin', 'doctor', 'staff']:
        return jsonify({'error': 'Unauthorized'}), 401

    query = request.args.get('q', '').lower()
    patients = Patient.query.all()

    if query:
        patients = [p for p in patients if query in p.name.lower() or query in p.patient_id.lower()]

    return jsonify([
        {
            'id': p.patient_id,
            'name': p.name,
            'age': p.age,
            'mobile': p.mobile
        } for p in patients])
# =====online user password forgot=====
# ===== ONLINE FORGOT PASSWORD =====

@app.route('/online-forgot-password', methods=['GET', 'POST'])
def online_forgot_password():

    if request.method == 'POST':

        step = request.form.get('step')

        # STEP 1 → FIND USER
        if step == 'find':

            login_id = request.form.get('login_id').strip()

            patient = OnlinePatient.query.filter_by(
                login_id=login_id
            ).first()

            if patient:

                otp = random.randint(100000, 999999)

                otp_storage[login_id] = otp

                print(f"\n🔑 ONLINE OTP for {login_id}: {otp}\n")

                return render_template(
                    'online/forgot_password.html',
                    step='otp',
                    login_id=login_id
                )

            flash('Login ID not found.', 'error')
```

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```
# STEP 2 → VERIFY OTP
elif step == 'otp':

    login_id = request.form.get('login_id')

    entered_otp = request.form.get('otp')

    saved_otp = otp_storage.get(login_id)

    if saved_otp and str(saved_otp) == str(entered_otp):

        return render_template(
            'online/forgot_password.html',
            step='reset',
            login_id=login_id
        )

    flash('Invalid OTP', 'error')

# STEP 3 → CHANGE PASSWORD
elif step == 'reset':

    login_id = request.form.get('login_id')

    new_password = request.form.get('new_password')

    patient = OnlinePatient.query.filter_by(
        login_id=login_id
    ).first()

    if patient:

        patient.password_hash = generate_password_hash(
            new_password
        )

        db.session.commit()

        otp_storage.pop(login_id, None)

        flash('Password Changed Successfully!', 'success')

        return redirect(url_for('online_login'))

    return render_template(
        'online/forgot_password.html',
        step='find'
    )

# ===== FORGOT PASSWORD =====
# ===== FORGOT PASSWORD =====

@app.route('/forgot-password', methods=['GET', 'POST'])
```

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```
def forgot_password():

    if request.method == 'POST':

        login_id = request.form.get('login_id', '').strip()

        role = request.form.get('role', '').strip()

        user = User.query.filter_by(
            login_id=login_id,
            role=role,
            is_active=True
        ).first()

        if user:

            otp = random.randint(100000, 999999)

            otp_storage[login_id] = otp

            print(f"\n🔑 OTP for {login_id}: {otp}\n")

            flash(
                'OTP Generated Successfully. Check Terminal.',
                'success'
            )

            return render_template(
                'forgot_password.html',
                step='otp',
                login_id=login_id,
                role=role
            )

        else:

            flash(
                'No account found with this Login ID and Role combination.',
                'error'
            )

            return render_template(
                'forgot_password.html',
                step='find'
            )

# ===== VERIFY OTP =====

@app.route('/verify-otp', methods=['POST'])
def verify_otp():

    login_id = request.form.get('login_id')

    role = request.form.get('role')
```

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```
entered_otp = request.form.get('otp')

saved_otp = otp_storage.get(login_id)

if saved_otp and str(saved_otp) == str(entered_otp):

    user = User.query.filter_by(
        login_id=login_id,
        role=role,
        is_active=True
    ).first()

    flash(
        'OTP Verified Successfully',
        'success'
    )

    return render_template(
        'forgot_password.html',
        step='reset',
        user=user,
        login_id=login_id,
        role=role
    )

    flash(
        'Invalid OTP.',
        'error'
    )

    return render_template(
        'forgot_password.html',
        step='otp',
        login_id=login_id,
        role=role
    )

# ===== RESET PASSWORD =====

@app.route('/reset-password', methods=['POST'])
def reset_password():

    login_id = request.form.get('login_id', '').strip()

    role = request.form.get('role', '').strip()

    new_password = request.form.get('new_password', '').strip()

    confirm_password = request.form.get('confirm_password', '').strip()

    if not new_password or len(new_password) < 4:

        flash(
            'Password must be at least 4 characters.',
            'error'
```

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```
)

user = User.query.filter_by(
    login_id=login_id
).first()

return render_template(
    'forgot_password.html',
    step='reset',
    user=user,
    login_id=login_id,
    role=role
)

if new_password != confirm_password:

    flash(
        'Passwords do not match.',
        'error'
    )

    user = User.query.filter_by(
        login_id=login_id
    ).first()

    return render_template(
        'forgot_password.html',
        step='reset',
        user=user,
        login_id=login_id,
        role=role
    )

user = User.query.filter_by(
    login_id=login_id,
    role=role,
    is_active=True
).first()

if user:

    user.password_hash = generate_password_hash(new_password)

    user.plain_password = new_password

    db.session.commit()

    otp_storage.pop(login_id, None)

    flash(
        'Password changed successfully! Please login with your new password.',
        'success'
    )

    return redirect(url_for('login'))
```

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```
flash(
    'Something went wrong. Please try again.',
    'error'
)

return redirect(url_for('forgot_password'))

# ===== INIT =====

def init_db():
    with app.app_context():
        db.create_all(bind_key=None)
        db.create_all(bind_key='online')

    # Create default admin if not exists
    admin = User.query.filter_by(role='admin').first()
    if not admin:
        admin = User(
            login_id='ADMIN001',
            password_hash=generate_password_hash('admin123'),
            plain_password='admin123',
            role='admin',
            name='Super Admin',
            email='admin@medigen.ai',
            mobile='9999999999'
        )
        db.session.add(admin)
        db.session.commit()
        print("✅ Default Admin Created: Login ID: ADMIN001, Password: admin123")

if __name__ == '__main__':
    init_db()
    print("\n🏠 MediGen-AI Server Starting...")
    print("🌐 Open in browser: http://127.0.0.1:5000")
    print("👤 Admin Login: ADMIN001 / admin123\n")
    app.run(debug=True, host='127.0.0.1', port=5000)
```

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Template/ index .html ----→ Home Page

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>MediGen-AI — Smart Healthcare Portal</title>
<link
href="https://fonts.googleapis.com/css2?family=Syne:wght@700;800&family=DM+Sans:wght@300;
400;500;600&display=swap" rel="stylesheet">
<link href="https://cdn.jsdelivr.net/npm/font-awesome/6.5.0/css/all.min.css"
rel="stylesheet">
<style>
:root{--teal:#0d9488;--tl:#14b8a6;--cy:#06b6d4;--navy:#0c1a2e;--card:#0f2642;--muted:#94a3b8;--
border:rgba(255,255,255,.07);}
*{margin:0;padding:0;box-sizing:border-box;}html{scroll-behavior:smooth;}
body{font-family:'DM Sans',sans-serif;background:var(--navy);color:#e2e8f0;overflow-x:hidden;}
nav{position:fixed;top:0;width:100%;z-index:999;padding:1rem 2.5rem;display:flex;align-
items:center;justify-content:space-between;backdrop-
filter:blur(20px);background:rgba(12,26,46,.92);border-bottom:1px solid rgba(13,148,136,.15);}
.logo{display:flex;align-items:center;gap:.6rem;text-decoration:none;}
.logo i{color:var(--tl);font-size:1.55rem;}
.logo-t{font-family:'Syne',sans-serif;font-weight:800;font-size:1.35rem;color:#fff;}
.logo-t span{color:var(--tl);}
.nav-links{display:flex;align-items:center;gap:1.8rem;list-style:none;}
.nav-links a{color:var(--muted);text-decoration:none;font-size:.9rem;font-weight:500;transition:.2s;}
.nav-links a:hover{color:var(--tl);}
.btn-nl{background:var(--teal);color:#fff;padding:.5rem 1.5rem;border-radius:8px;font-
weight:600;cursor:pointer;border:none;font-family:'DM Sans',sans-serif;font-size:.9rem;text-
decoration:none;display:inline-flex;align-items:center;gap:.4rem;transition:.2s;}
.btn-nl:hover{background:#0f766e;transform:translateY(-1px);}
.hero{min-height:100vh;display:flex;align-items:center;justify-
content:center;position:relative;padding:7rem 2rem 4rem;overflow:hidden;}
.hbg{position:absolute;inset:0;background:radial-gradient(ellipse 80% 60% at 50%
0%,rgba(13,148,136,.13),transparent 70%),radial-gradient(ellipse 40% 40% at 85%
80%,rgba(6,182,212,.07),transparent 60%);pointer-events:none;}
.hgrid{position:absolute;inset:0;background-image:linear-gradient(rgba(13,148,136,.04)
1px,transparent 1px),linear-gradient(90deg,rgba(13,148,136,.04) 1px,transparent 1px);background-
size:55px 55px;pointer-events:none;}
.hcont{max-width:900px;text-align:center;position:relative;z-index:1;}
.badge{display:inline-flex;align-
items:center;gap:.5rem;background:rgba(13,148,136,.12);border:1px solid
rgba(13,148,136,.3);padding:.4rem 1.3rem;border-radius:100px;font-size:.78rem;color:var(--tl);font-
weight:600;letter-spacing:.08em;text-transform:uppercase;margin-bottom:2rem;animation:fu .6s
ease both;}
h1{font-family:'Syne',sans-serif;font-size:clamp(2.5rem,6vw,4.5rem);font-weight:800;line-
height:1.1;margin-bottom:1.5rem;animation:fu .6s .15s ease both;}
.hl{color:var(--tl);position:relative;}
.hl::after{content:'';position:absolute;bottom:-4px;left:0;right:0;height:3px;background:linear-
gradient(90deg,var(--teal),var(--cy));border-radius:2px;}
.hdesc{font-size:1.1rem;color:var(--muted);max-width:600px;margin:0 auto 3rem;line-
height:1.75;font-weight:300;animation:fu .6s .3s ease both;}
.hbtns{display:flex;gap:1rem;justify-content:center;flex-wrap:wrap;animation:fu .6s .45s ease both;}
.bp{background:linear-gradient(135deg,var(--teal),var(--cy));color:#fff;padding:.9rem
2.3rem;border-radius:10px;text-decoration:none;font-weight:600;font-
```

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```
size:1rem;transition:.3s;display:inline-flex;align-items:center;gap:.5rem;box-shadow:0 0 30px
rgba(13,148,136,.25);}
.bp:hover{transform:translateY(-2px);box-shadow:0 0 45px rgba(13,148,136,.45);}
.bs{background:transparent;color:var(--tl);padding:.9rem 2.3rem;border-radius:10px;text-
decoration:none;font-weight:600;font-size:1rem;border:1.5px solid
rgba(13,148,136,.5);transition:.3s;display:inline-flex;align-items:center;gap:.5rem;}
.bs:hover{background:rgba(13,148,136,.1);transform:translateY(-2px);}
.stats{display:flex;gap:3rem;justify-content:center;margin-top:4.5rem;flex-wrap:wrap;animation:fu
.6s .6s ease both;}
.stat{text-align:center;}
.sn{font-family:'Syne',sans-serif;font-size:2.3rem;font-weight:800;color:var(--tl);}
.sl{font-size:.74rem;color:var(--muted);text-transform:uppercase;letter-spacing:.1em;margin-
top:.3rem;}
.sec{padding:5.5rem 2rem;}.sec-dark{background:rgba(0,0,0,.2);}
.cont{max-width:1100px;margin:0 auto;}
.stitle{font-family:'Syne',sans-serif;font-size:clamp(1.8rem,4vw,2.5rem);font-weight:800;text-
align:center;margin-bottom:.6rem;}
.ssub{text-align:center;color:var(--muted);font-size:1rem;max-width:550px;margin:0 auto
3.5rem;line-height:1.65;}
/* ROLES */
.rgrid{display:grid;grid-template-columns:repeat(auto-fit,minmax(225px,1fr));gap:1.4rem;}
.rcard{background:var(--card);border:1px solid var(--border);border-radius:18px;padding:2rem
1.7rem;text-align:center;cursor:pointer;transition:.3s;position:relative;overflow:hidden;text-
decoration:none;display:block;color:#e2e8f0;}
.rcard::before{content:'';position:absolute;inset:0;background:linear-
gradient(135deg,rgba(13,148,136,.07),transparent);opacity:0;transition:.3s;}
.rcard:hover{transform:translateY(-6px);border-color:rgba(13,148,136,.4);box-shadow:0 22px 45px
rgba(0,0,0,.3);}
.rcard:hover::before{opacity:1;}
.rico{width:66px;height:66px;border-radius:18px;display:flex;align-items:center;justify-
content:center;font-size:1.65rem;margin:0 auto 1.2rem;}
.rcard h3{font-family:'Syne',sans-serif;font-size:1.15rem;font-weight:700;margin-bottom:.45rem;}
.rcard p{font-size:.85rem;color:var(--muted);line-height:1.6;margin-bottom:1.2rem;}
.rlb{display:inline-block;color:var(--tl);font-size:.83rem;font-weight:600;border:1px solid
rgba(13,148,136,.35);padding:.42rem 1.2rem;border-radius:8px;transition:.25s;}
.rcard:hover .rlb{background:rgba(13,148,136,.14);}
.ia{background:rgba(139,92,246,.13);color:#a78bfa;}
.id{background:rgba(16,185,129,.13);color:#34d399;}
.is{background:rgba(6,182,212,.13);color:#22d3ee;}
.ip{background:rgba(249,115,22,.13);color:#fb923c;}
/* STEPS */
.steps{max-width:750px;margin:0 auto;}
.step{display:flex;gap:2rem;align-items:flex-start;padding:1.8rem 0;position:relative;}
.step:not(:last-child)::after{content:'';position:absolute;left:23px;top:66px;bottom:-
8px;width:2px;background:linear-gradient(to bottom,var(--teal),transparent);z-index:0;}
.snum{flex-shrink:0;width:48px;height:48px;background:linear-gradient(135deg,var(--teal),var(--
cy));border-radius:50%;display:flex;align-items:center;justify-content:center;font-
family:'Syne',sans-serif;font-weight:700;font-size:1.1rem;position:relative;z-index:1;box-shadow:0 0
20px rgba(13,148,136,.3);}
.sbody h3{font-family:'Syne',sans-serif;font-size:1.05rem;font-weight:700;margin-
bottom:.35rem;padding-top:.25rem;}
.sbody p{color:var(--muted);font-size:.88rem;line-height:1.65;}
/* FEATURES */
.fgrid{display:grid;grid-template-columns:repeat(auto-fit,minmax(280px,1fr));gap:1.3rem;}
```

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```
.fcard{background:var(--card);border:1px solid var(--border);border-
radius:14px;padding:1.6rem;display:flex;gap:1.1rem;align-items:flex-start;transition:.3s;}
.fcard:hover{border-color:rgba(13,148,136,.28);transform:translateY(-3px);}
.fico{width:44px;height:44px;border-radius:11px;flex-shrink:0;display:flex;align-
items:center;justify-content:center;background:rgba(13,148,136,.13);color:var(--tl);font-
size:1.1rem;}
.fcard h4{font-family:'Syne',sans-serif;font-size:.97rem;font-weight:700;margin-bottom:.3rem;}
.fcard p{font-size:.83rem;color:var(--muted);line-height:1.55;}
footer{background:rgba(0,0,0,.4);border-top:1px solid var(--border);padding:2.2rem 2rem;text-
align:center;}
footer p{color:var(--muted);font-size:.84rem;line-height:1.7;}
footer strong{color:var(--tl);}
@keyframes
fu{from{opacity:0;transform:translateY(26px);}to{opacity:1;transform:translateY(0);}}
@keyframes fl{0%{transform:translateY(100vh)
scale(0);opacity:0;}10%{opacity:.5;}90%{opacity:.2;}100%{transform:translateY(-80px)
scale(1.3);opacity:0;}}
.pt{position:absolute;border-radius:50%;pointer-events:none;animation:fl linear infinite;}
@media(max-width:768px){.nav-links li:not(:last-child){display:none;}.stats{gap:1.5rem;}.step{flex-
direction:column;gap:1rem;}.step::after{display:none!important;}}
</style>
</head>
<body>

<nav id="nav">
  <a href="/" class="logo"><i class="fas fa-heartbeat"></i><div class="logo-t"><img alt="MediGen-AI logo" data-bbox="738 421 760 436"/>
MediGen<span>-AI</span></div></a>
  <ul class="nav-links">
    <li><a href="#roles">Roles</a></li>
    <li><a href="#workflow">Workflow</a></li>
    <li><a href="#features">Features</a></li>
    <li>
      <a href="/online-login" class="btn-nl"
        style="background:linear-gradient(135deg,#1bf0de,#0ad6fa);">
        <i class="fas fa-user-plus"></i>
        direct login Patient
      </a>
    </li>
  </ul>
</nav>

<section class="hero">
  <div class="hbg"></div><div class="hgrid"></div><div id="pc"></div>
  <div class="hcont">
    <div class="badge"><i class="fas fa-robot"></i>AI-Powered Healthcare Portal</div>
    <h1>Smart Healthcare,<br><span class="hl">Seamless Delivery</span></h1>
    <p class="hdesc">MediGen-AI digitizes hospital OPD workflows with AI-driven prescriptions
and instant medicine delivery — from <strong style="color:#e2e8f0">Symptom to
Doorstep</strong> in minutes.</p>
    <div class="hbtns">
      <a href="/login" class="bp"><i class="fas fa-sign-in-alt"></i> Get Started </a>
      <a href="#workflow" class="bs"><i class="fas fa-play-circle"></i> How it Works</a>
    </div>
  </div>
</section>
```


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```
<div class="stats">
  <div class="stat"><div class="sn">3x</div><div class="sl">Faster OPD</div></div>
  <div class="stat"><div class="sn">0</div><div class="sl">Rx Errors</div></div>
  <div class="stat"><div class="sn">10<small style="font-size:1.1rem">min</small></div><div
class="sl">Delivery</div></div>
  <div class="stat"><div class="sn">AI</div><div class="sl">Diagnosis</div></div>
</div>
</div>
</section>

<section class="sec" id="roles">
  <div class="cont">
    <h2 class="stitle">Choose Your Role</h2>
    <p class="ssub">MediGen-AI has 4 dedicated portals — each tailored for a different user.</p>
    <div class="rgrid">
      <a href="/login" class="rcard"><div class="rico ia"><i class="fas fa-user-
shield"></i></div><h3>Admin</h3><p>Manage users, doctors, staff and monitor all hospital
activity from a central dashboard.</p><span class="rlb"><i class="fas fa-sign-in-alt"></i> Admin
Login</span></a>
      <a href="/login" class="rcard"><div class="rico id"><i class="fas fa-user-
md"></i></div><h3>Doctor</h3><p>View patient queue, generate AI-assisted prescriptions and
digitally sign them with one click.</p><span class="rlb"><i class="fas fa-sign-in-alt"></i> Doctor
Login</span></a>
      <a href="/login" class="rcard"><div class="rico is"><i class="fas fa-id-
card"></i></div><h3>OPD Staff</h3><p>Register incoming patients, record symptoms, and
manage the smart digital queue system.</p><span class="rlb"><i class="fas fa-sign-in-alt"></i>
Staff Login</span></a>
      <a href="/login" class="rcard"><div class="rico ip"><i class="fas fa-user-
injured"></i></div><h3>Patient</h3><p>View your prescription, download PDF, and order
medicines via Blinkit — 10-min delivery.</p><span class="rlb"><i class="fas fa-sign-in-alt"></i>
Patient Login</span></a>
    </div>
  </div>
</section>

<section class="sec sec-dark" id="workflow">
  <div class="cont">
    <h2 class="stitle">How MediGen-AI Works</h2>
    <p class="ssub">Seamless end-to-end flow — from hospital registration to doorstep medicine
delivery.</p>
    <div class="steps">
      <div class="step"><div class="snum">1</div><div class="sbody"><h3><img alt="Hospital icon" data-bbox="712 634 731 648"/> Patient Arrives at
Hospital</h3><p>OPD Staff registers the patient — name, age, weight, mobile, ABHA Address. A
unique <strong style="color:var(--tl)">Patient UID</strong> is auto-generated and patient joins the
smart digital queue.</p></div></div>
      <div class="step"><div class="snum">2</div><div class="sbody"><h3><img alt="Doctor icon" data-bbox="712 688 731 702"/> Doctor Reviews
& Prescribes</h3><p>Doctor logs in, sees waiting list, selects patient. Clicks <strong
style="color:#a78bfa">AI Suggest</strong> — AI instantly generates a prescription draft. Doctor
reviews, edits, and finalizes with digital signature.</p></div></div>
      <div class="step"><div class="snum">3</div><div class="sbody"><h3><img alt="Download icon" data-bbox="712 742 731 756"/> Patient Downloads
Prescription</h3><p>Patient logs in with email. The finalized prescription appears immediately.
They can <strong style="color:var(--tl)">Print / PDF</strong> for records or pharmacy
use.</p></div></div>
      <div class="step"><div class="snum">4</div><div class="sbody"><h3><img alt="Lightning bolt icon" data-bbox="712 796 731 810"/> Instant Medicine
Delivery via Blinkit</h3><p>Every medicine has a direct <strong style="color:#FFD700"> 1mg
```

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order link. One click — medicines ordered from nearest pharmacy. Delivered in <strong style="color:var(--tl)">10 minutes.</p></div></div>

</div>
</div>
</section>

<section class="sec" id="features">

<div class="cont">

<h2 class="stitle">Key Features</h2>

<p class="ssub">Everything needed for a complete, intelligent digital healthcare experience.</p>

<div class="fgrid2">

<div class="fcard"><div class="fico"><i class="fas fa-brain"></i></div><div><h4>AI Symptom Checker</h4><p>NLP-powered symptom analysis using Google Gemini / OpenAI to identify conditions and suggest treatment.</p></div></div>

<div class="fcard"><div class="fico"><i class="fas fa-file-prescription"></i></div><div><h4>Smart Prescription</h4><p>Auto-generated digital prescriptions with medicines, dosages, and follow-up advice — fully editable by doctor.</p></div></div>

<div class="fcard"><div class="fico"><i class="fas fa-signature"></i></div><div><h4>Digital Signature</h4><p>Every prescription verified by a licensed doctor before delivery — Human-in-the-Loop safety.</p></div></div>

<div class="fcard"><div class="fico"><i class="fas fa-file-pdf"></i></div><div><h4>PDF / Print</h4><p>Professionally formatted prescription printable by patients for records or pharmacy use.</p></div></div>

<div class="fcard"><div class="fico"><i class="fas fa-bolt"></i></div><div><h4>Medician portal Integration</h4><p>One-click Quick commerce medicine ordering per prescription line — 10-minute doorstep delivery.</p></div></div>

<div class="fcard"><div class="fico"><i class="fas fa-shield-alt"></i></div><div><h4>Role-Based Auth</h4><p>4 roles (Admin/Doctor/Staff/Patient) with SHA-256 encrypted passwords and Flask session management.</p></div></div>

</div>
</div>
</section>

<footer>

<p style="margin-bottom:.35rem;">MediGen-AI — A Generative AI-Powered Healthcare Portal</p>

<p>Built by Ravi Ranjan Kumar | Roll: 05 | MCA 4th Sem | Aryabhata Knowledge University, Patna | 2024–26</p>

<p style="margin-top:.4rem;font-size:.76rem;opacity:.5;">Under guidance of Dr. Amit Kumar Shukla (HOD-IT) | Cimage College</p>

</footer>

<script>

```
const pc=document.getElementById('pc');
for(let i=0;i<15;i++){const p=document.createElement('div');p.className='pt';const
s=Math.random()*4+1.5;p.style.cssText=`width:${s}px;height:${s}px;background:rgba(13,148,136,${
Math.random()*4+.15});left:${Math.random()*100}%`;animation-
duration:${Math.random()*13+8}s;animation-delay:-${Math.random()*13}s`;pc.appendChild(p);}
window.addEventListener('scroll',()=>{document.getElementById('nav').style.background=window.
scrollY>60?'rgba(9,21,37,.97)':'rgba(12,26,46,.92)'});
document.querySelectorAll('a[href^="#"]').forEach(a=>a.addEventListener('click',e=>{const
t=document.querySelector(a.getAttribute('href'));if(t){e.preventDefault();t.scrollIntoView({behavior:
'smooth'}})}));
</script></body></html>
```

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Template/login.html-----> login dashboard

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Login | MediGen-AI</title>
  <link
href="https://fonts.googleapis.com/css2?family=Syne:wght@400;600;700;800&family=DM+Sans:wght@300;400;500&display=swap" rel="stylesheet">
  <style>
    :root {
      --teal: #0d9488; --teal-dark: #0f766e; --teal-light: #99f6e4;
      --navy: #0f172a; --slate: #1e293b; --white: #f8fafc; --accent: #f59e0b;
    }
    * { margin: 0; padding: 0; box-sizing: border-box; }
    body {
      font-family: 'DM Sans', sans-serif;
      background: var(--navy);
      color: var(--white);
      min-height: 100vh;
      display: flex;
      align-items: center;
      justify-content: center;
    }
    body::before {
      content: '';
      position: fixed;
      inset: 0;
      background: radial-gradient(ellipse at 30% 50%, rgba(13,148,136,0.12) 0%, transparent
60%);
      pointer-events: none;
    }
    .container {
      width: 100%;
      max-width: 460px;
      padding: 2rem;
      position: relative;
      z-index: 1;
    }
    .back-link {
      display: inline-flex;
      align-items: center;
      gap: 0.4rem;
      color: rgba(248,250,252,0.4);
      text-decoration: none;
      font-size: 0.9rem;
      margin-bottom: 2rem;
      transition: color 0.2s;
    }
    .back-link:hover { color: var(--teal-light); }
    .card {
      background: rgba(255,255,255,0.04);
      border: 1px solid rgba(255,255,255,0.08);
```

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```
border-radius: 24px;
padding: 2.5rem;
backdrop-filter: blur(20px);
}
.logo {
  font-family: 'Syne', sans-serif;
  font-size: 1.5rem;
  font-weight: 800;
  color: var(--teal-light);
  text-align: center;
  margin-bottom: 0.3rem;
}
.logo span { color: var(--accent); }
.subtitle {
  text-align: center;
  color: rgba(248,250,252,0.4);
  font-size: 0.9rem;
  margin-bottom: 2rem;
}
/* Role selector */
.role-tabs {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 0.6rem;
  margin-bottom: 1.8rem;
}
.role-tab {
  padding: 0.7rem;
  border-radius: 12px;
  border: 2px solid rgba(255,255,255,0.08);
  background: transparent;
  color: rgba(248,250,252,0.5);
  cursor: pointer;
  font-family: 'DM Sans', sans-serif;
  font-size: 0.85rem;
  font-weight: 500;
  transition: all 0.2s;
  text-align: center;
}
.role-tab:hover { border-color: var(--teal); color: var(--white); }
.role-tab.active {
  border-color: #0f766e;
  background: rgba(13,148,136,0.15);
  color: #0f766e;
}
.form-group { margin-bottom: 1.2rem; }
label { display: block; font-size: 0.85rem; font-weight: 500; color: rgba(248,250,252,0.7);
margin-bottom: 0.4rem; }
input, select {
  width: 100%;
  background: rgba(255,255,255,0.05);
  border: 1px solid rgba(255,255,255,0.1);
  border-radius: 12px;
  padding: 0.8rem 1rem;
  color: var(--white);
```

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```
    font-family: 'DM Sans', sans-serif;
    font-size: 0.95rem;
    transition: border-color 0.2s;
    outline: none;
}
input:focus, select:focus { border-color: var(--teal); }
select option { background: var(--slate); }
.btn-login {
    width: 100%;
    padding: 0.9rem;
    background: var(--teal);
    color: white;
    border: none;
    border-radius: 12px;
    font-family: 'DM Sans', sans-serif;
    font-size: 1rem;
    font-weight: 600;
    cursor: pointer;
    transition: all 0.2s;
    margin-top: 0.5rem;
}
.btn-login:hover { background: var(--teal-dark); transform: translateY(-1px); }
.alert {
    padding: 0.8rem 1rem;
    border-radius: 10px;
    margin-bottom: 1.2rem;
    font-size: 0.9rem;
}
.alert-error { background: rgba(239,68,68,0.1); border: 1px solid rgba(239,68,68,0.3); color:
#fca5a5; }
.alert-success { background: rgba(34,197,94,0.1); border: 1px solid rgba(34,197,94,0.3); color:
#86efac; }
.demo-info {
    margin-top: 1.5rem;
    padding: 1rem;
    background: rgba(245,158,11,0.08);
    border: 1px solid rgba(245,158,11,0.2);
    border-radius: 12px;
    font-size: 0.82rem;
    color: rgba(248,250,252,0.6);
}
.demo-info strong { color: var(--accent); }
</style>
</head>
<body>
<div class="container">
    <a href="/" class="back-link" style="color: #08face;"><← Back to Home</a>

    <div class="card">
        <div class="logo"><img alt="MediGen logo" data-bbox="344 741 364 754"/> MediGen<span>-AI</span></div>
        <p class="subtitle">Sign in to your account</p>

        {% with messages = get_flashed_messages(with_categories=true) %}
            {% for category, message in messages %}
                <div class="alert alert-{{ 'error' if category == 'error' else 'success' }}">{{ message }}</div>
            {% endfor %}
        {% endwith %}
    </div>
</div>
</body>
</html>
```

Project Title:- MediGen-AI

```
{% endfor %}
{% endwith %}

<div class="role-tabs">
  <button class="role-tab" onclick="selectRole('admin')" id="tab-admin"><img alt="Admin icon" data-bbox="745 142 760 157"/>
Admin</button>
  <button class="role-tab" onclick="selectRole('doctor')" id="tab-doctor"><img alt="Doctor icon" data-bbox="745 172 760 187"/>
Doctor</button>
  <button class="role-tab" onclick="selectRole('staff')" id="tab-staff"><img alt="Staff icon" data-bbox="715 202 730 217"/> OPD Staff</button>
  <button class="role-tab" onclick="selectRole('patient')" id="tab-patient"><img alt="Patient icon" data-bbox="755 217 770 232"/>
Patient</button>
</div>

<form method="POST" action="/login">
  <input type="hidden" name="role" id="role-input" value="admin">

  <div class="form-group">
    <label>Login ID</label>
    <input type="text" name="login_id" placeholder="Enter your Login ID" required
autocomplate="off">
  </div>

  <div class="form-group">
    <label>Password</label>
    <input type="password" name="password" placeholder="Enter your password"
required>
  </div>

  <button type="submit" class="btn-login">Sign In →</button>
</form>

<div style="text-align:center;margin-top:1.2rem">
  <a href="/forgot-password" style="color:rgb(8, 222, 245);font-size:0.85rem;text-
decoration:none;">
<img alt="Forgot Password icon" data-bbox="245 552 260 567"/> Forgot Password?
  </a>
</div>
</div>
</div>

<script>
function selectRole(role) {
  document.querySelectorAll('.role-tab').forEach(t => t.classList.remove('active'));
  document.getElementById('tab-' + role).classList.add('active');
  document.getElementById('role-input').value = role;
}

// Auto-select from URL param
const urlParams = new URLSearchParams(window.location.search);
const role = urlParams.get('role') || 'admin';
selectRole(role);
</script>
</body>
</html>
```

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Template/base.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>{% block title %}MediGen-AI{% endblock %}</title>
  <link
href="https://fonts.googleapis.com/css2?family=Syne:wght@400;600;700;800&family=DM+Sans:wght@300;400;500&display=swap" rel="stylesheet">
  <style>
    :root {
      --teal: #0d9488; --teal-dark: #0f766e; --teal-light: #99f6e4;
      --navy: #0f172a; --slate: #1e293b; --white: #f8fafc;
      --accent: #f59e0b; --danger: #ef4444; --success: #22c55e;
      --sidebar-w: 260px;
    }
    * { margin: 0; padding: 0; box-sizing: border-box; }
    body { font-family: 'DM Sans', sans-serif; background: #0a1628; color: var(--white); display:
flex; min-height: 100vh; }

    /* SIDEBAR */
    .sidebar {
      width: var(--sidebar-w);
      background: var(--navy);
      border-right: 1px solid rgba(255,255,255,0.06);
      display: flex;
      flex-direction: column;
      position: fixed;
      top: 0; left: 0; bottom: 0;
      z-index: 50;
    }
    .sidebar-logo {
      padding: 1.5rem 1.5rem 1rem;
      font-family: 'Syne', sans-serif;
      font-size: 1.2rem;
      font-weight: 800;
      color: var(--teal-light);
      border-bottom: 1px solid rgba(255,255,255,0.06);
      white-space: nowrap;
    }
    .sidebar-logo span { color: var(--accent); }
    .sidebar-user {
      padding: 1rem 1.5rem;
      background: rgba(255,255,255,0.03);
      border-bottom: 1px solid rgba(255,255,255,0.06);
    }
    .user-name { font-weight: 600; font-size: 0.9rem; }
    .user-role {
      font-size: 0.75rem;
      color: rgba(255,255,255,0.4);
      text-transform: uppercase;
      letter-spacing: 0.05em;
    }
    .sidebar-nav { flex: 1; padding: 1rem 0; }
```

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```
.nav-item {
  display: flex;
  align-items: center;
  gap: 0.7rem;
  padding: 0.7rem 1.5rem;
  color: rgba(255,255,255,0.5);
  text-decoration: none;
  font-size: 0.9rem;
  transition: all 0.2s;
  border-left: 3px solid transparent;
}
.nav-item:hover { color: var(--white); background: rgba(255,255,255,0.04); }
.nav-item.active { color: var(--teal-light); background: rgba(13,148,136,0.08); border-left-color:
var(--teal); }
.nav-item .icon { font-size: 1.1rem; width: 1.5rem; text-align: center; }
.sidebar-footer {
  padding: 1rem 1.5rem;
  border-top: 1px solid rgba(255,255,255,0.06);
}
.logout-btn {
  display: flex;
  align-items: center;
  gap: 0.5rem;
  color: rgba(255,255,255,0.4);
  text-decoration: none;
  font-size: 0.85rem;
  transition: color 0.2s;
  padding: 0.5rem 0;
}
.logout-btn:hover { color: #fca5a5; }

/* MAIN */
.main {
  margin-left: var(--sidebar-w);
  flex: 1;
  display: flex;
  flex-direction: column;
  min-height: 100vh;
}
.topbar {
  background: var(--navy);
  border-bottom: 1px solid rgba(255,255,255,0.06);
  padding: 1rem 2rem;
  display: flex;
  align-items: center;
  gap: 4px;
  position: sticky;
  top: 0;
  z-index: 40;
}
.page-title {
  font-family: 'Syne', sans-serif;
  font-size: 1.2rem;
  font-weight: 700;
}
```

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```
.content { padding:0.1rem 2rem 2rem 2rem; flex: 1; }

/* CARDS */
.card {
  background: rgba(255,255,255,0.03);
  border: 1px solid rgba(255,255,255,0.07);
  border-radius: 16px;
  padding: 1.5rem;
  margin-bottom: 1.5rem;
}
.card-title {
  font-family: 'Syne', sans-serif;
  font-size: 1rem;
  font-weight: 700;
  margin-bottom: 1.2rem;
  color: var(--teal-light);
}

/* STATS */
.stats-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(180px, 1fr)); gap: 1rem; margin-bottom: 1.5rem; }
.stat-card {
  background: rgba(255,255,255,0.03);
  border: 1px solid rgba(255,255,255,0.07);
  border-radius: 16px;
  padding: 1.5rem;
  text-align: center;
}
.stat-num { font-family: 'Syne', sans-serif; font-size: 2rem; font-weight: 800; color: var(--teal-light); }
.stat-label { font-size: 0.85rem; color: rgba(255,255,255,0.5); margin-top: 0.3rem; }

/* TABLE */
table { width: 100%; border-collapse: collapse; font-size: 0.9rem; }
th { text-align: left; padding: 0.8rem 1rem; color: rgba(255,255,255,0.4); font-weight: 500; font-size: 0.8rem; text-transform: uppercase; letter-spacing: 0.05em; border-bottom: 1px solid rgba(255,255,255,0.06); }
td { padding: 0.9rem 1rem; border-bottom: 1px solid rgba(255,255,255,0.04); color: rgba(255,255,255,0.8); }
tr:hover td { background: rgba(255,255,255,0.02); }

/* FORMS */
.form-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 1rem; }
.form-group { margin-bottom: 1.2rem; }
.form-group.full { grid-column: 1 / -1; }
label { display: block; font-size: 0.85rem; font-weight: 500; color: rgba(255,255,255,0.6); margin-bottom: 0.4rem; }
input[type=text], input[type=email], input[type=number], input[type=password], textarea, select {
  width: 100%;
  background: rgba(255,255,255,0.05);
  border: 1px solid rgba(255,255,255,0.1);
  border-radius: 10px;
  padding: 0.75rem 1rem;
  color: var(--white);
}
```

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```
font-family: 'DM Sans', sans-serif;
font-size: 0.9rem;
transition: border-color 0.2s;
outline: none;
}
input:focus, textarea:focus, select:focus { border-color: var(--teal); }
select option { background: var(--slate); }
textarea { resize: vertical; min-height: 80px; }

/* BUTTONS */
.btn { padding: 0.65rem 1.5rem; border-radius: 10px; border: none; cursor: pointer; font-
family: 'DM Sans', sans-serif; font-size: 0.9rem; font-weight: 500; transition: all 0.2s; text-
decoration: none; display: inline-flex; align-items: center; gap: 0.4rem; }
.btn-primary { background: var(--teal); color: white; }
.btn-primary:hover { background: var(--teal-dark); }
.btn-danger { background: rgba(239,68,68,0.15); color: #fca5a5; border: 1px solid
rgba(239,68,68,0.2); }
.btn-danger:hover { background: rgba(239,68,68,0.25); }
.btn-outline { background: transparent; color: var(--teal-light); border: 1px solid
rgba(13,148,136,0.4); }
.btn-outline:hover { background: rgba(13,148,136,0.1); }
.btn-sm { padding: 0.4rem 0.9rem; font-size: 0.82rem; }

/* BADGE */
.badge { display: inline-block; padding: 0.2rem 0.7rem; border-radius: 50px; font-size: 0.75rem;
font-weight: 500; }
.badge-teal { background: rgba(13,148,136,0.15); color: var(--teal-light); border: 1px solid
rgba(13,148,136,0.3); }
.badge-amber { background: rgba(245,158,11,0.15); color: #fde68a; border: 1px solid
rgba(245,158,11,0.3); }
.badge-green { background: rgba(34,197,94,0.15); color: #86efac; border: 1px solid rgba(34,197,94,0.3); }
.badge-red { background: rgba(239,68,68,0.15); color: #fca5a5; border: 1px solid rgba(239,68,68,0.3); }

/* ALERTS */
.alert { padding: 0.8rem 1rem; border-radius: 10px; margin-bottom: 1rem; font-size: 0.9rem; }
.alert-error { background: rgba(239,68,68,0.1); border: 1px solid rgba(239,68,68,0.3); color: #fca5a5; }
.alert-success { background: rgba(34,197,94,0.1); border: 1px solid rgba(34,197,94,0.3); color: #86efac; }

.empty-state { text-align: center; padding: 3rem 1rem; color: rgba(255,255,255,0.3); }
.empty-state .icon { font-size: 3rem; margin-bottom: 1rem; }
</style>
{% block extra_css %}{% endblock %}
</head>
<body>
<aside class="sidebar">
  <div class="sidebar-logo"><img alt="MediGen-AI logo" data-bbox="385 683 405 698" /> MediGen<span>-AI</span></div>
  <div class="sidebar-user">
    <div class="user-name">{{ session.name }}</div>
    <div class="user-role">{{ session.role }}</div>
  </div>
  <nav class="sidebar-nav">
    {% block sidebar_nav %}{% endblock %}
  </nav>
</aside>

<main class="main">
```

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```
<div class="topbar">
  <div class="page-title" style="margin-right:auto;">
    {% block page_title %}{% endblock %}
  </div>

</a>
  <a href="/logout" style="
    background: var(--danger);
    color: white;
    padding: 8px 12px;
    border-radius: 8px;
    text-decoration: none;
    font-size: 0.85rem;">
    Sign Out
  </a>
</div>
<div class="content">
  {% with messages = get_flashed_messages(with_categories=True) %}
  {% for category, message in messages %}
    <div class="alert alert-{{ 'error' if category == 'error' else 'success' }}">{{ message }}</div>
  {% endfor %}
  {% endwith %}

  {% block content %}{% endblock %}
</div>
</main>

{% block extra_js %}{% endblock %}
</body>
</html>
```

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Template/forget_password.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Forgot Password | MediGen-AI</title>
  <link
href="https://fonts.googleapis.com/css2?family=Syne:wght@400;600;700;800&family=DM+Sans:wght@300;400;500&display=swap" rel="stylesheet">
  <style>
    :root {
      --teal: #0d9488; --teal-dark: #0f766e; --teal-light: #99f6e4;
      --navy: #0f172a; --slate: #1e293b; --white: #f8fafc; --accent: #f59e0b;
    }
    * { margin: 0; padding: 0; box-sizing: border-box; }
    body {
      font-family: 'DM Sans', sans-serif;
      background: var(--navy);
      color: var(--white);
      min-height: 100vh;
      display: flex;
      align-items: center;
      justify-content: center;
    }
    body::before {
      content: "";
      position: fixed;
      inset: 0;
      background: radial-gradient(ellipse at 70% 40%, rgba(124,58,237,0.1) 0%, transparent
60%);
      pointer-events: none;
    }
    .container { width: 100%; max-width: 460px; padding: 2rem; position: relative; z-index: 1; }
    .back-link {
      display: inline-flex; align-items: center; gap: 0.4rem;
      color: rgba(248,250,252,0.4); text-decoration: none; font-size: 0.9rem;
      margin-bottom: 2rem; transition: color 0.2s;
    }
    .back-link:hover { color: var(--teal-light); }
    .card {
      background: rgba(255,255,255,0.04);
      border: 1px solid rgba(255,255,255,0.08);
      border-radius: 24px;
      padding: 2.5rem;
      backdrop-filter: blur(20px);
    }
    .icon-circle {
      width: 64px; height: 64px; border-radius: 50%;
      background: rgba(124,58,237,0.15);
      border: 1px solid rgba(124,58,237,0.3);
      display: flex; align-items: center; justify-content: center;
      font-size: 1.8rem; margin: 0 auto 1.2rem;
    }
    h2 { font-family: 'Syne', sans-serif; font-size: 1.4rem; font-weight: 700; text-align: center;
margin-bottom: 0.3rem; }
```

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```
.subtitle { text-align: center; color: rgba(248,250,252,0.4); font-size: 0.88rem; margin-bottom: 2rem; line-height: 1.5; }

.role-tabs { display: grid; grid-template-columns: repeat(3,1fr); gap: 0.5rem; margin-bottom: 1.5rem; }
.role-tab {
  padding: 0.6rem 0.5rem; border-radius: 10px;
  border: 2px solid rgba(255,255,255,0.08);
  background: transparent; color: rgba(248,250,252,0.5);
  cursor: pointer; font-family: 'DM Sans', sans-serif;
  font-size: 0.82rem; font-weight: 500; transition: all 0.2s; text-align: center;
}
.role-tab:hover { border-color: #7c3aed; color: var(--white); }
.role-tab.active { border-color: #7c3aed; background: rgba(124,58,237,0.12); color: #c4b5fd; }

.form-group { margin-bottom: 1.2rem; }
label { display: block; font-size: 0.85rem; font-weight: 500; color: rgba(248,250,252,0.7); margin-bottom: 0.4rem; }
input {
  width: 100%; background: rgba(255,255,255,0.05);
  border: 1px solid rgba(255,255,255,0.1); border-radius: 12px;
  padding: 0.8rem 1rem; color: var(--white);
  font-family: 'DM Sans', sans-serif; font-size: 0.95rem;
  transition: border-color 0.2s; outline: none;
}
input:focus { border-color: #7c3aed; }

.btn-submit {
  width: 100%; padding: 0.9rem;
  background: linear-gradient(135deg, #7c3aed, #4f46e5);
  color: white; border: none; border-radius: 12px;
  font-family: 'DM Sans', sans-serif; font-size: 1rem; font-weight: 600;
  cursor: pointer; transition: all 0.2s; margin-top: 0.5rem;
}
.btn-submit:hover { opacity: 0.9; transform: translateY(-1px); }

.alert { padding: 0.8rem 1rem; border-radius: 10px; margin-bottom: 1.2rem; font-size: 0.9rem; }
.alert-error { background: rgba(239,68,68,0.1); border: 1px solid rgba(239,68,68,0.3); color: #fca5a5; }
.alert-success { background: rgba(34,197,94,0.1); border: 1px solid rgba(34,197,94,0.3); color: #86efac; }

.user-found-box {
  background: rgba(34,197,94,0.08);
  border: 1px solid rgba(34,197,94,0.2);
  border-radius: 14px;
  padding: 1rem 1.2rem;
  margin-bottom: 1.2rem;
}
.user-found-box .name { font-weight: 600; color: #86efac; font-size: 1rem; }
.user-found-box .role { font-size: 0.8rem; color: rgba(255,255,255,0.4); margin-top: 0.2rem; text-transform: uppercase; }

.divider { text-align: center; color: rgba(255,255,255,0.2); margin: 1rem 0; font-size: 0.85rem; }
```

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```
</style>
</head>
<body>
<div class="container">
  <a href="/login" class="back-link"><- Back to Login</a>

  <div class="card">
    <div class="icon-circle"><img alt="lock icon" data-bbox="388 183 405 198"/></div>

    {% if step == 'find' %}
    <h2>Forgot Password?</h2>
    <p class="subtitle">Enter your Login ID and select your role to find your account.</p>

    {% with messages = get_flashed_messages(with_categories=true) %}
      {% for category, message in messages %}
        <div class="alert alert-{{ 'error' if category == 'error' else 'success' }}">{{ message }}</div>
      {% endfor %}
    {% endwith %}

    <div class="role-tabs">
      <button class="role-tab" onclick="selectRole('doctor')" id="tab-doctor"><img alt="doctor icon" data-bbox="745 358 762 373"/>
      Doctor</button>
      <button class="role-tab" onclick="selectRole('staff')" id="tab-staff"><img alt="staff icon" data-bbox="715 388 732 403"/> Staff</button>
      <button class="role-tab" onclick="selectRole('patient')" id="tab-patient"><img alt="patient icon" data-bbox="752 403 769 418"/>
      Patient</button>
    </div>

    <form method="POST" action="/forgot-password">
      <input type="hidden" name="role" id="role-input" value="doctor">
      <div class="form-group">
        <label>Your Login ID</label>
        <input type="text" name="login_id" placeholder="Enter your Login ID" required
autocomple="off">
      </div>
      <button type="submit" class="btn-submit">Find My Account →</button>
    </form>

    {% elif step == 'otp' %}
    <h2>OTP Verification</h2>
    <p class="subtitle">
    Enter the OTP shown in terminal/console.
    </p>
    {% with messages = get_flashed_messages(with_categories=true) %}
      {% for category, message in messages %}
        <div class="alert alert-{{ 'error' if category == 'error' else 'success' }}">
          {{ message }}
        </div>
      {% endfor %}
    {% endwith %}
    <form method="POST" action="/verify-otp">
      <input type="hidden" name="login_id" value="{{ login_id }}">
      <input type="hidden" name="role" value="{{ role }}">
      <div class="form-group">
        <label>Enter OTP</label>
        <input
          type="text"

```

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```
        name="otp"
        placeholder="6-digit OTP"
        required>
    </div>
    <button type="submit" class="btn-submit">
        Verify OTP →
    </button>
</form>
    {% elif step == 'reset' %}
    <h2>Reset Password</h2>
    <p class="subtitle">Account found! Set your new password below.</p>

    {% with messages = get_flashed_messages(with_categories=true) %}
    {% for category, message in messages %}
        <div class="alert alert-{{ 'error' if category == 'error' else 'success' }}">{{ message }}</div>
    {% endfor %}
    {% endwith %}
    <div class="user-found-box">
        <div class="name">✔ {{ user.name }}</div>
        <div class="role">{{ user.role }} · {{ user.login_id }}</div>
    </div>
    <form method="POST" action="/reset-password">
        <input type="hidden" name="login_id" value="{{ login_id }}">
        <input type="hidden" name="role" value="{{ role }}">
        <div class="form-group">
            <label>New Password *</label>
            <input type="password" name="new_password" placeholder="Enter new password (min
4 chars)" required>
        </div>
        <div class="form-group">
            <label>Confirm New Password *</label>
            <input type="password" name="confirm_password" placeholder="Re-enter new
password" required>
        </div>
        <button type="submit" class="btn-submit">🔒 Set New Password</button>
    </form>

    <div class="divider">or</div>
    <a href="/forgot-password" style="display:block;text-align:center;color:rgba(255,255,255,0.4);font-size:0.85rem;text-decoration:none">← Search
again</a>
    {% endif %}
</div>
</div>
<script>
function selectRole(role) {
    document.querySelectorAll('.role-tab').forEach(t => t.classList.remove('active'));
    const tab = document.getElementById('tab-' + role);
    if (tab) tab.classList.add('active');
    const input = document.getElementById('role-input');
    if (input) input.value = role;
}
selectRole('doctor');
</script>
</body></html>
```

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Admin/dashboard.html

```
{% extends 'base.html' %}
{% block title %}Admin Dashboard | MediGen-AI{% endblock %}
{% block page_title %}Admin Dashboard{% endblock %}
{% block sidebar_nav %}
<a href="/admin" class="nav-item active" style="background:#0f766e;color:white;border-
radius:10px;"><span class="icon"><img alt="Dashboard icon" data-bbox="428 173 448 188"/></span> Dashboard</a>&NonBreakingSpace;
<a href="/admin/register-doctor" class="nav-item" style="background:#0f766e;color:white;border-
radius:10px;"><span class="icon"><img alt="Add Doctor icon" data-bbox="428 201 448 216"/></span> <strong>Add Doctor/Staff</strong></a>
{% endblock %}
{% block content %}
<div class="stats-grid">
  <div class="stat-card">
    <div class="stat-num">{{ doctors|length }}</div>
    <div class="stat-label"><img alt="Doctors icon" data-bbox="381 283 396 298"/> Doctors</div>
  </div>
  <div class="stat-card">
    <div class="stat-num">{{ staff|length }}</div>
    <div class="stat-label"><img alt="Medical Staff icon" data-bbox="381 338 396 353"/> Medical Staff</div>
  </div>
  <div class="stat-card">
    <div class="stat-num">{{ patients|length }}</div>
    <div class="stat-label"><img alt="Patients icon" data-bbox="381 393 396 408"/> Patients</div>
  </div>
  <div class="stat-card">
    <div class="stat-num">{{ prescriptions|length }}</div>
    <div class="stat-label"><img alt="Prescriptions icon" data-bbox="381 448 396 463"/> Prescriptions</div>
  </div>
</div>
<!-- STAFF -->
<div class="card">
  <div style="display:flex;align-items:center;justify-content:space-between;margin-
bottom:1.2rem">
    <div class="card-title" style="margin-bottom:0"><img alt="Medical Staff icon" data-bbox="563 543 578 558"/> Medical Staff</div>
    <a href="/admin/register-doctor" class="btn btn-outline btn-sm"
style="background:#0f766e;color:white;border-radius:10px;">+ Add Staff</a>
  </div>
  {% if staff %}
  <table>
    <thead>
      <tr>
        <th>Login ID</th>
        <th>Password</th>
        <th>Name</th>
        <th>Mobile</th>
        <th>Actions</th>
      </tr>
    </thead>
    <tbody>
      {% for s in staff %}
      <tr>
        <td><span class="badge badge-amber">{{ s.login_id }}</span></td>
        <td></td>
```


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```
<span class="pwd-mask" id="pwd-s-{{ s.id }}" style="font-family:monospace;letter-
spacing:2px">.....</span>
<span class="pwd-plain" id="pwdp-s-{{ s.id }}" style="display:none;font-
family:monospace;color:#99f6e4">{{ s.plain_password or '(set at registration)' }}</span>
<button onclick="togglePwd('s-{{ s.id }}')"
style="background:none;border:none;cursor:pointer;color:rgba(255,255,255,0.4);font-
size:0.8rem;margin-left:4px">👁</button>
</td>
<td>{{ s.name }}</td>
<td>{{ s.mobile or '—' }}</td>
<td>
<a href="/admin/edit-user/{{ s.id }}"
class="btn btn-outline btn-sm">
🖋 Edit
</a>
<a href="/admin/delete-user/{{ s.id }}"
class="btn btn-danger btn-sm"
onclick="return confirm('Delete this staff permanently?')">
🗑 Delete
</a>
</td>
</tr>
{% endfor %}
</tbody>
</table>
{% else %}
<div class="empty-state"><div class="icon">👤</div><p>No staff registered yet.</p></div>
{% endif %}
</div>
<!-- DOCTORS -->
<div class="card">
<div style="display:flex;align-items:center;justify-content:space-between;margin-
bottom:1.2rem">
<div class="card-title" style="margin-bottom:0">👨 Registered Doctors</div>
<a href="/admin/register-doctor" class="btn btn-primary btn-sm">+ Add Doctor</a>
</div>
{% if doctors %}
<table>
<thead>
<tr>
<th>Login ID</th>
<th>Password</th>
<th>Name</th>
<th>Qualification</th>
<th>Mobile</th>
<th>Actions</th>
</tr>
</thead>
<tbody>
{% for d in doctors %}
<tr>
<td><span class="badge badge-teal">{{ d.login_id }}</span></td>
<td>
<span class="pwd-mask" id="pwd-d-{{ d.id }}" style="font-family:monospace;letter-
spacing:2px">.....</span>
```

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```
<span class="pwd-plain" id="pwdp-d-{{ d.id }}" style="display:none;font-family:monospace;color:#99f6e4">{{ d.plain_password or '(set at registration)' }}</span>
<button onclick="togglePwd('d-{{ d.id }}')"
style="background:none;border:none;cursor:pointer;color:rgba(255,255,255,0.4);font-size:0.8rem;margin-left:4px">👁</button>
</td>
<td>{{ d.name }}</td>
<td>{{ d.qualification or '—' }}</td>
<td>{{ d.mobile or '—' }}</td>
<td>
<a href="/admin/edit-user/{{ d.id }}"
class="btn btn-outline btn-sm">
🖋 Edit
</a>
<a href="/admin/delete-user/{{ d.id }}" class="btn btn-danger btn-sm"
onclick="return confirm('Delete this doctor permanently?')">
🗑 Delete
</a>
</td>
</tr>
{% endfor %}
</tbody>
</table>
{% else %}
<div class="empty-state"><div class="icon">👤</div><p>No doctors registered yet.</p></div>
{% endif %}
</div>
{% endblock %}
{% block extra_js %}
<script>
function togglePwd(id) {
  const mask = document.getElementById('pwd-' + id);
  const plain = document.getElementById('pwdp-' + id);
  if (mask.style.display === 'none') {
    mask.style.display = ''; plain.style.display = 'none';
  } else {
    mask.style.display = 'none'; plain.style.display = '';
  }
}
</script>
<style>

table{
  width:100%;
  border-collapse:collapse;
}
table td,
table th{
  white-space:nowrap;
  vertical-align:middle;
  padding:14px;
}
</style>
{% endblock %}
```

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Admin/edit_user.html

```
{% extends 'base.html' %}
{% block title %}Edit User | MediGen-AI{% endblock %}
{% block page_title %}Edit User{% endblock %}

{% block sidebar_nav %}
<a href="/admin" class="nav-item active"><span class="icon"><img alt="Dashboard icon" data-bbox="628 173 648 188"/></span> Dashboard</a>
<a href="/admin/register-doctor" class="nav-item"><span class="icon">+</span> Add
Doctor/Staff</a>
{% endblock %}

{% block content %}
<div class="card" style="max-width:620px">
  <div class="card-title">Edit: {{ user.name }}</div>
  <div style="margin-bottom:1rem"><span class="badge badge-teal">{{ user.login_id }}</span>
<span class="badge badge-amber">{{ user.role }}</span></div>

  <form method="POST">
    <div class="form-grid">
      <div class="form-group full">
        <label>Full Name *</label>
        <input type="text" name="name" value="{{ user.name }}" required>
      </div>
      <div class="form-group">
        <label>Qualification</label>
        <input type="text" name="qualification" value="{{ user.qualification or '' }}">
      </div>
      <div class="form-group">
        <label>Mobile</label>
        <input type="text" name="mobile" value="{{ user.mobile or '' }}">
      </div>
      <div class="form-group full">
        <label>Email</label>
        <input type="email" name="email" value="{{ user.email or '' }}">
      </div>
    </div>

    <!-- CREDENTIALS CHANGE -->
    <div style="background:rgba(245,158,11,0.07);border:1px solid rgba(245,158,11,0.25);border-
radius:14px;padding:1.2rem 1.5rem;margin:0.5rem 0 1.2rem">
      <div style="font-family:'Syne',sans-serif;font-weight:700;color:#fde68a;margin-
bottom:1rem;font-size:0.95rem">
        <img alt="Change Login Credentials icon" data-bbox="243 653 263 668"/> Change Login Credentials
      </div>
      <div class="form-grid">
        <div class="form-group">
          <label>Login ID</label>
          <input type="text" name="login_id" value="{{ user.login_id }}" autocomplete="off">
          <div style="font-size:0.78rem;color:rgba(255,255,255,0.35);margin-top:0.3rem">Change
login ID if needed.</div>
        </div>
        <div class="form-group">
          <label>New Password</label>
          <div style="position:relative">
```

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```
<input type="text" name="new_password" id="pwd-field" placeholder="Leave
blank to keep current" autocomplete="off">
  <button type="button" onclick="genPassword()"
style="position:absolute;right:8px;top:50%;transform:translateY(-
50%);background:rgba(245,158,11,0.2);border:none;color:#fde68a;padding:0.3rem 0.6rem;border-
radius:6px;cursor:pointer;font-size:0.75rem">Auto</button>
</div>
{% if user.plain_password %}
  <div style="font-size:0.78rem;color:rgba(255,255,255,0.35);margin-top:0.3rem">
    Current: <span style="font-family:monospace;color:#fde68a" id="cur-
pwd">.....</span>
    <button type="button" onclick="toggleCurrent()"
style="background:none;border:none;cursor:pointer;color:rgba(255,255,255,0.4);font-
size:0.75rem">⦿</button>
    <span id="cur-pwd-val" style="display:none">{{ user.plain_password }}</span>
  </div>
{% endif %}
</div>
</div>
</div>

<div style="display:flex;gap:1rem">
  <button type="submit" class="btn btn-primary">💾 Save Changes</button>
  <a href="/admin" class="btn btn-outline">Cancel</a>
</div>
</form>
</div>

<script>
function genPassword() {
  const chars = 'ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz23456789';
  let pwd = '';
  for (let i = 0; i < 8; i++) pwd += chars[Math.floor(Math.random() * chars.length)];
  document.getElementById('pwd-field').value = pwd;
}
function toggleCurrent() {
  const cur = document.getElementById('cur-pwd');
  const val = document.getElementById('cur-pwd-val');
  if (val) cur.textContent = cur.textContent === '.....' ? val.textContent : '.....';
}
</script>
{% endblock %}
```

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Admin/register_user.html

```
{% extends 'base.html' %}
{% block title %}Register Doctor/Staff | MediGen-AI{% endblock %}
{% block page_title %}Register Doctor / Staff{% endblock %}
{% block sidebar_nav %}
<a href="/admin" class="nav-item"><span class="icon"><img alt="Dashboard icon" data-bbox="585 158 605 173"/></span> Dashboard</a>
<a href="/admin/register-doctor" class="nav-item active"><span class="icon"><img alt="Add Doctor/Staff icon" data-bbox="740 173 755 188"/></span> Add
Doctor/Staff</a>
{% endblock %}
{% block content %}
<div class="card" style="max-width:680px">
  <div class="card-title">Register New Doctor / Medical Staff</div>
  <form method="POST">
    <div class="form-group">
      <label>Role *</label>
      <select name="role" required>
        <option value="staff"><img alt="Medical Staff icon" data-bbox="405 308 425 323"/> Medical Staff</option>
        <option value="doctor"><img alt="Doctor icon" data-bbox="418 323 438 338"/> Doctor</option>
      </select>
    </div>
    <div class="form-grid">
      <div class="form-group full">
        <label>Full Name *</label>
        <input type="text" name="name" placeholder="Dr. Rajan Sharma" required>
      </div>
      <div class="form-group">
        <label>Qualification</label>
        <input type="text" name="qualification" placeholder="MBBS, MD - General Medicine">
      </div>
      <div class="form-group">
        <label>Mobile Number</label>
        <input type="text" name="mobile" placeholder="9876543210" maxlength="10">
      </div>
      <div class="form-group full">
        <label>Email Address</label>
        <input type="email" name="email" placeholder="doctor@hospital.com">
      </div>
    </div>

    <!-- LOGIN CREDENTIALS BOX -->
    <div style="background:rgba(13,148,136,0.07);border:1px solid rgba(13,148,136,0.25);border-
radius:14px;padding:1.2rem 1.5rem;margin:0.5rem 0 1.2rem">
      <div style="font-family:'Syne',sans-serif;font-weight:700;color:var(--teal-light);margin-
bottom:1rem;font-size:0.95rem">
        <img alt="Key icon" data-bbox="240 683 260 698"/> Set Login Credentials
      </div>
      <div class="form-grid">
        <div class="form-group">
          <label>Login ID *</label>
          <input type="text" name="login_id" placeholder="e.g. DOC001" required
autocomplete="off">
          <div style="font-size:0.78rem;color:rgba(255,255,255,0.35);margin-top:0.3rem">Must
be unique. No spaces.</div>
        </div>
      </div>
    </div>
  </form>
</div>
</div>
```

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```
<div class="form-group">
  <label>Password *</label>
  <div style="position:relative">
    <input type="text" name="password" id="pwd-field" placeholder="Set a password"
required autocomplete="off">
    <button type="button" onclick="genPassword()"
style="position:absolute;right:8px;top:50%;transform:translateY(-
50%);background:rgba(13,148,136,0.2);border:none;color:var(--teal-light);padding:0.3rem
0.6rem;border-radius:6px;cursor:pointer;font-size:0.75rem">Auto</button>
  </div>
  <div style="font-size:0.78rem;color:rgba(255,255,255,0.35);margin-top:0.3rem">Min 4
characters.</div>
</div>
</div>
</div>
</div>

<div style="display:flex;gap:1rem;margin-top:0.5rem">
  <button type="submit" class="btn btn-primary">✔ Register</button>
  <a href="/admin" class="btn btn-outline">Cancel</a>
</div>
</form>
</div>

<script>
function genPassword() {
  const chars = 'ABCDEFGHJKLMNPQRSTUVWXYZabcdefghjkmnpqrstuvwxyz23456789';
  let pwd = '';
  for (let i = 0; i < 8; i++) pwd += chars[Math.floor(Math.random() * chars.length)];
  document.getElementById('pwd-field').value = pwd;
}
</script>
{% endblock %}
```

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Staff/dashboard.html

```
{% extends 'base.html' %}
{% block title %}Staff Dashboard | MediGen-AI{% endblock %}
{% block page_title %}Medical Staff Dashboard{% endblock %}

{% block sidebar_nav %}
<a href="/staff" class="nav-item" style="background:#0f766e;color:white;border-
radius:10px;"><span>🏠</span> Dashboard</a><br>
<a href="/staff/register-patient" class="nav-item"
style="background:#0f766e;color:white;border-radius:10px;"><span class="icon"> + </span>
<strong>Register Patient</strong></a>
{% endblock %}

{% block content %}

<div class="stats-grid">
  <div class="stat-card">
    <div class="stat-num">{{ patients|length }}</div>
    <div class="stat-label">👤 Patients Registered</div>
  </div>
  <div class="stat-card">
    <div class="stat-num">{{ patients|selectattr('registered_at')|list|length }}</div>
    <div class="stat-label">📄 Total Records</div>
  </div>
</div>

<div class="card">
  <div style="display:flex;align-items:center;justify-content:space-between;margin-
bottom:1.2rem">
    <div class="card-title" style="margin-bottom:0">👤 Registered Patients</div>
    <a href="/staff/clear-dashboard"
class="btn btn-danger btn-sm"
style="
background:#dc2626;
color:white;
padding:2px 6px;
border-radius:4px;
text-decoration:none;
font-size:9px;
font-weight:500;
line-height:1;"
onclick="return confirm('Clear all dashboard records?')">
🗑️ Clear<br>Dashboard
  </a>
</div>
</div>

{% if patients %}
<table>
  <thead>
    <tr>
      <th>Patient ID</th>
      <th>Name</th>
      <th>gender</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>{{ patient.id }}</td>
      <td>{{ patient.name }}</td>
      <td>{{ patient.gender }}</td>
    </tr>
  </tbody>
</table>
</div>
</div>
```

Project Title:- MediGen-AI

```
<th>Age</th>
<th>weight</th>
<th>Mobile</th>
<th>Abha Address</th>
<th>Address</th>
<th>Registered On</th>
<th>Token Number</th>
<th>Action</th>
</tr>
</thead>
<tbody>
{% for p in patients %}
<tr>
<td><span class="badge badge-green">{{ p.patient_id }}</span></td>
<td>{{ p.name }}</td>
<td>{{ p.gender or '-' }}</td>
<td>{{ p.age or '—' }}</td>
<td>{{ p.weight or '—' }}</td>
<td>{{ p.mobile or '—' }}</td>
<td>{{ p.abha_address or '—' }}</td>
<td>{{ p.address or '—' }}</td>
<td>{{ p.registered_at.strftime('%d %b %Y %I: %M %p') if p.registered_at else '—'
}}</td>
<td>
<span class="badge badge-green">
OPD-{{ p.id }}
</span>
</td>
<td>
<a href="/edit_patient/{{ p.id }}"
class="btn btn-primary btn-sm">
Edit
</a>
<a href="/print-opd-slip/{{ p.id }}"
class="btn btn-success btn-sm"
target="_blank"
style="background:#0f766e;color:white;">
Print Slip
</a>
</td>
</tr>
{% endfor %}
</tbody>
</table>
{% else %}
<div class="empty-state">
<div class="icon"><img alt="No patients icon" data-bbox="345 740 365 755"/></div>
<p>No patients registered yet.</p>
<a href="/staff/register-patient" class="btn btn-primary" style="margin-top:1rem">Register
First Patient</a>
</div>
{% endif %}
```


Poject Title:- MediGen-AI

```
</div>
<style>

.top-right{
  display:flex;
  justify-content:flex-end;
  margin-bottom:-10px;
}

.profile-dropdown{
  position:relative;
}

.profile-btn{
  display:flex;
  align-items:center;
  gap:10px;

  background:rgba(255,255,255,0.05);

  border:1px solid rgba(255,255,255,0.1);

  color:white;

  padding:10px 16px;

  border-radius:10px;

  cursor:pointer;

  font-size:15px;
}

.profile-name{
  color:#99f6e4;
  font-weight:600;
}

.dropdown-content{
  display:none;

  position:absolute;

  right:0;
  top:55px;

  background:#111827;

  border-radius:10px;

  min-width:170px;

  overflow:hidden;

  z-index:1000;
```

Project Title:- MediGen-AI

```
}

.dropdown-content.show{
  display:block;
}

.dropdown-item{
  display:block;

  padding:12px 15px;

  color:white;

  text-decoration:none;
}

.logout{
  color:#f42c2c;
}

</style>

<script>

function toggleDropdown(){

  document
    .getElementById("profileDropdown")
    .classList.toggle("show");

}

window.onclick = function(event){

  if(!event.target.closest('.profile-btn')){

    let dropdown = document.getElementById("profileDropdown");

    if(dropdown.classList.contains('show')){

      dropdown.classList.remove('show');

    }

  }

}

</script>
{% endblock %}
```

Project Title:- MediGen-AI

```
<th>Age</th>
<th>weight</th>
<th>Mobile</th>
<th>Abha Address</th>
<th>Address</th>
<th>Registered On</th>
<th>Token Number</th>
<th>Action</th>
</tr>
</thead>
<tbody>
{% for p in patients %}
<tr>
<td><span class="badge badge-green">{{ p.patient_id }}</span></td>
<td>{{ p.name }}</td>
<td>{{ p.gender or '-' }}</td>
<td>{{ p.age or '—' }}</td>
<td>{{ p.weight or '—' }}</td>
<td>{{ p.mobile or '—' }}</td>
<td>{{ p.abha_address or '—' }}</td>
<td>{{ p.address or '—' }}</td>
<td>{{ p.registered_at.strftime('%d %b %Y %I: %M %p') if p.registered_at else '—'
}}</td>
<td>
<span class="badge badge-green">
OPD-{{ p.id }}
</span>
</td>
<td>
<a href="/edit_patient/{{ p.id }}"
class="btn btn-primary btn-sm">
Edit
</a>
<a href="/print-opd-slip/{{ p.id }}"
class="btn btn-success btn-sm"
target="_blank"
style="background:#0f766e;color:white;">
Print Slip
</a>
</td>
</tr>
{% endfor %}
</tbody>
</table>
{% else %}
<div class="empty-state">
<div class="icon">🤖</div>
<p>No patients registered yet.</p>
<a href="/staff/register-patient" class="btn btn-primary" style="margin-top:1rem">Register
First Patient</a>
</div>
{% endif %}
```

Project Title:- MediGen-AI

```
</div>
<style>

.top-right{
  display:flex;
  justify-content:flex-end;
  margin-bottom:-10px;
}

.profile-dropdown{
  position:relative;
}

.profile-btn{
  display:flex;
  align-items:center;
  gap:10px;

  background:rgba(255,255,255,0.05);

  border:1px solid rgba(255,255,255,0.1);

  color:white;

  padding:10px 16px;

  border-radius:10px;

  cursor:pointer;

  font-size:15px;
}

.profile-name{
  color:#99f6e4;
  font-weight:600;
}

.dropdown-content{
  display:none;

  position:absolute;

  right:0;
  top:55px;

  background:#111827;

  border-radius:10px;

  min-width:170px;

  overflow:hidden;

  z-index:1000;
```

Project Title:- MediGen-AI

```
}

.dropdown-content.show{
  display:block;
}

.dropdown-item{
  display:block;

  padding:12px 15px;

  color:white;

  text-decoration:none;
}

.logout{
  color:#f42c2c;
}

</style>

<script>

function toggleDropdown(){

  document
    .getElementById("profileDropdown")
    .classList.toggle("show");

}

window.onclick = function(event){

  if(!event.target.closest('.profile-btn')){

    let dropdown = document.getElementById("profileDropdown");

    if(dropdown.classList.contains('show')){

      dropdown.classList.remove('show');

    }

  }

}

</script>
{% endblock %}
```

Project Title:- MediGen-AI

Staff /opd_slip.html

```
<!DOCTYPE html>
<html>
<head>
  <title>OPD Slip</title>

  <style>

    body{
      font-family:Arial;
      background:#f3f4f6;
      padding:20px;
    }

    .slip{
      max-width:700px;
      margin:auto;
      background:white;
      border-radius:12px;
      padding:30px;
      border:2px solid #0f766e;
    }

    .header{
      text-align:center;
      border-bottom:2px solid #0f766e;
      padding-bottom:15px;
      margin-bottom:20px;
    }

    .hospital{
      font-size:32px;
      font-weight:bold;
      color:#0f766e;
    }

    .tag{
      color:#555;
      margin-top:5px;
    }

    .token{
      margin-top:20px;
      text-align:center;
      font-size:28px;
      font-weight:bold;
      color:#dc2626;
    }

    table{
      width:100%;
      border-collapse:collapse;
      margin-top:20px;
    }
```

Project Title:- MediGen-AI

```
td{
  padding:12px;
  border:1px solid #ddd;
}

.label{
  font-weight:bold;
  background:#f9fafb;
  width:35%;
}

.footer{
  margin-top:30px;
  text-align:center;
  color:#666;
  font-size:14px;
}

.print-btn{
  margin-top:25px;
  text-align:center;
}

button{
  background:#0f766e;
  color:white;
  border:none;
  padding:12px 24px;
  border-radius:8px;
  cursor:pointer;
  font-size:16px;
}

@media print{
  .action-buttons{
    display:none !important;
  }
  .print-btn{
    display:none;
  }

  body{
    background:white;
    padding:0;
  }

  .slip{
    border:none;
  }
}

.action-buttons{
```

Project Title:- MediGen-AI

```
margin-top:30px;

display:flex;

justify-content:center;

align-items:center;

gap:10px;

flex-wrap:wrap;

width:100%;


height:42px;

border:none;

border-radius:10px;

background: color #0f766e;

color:white;

font-size:14px;

font-weight:700;

cursor:pointer;

transition:0.3s;

padding:0 14px;

}
</style>
</head>

<body>

<div class="slip">

  <div class="header">

    <div class="hospital">
       MediGen-AI Hospital
    </div>

    <div class="tag">
      Smart OPD Registration System
    </div>

  </div>

</div>
```

Project Title:- MediGen-AI

```
<div class="token">
  TOKEN NO: OPD-{{ patient.id }}
</div>

<table>

  <tr>
    <td class="label">Patient ID</td>
    <td>{{ patient.patient_id }}</td>
  </tr>

  <tr>
    <td class="label">Patient Name</td>
    <td>{{ patient.name }}</td>
  </tr>

  <tr>
    <td class="label">Gender</td>
    <td>{{ patient.gender or '-' }}</td>
  </tr>

  <tr>
    <td class="label">Age</td>
    <td>{{ patient.age or '-' }}</td>
  </tr>

  <tr>
    <td class="label">Weight</td>
    <td>{{ patient.weight or '-' }} KG</td>
  </tr>

  <tr>
    <td class="label">Mobile</td>
    <td>{{ patient.mobile or '-' }}</td>
  </tr>

  <tr>
    <td class="label">ABHA Address</td>
    <td>{{ patient.abha_address or '-' }}</td>
  </tr>

  <tr>
    <td class="label">Assigned Doctor</td>
    <td>
      {% if doctor %}
        {{ doctor.name }}
      {% else %}
        Not Assigned
      {% endif %}
    </td>
  </tr>

  <tr>
    <td class="label">Registration Time</td>
```

Poject Title:- MediGen-AI

```
<td>
    {{ patient.registered_at.strftime('%d %b %Y %I:%M %p') }}
</td>
</tr>

</table>

<div class="footer">
    Please keep this OPD slip safely and wait for your token number.
</div>

<div class="action-buttons no-print">
<button onclick="window.close()" class="action-btn">
    ← Back to Dashboard
</button>

<button onclick="window.print()" class="action-btn">
    🖨️ Print OPD Slip
</button>

</div>

</div>

</body>
</html>
```

Project Title:- MediGen-AI

Staff / register_patient.html

```
{% extends 'base.html' %}
{% block title %}Register Patient | MediGen-AI{% endblock %}
{% block page_title %}Register New Patient{% endblock %}
{% block sidebar_nav %}
<a href="/staff" class="nav-item"><span class="icon"><img alt="Dashboard icon" data-bbox="568 171 588 184"/></span> Dashboard</a>
<a href="/staff/register-patient" class="nav-item active"><span class="icon"><img alt="Register Patient icon" data-bbox="568 186 588 199"/></span> Register Patient</a>
{% endblock %}
{% block content %}
<div class="card" style="max-width:700px">
  <div class="card-title">New Patient Registration</div>
  <form method="POST">
    <div class="form-grid">
      <div class="form-group full">
        <label>Full Name *</label>
        <input type="text" name="name" placeholder="Patient's full name" required>
      </div>
      <div class="form-group">
        <label>Gender</label>
        <input type="text" name="gender" placeholder="male/female/Other">
      </div>
      <div class="form-group">
        <label>Age *</label>
        <input type="number" name="age" placeholder="25" min="0" max="150" required>
      </div>
      <div class="form-group">
        <label>Weight (kg) *</label>
        <input type="number" name="weight" placeholder="65" step="0.1">
      </div>
      <div class="form-group">
        <label>Mobile Number *</label>
        <input type="text" name="mobile" placeholder="9876543210" required
maxlength="10">
      </div>
      <div class="form-group">
        <label>Email Address</label>
        <input type="email" name="email" placeholder="patient@email.com">
      </div>
      <div class="form-group">
        <label>Abha Address *</label>
        <input type="text" name="abha_address" placeholder="XXXX XXXX XXXX"
maxlength="30">
      </div>
      <div class="form-group full">
        <label>Address *</label>
        <textarea name="address" placeholder="Full residential address..."></textarea>
      </div>
    </div>
    <!-- LOGIN CREDENTIALS -->
    <div style="background:rgba(34,197,94,0.07);border:1px solid rgba(34,197,94,0.25);border-
radius:14px;padding:1.2rem 1.5rem;margin:0.5rem 0 1.2rem">
      <div style="font-family:'Syne',sans-serif;font-weight:700;color:#86efac;margin-
bottom:1rem;font-size:0.95rem">
        <img alt="Key icon" data-bbox="243 811 263 824" /> Set Patient Login Credentials
```

Project Title:- MediGen-AI

```
</div>
<div class="form-grid">
  <div class="form-group">
    <label>Login ID / Patient ID *</label>
    <input type="text" name="login_id" placeholder="e.g. PAT001 or mobile number"
required autocomplete="off">
    <div style="font-size:0.78rem;color:rgba(255,255,255,0.35);margin-top:0.3rem">This
will be the Patient's unique ID. No spaces.</div>
  </div>
  <div class="form-group">
    <label>Password *</label>
    <div style="position:relative">
      <input type="text" name="password" id="pwd-field" placeholder="Set a password
for patient" required autocomplete="off">
      <button type="button" onclick="genPassword()"
style="position:absolute;right:8px;top:50%;transform:translateY(-
50%);background:rgba(34,197,94,0.2);border:none;color:#86efac;padding:0.3rem 0.6rem;border-
radius:6px;cursor:pointer;font-size:0.75rem">Auto</button>
    </div>
    <div style="font-size:0.78rem;color:rgba(255,255,255,0.35);margin-top:0.3rem">Share
this password with the patient.</div>
  </div>
</div>
<div>
  <div class="form-group">
    <label>Select Doctor</label>
    <select name="assigned_doctor_id" required">
      <option value="">-- Select Doctor --</option>

      {% for d in doctors %}
      <option value="{{ d.id }}">
        {{ d.name }}
        {% if d.qualification %}
        ({{ d.qualification }})
        {% endif %}
      </option>
      {% endfor %}
    </select>
  </div>
  <div style="display:flex;gap:1rem;margin-top:0.5rem">
    <button type="submit" class="btn btn-primary">📌 Register Patient</button>
    <a href="/staff" class="btn btn-outline">Cancel</a>
  </div>
</form>
</div>

<script>
function genPassword() {
  const chars = 'ABCDEFGHJKLMNPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz23456789';
  let pwd = '';
  for (let i = 0; i < 8; i++) pwd += chars[Math.floor(Math.random() * chars.length)];
  document.getElementById('pwd-field').value = pwd;
}
</script>
{% endblock %}
```

Project Title:- MediGen-AI

Doctor /dashboard.html

```
{% extends 'base.html' %}
{% block title %}Doctor Dashboard | MediGen-AI{% endblock %}
{% block page_title %}Doctor Dashboard{% endblock %}
{% block sidebar_nav %}
<a href="/doctor" class="nav-item active" style="background:#0f766e;color:white;border-radius:10px;>
    <span class="icon"><img alt="Dashboard icon" data-bbox="338 198 358 213"/></span> Dashboard
</a><br>

<a href="/staff/register-patient"
    class="nav-item"
    style="background:#0f766e;color:white;border-radius:10px;">
    <span class="icon"><img alt="Register Patient icon" data-bbox="338 278 358 293"/></span> Register Patient
</a>
{% endblock %}
{% block content %}
<style>
#patient-table {
    width: 100%;
    font-size: 10px;
}
#patient-table th,
#patient-table td {
    padding: 4px;
    white-space: nowrap;
    text-align: center;
}
.btn-sm {
    padding: 4px 8px;
    font-size: 11px;
}
</style>
<div class="stats-grid">
    <div class="stat-card">
        <div class="stat-num">{{ patients|length }}</div>
        <div class="stat-label"><img alt="Total Patients icon" data-bbox="383 588 403 603"/> Total Patients</div>
    </div>
    <div class="stat-card">
        <div class="stat-num">{{ prescriptions|length }}</div>
        <div class="stat-label"><img alt="My Prescriptions icon" data-bbox="383 643 403 658"/> My Prescriptions</div>
    </div>
</div>
<!-- Online Patient Search -->
<!-- Online Patient Search -->
<div class="card" style="
margin-bottom:1.5rem;
background:linear-gradient(135deg,#0f172a,#111827);
border:1px solid rgba(45,212,191,.15);
">

<div style="
display:flex;
justify-content:space-between;
```

Project Title:- MediGen-AI

```
align-items:center;
flex-wrap:wrap;
gap:1rem;
width:100%;
">

<!-- LEFT SIDE -->
<div>
  <div class="card-title" style="margin-bottom:4px;">
    🌐 Online Patient Search
  </div>

  <div style="
font-size:13px;
color:rgba(255,255,255,.5);
">
    Search online registered patient by Patient ID
  </div>
</div>

<!-- RIGHT SIDE SEARCH -->
<div style="
margin-left:auto;
display:flex;
justify-content:flex-end;
">

  <form method="GET"
    action="/doctor/search-online-patient"
    style="
display:flex;
gap:.5rem;
align-items:center;
">

    <input type="text"
      name="patient_id"
      placeholder="Enter Online Patient ID"
      required
      style="
width:240px;
background:rgba(255,255,255,.06);
border:1px solid rgba(255,255,255,.08);
color:white;
border-radius:10px;
padding:.7rem 1rem;
">

    <button type="submit"
      class="btn btn-primary"
      style="
padding:.7rem 1rem;
white-space:nowrap;
">
```

Project Title:- MediGen-AI

```

         Search
    </button>

</form>

</div>

</div>
</div>
{% if online_patient %}

<div class="card" style="
margin-bottom:1.5rem;
border:1px solid rgba(45,212,191,.2);
background:linear-gradient(135deg,#0f172a,#111827);
">
    <div class="card-title">
         Online Patient Found
    </div>

    <table id="patient-table">
        <thead>
            <tr>
                <th>Patient ID</th>
                <th>Name</th>
                <th>Gender</th>
                <th>Age</th>
                <th>Weight</th>
                <th>Mobile</th>
                <th>Email</th>
                <th>ABHA Address</th>
                <th>Address</th>
                <th>Symptoms</th>
                <th>Department</th>
                <th>Doctor Type</th>
                <th>Action</th>
            </tr>
        </thead>

        <tbody>
            <tr>
                <td>
                    <span class="badge badge-green">
                        {{ online_patient.patient_id }}
                    </span>
                </td>

                <td>{{ online_patient.name }}</td>

                <td>{{ online_patient.gender or '—' }}</td>

                <td>{{ online_patient.age or '—' }}</td>
                <td>{{ online_patient.weight or '—' }}</td>
                <td>{{ online_patient.mobile or '—' }}</td>
            </tr>
        </tbody>
    </table>
</div>
</div>
</div>

```


Project Title:- MediGen-AI

```
<td>{{ online_patient.email or '—' }}</td>
<td>{{ online_patient.symptoms or '—' }}</td>
<td>
    {{ online_patient.suggested_department or '—' }}
</td>
<td>
    {{ online_patient.suggested_doctor_type or '—' }}
</td>
<td>
    <a href="/doctor/write-online-prescription/{{ online_patient.patient_id }}"
    class="btn btn-primary btn-sm">
         Write Prescription
    </a>
</td>
</tr>
</tbody>
</table>
</div>
{% endif %}
<!-- Patient Queue -->
<div class="card" style="max-width:100%;
margin:auto;
padding:10px;">
<div class="card-title">🩺 Patient Queue — Write Prescription</div>
{% if patients %}
<table id="patient-table">
    <thead>
<tr>
    <th>Patient ID</th>
    <th>Name</th>
    <th>Gender</th>
    <th>Age</th>
    <th>Weight</th>
    <th>Mobile</th>
    <th>Email</th>
    <th>ABHA Address</th>
    <th>Address</th>
    <th>Action</th>
</tr>
</thead>
<tbody>
    {% for p in patients %}
    <tr class="patient-row" data-name="{{ p.name|lower }}" data-id="{{ p.patient_id|lower }}">
        <td><span class="badge badge-green">{{ p.patient_id }}</span></td>
        <td>{{ p.name }}</td>
        <td>{{ p.gender or '—' }}</td>
        <td>{{ p.age or '—' }}</td>
        <td>{{ p.weight or '—' }}</td>
        <td>{{ p.mobile or '—' }}</td>
        <td>{{ p.email or '—' }}</td>
        <td>{{ p.abha_address or '—' }}</td>
        <td>{{ p.address or '—' }}</td>
        <td>
            <a href="/doctor/write-prescription/{{ p.patient_id }}"
```

Project Title:- MediGen-AI

```
class="btn btn-primary btn-sm">

     Write Prescription
</a>
</td>
    </tr>
    {% endfor %}
</tbody>
</table>
{% else %}
<div class="empty-state">
    <div class="icon"><img alt="Empty state icon" data-bbox="345 235 365 250"/></div>
    <p>No patients in queue yet. Ask medical staff to register patients.</p>
</div>
{% endif %}
</div>

<!-- My Prescriptions -->
<!-- My Prescriptions -->
<div class="card">

    <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px;">

        <div class="card-title" style="margin-bottom:0;">
             My Recent Prescriptions
        </div>

        <a href="/doctor/clear-dashboard"
            onclick="return confirm('Clear dashboard?')"
            style="
                background:#dc2626;
                color:white;
                padding:2px 6px;
                border-radius:4px;
                text-decoration:none;
                font-size:9px;
                font: weight 400px;
                line-height:1;">
             Clear<br>Dashboard
        </a>
    </div>
</div>
{% if prescriptions %}
<table>
    <thead>
        <tr>
            <th>Prescription ID</th>
            <th>Patient ID</th>
            <th>Diagnosis</th>
            <th>Date</th>
            <th>Status</th>
        </tr>
    </thead>
    <tbody>
        {% for rx in prescriptions[-10:]|reverse %}
        <tr>
```

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```
<td>
  <span class="badge badge-teal">
    {{ rx.prescription_id }}
  </span>
</td>
<td>{{ rx.patient_id }}</td>
<td>
  {{ rx.diagnosis[:40] + '...' if rx.diagnosis and rx.diagnosis|length > 40 else rx.diagnosis or '—' }}
</td>
<td>
  {{ rx.created_at.strftime('%d %b %Y %I:%M %p') }}
</td>
<td>
  <span class="badge badge-green">
    Finalized
  </span>
</td>
<td>
  <a href="/doctor/prescription/{{ rx.prescription_id }}"
    class="btn btn-sm"
    style="background:#0f766e;color:white;">
    📄 View
  </a>
  <a href="/doctor/edit-prescription/{{ rx.prescription_id }}"
    class="btn btn-outline btn-sm">
    ✎ Edit
  </a>
</td>
</tr>
    {% endfor %}
  </tbody>
</table>
{% else %}
  <div class="empty-state"><div class="icon">📄</div><p>No prescriptions written yet.</p></div>
  {% endif %}
</div>
{% endblock %}
{% block extra_js %}
<script>
function filterPatients() {
  const query = document.getElementById('patient-search').value.toLowerCase();
  document.querySelectorAll('.patient-row').forEach(row => {
    const name = row.dataset.name;
    const id = row.dataset.id;
    row.style.display = (name.includes(query) || id.includes(query)) ? '' : 'none';
  });
}
</script>
{% endblock %}
```

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write_prescription.html

```
{% extends 'base.html' %}
{% block title %}Write Prescription | MediGen-AI{% endblock %}
{% block page_title %}Write Prescription{% endblock %}

{% block sidebar_nav %}
<a href="/doctor" class="nav-item"><span class="icon"><img alt="Dashboard icon" data-bbox="585 185 605 198"/></span> Dashboard</a>
{% endblock %}

{% block extra_css %}
<style>
.patient-info-bar {
  background: rgba(13,148,136,0.08);
  border: 1px solid rgba(13,148,136,0.2);
  border-radius: 14px;
  padding: 1.2rem 1.5rem;
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(140px, 1fr));
  gap: 1rem;
  margin-bottom: 1.5rem;
}
.info-item label { font-size: 0.75rem; color: rgba(255,255,255,0.4); text-transform: uppercase; }
.info-item span { display: block; font-size: 0.95rem; font-weight: 500; color: var(--white); margin-top: 0.2rem; }

.medicine-row {
  display: grid;
  grid-template-columns: 2fr 1fr 1.5fr 1fr auto;
  gap: 0.6rem;
  align-items: center;
  margin-bottom: 0.6rem;
  padding: 0.8rem;
  background: rgba(255,255,255,0.03);
  border: 1px solid rgba(255,255,255,0.06);
  border-radius: 10px;
}
.medicine-row input { margin: 0; }
.remove-btn {
  background: rgba(239,68,68,0.1);
  border: 1px solid rgba(239,68,68,0.2);
  color: #fca5a5;
  border-radius: 8px;
  padding: 0.5rem 0.7rem;
  cursor: pointer;
  font-size: 0.85rem;
  transition: all 0.2s;
}
.remove-btn:hover { background: rgba(239,68,68,0.2); }

.ai-btn {
  background: linear-gradient(135deg, #7c3aed, #4f46e5);
  color: white;
  border: none;
  border-radius: 10px;
}
```

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```
padding: 0.65rem 1.2rem;
cursor: pointer;
font-family: 'DM Sans', sans-serif;
font-size: 0.9rem;
font-weight: 500;
transition: all 0.2s;
display: inline-flex;
align-items: center;
gap: 0.5rem;
}
.ai-btn:hover { transform: translateY(-1px); box-shadow: 0 4px 15px rgba(124,58,237,0.4); }
.ai-btn:disabled { opacity: 0.6; cursor: not-allowed; transform: none; }
.ai-loading { display: none; align-items: center; gap: 0.5rem; color: #a78bfa; font-size: 0.85rem; }
.ai-loading.show { display: flex;
@keyframes spin { to { transform: rotate(360deg); } }
.spinner { width: 16px; height: 16px; border: 2px solid rgba(167,139,250,0.3); border-top-color:
#a78bfa; border-radius: 50%; animation: spin 0.8s linear infinite;
.col-header { font-size: 0.75rem; color: rgba(255,255,255,0.4); text-transform: uppercase; padding:
0 0.3rem; }
.med-cols { display: grid; grid-template-columns: 2fr 1fr 1.5fr 1fr auto; gap: 0.6rem; margin-
bottom: 0.4rem; padding: 0 0.8rem; }
</style>
{% endblock %}
{% block content %}
<div class="patient-info-bar">
  <div class="info-item">
    <label>Patient ID</label>
    <span>{{ patient.patient_id }}</span>
  </div>
  <div class="info-item">
    <label>Name</label>
    <span>{{ patient.name }}</span>
  </div>
  <div class="info-item">
    <label>Gender</label>
    <span>{{ patient.gender or '—' }}</span>
  </div>
  <div class="info-item">
    <label>Age</label>
    <span>{{ patient.age or '—' }} yrs</span>
  </div>
  <div class="info-item">
    <label>Weight</label>
    <span>{{ patient.weight or '—' }} kg</span>
  </div>
  <div class="info-item">
    <label>Mobile</label>
    <span>{{ patient.mobile or '—' }}</span>
  </div>
  <div class="info-item">
    <label>Email</label>
    <span>{{ patient.email or '—' }}</span>
  </div>&nbsp;
  <div class="info-item">
    <label>ABHA Address</label>
```

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```
<span>{{ patient.abha_address or '-' }}</span>
</div>
<div class="info-item">
<label>Address</label>
<span>{{ patient.address or '-' }}</span>
</div>
</div>
<form method="POST" onsubmit="prepareSubmit()">
<input type="hidden" name="medicines_json" id="medicines-json">
<!-- Symptoms -->
<div class="card">
<div class="card-title">🩺 Clinical Details</div>
<div class="form-grid">
<div class="form-group full">
<label>Chief Complaints / Symptoms *</label>
<textarea name="symptoms" id="symptoms-input" placeholder="Patient complaints:
fever since 3 days, headache, body ache..." required></textarea>
</div>
<div class="form-group full">
<label>Diagnosis / Assessment</label>
<textarea name="diagnosis" placeholder="Diagnosis: Viral Fever / Upper Respiratory
Tract Infection..."></textarea>
</div>
</div>
</div>
</div>
<!-- Medicines -->
<div class="card">
<div style="display:flex;align-items:center;justify-content:space-between;flex-
wrap:wrap;gap:1rem;margin-bottom:1rem">
<div class="card-title" style="margin-bottom:0">💊 Medicines</div>
<div style="display:flex;align-items:center;gap:0.8rem;flex-wrap:wrap">
<div class="ai-loading" id="ai-loading"><div class="spinner"></div> AI thinking...</div>
<button type="button" class="ai-btn" id="ai-btn" onclick="getAISuggestions()">🧠 AI
Suggest Medicines</button>
<button type="button" class="btn btn-outline btn-sm" onclick="addMedicineRow()">+
Add Medicine</button>
</div>
</div>
<div class="med-cols">
<div class="col-header">Medicine Name</div>
<div class="col-header">Dosage</div>
<div class="col-header">Frequency</div>
<div class="col-header">Duration</div>
<div class="col-header"></div>
</div>
<div id="medicines-container"></div>
<p style="font-size:0.82rem;color:rgba(255,255,255,0.3);margin-top:0.5rem">
💡 Enter symptoms above and click "AI Suggest Medicines" for intelligent recommendations
</p>
</div>
<!-- Lab Reports & Notes -->
<div class="card">
```

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```
<div class="card-title">🩺 Lab Reports & Notes</div>
<div class="form-grid">
  <div class="form-group full">
    <label>Lab Reports Advised</label>
    <textarea name="lab_reports" placeholder="CBC, Blood Sugar Fasting, Urine R/E, Chest
X-Ray..."></textarea>
  </div>
  <div class="form-group full">
    <label>Doctor's Notes / Advice</label>
    <textarea name="notes" placeholder="Rest for 3 days. Plenty of fluids. Follow up after 5
days..."></textarea>
  </div>
</div>
<div style="display:flex;gap:1rem;align-items:center">
  <button type="submit" class="btn btn-primary" style="font-size:1rem;padding:0.8rem
2rem">
    ☒ Finalize & Sign Prescription
  </button>
  <a href="/doctor" class="btn btn-outline">Cancel</a>
  <span style="font-size:0.82rem;color:rgba(255,255,255,0.3)">Digital signature will be applied
automatically</span>
</div>
</form>
{% endblock %}

{% block extra_js %}
<script>
let medicines = [];

function addMedicineRow(med = {}) {
  const idx = medicines.length;
  medicines.push(med);

  const container = document.getElementById('medicines-container');
  const row = document.createElement('div');
  row.className = 'medicine-row';
  row.id = 'med-row-' + idx;
  row.innerHTML = `
    <input type="text" placeholder="Medicine name" value="${med.name || ''}"
onchange="updateMed(${idx}, 'name', this.value)">
    <input type="text" placeholder="1 tab / 5ml" value="${med.dosage || ''}"
onchange="updateMed(${idx}, 'dosage', this.value)">
    <input type="text" placeholder="OD / BD / TDS" value="${med.frequency || ''}"
onchange="updateMed(${idx}, 'frequency', this.value)">
    <input type="text" placeholder="5 days" value="${med.duration || ''}"
onchange="updateMed(${idx}, 'duration', this.value)">
    <button type="button" class="remove-btn" onclick="removeMed(${idx})">X</button>
  `;
  container.appendChild(row);
}

function updateMed(idx, field, value) {
  medicines[idx][field] = value;
}
```

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```
function removeMed(idx) {
  const row = document.getElementById('med-row-' + idx);
  if (row) row.remove();
  medicines[idx] = null;
}

function prepareSubmit() {
  const validMeds = medicines.filter(m => m && m.name && m.name.trim());
  document.getElementById('medicines-json').value = JSON.stringify(validMeds);
}

async function getAISuggestions() {
  const symptoms = document.getElementById('symptoms-input').value.trim();
  if (!symptoms) {
    alert('Please enter patient symptoms first.');
```

return;

```
  }
  const btn = document.getElementById('ai-btn');
  const loading = document.getElementById('ai-loading');
  btn.disabled = true;
  loading.classList.add('show');
  try {
    const response = await fetch('/doctor/ai-suggest', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ symptoms })
    });
    const data = await response.json();
    if (data.success && data.medicines.length > 0) {
      // Clear and add AI suggestions
      document.getElementById('medicines-container').innerHTML = '';
      medicines = [];
      data.medicines.forEach(med => addMedicineRow(med));
      // Show success indicator
      btn.textContent = '✓ AI Suggestions Applied';
      btn.style.background = 'linear-gradient(135deg, #059669, #047857)';
      setTimeout(() => {
        btn.textContent = '🩺 AI Suggest Medicines';
        btn.style.background = '';
      }, 3000);
    }
  } catch (error) {
    alert('AI suggestion failed. Please add medicines manually.');
```

finally {

```
    btn.disabled = false;
    loading.classList.remove('show');
  }
}

// Start with 2 empty medicine rows
addMedicineRow();
addMedicineRow();
</script>
{% endblock %}
```

patient dashboard.html

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```
</div>
{% endblock %}
{% block sidebar_nav %}
<a href="/patient" class="nav-item active" style="background:#0f766e;color:white;border-
radius:10px;"><span class="icon"><img alt="Prescriptions icon" data-bbox="428 143 448 158"/></span> My Prescriptions</a>
<br><a href="#"
onclick="
window.open('https://www.cowin.gov.in/', '_blank');
window.open('https://abha.abdm.gov.in/abha/v3/', '_blank');
return false;
class="nav-item"
style="
background:#0f766e;
color:white;
border-radius:10px;
font-weight:600;
">
&#129658; Create ABHA Account
</a><br>
<a href="#"
onclick="
window.open('https://pmjay.gov.in/', '_blank');
window.open('https://hem.nha.gov.in/search', '_blank');
window.open('https://beneficiary.nha.gov.in/', '_blank');
return false;
class="nav-item"
style="
background:#0f766e;
color:white;
border-radius:10px;
font-weight:600;
">
&#129658; Ayushman Card Services
</a><br>
<a href="#"
onclick="
window.open('https://janaushadhi.gov.in/near-by-kendra');
window.open('https://janaushadhi.gov.in/productportfolio/ProductmrpList');
return false;
"
class="nav-item"
style="
background:#0f766e;
color:white;
border-radius:10px;
font-weight:600;
">
&#128138; Jan Aushadhi Services
</a>
<br>
<a href="#"
onclick="
window.open('https://esanjeevani.mohfw.gov.in/', '_blank');
return false;"
class="nav-item"
```

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```
style="
background:#0f766e;
color:white;
border-radius:10px;
font-weight:600;
">
&#128657; eSanjeevani Services
</a><br>
<a href="#"
onclick="
window.open('https://pmjay.gov.in/', '_blank');
window.open('https://irdai.gov.in/', '_blank');
window.open('https://policybazar.com/', '_blank');
window.open('https://www.jansuraksha.gov.in/', '_blank');
return false;
"
class="nav-item"
style="
background:#0f766e;
color:white;
border-radius:10px;
font-weight:600;
">
&#128737; Insurance Services
</a><br>
{% endblock %}
{% block content %}
{% if patient %}
<div style="background:rgba(13,148,136,0.08);border:1px solid rgba(13,148,136,0.2);border-
radius:14px;padding:1.2rem 1.5rem;margin-bottom:1.5rem;display:grid;grid-template-
columns:repeat(auto-fit,minmax(140px,1fr));gap:1rem">
  <div>
    <div style="font-size:0.75rem;color:rgba(255,255,255,0.4);text-transform:uppercase">Patient
ID</div>
    <div style="font-size:0.95rem;font-weight:500;margin-top:0.2rem">{{ patient.patient_id
}}</div>
  </div>
  <div>
    <div style="font-size:0.75rem;color:rgba(255,255,255,0.4);text-transform:uppercase">Patient Name</div>
    <div style="font-size:0.95rem;font-weight:500;margin-top:0.2rem">{{ patient.name }}</div>
  </div>
  <div>
    <div style="font-size:0.75rem;color:rgba(255,255,255,0.4);text-transform:uppercase">Gender</div>
    <div style="font-size:0.95rem;font-weight:500;margin-top:0.2rem">{{ patient.gender or '—' }}</div>
  </div>
  <div>
    <div style="font-size:0.75rem;color:rgba(255,255,255,0.4);text-transform:uppercase">Age / Weight</div>
    <div style="font-size:0.95rem;font-weight:500;margin-top:0.2rem">{{ patient.age or '—' }} yrs / {{ patient.weight or '—'
}} kg</div>
  </div>
  <div>
    <div style="font-size:0.75rem;color:rgba(255,255,255,0.4);text-
transform:uppercase">Mobile</div>
    <div style="font-size:0.95rem;font-weight:500;margin-top:0.2rem">{{ patient.mobile or '—'
}}</div>
  </div>
</div>
```

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```
<div>
  <div style="font-size:0.75rem;color:rgba(255,255,255,0.4);text-transform:uppercase">ABHA
Address</div>
  <div style="font-size:0.95rem;font-weight:500;margin-top:0.2rem">{{ patient.abha_address or
'—' }}</div>
</div>
<div>
  <div style="font-size:0.75rem;color:rgba(255,255,255,0.4);text-
transform:uppercase">Address</div>
  <div style="font-size:0.95rem;font-weight:500;margin-top:0.2rem">{{ patient.address or '—'
}}</div>
</div>
</div>
{% endif %}
<div class="card">
  <div class="card-title"><img alt="Prescription icon" data-bbox="365 288 380 303"/> My Prescriptions</div>
  {% if prescriptions %}
  <div style="display:grid;gap:1rem">
    {% for rx in prescriptions %}
    <div style="background:rgba(255,255,255,0.03);border:1px solid rgba(255,255,255,0.07);border-
radius:14px;padding:1.2rem;display:flex;align-items:center;justify-content:space-between;flex-wrap:wrap;gap:1rem">
      <div>
        <div style="font-family:'Syne',sans-serif;font-weight:700;margin-bottom:0.3rem">{{ rx.diagnosis|:60| if rx.diagnosis
else 'General Consultation' }}</div>
        <div style="font-size:0.85rem;color:rgba(255,255,255,0.4)">
          By Dr. {{ rx.doctor_name }} | {{ rx.created_at.strftime('%d %B %Y') }}
        </div>
        <div style="margin-top:0.5rem">
          {% for med in rx.medicines_list|:3| %}
          <span class="badge badge-teal" style="margin-right:0.3rem">{{ med.name }}</span>
          {% endfor %}
          {% if rx.medicines_list|length > 3 %}
            <span style="font-size:0.8rem;color:rgba(255,255,255,0.4)">+{{ rx.medicines_list|length - 3 }}
more</span>
          {% endif %}
        </div>
      </div>
    </div>
    <div style="display:flex;gap:0.7rem;flex-wrap:wrap">
      <span class="badge badge-teal">{{ rx.prescription_id }}</span>
      <a href="/patient/prescription/{{ rx.prescription_id }}" class="btn btn-primary btn-sm"><img alt="Eye icon" data-bbox="735 605 750 620"/> View</a>
    </div>
  </div>
  {% endfor %}
</div>
{% else %}
<div class="empty-state">
  <div class="icon"><img alt="Empty state icon" data-bbox="310 695 325 710"/></div>
  <p>No prescriptions yet. Please visit the doctor.</p>
</div>
{% endif %}
</div>
{% endblock %}
```

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Patient prescription.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Prescription {{ prescription.prescription_id }} | MediGen-AI</title>
  <link
href="https://fonts.googleapis.com/css2?family=Syne:wght@400;600;700;800&family=DM+Sans:wght@300;400;500&display=swap" rel="stylesheet">
  <style>
    :root {
      --teal: #0d9488; --teal-dark: #0f766e; --teal-light: #14b8a6;
      --navy: #0f172a; --white: #ffffff;
    }
    * { margin: 0; padding: 0; box-sizing: border-box; }

    body {
      font-family: 'DM Sans', sans-serif;
      background: #f0f4f8;
      color: #1a202c;
      padding: 1rem;
    }

    .action-bar {
      max-width: 850px;
      margin: 0 auto 1rem;
      display: flex;
      justify-content: flex-end;
      gap: 0.8rem;
      align-items: center;
      flex-wrap: wrap;
    }

    .btn {
      padding: 0.6rem 1.3rem;
      border-radius: 8px;
      border: none;
      cursor: pointer;
      font-family: 'DM Sans', sans-serif;
      font-size: 0.9rem;
      font-weight: 500;
      text-decoration: none;
      display: inline-flex;
      align-items: center;
      gap: 0.4rem;
      transition: all 0.2s;
    }

    .btn-teal { background: var(--teal); color: white; }
    .btn-teal:hover { background: var(--teal-dark); }
    .btn-gray { background: #e2e8f0; color: #475569; }
    .btn-gray:hover { background: #cbd5e1; }

    /* PRESCRIPTION PAPER */
    .rx-paper {
      max-width: 850px;
```

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```
margin: 0 auto;
background: white;
border-radius: 8px;
overflow: hidden;
box-shadow: 0 4px 20px rgba(0,0,0,0.1);
}

/* HEADER */
.rx-header {
background: linear-gradient(135deg, #0f766e, #0d9488);
color: white;
padding: 1.5rem 2rem;
display: flex;
justify-content: space-between;
align-items: flex-start;
}
.hospital-info h1 {
font-family: 'Syne', sans-serif;
font-size: 1.4rem;
font-weight: 800;
margin-bottom: 0.2rem;
}
.hospital-info p { font-size: 0.8rem; opacity: 0.85; line-height: 1.5; }
.rx-id-box {
text-align: right;
background: rgba(255,255,255,0.15);
padding: 0.8rem 1.2rem;
border-radius: 8px;
}
.rx-id-box .label { font-size: 0.7rem; opacity: 0.7; text-transform: uppercase; }
.rx-id-box .value { font-family: 'Syne', sans-serif; font-size: 1rem; font-weight: 700; }

/* DOCTOR INFO */
.doctor-bar {
background: #f8fafc;
border-bottom: 2px solid #4a92f0;
padding: 1rem 2rem;
display: flex;
justify-content: space-between;
align-items: center;
}
.doctor-name { font-family: 'Syne', sans-serif; font-size: 1.1rem; font-weight: 700; color: var(--teal-dark); }
.doctor-qual { font-size: 0.82rem; color: #64748b; }
.rx-date { font-size: 0.85rem; color: #64748b; text-align: right; }

/* PATIENT INFO */
.patient-section {
padding: 1rem 2rem;
border-bottom: 3px #3e8bf0;
display: grid;
grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
gap: 0.8rem;
}
.pinfo { }
```

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```
.pinfo .key { font-size: 0.72rem; text-transform: uppercase; color: #94a3b8; letter-spacing: 0.05em; }
.pinfo .val { font-size: 0.9rem; font-weight: 500; color: #1e293b; margin-top: 0.1rem; }

/* RX SYMBOL */
.rx-body { padding: 1.5rem 2rem; }
.rx-symbol {
  font-family: 'Syne', sans-serif;
  font-size: 2.5rem;
  color: var(--teal);
  font-weight: 800;
  margin-bottom: 0.5rem;
}

.section-head {
  font-family: 'Syne', sans-serif;
  font-size: 0.85rem;
  font-weight: 700;
  text-transform: uppercase;
  color: #475569;
  letter-spacing: 0.1em;
  margin: 1.2rem 0 0.7rem;
  padding-bottom: 0.4rem;
  border-bottom: 2px solid #297ff0;
}

/* MEDICINES TABLE */
.med-table { width: 100%; border-collapse: collapse; font-size: 0.88rem; }
.med-table th {
  background: #f1f5f9;
  padding: 0.6rem 0.8rem;
  text-align: left;
  font-weight: 600;
  font-size: 0.78rem;
  text-transform: uppercase;
  color: #64748b;
  letter-spacing: 0.05em;
}
.med-table td { padding: 0.8rem 0.8rem; border-bottom: 1px solid #f1f5f9; vertical-align: middle; }
.med-table tr:last-child td { border-bottom: none; }
.med-name { font-weight: 600; color: #0f172a; }

/* PHARMACY LINK */
.pharmacy-btn {
  display: inline-flex;
  align-items: center;
  gap: 0.3rem;
  background: #fff7ed;
  border: 1px solid #fed7aa;
  color: #ea580c;
  padding: 0.3rem 0.7rem;
  border-radius: 20px;
  font-size: 0.75rem;
  font-weight: 600;
}
```

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```
    text-decoration: none;
    transition: all 0.2s;
}
.pharmacy-btn:hover { background: #ffedd5; }

.info-box {
    background: #f8fafc;
    border-radius: 8px;
    padding: 0.8rem 1rem;
    font-size: 0.88rem;
    color: #475569;
    line-height: 1.6;
    margin-top: 0.5rem;
}

/* SIGNATURE */
.signature-section {
    margin-top: 2rem;
    padding-top: 1rem;
    border-top: 1px dashed #e2e8f0;
    display: flex;
    justify-content: flex-end;
}
.sig-box {
    text-align: center;
    min-width: 180px;
}
.sig-area {
    border-bottom: 2px solid #0d9488;
    height: 50px;
    margin-bottom: 0.3rem;
    display: flex;
    align-items: flex-end;
    justify-content: center;
    padding-bottom: 0.3rem;
}
.sig-text {
    font-family: 'Syne', sans-serif;
    font-size: 1.1rem;
    font-weight: 800;
    color: var(--teal-dark);
    font-style: italic;
}
.sig-label { font-size: 0.78rem; color: #2278f1; }
.sig-name { font-size: 0.9rem; font-weight: 600; color: #1e293b; margin-top: 0.2rem; }
.sig-qual { font-size: 0.78rem; color: #64748b; }

/* FOOTER */
.rx-footer {
    background: #f1f5f9;
    padding: 0.8rem 2rem;
    display: flex;
    justify-content: space-between;
    font-size: 0.78rem;
    color: #5966f2;
}
```

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```
}

/* PRINT STYLES */
@media print {
  body { background: white; padding: 0; }
  .action-bar { display: none; }
  .rx-paper { box-shadow: none; border-radius: 0; }
  .pharmacy-btn { -webkit-print-color-adjust: exact; print-color-adjust: exact; }
}
@media print {
th, td {
padding: 4px !important; /* टेबल की रो को पतला करने के लिए */
}
}
</style>
</head>
<body>

<div class="action-bar">
{% if session.get('role') == 'doctor' %}
<a href="/doctor"
class="btn btn-sm"
style="background:#0b9c90;color:white;">
 Back
</a>
{% elif session.get('online_patient_id') %}
<a href="/online-dashboard"
class="btn btn-sm"
style="background:#099b8f;color:white;">
 Dashboard
</a>

{% else %}
<a href="/patient"
class="btn btn-sm"
style="background:#099b8f;color:white;">
 Dashboard
</a>
{% endif %}
<button onclick="window.print()" class="btn btn-teal"> Print / Save PDF</button>
</div>

<div class="rx-paper" id="prescription">
<!-- HEADER -->
<div class="rx-header">
<div class="hospital-info">
<h1> MediGen-AI Hospital</h1>
<p><strong>AI-Powered Smart Healthcare Portal</strong><br>
<strong>Aryabhatta Knowledge University, Patna | Contact: 0612-2451070</strong></p>
</div>
<div class="rx-id-box">
<div class="label">Prescription No.</div>
<div class="value">{{ prescription.prescription_id }}</div>
</div>
```

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```
<div class="label" style="margin-top:0.3rem">{{ prescription.created_at.strftime('%d %B
%Y') }}</div>
</div>
</div>

<!-- DOCTOR INFO -->
<div class="doctor-bar">
  <div>
    <div class="doctor-name">{{ doctor.name if doctor else 'Unknown' }}</div>
    <div class="doctor-qual"><strong>{{ doctor.qualification if doctor else '' }}<br>Address: 123
Healthcare Street, Patna 800001<br>Regd.no.384/28</strong></div>
  </div>
  <div class="rx-date">
    <div style="font-size:0.78rem;color:#94a3b8">Date of Issue</div>
    <div style="font-weight:600">{{ prescription.created_at.strftime('%d/%m/%Y') }}</div>
  </div>
</div>

<!-- PATIENT INFO -->
<div class="patient-section">
  <div class="pinfo">
    <div class="key">Patient ID</div>
    <div class="val">{{ patient.patient_id }}</div>
  </div>
  <div class="pinfo">
    <div class="key">Patient Name</div>
    <div class="val">{{ patient.name }}</div>
  </div>
  <div class="pinfo">
    <div class="key">Gender</div>
    <div class="val">{{ patient.gender or '—' }}</div>
  </div>
  <div class="pinfo">
    <div class="key">Age / Weight</div>
    <div class="val">{{ patient.age or '—' }} yrs / {{ patient.weight or '—' }} kg</div>
  </div>
  <div class="pinfo">
    <div class="key">Email</div>
    <div class="val">{{ patient.email or '—' }}</div>
  </div>
  <div class="pinfo">
    <div class="key">ABHA Address</div>
    <div class="val">{{ patient.abha_address or '—' }}</div>
  </div>
  <div class="pinfo">
    <div class="key">Address</div>
    <div class="val">{{ patient.address or '—' }}</div>
  </div>
  <div class="pinfo">
    <div class="key">Mobile Number</div>
    <div class="val">{{ patient.mobile or '—' }}</div>
  </div>
</div>

<!-- BODY -->
```

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```
<div class="rx-body">
  <div class="rx-symbol">R</div>

  {% if prescription.symptoms %}
  <div class="section-head">Chief Complaints</div>
  <div class="info-box">{{ prescription.symptoms }}</div>
  {% endif %}

  {% if prescription.diagnosis %}
  <div class="section-head">Diagnosis</div>
  <div class="info-box" style="font-weight:600;color:#0f172a">{{ prescription.diagnosis }}</div>
  {% endif %}

  <!-- MEDICINES -->
  <div class="section-head">Prescribed Medicines</div>
  {% if medicines %}
  <table class="med-table">
    <thead>
      <tr>
        <th>#</th>
        <th>Medicine</th>
        <th>Dosage</th>
        <th>Frequency</th>
        <th>Duration</th>
        <th>Order Online</th>
      </tr>
    </thead>
    <tbody>
      {% for med in medicines %}
      {% if med.name %}
      <tr>
        <td style="color:#94a3b8;font-weight:600">{{ loop.index }}</td>
        <td><div class="med-name">{{ med.name }}</div></td>
        <td>{{ med.dosage or '—' }}</td>
        <td>{{ med.frequency or '—' }}</td>
        <td>{{ med.duration or '—' }}</td>
        <td>
          <a href="https://www.netmeds.com/catalogsearch/result?q={{ med.name|urlencode }}"
            target="_blank" class="pharmacy-btn">
 NetMeds
          </a>
          <a href="https://www.apollopharmacy.in/"
            target="_blank" class="pharmacy-btn" style="margin-left:4px;background:#f0fdf4;border-color:#86efac;color:#16a34a">
 apollopharmacy
          </a>
          <a href="https://1mg.in/search/all?name={{ med.name|urlencode }}"
            target="_blank" class="pharmacy-btn" style="margin-left:4px;background:#f0fdf4;border-color:#86efac;color:#16a34a">
 1mg
          </a>
        </td>
      </tr>
      {% endif %}
    </tbody>
  </table>
  {% endif %}
```

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```
{% endfor %}
</tbody>
</table>
{% else %}
<p style="color:#94a3b8;font-size:0.88rem">No medicines prescribed.</p>
{% endif %}

{% if prescription.lab_reports %}
<div class="section-head">Lab Reports Advised</div><a
href="https://www.pathkindlabs.com/"
target="_blank" class="pharmacy-btn" style="margin-
left:4px;background:#f0fdf4;border-color:#86efac;color:#16a34a">
 pathlabs
</a>
<div class="info-box">{{ prescription.lab_reports }}</div>
{% endif %}

{% if prescription.notes %}
<div class="section-head">Doctor's Advice & Notes</div>
<div class="info-box">{{ prescription.notes }}</div>
{% endif %}

<!-- DIGITAL SIGNATURE -->
<div class="signature-section">
<div class="sig-box">
<div class="sig-area">
<div class="sig-text">{{ doctor.name.split()[0] if doctor else 'Doctor' }}</div>
</div>
<div class="sig-label"><strong> Digital Signature</strong></div>
<div class="sig-name">{{ doctor.name if doctor else '-' }}</div>
<div class="sig-qual"><strong>{{ doctor.qualification if doctor else '' }}</div>
</div>
</div>
</div>

<!-- FOOTER -->
<div class="rx-footer">
<span><strong>  MediGen-AI | AI-Powered Smart Prescription</strong></span>
<span><strong>Generated: {{ prescription.created_at.strftime('%d/%m/%Y %H:%M') }} |
Valid for 14 days</strong></span>
</div>
</div>

<div style="text-align:center;margin:1.5rem;font-size:0.82rem;color:#94a3b8">
<strong>⚠ This is a digitally generated prescription. Order medicines only from licensed
pharmacies.</strong>
</div>

</body>
</html>
```

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Online / login.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>MediGen-AI | Online Patient Login</title>

  <link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap"
rel="stylesheet">

  <style>
    *{
      margin:0;
      padding:0;
      box-sizing:border-box;
      font-family:'Poppins',sans-serif;
    }

    body{
      min-height:100vh;
      display:flex;
      justify-content:center;
      align-items:center;
      background:
        linear-gradient(rgba(0,0,0,0.55), rgba(0,0,0,0.55)),
        url('https://images.unsplash.com/photo-1584982751601-
97dcc096659c?q=80&w=1470&auto=format&fit=crop');
      background-size:cover;
      background-position:center;
      overflow:hidden;
    }
    .main-container{
      width:100%;
      display:flex;
      justify-content:center;
      align-items:center;
      padding:20px;
    }
    .login-card{
      width:100%;
      max-width:430px;
      background:rgba(255,255,255,0.12);
      backdrop-filter:blur(12px);
      border:1px solid rgba(255,255,255,0.2);
      border-radius:24px;
      padding:40px 35px;
      box-shadow:0 10px 40px rgba(0,0,0,0.35);
      animation:fadeIn 0.8s ease;
    }

    @keyframes fadeIn{
      from{
```

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```
        opacity:0;
        transform:translateY(25px);
    }
    to{
        opacity:1;
        transform:translateY(0);
    }
}
.logo{
    text-align:center;
    margin-bottom:15px;
}
.logo h1{
    color:#fff;
    font-size:34px;
    font-weight:700;
}

.logo span{
    color:#00ffd5;
}

.subtitle{
    text-align:center;
    color:#e7e7e7;
    font-size:14px;
    margin-bottom:30px;
}

.medical-icon{
    font-size:55px;
    text-align:center;
    margin-bottom:12px;
}

.input-group{
    margin-bottom:22px;
}

.input-group label{
    color:#fff;
    display:block;
    margin-bottom:8px;
    font-size:14px;
    font-weight:500;
}

.input-group input{
    width:100%;
    padding:14px 15px;
    border:none;
    outline:none;
    border-radius:12px;
    background:rgba(255,255,255,0.15);
    color:#fff;
}
```

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```
    font-size:15px;
    transition:0.3s;
    border:1px solid transparent;
  }
  .input-group input::placeholder{
    color:#ddd;
  }

  .input-group input:focus{
    border:1px solid #00ffd5;
    background:rgba(255,255,255,0.22);
  }
  .login-btn{
    width:100%;
    padding:14px;
    border:none;
    border-radius:14px;
    background:#0fac9f;
    color:white;
    font-size:16px;
    font-weight:600;
    cursor:pointer;
    transition:0.3s;
    margin-top:10px;
  }
  .login-btn:hover{
    transform:translateY(-2px);
    box-shadow:0 8px 20px rgba(15,118,110,0.4);
  }
  .bottom-links{
    margin-top:22px;
    text-align:center;
  }
  .bottom-links a{
    color:#00ffd5;
    text-decoration:none;
    font-size:14px;
    transition:0.3s;
  }
  .bottom-links a:hover{
    color:#fff;
  }
  .flash-message{
    padding:12px;
    border-radius:10px;
    margin-bottom:18px;
    text-align:center;
    font-size:14px;
  }
  .success{
    background:rgba(0,255,140,0.2);
    color:#7CFFB2;
  }
  .error{
    background:rgba(255,0,0,0.18);
```

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```
        color:#ffb0b0;
    }
    .footer-text{
        text-align:center;
        color:#ddd;
        margin-top:20px;
        font-size:12px;
    }
    @media(max-width:500px){
        .login-card{
            padding:30px 22px;
        }
        .logo h1{
            font-size:28px;
        }
    }
}
</style>
</head>
<body>
    <div class="main-container">
        <div class="login-card">
            <div class="medical-icon"><img alt="Medical icon" data-bbox="418 368 438 383"/></div>
            <div class="logo">
                <h1>MediGen<span>-AI</span></h1>
            </div>
            <p class="subtitle">
                Smart Healthcare & Online Patient Portal
            </p>
            {% with messages = get_flashed_messages(with_categories=true) %}
            {% if messages %}
                {% for category, message in messages %}
                    <div class="flash-message {{ category }}">
                        {{ message }}
                    </div>
                {% endfor %}
            {% endif %}
            {% endwith %}
            <form method="POST">
                <div class="input-group">
                    <label>Login ID</label>
                    <input type="text"
                        name="login_id"
                        placeholder="Enter Login ID"
                        required>
                </div>
                <div class="input-group">
                    <label>Password</label>
                    <input type="password"
                        name="password"
                        placeholder="Enter Password"
                        required>
                </div>
                <button type="submit" class="login-btn">
                    <img alt="Login icon" data-bbox="258 795 278 810"/> Login Now
                </button>
            </form>
        </div>
    </div>
</body>
</html>
```

Project Title:- MediGen-AI

```
</form>
<div class="bottom-links">
  <p style="margin-bottom:10px;">
    <a href="{{ url_for('online_register') }}">
       New Patient Registration
    </a>
  </p>
  <p style="margin-bottom:10px;">
    <a href="{{ url_for('online_forgot_password') }}">
       Forgot Password?
    </a>
  </p>
  <p>
    <a href="{{ url_for('index') }}">
       Back To Home
    </a>
  </p>
</div>

<div class="footer-text">
  © 2026 MediGen-AI Healthcare System
</div>
</div>
</div>
</body>
</html>
```

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Online / register.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Online Patient Registration</title>

  <link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap"
rel="stylesheet">

  <style>

    *{
      margin:0;
      padding:0;
      box-sizing:border-box;
    }

    body{
      font-family:'Poppins',sans-serif;
      min-height:100vh;
      display:flex;
      justify-content:center;
      align-items:center;
      padding:30px;
      background:
        linear-gradient(rgba(15,23,42,0.75),rgba(15,23,42,0.75)),
        url('https://images.unsplash.com/photo-1587351021759-
3e566b6af7cc?q=80&w=1600&auto=format&fit=crop');
      background-size:cover;
      background-position:center;
      background-attachment:fixed;
    }

    .main-box{
      width:100%;
      max-width:1100px;
      display:grid;
      grid-template-columns:1fr 1fr;
      background:rgba(255,255,255,0.12);
      backdrop-filter:blur(14px);
      border-radius:24px;
      overflow:hidden;
      box-shadow:0 10px 40px rgba(0,0,0,0.35);
      border:1px solid rgba(255,255,255,0.15);
    }

    .left-side{
      padding:60px 50px;
      color:white;
      background:linear-gradient(135deg,#0f766e,#0891b2);
      display:flex;
```

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```
    flex-direction:column;
    justify-content:center;
  }

  .logo{
    font-size:42px;
    margin-bottom:15px;
  }

  .left-side h1{
    font-size:38px;
    line-height:1.2;
    margin-bottom:18px;
    font-weight:700;
  }

  .left-side p{
    font-size:15px;
    line-height:1.8;
    color:rgba(255,255,255,0.9);
    margin-bottom:30px;
  }

  .feature-box{
    background:rgba(255,255,255,0.12);
    padding:14px 18px;
    border-radius:12px;
    margin-bottom:14px;
    font-size:14px;
  }

  .right-side{
    background:white;
    padding:45px;
    overflow-y:auto;
    max-height:95vh;
  }

  .form-title{
    font-size:32px;
    color:#0f172a;
    margin-bottom:8px;
    font-weight:700;
  }

  .sub-title{
    color:#64748b;
    margin-bottom:28px;
    font-size:14px;
  }

  .input-group{
    margin-bottom:18px;
  }
```

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```
label{
  display:block;
  margin-bottom:7px;
  color:#0f172a;
  font-size:14px;
  font-weight:600;
}

input,
select,
textarea{
  width:100%;
  padding:13px 15px;
  border:1px solid #cbd5e1;
  border-radius:12px;
  font-size:15px;
  transition:0.3s;
  background:#f8fafc;
}

input:focus,
select:focus,
textarea:focus{
  outline:none;
  border-color:#0891b2;
  box-shadow:0 0 4px rgba(8,145,178,0.15);
  background:white;
}

textarea{
  min-height:100px;
  resize:vertical;
}

.grid-2{
  display:grid;
  grid-template-columns:1fr 1fr;
  gap:15px;
}

.register-btn{
  width:100%;
  border:none;
  padding:15px;
  border-radius:14px;
  background:linear-gradient(135deg,#0f766e,#0891b2);
  color:white;
  font-size:16px;
  font-weight:600;
  cursor:pointer;
  transition:0.3s;
  margin-top:8px;
}

.register-btn:hover{
```

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```
transform:translateY(-2px);
box-shadow:0 10px 20px rgba(15,118,110,0.25);
}

.bottom-text{
text-align:center;
margin-top:22px;
font-size:14px;
color:#64748b;
}

.bottom-text a{
color:#0f766e;
text-decoration:none;
font-weight:600;
}

@media(max-width:900px){

.main-box{
grid-template-columns:1fr;
}

.left-side{
display:none;
}

.right-side{
padding:30px 22px;
}

.grid-2{
grid-template-columns:1fr;
}
}

</style>
</head>
<body>

<div class="main-box">

<div class="left-side">

<div class="logo"><img alt="MediGen-AI logo" data-bbox="343 674 363 688"/></div>

<h1>MediGen-AI<br>Online OPD Portal</h1>

<p>
Register online from home, get AI-based department suggestions,
book appointments instantly and manage your digital health records securely.
</p>

<div class="feature-box"><img alt="AI Suggested Department & Doctor icon" data-bbox="395 795 415 810"/> AI Suggested Department & Doctor</div>
```

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```
<div class="feature-box"><img alt="Digital OPD Slip & Prescription icon" data-bbox="395 88 412 105"/> Digital OPD Slip & Prescription</div>

<div class="feature-box"><img alt="ABHA & Ayushman Bharat Support icon" data-bbox="395 118 412 135"/> ABHA & Ayushman Bharat Support</div>

<div class="feature-box"><img alt="Jan Aushadhi & Lab Test Services icon" data-bbox="395 148 412 165"/> Jan Aushadhi & Lab Test Services</div>

</div>

<div class="right-side">

  <div class="form-title">Online Patient Registration</div>

  <div class="sub-title">
    Fill your medical details to create your online patient account.
  </div>

  {% with messages = get_flashed_messages(with_categories=true) %}
  {% if messages %}
  {% for category, message in messages %}
  <div style="
padding:10px;
margin-bottom:15px;
border-radius:5px;
background-color:#d4edda;
color:#155724;
font-weight:bold;
">
    {{ message }}
  </div>
  {% endfor %}
  {% endif %}
  {% endwith %}
  <form method="POST">

    <div class="input-group">
      <label>Full Name</label>
      <input type="text" name="name" placeholder="Enter Full Name" required>
    </div>

    <div class="grid-2">

      <div class="input-group">
        <label>Gender</label>
        <select name="gender" required>
          <option value="">Select Gender</option>
          <option>Male</option>
          <option>Female</option>
          <option>Other</option>
        </select>
      </div>

      <div class="input-group">
        <label>Age</label>
        <input type="number" name="age" placeholder="Enter Age">
      </div>

    </div>

  </form>

</div>
```

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```
</div>

<div class="grid-2">

  <div class="input-group">
    <label>Weight</label>
    <input type="number" step="0.1" name="weight" placeholder="Weight in KG">
  </div>

  <div class="input-group">
    <label>Mobile Number</label>
    <input type="text" name="mobile" placeholder="Enter Mobile Number"
      maxlength="10" minlength="10" pattern="[0-9]{10}" input="numeric" required>
  </div>

</div>

<div class="input-group">
  <label>Email Address</label>
  <input type="email" name="email" placeholder="Enter Email Address">
</div>

<div class="input-group">
  <label>ABHA Address</label>
  <input type="text" name="abha_address" placeholder="Enter ABHA Address">
</div>
<div class="input-group">
  <label>Address</label>
  <input type="text" name="address" placeholder="Enter Address">
</div>
<div class="input-group">
<label>Symptoms</label>

<div style="display:flex; gap:10px;">

  <textarea
    id="symptoms"
    name="symptoms"
    placeholder="Enter symptoms..."
    style="min-height:70px;"
  ></textarea>

  <button
    type="button"
    onclick="suggestDoctor()"
    style="
      border:none;
      background:#0891b2;
      color:white;
      padding:12px 16px;
      border-radius:12px;
      cursor:pointer;
      font-weight:600;
    "
  >

</div>
```

Project Title:- MediGen-AI

```
        AI Suggest
      </button>

    </div>
  </div>

  <div class="grid-2">

    <div class="input-group">
      <label>Suggested Department</label>

      <input
        type="text"
        id="department"
        readonly
        placeholder="AI Suggested Department"
      >
    </div>

    <div class="input-group">
      <label>Doctor Type</label>

      <input
        type="text"
        id="doctor_type"
        readonly
        placeholder="Suggested Doctor"
      >
    </div>

  </div>

  <div class="grid-2">

    <div class="input-group">
      <label>Create Login ID</label>

      <input
        type="text"
        name="login_id"
        placeholder="Create Login ID"
        maxlength="10"
        required
      >
    </div>

    <div class="input-group">
      <label>Create Password</label>

      <input
        type="password"
        name="password"
        placeholder="Create Password"
        maxlength="10"
        required
      >
    </div>

  </div>
</div>
```

Project Title:- MediGen-AI

```
>
</div>

</div>

<button type="submit" class="register-btn">
   Register Online Patient
</button>
</form>

  <div class="bottom-text">
    Already have an account?
    <a href="{{ url_for('online_login') }}">
      Login Here
    </a>
    <p>
      <a href="/online-login">
         Back To Home
      </a>
    </p>
  </div>

</div>

</div>
<script>

function suggestDoctor(){

  let symptoms =
  document.getElementById('symptoms').value.toLowerCase();

  let department = "General Medicine";
  let doctor = "MBBS Doctor";

  if(symptoms.includes('heart') || symptoms.includes('chest')){
    department = "Cardiology";
    doctor = "Cardiologist";
  }

  else if(symptoms.includes('skin') || symptoms.includes('allergy')){
    department = "Dermatology";
    doctor = "Skin Specialist";
  }

  else if(symptoms.includes('eye')){
    department = "Ophthalmology";
    doctor = "Eye Specialist";
  }

  else if(symptoms.includes('bone') || symptoms.includes('joint')){
    department = "Orthopedic";
    doctor = "Orthopedic Doctor";
  }
}
```

Project Title:- MediGen-AI

```
    else if(symptoms.includes('child') || symptoms.includes('baby')){
        department = "Pediatrics";
        doctor = "Child Specialist";
    }

    document.getElementById('department').value = department;

    document.getElementById('doctor_type').value = doctor;
}

</script>
</body>
</html>
```

Project Title:- MediGen-AI

online / dashboard.html

```
{% extends 'base.html' %}
```

```
{% block title %}Patient Dashboard | MediGen-AI{% endblock %}
```

```
{% block page_title %}
```

```
<div style="
display:flex;
justify-content:space-between;
align-items:center;
width:100%;
flex-wrap:wrap;
gap:10px;
">
```

```
<div style="
font-size:28px;
font-weight:780;
background:linear-gradient(90deg,#11f1df,#06b6d4);
-webkit-background-clip:text;
-webkit-text-fill-color:transparent;
">
```

```
 My Health Dashboard
</div>&nbsp;
```

```
<div style="
display:flex;
align-items:center;
gap:10px;
">
```

```
<a href="https://www.india.gov.in/directory/helpline/"
target="_blank"
style="
background:linear-gradient(135deg,#ef4444,#dc2626);
padding:9px 16px;
border-radius:12px;
color:white;
text-decoration:none;
font-size:13px;
font-weight:700;
box-shadow:0 4px 15px rgba(239,68,68,0.35);
">
```

```
 Emergency
</a>
```

```
<a href="https://www.google.com/maps/search/nearby+hospitals/"
target="_blank"
style="
background:linear-gradient(135deg,#0f766e,#14b8a6);
padding:9px 16px;
border-radius:12px;
color:white;
```

Project Title:- MediGen-AI

```
text-decoration:none;
font-size:13px;
font-weight:700;
box-shadow:0 4px 15px rgba(20,184,166,0.35);
">
 Nearby Hospital
</a>

</div>

</div>

{% endblock %}

{% block sidebar_nav %}

<a href="/patient" class="nav-item modernSideBtn">
 Dashboard
</a>

<br>

<a href="#" onclick="
window.open('https://abha.abdm.gov.in/');
window.open('https://www.cowin.gov.in/', '_blank');
return false;"
class="nav-item modernSideBtn">
☐ Create ABHA
</a>

<br>

<a href="#" onclick="
window.open('https://pmjay.gov.in/');
window.open('https://hem.nha.gov.in/search', '_blank');
window.open('https://beneficiary.nha.gov.in/', '_blank');
return false;"
class="nav-item modernSideBtn">
 Ayushman Card
</a>

<br>

<a href="#" onclick="
window.open('https://janaushadhi.gov.in/near-by-kendra');
window.open('https://janaushadhi.gov.in/productportfolio/ProductmrpList');
return false;"
class="nav-item modernSideBtn">
 Jan Aushadhi
</a>

<br>
```

Project Title:- MediGen-AI

```
<a href="#" onclick="
window.open('https://esanjeevani.mohfw.gov.in/');
return false;"
class="nav-item modernSideBtn">
 eSanjeevani
</a>
<br>
<a href="#" onclick="
window.open('https://pmjay.gov.in/', '_blank');
window.open('https://irdai.gov.in/', '_blank');
window.open('https://policybazar.com/', '_blank');
window.open('https://www.jansuraksha.gov.in/', '_blank');
return false;"
class="nav-item modernSideBtn">
 Insurance
</a>
<style>
.modernSideBtn{
background:linear-gradient(135deg,#0f766e,#14b8a6);
color:white !important;
border-radius:14px;
padding:12px 14px;
font-weight:700;
display:block;
margin-bottom:10px;
box-shadow:0 6px 18px rgba(20,184,166,0.22);
transition:0.3s;
}

.modernSideBtn:hover{
transform:translateY(-2px) scale(1.02);
}
</style>
{% endblock %}
{% block content %}
{% if patient %}
<div style="
background:
linear-gradient(rgba(255,255,255,0.88), rgba(255,255,255,0.92)),
url('https://images.unsplash.com/photo-1587351021759-
3e566b6af7cc?q=80&w=1600&auto=format&fit=crop');
background-size:cover;
background-position:center;
border-radius:28px;
padding:28px;
border:1px solid rgba(255,255,255,0.5);

box-shadow:
0 10px 40px rgba(15,118,110,0.15),
0 4px 15px rgba(0,0,0,0.05);
backdrop-filter:blur(12px);

">
<!-- TOP PROFILE -->
<div style="
```

Project Title:- MediGen-AI

```
display:flex;
justify-content:space-between;
align-items:center;
flex-wrap:wrap;
gap:15px;
margin-bottom:25px;
">
<div>
<div style="
font-size:30px;
font-weight:800;
color:#0f172a;
margin-bottom:6px;
">
👤 {{ patient.name }}
</div>
<div style="
font-size:14px;
color:#475569;
font-weight:500;
">
Smart Digital Patient Profile
</div>
</div>

<div style="
background:linear-gradient(135deg,#0f766e,#14b8a6);
padding:14px 22px;
border-radius:18px;
color:white;
font-weight:800;
font-size:18px;
box-shadow:0 8px 20px rgba(20,184,166,0.35);
">
🆔 {{ patient.opd_number if patient.opd_number else 'OPD-' ~ patient.id }}
</div>
</div>
<!-- PATIENT INFO -->
<div style="
display:grid;
grid-template-columns:repeat(auto-fit,minmax(190px,1fr));
gap:16px;
">
<div class="modernCard">
<label>Login ID</label>
<span>{{ patient.mobile }}</span>
</div>
<div class="modernCard">
<label>Patient ID</label>
<span>{{ patient.patient_id }}</span>
</div>

<div class="modernCard">
<label>Gender</label>
<span>{{ patient.gender or 'N/A' }}</span>
```

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```
</div>

<div class="modernCard">
<label>Age</label>
<span>{{ patient.age }} Years</span>
</div>
<div class="modernCard">
<label>Weight</label>
<span>{{ patient.weight }} KG</span>
</div>
<div class="modernCard">
<label>Mobile</label>
<span>{{ patient.mobile }}</span>
</div>
<div class="modernCard">
<label> Email Address</label>
<span>{{ patient.email or 'N/A' }}</span>
</div>
<div class="modernCard">
<label>ABHA Address</label>
<span>{{ patient.abha_address or 'N/A' }}</span>
</div>

<div class="modernCard">
<label> Address</label>
<span>{{ patient.address or 'N/A' }}</span>
</div>

<div class="modernCard">
<label>Symptoms</label>
<span>{{ patient.symptoms or 'N/A' }}</span>
</div>
<div class="modernCard">
<label>Department</label>
<span>{{ patient.department or 'General OPD' }}</span>
</div>
<div class="modernCard">
<label>Doctor Type</label>
<span>{{ patient.doctor_type or 'General Physician' }}</span>
</div>
<div class="modernCard">
<label>Registration Date</label>
<span>
{{ patient.created_at.strftime('%d %b %Y %I:%M %p') if patient.created_at else 'N/A' }}
</span>
</div>
</div>
<!-- BUTTONS -->

<div style="
margin-top:28px;
display:flex;
gap:14px;
flex-wrap:wrap;
">
```

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```
<button onclick="window.print()" class="print-btn no-print"
style="
background:linear-gradient(135deg,#2563eb,#06b6d4);
padding:14px 24px;
border-radius:14px;
color:white;
text-decoration:none;
font-weight:800;
font-size:14px;
box-shadow:0 6px 18px rgba(37,99,235,0.3);
">
 Print OPD Slip
</button>
</a>
</div>
</div>
<style>
.modernCard{
background:rgba(255,255,255,0.72);
border:1px solid rgba(255,255,255,0.6);
padding:15px;
border-radius:18px;
box-shadow:0 4px 12px rgba(0,0,0,0.04);
backdrop-filter:blur(10px);
transition:0.3s;
}
.modernCard:hover{
transform:translateY(-3px);
box-shadow:0 10px 20px rgba(15,118,110,0.12);
}
.modernCard label{
display:block;
font-size:11px;
font-weight:800;
letter-spacing:1px;
text-transform:uppercase;
color:#0f766e;
margin-bottom:7px;
}
.modernCard span{
font-size:15px;
font-weight:700;
color:#0f172a;
word-break:break-word;
}
.print-btn{
background:linear-gradient(135deg,#14b8a6,#0f766e);
color:white;
border:none;
padding:10px 18px;
border-radius:12px;
font-size:14px;
font-weight:700;
cursor:pointer;
}
```

Project Title:- MediGen-AI

```
}
.print-btn{
background:linear-gradient(135deg,#14b8a6,#0f766e);
color:white;
border:none;
padding:10px 18px;
border-radius:12px;
font-size:14px;
font-weight:700;
cursor:pointer;
font-weight:700;
}
/* PRINT MODE */
@media print{
body{
background:#f4fffd !important;
color:#111827 !important;
font-family:'Segoe UI',sans-serif;
}
}
/* HIDE BUTTONS */
.no-print,
.print-btn{
display:none !important;
}
/* TOPBAR */
.topbar,
header,
nav{
background:#e6fffb !important;
color:#065f46 !important;
border-bottom:1px solid #b6f0e5 !important;
box-shadow:none !important;
}
/* SIDEBAR */
.sidebar{
background:#ecfffb !important;
border-right:1px solid #b6f0e5 !important;
}
/* CARDS */
.card,
.profile-card,
.info-card,
.dashboard-card,
.modernCard{
background:white !important;
border:1px solid #d1fae5 !important;
box-shadow:none !important;
}
/* LINKS CLICKABLE */
a{
color:#0f766e !important;
text-decoration:underline !important;
pointer-events:auto !important;
cursor:pointer !important;
}
```

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```
}
/* hyperlink preserve in PDF */

a[href]:after{

content:"";
}
/* headings */

h1,h2,h3,h4{

color:#065f46 !important;

}
/* PROFESSIONAL PRINT */

*{
-webkit-print-color-adjust:exact;

print-color-adjust:exact;

}

}
</style>
{% if prescriptions %}
<div style="
margin-top:30px;
background:white;
border-radius:24px;
padding:24px;
box-shadow:0 8px 30px rgba(15,118,110,.08);
border:1px solid rgba(15,118,110,.08);
">
<div style="
font-size:24px;
font-weight:800;
color:#0f172a;
margin-bottom:20px;
">
📋 My Prescriptions
</div>
{% for rx in prescriptions %}
<div style="
border:1px solid rgba(15,118,110,.12);
border-radius:20px;
padding:20px;
margin-bottom:20px;
background:linear-gradient(135deg,#f8fffe,#f0fdfa);
">
<div style="
display:flex;
justify-content:space-between;
align-items:center;
flex-wrap:wrap;
```

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```
gap:10px;
margin-bottom:18px;
">
<div>
<div style="
font-size:18px;
font-weight:800;
color:#0f172a;
">
📄 {{ rx.prescription_id }}
</div>
<div style="
font-size:13px;
color:#64748b;
margin-top:4px;
">
{{ rx.created_at.strftime('%d %b %Y %I:%M %p') }}
</div>

</div>
<div style="
background:#dcfce7;
color:#166534;
padding:8px 14px;
border-radius:12px;
font-size:13px;
font-weight:700;
">
✅ Finalized
</div>
<a href="/online/prescription/{{ rx.prescription_id }}"
class="print-btn">

👁 View Prescription

</a>
</div>
<div style="
display:grid;
grid-template-columns:repeat(auto-fit,minmax(240px,1fr));
gap:16px;
">
<div class="modernCard">
<label>Doctor</label>
<span>
{{ rx.doctor_name }}
</span>
</div>
<div class="modernCard">
<label>Qualification</label>
<span>
{{ rx.doctor_qualification or 'N/A' }}
</span>
</div>
<div class="modernCard">
```

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```
<label>Symptoms</label>
<span>
{{ rx.symptoms or 'N/A' }}
</span>
</div>
<div class="modernCard">
<label>Diagnosis</label>
<span>
{{ rx.diagnosis or 'N/A' }}
</span>
</div>
<div class="modernCard">
<label>Lab Reports</label>
<span>
{{ rx.lab_reports or 'N/A' }}
</span>
</div>
<div class="modernCard">
<label>Doctor Notes</label>
<span>
{{ rx.notes or 'N/A' }}
</span>
</div>

</div>
{% endfor %}
{% endif %}
{% endif %}
{% endblock %}
```

Project Title:- MediGen-AI

Online / forget_password.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>MediGen-AI | Secure Recovery</title>
<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
<link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"
rel="stylesheet">
<style>
*{
margin:0;
padding:0;
box-sizing:border-box;
}
body{
font-family:'Poppins',sans-serif;
height:100vh;
display:flex;
justify-content:center;
align-items:center;
overflow:hidden;
background:
linear-gradient(rgba(7,12,25,0.82), rgba(7,12,25,0.82)),
url('https://images.unsplash.com/photo-1580281657527-47f249e8f4df?q=80&w=1600&auto=format&fit=crop');
background-size:cover;
background-position:center;
background-repeat:no-repeat;

position:relative;
}
/* Floating Glow */
body::before{
content:'';
position:absolute;
width:500px;
height:500px;
background:rgba(20,184,166,0.18);
border-radius:50%;
top:-120px;
left:-120px;
filter:blur(100px);
}
body::after{
content:'';
position:absolute;
width:450px;
height:450px;
background:rgba(6,182,212,0.16);
border-radius:50%;
bottom:-120px;
```

Project Title:- MediGen-AI

```
right:-120px;
filter:blur(100px);
}
/* Card */
.card{
position:relative;
z-index:2;
width:100%;
max-width:430px;
padding:42px 34px;
border-radius:28px;
background:rgba(255,255,255,0.12);
border:1px solid rgba(255,255,255,0.15);
backdrop-filter:blur(18px);
box-shadow:
0 10px 45px rgba(0,0,0,0.35),
0 0 60px rgba(20,184,166,0.08);
animation:popup .45s ease;
color:white;
}
@keyframes popup{
from{
opacity:0;
transform:translateY(25px) scale(.96);
}
to{
opacity:1;
transform:translateY(0) scale(1);
}
}
/* Logo */
.logo{
width:85px;
height:85px;
margin:auto;
border-radius:24px;
display:flex;
justify-content:center;
align-items:center;
background:linear-gradient(135deg,#14b8a6,#06b6d4);
box-shadow:0 0 35px rgba(20,184,166,.45);
margin-bottom:22px;
}
.logo i{
font-size:38px;
color:white;
}
h1{
text-align:center;
font-size:30px;
font-weight:700;
margin-bottom:10px;
}
.sub{
text-align:center;
```

Project Title:- MediGen-AI

```
font-size:14px;
line-height:1.7;
color:#dbeafe;
margin-bottom:28px;
}
/* Inputs */
.field{
margin-bottom:18px;
position:relative;
}
.field i{
position:absolute;
left:16px;
top:16px;
color:#cbd5e1;
font-size:15px;
}
input{
width:100%;
padding:15px 15px 15px 48px;
border:none;
border-radius:14px;
background:rgba(255,255,255,0.12);
border:1px solid rgba(255,255,255,0.14);
color:white;
font-size:15px;
outline:none;
transition:.3s;
}
input:focus{
border-color:#14b8a6;
background:rgba(255,255,255,0.18);
box-shadow:0 0 18px rgba(20,184,166,.2);
}
input::placeholder{
color:#dbeafe;
}
/* Button */
button{
width:100%;
padding:15px;
border:none;
border-radius:14px;
background:linear-gradient(135deg,#14b8a6,#06b6d4);
color:white;
font-size:16px;
font-weight:600;
cursor:pointer;
transition:.3s;
}
button:hover{
transform:translateY(-2px);
box-shadow:0 8px 25px rgba(20,184,166,.4);
}
/* Message */
```

Project Title:- MediGen-AI

```
.msg{
padding:13px;
border-radius:12px;
margin-bottom:18px;
text-align:center;
font-size:14px;
background:rgba(255,255,255,0.12);
border:1px solid rgba(255,255,255,0.1);
}
/* Back */
.back{
text-align:center;
margin-top:22px;
}
.back a{
color:#5eead4;
text-decoration:none;
font-size:14px;
font-weight:500;
}
.back a:hover{
text-decoration:underline;
}
</style>
</head>
<body>
<div class="card">
<div class="logo">
<i class="fas fa-user-shield"></i>
</div>
<h1>Password Recovery</h1>
<div class="sub">
Securely recover your MediGen-AI patient account using OTP verification.
</div>
{% with messages = get_flashed_messages(with_categories=true) %}
{% if messages %}
{% for category, message in messages %}
<div class="msg">
{{ message }}
</div>
{% endfor %}
{% endif %}
{% endwith %}
<!-- STEP 1 -->
{% if step == 'find' %}
<form method="POST">
<input type="hidden" name="step" value="find">
<div class="field">
<i class="fas fa-user"></i>
<input type="text"
name="login_id"
placeholder="Enter Login ID"
required>
</div>
<button type="submit">
```

Project Title:- MediGen-AI

```
<i class="fas fa-paper-plane"></i>
Generate OTP
</button>
</form>
{% endif %}
<!-- STEP 2 -->
{% if step == 'otp' %}
<form method="POST">
<input type="hidden" name="step" value="otp">
<input type="hidden"
name="login_id"
value="{{ login_id }}">
<div class="field">
<i class="fas fa-key"></i>
<input type="text"
name="otp"
placeholder="Enter OTP"
required>
</div>
<button type="submit">
<i class="fas fa-check-circle"></i>
Verify OTP
</button>
</form>
{% endif %}
<!-- STEP 3 -->
{% if step == 'reset' %}
<form method="POST">
<input type="hidden" name="step" value="reset">
<input type="hidden"
name="login_id"
value="{{ login_id }}">
<div class="field">
<i class="fas fa-lock"></i>
<input type="password"
name="new_password"
placeholder="Create New Password"
required>
</div>
<button type="submit">
<i class="fas fa-sync-alt"></i>
Change Password
</button>
</form>
{% endif %}
<div class="back">
<a href="/online-login">
<i class="fas fa-arrow-left"></i>
Back to Login
</a>
</div>
</div>
</body>
</html>
```

Project Title:- MediGen-AI

Online / opd_slip.html

```
@app.route('/patient/opd-slip/<patient_id>')
def patient_opd_slip(patient_id):

    patient = Patient.query.filter_by(patient_id=patient_id).first()

    if not patient:
        flash('Patient not found')
        return redirect('/patient')

    return render_template(
        'patient_opd_slip.html',
        patient=patient
    )
patient_opd_slip.html
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>OPD Slip</title>
<style>
body{
font-family:Arial,sans-serif;
background:#f1f5f9;
padding:30px;
}
.slip{
max-width:900px;
margin:auto;
background:white;
border-radius:20px;
padding:35px;
box-shadow:0 10px 30px rgba(0,0,0,0.08);
border-top:8px solid #0f766e;
}
.top{
display:flex;
justify-content:space-between;
align-items:center;
margin-bottom:30px;
border-bottom:2px dashed #cbd5e1;
padding-bottom:20px;
}
.hospital{
font-size:32px;
font-weight:800;
color:#0f766e;
}
.sub{
font-size:14px;
color:#64748b;
margin-top:5px;
}
.opd{
```

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```
background:#0f766e;
color:white;
padding:12px 18px;
border-radius:14px;
font-weight:800;
font-size:20px;
}
.grid{
display:grid;
grid-template-columns:repeat(2,1fr);
gap:18px;
margin-top:20px;
}
.card{
background:#f8fafc;
padding:16px;
border-radius:14px;
border:1px solid #e2e8f0;
}
.label{
font-size:12px;
font-weight:800;
color:#0f766e;
margin-bottom:6px;
text-transform:uppercase;
}
.value{
font-size:16px;
font-weight:700;
color:#0f172a;
}
.printBtn{
margin-top:35px;
text-align:center;
}
button{
background:#0f766e;
color:white;
border:none;
padding:14px 30px;
border-radius:14px;
font-size:16px;
font-weight:700;
cursor:pointer;
}

button:hover{
background:#115e59;
}
@media print{
button{
display:none;
}
}
body{
background:white;
```

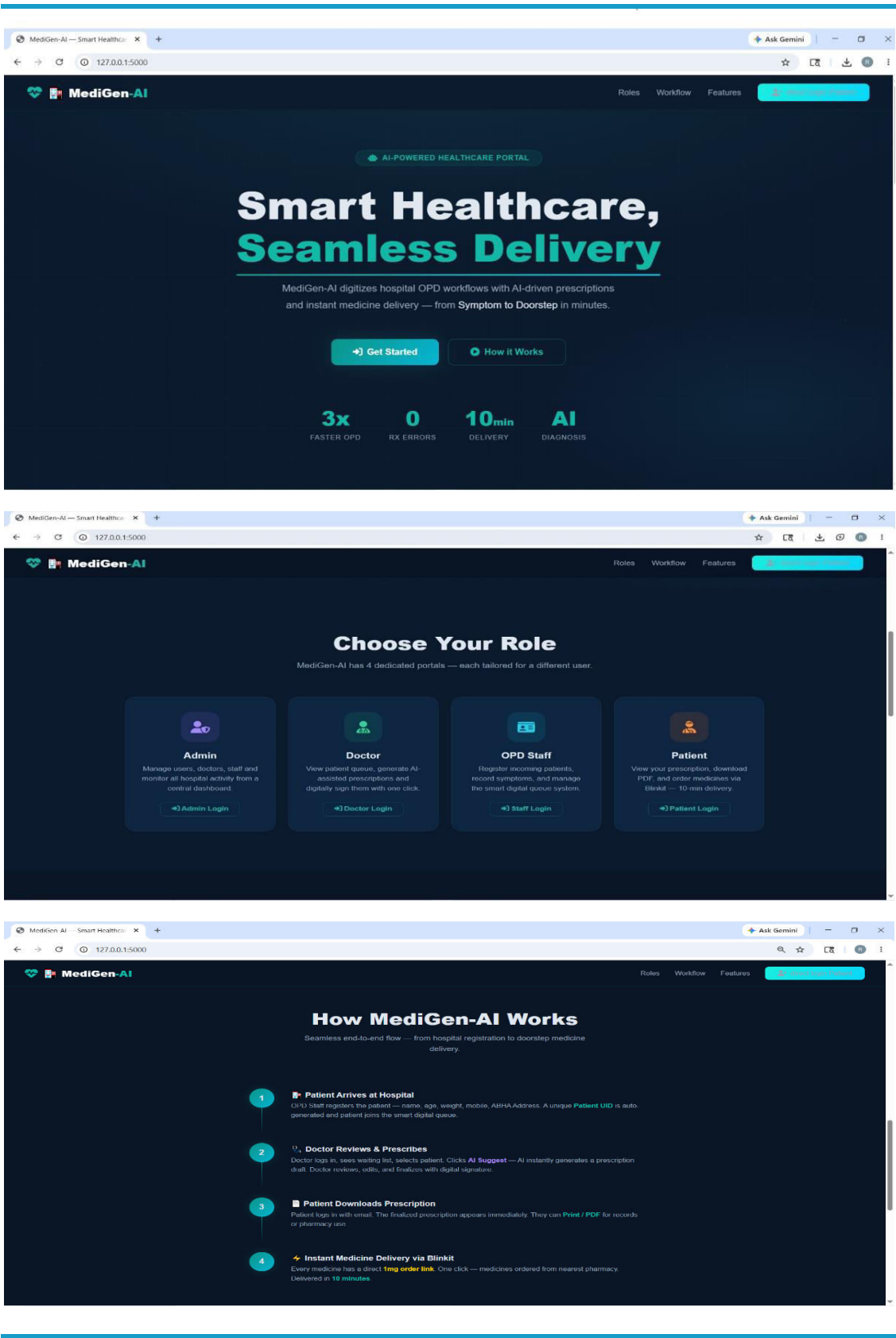
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```
padding:0;
}
.slip{
box-shadow:none;
border:none;
}
}
</style>
</head>
<body>
<div class="slip">
<div class="top">
<div>
<div class="hospital">
 MediGen-AI
</div>
<div class="sub">
Smart Digital Healthcare OPD Slip
</div>
</div>
<div class="opd">
OPD-{{ patient.id }}
</div>
</div>
<div class="grid">
<div class="card">
<div class="label">Patient Name</div>
<div class="value">{{ patient.name }}</div>
</div>
<div class="card">
<div class="label">Patient ID</div>
<div class="value">{{ patient.patient_id }}</div>
</div>
<div class="card">
<div class="label">Login ID</div>
<div class="value">{{ patient.mobile }}</div>
</div>
<div class="card">
<div class="label">Gender</div>
<div class="value">{{ patient.gender or 'N/A' }}</div>
</div>
<div class="card">
<div class="label">Age</div>
<div class="value">{{ patient.age }} Years</div>
</div>
<div class="card">
<div class="label">Weight</div>
<div class="value">{{ patient.weight }} KG</div>
</div>
<div class="card">
<div class="label">Mobile</div>
<div class="value">{{ patient.mobile }}</div>
</div>
<div class="card">
<div class="label">ABHA Address</div>
```

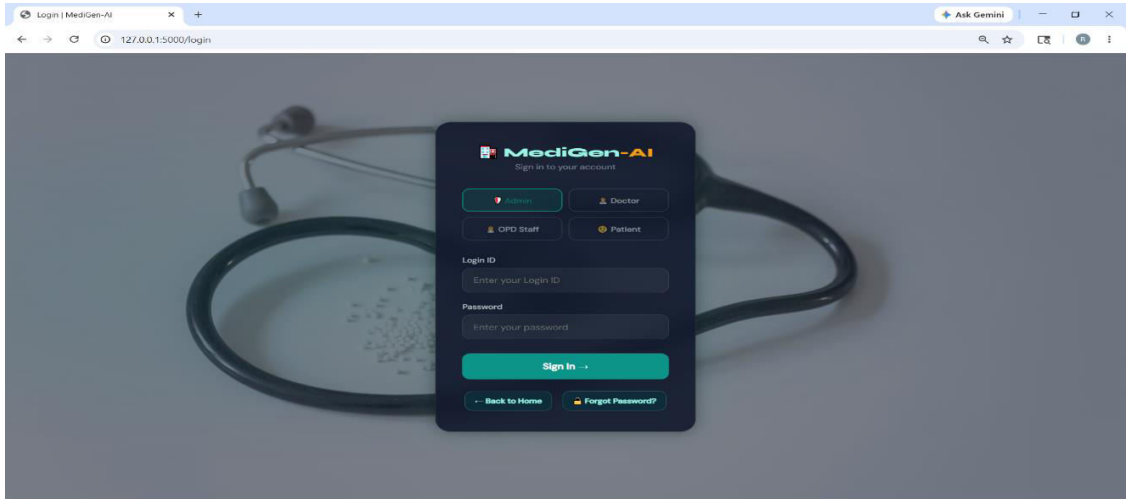
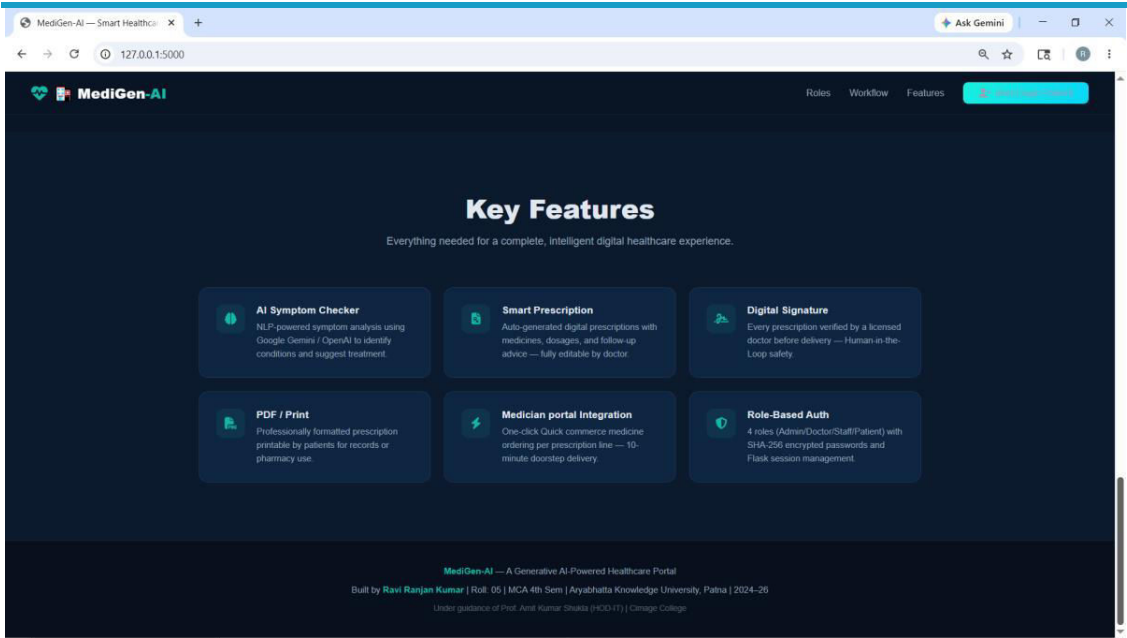
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```
<div class="value">{{ patient.abha_address or 'N/A' }}</div>
</div>
<div class="card">
<div class="label">Department</div>
<div class="value">{{ patient.department or 'General OPD' }}</div>
</div>
<div class="card">
<div class="label">Doctor Type</div>
<div class="value">{{ patient.doctor_type or 'General Physician' }}</div>
</div>
<div class="card">
<div class="label">Symptoms</div>
<div class="value">{{ patient.symptoms or 'N/A' }}</div>
</div>
<div class="card">
<div class="label">Address</div>
<div class="value">{{ patient.address or 'N/A' }}</div>
</div>
<div class="card">
<div class="label">Date & Time</div>
<div class="value">
{{ patient.created_at.strftime('%d %B %Y %I:%M %p') if patient.created_at else 'N/A' }}
</div>
</div>
</div>
<div class="printBtn">
<button onclick="window.print()">
 Print OPD Slip
</button>
</div>
</div>
</body>
</html>
```

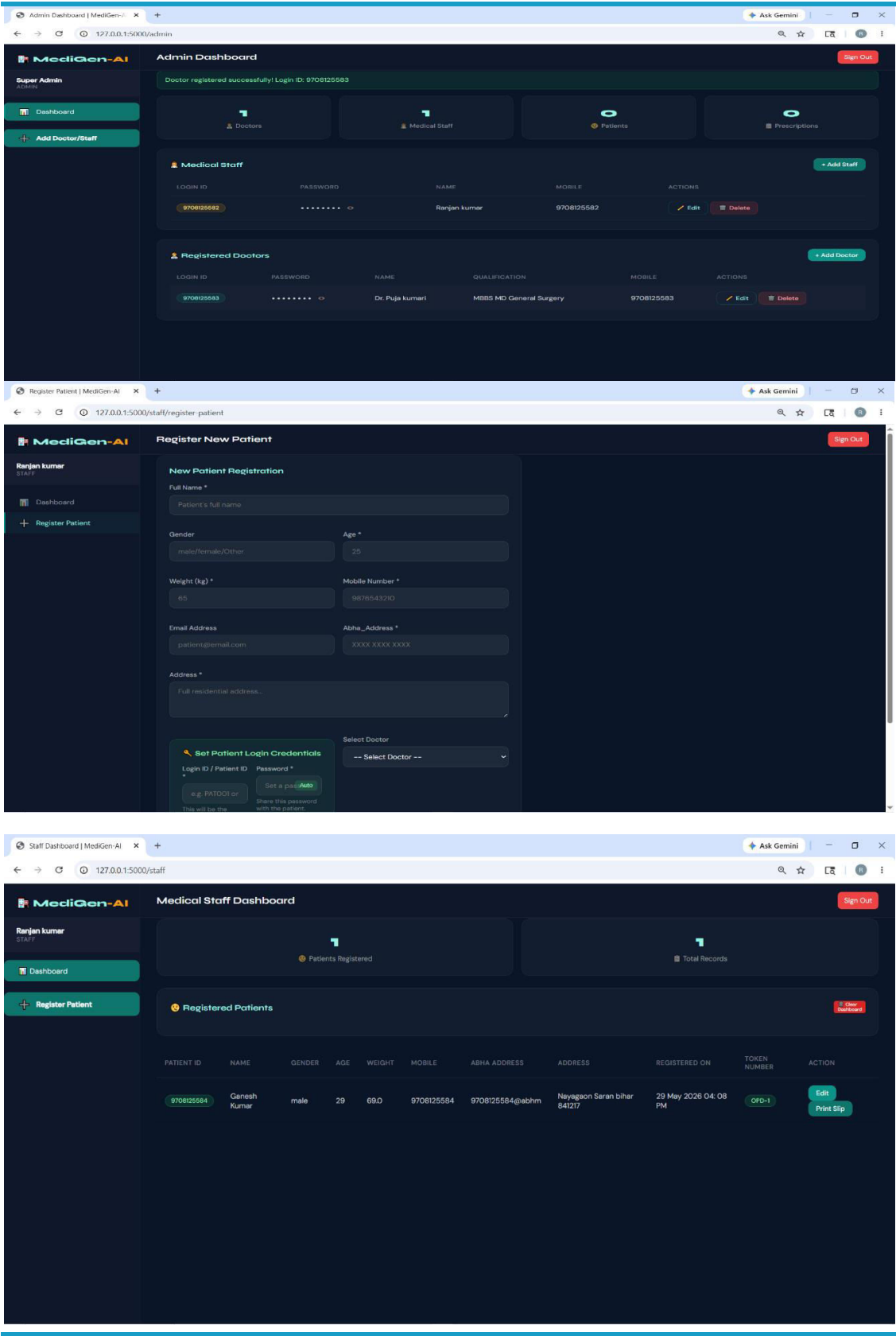
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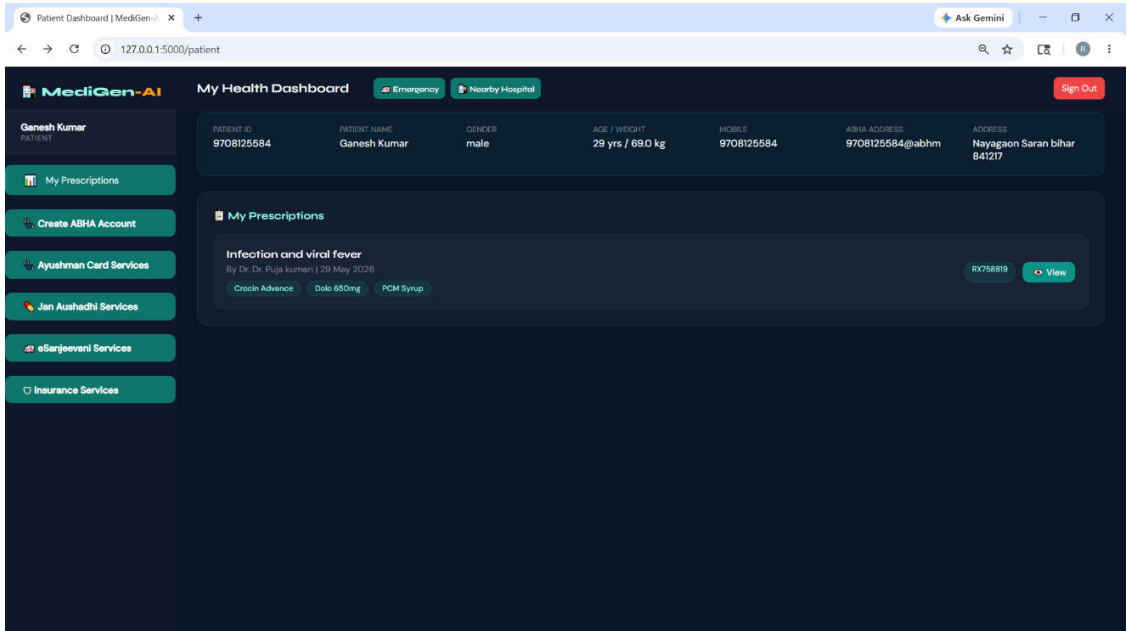
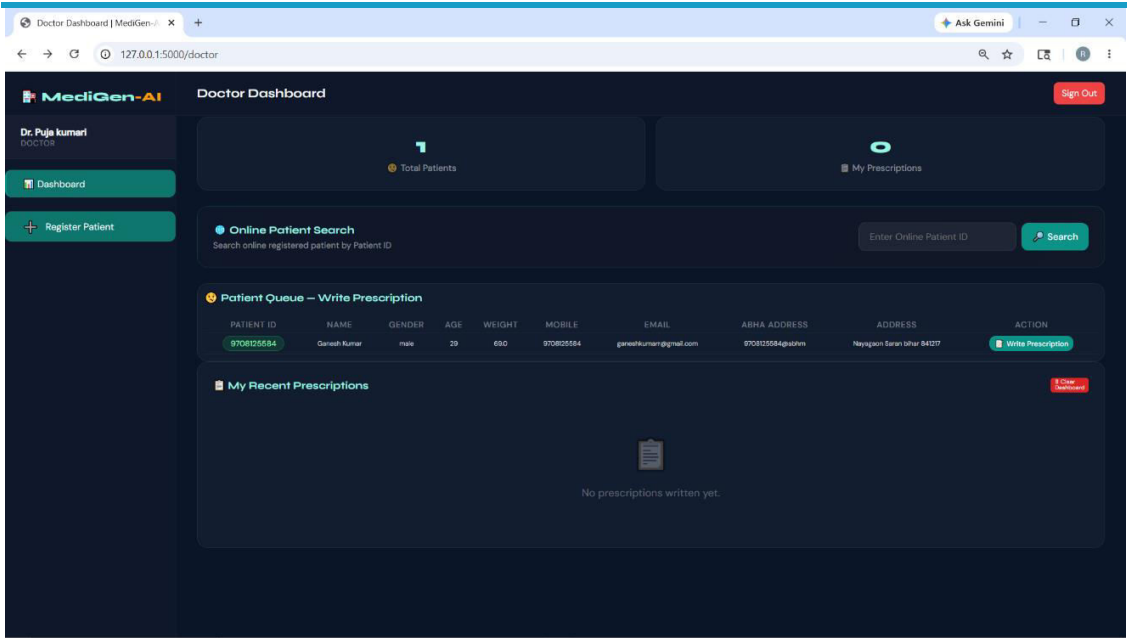
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Online Patient Registration

127.0.0.1:5000/online-register

MediGen-AI Online OPD Portal

Register online from home, get AI-based department suggestions, book appointments instantly and manage your digital health records securely.

- AI Suggested Department & Doctor
- Digital OPD Slip & Prescription
- ABHA & Ayushman Bharat Support
- Jan Aushadhi & Lab Test Services

Online Patient Registration

Fill your medical details to create your online patient account.

Full Name
Enter Full Name

Gender
Select Gender: Raunak Ranjan Kumar raunak563737@gmail.com

Weight
Weight in KG: Enter Mobile Number

Email Address
Enter Email Address

ABHA Address
Enter ABHA Address

Address
Enter Address

Symptoms
Enter symptoms... AI Suggest

MediGen-AI | Online Patient Login

127.0.0.1:5000/online-login

MediGen-AI

Smart Healthcare & Online Patient Portal

Login ID
Enter Login ID

Password
Enter Password

[Login Now](#)

[New Patient Registration](#)

[Forgot Password](#)

[Back to Home](#)

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Patient Dashboard | MediGen-AI

127.0.0.1:5000/online-dashboard

My Health Dashboard

[Emergency](#) [Nearby Hospital](#) [Sign Out](#)

Login Successful

Ramesh Kumar
Smart Digital Patient Profile

[OPD-1](#)

LOGIN ID 9708125585	PATIENT ID ONL591900	GENDER Male	AGE 49 Years	WEIGHT 80.0 KG	MOBILE 9708125585
EMAIL ADDRESS rameshkumarr@gmail.com	ABHA ADDRESS 9708125585@abhm	ADDRESS hajipur gandhichok	SYMPTOMS heart surgery	DEPARTMENT General OPD	DOCTOR TYPE General Physician

REGISTRATION DATE
29 May 2026 04:39 PM

[Print OPD Slip](#)

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